

GRADE 5 COMMON CORE MATH

Question Collection

Main • Backward Branches • Forward 1 (Grade 6) • Forward 2 (Grade 7)
71 question sets • 1930 questions

CONTENTS

Set 1	5.OA.A.1	Evaluate expressions that use parentheses	7
Set 2	5.OA.A.1	Evaluate expressions with brackets and braces	44
Set 3	5.OA.A.1	Place grouping symbols to make an equation true	81
Set 4	5.OA.A.2	Write numerical expressions from words	118
Set 5	5.OA.A.2	Interpret numerical expressions without evaluating them	155
Set 6	5.OA.B.3	Generate two numerical patterns and compare their terms	192
Set 58	5.OA.B.3	Explain why two patterns have the relationship they do	223
Set 7	5.OA.B.3	Form ordered pairs from two patterns and graph them	254
Set 8	5.NBT.A.1	A digit is worth 10 times as much as the same digit one place to its right	291
Set 9	5.NBT.A.1	A digit is worth 1/10 of the same digit one place to its left	328
Set 10	5.NBT.A.2	Multiply and divide whole numbers by powers of 10; write powers of 10 with expon	365
Set 11	5.NBT.A.2	Place the decimal point when multiplying or dividing a decimal by a power of 10	396
Set 59	5.NBT.A.2	Explain patterns in the number of zeros and the placement of the decimal point	427
Set 12	5.NBT.A.3.a	Read and write decimals to thousandths using numerals and words	458
Set 13	5.NBT.A.3.a	Write decimals to thousandths in expanded form	495
Set 14	5.NBT.A.3.b	Compare decimals to thousandths	526
Set 15	5.NBT.A.4	Round decimals to any place	557

CONTENTS (continued)

Set 16	5.NBT.B.5	Multiply multi-digit whole numbers using the standard algorithm	588
Set 17	5.NBT.B.6	Divide with up to four-digit dividends and two-digit divisors	625
Set 18	5.NBT.B.6	Explain division using area models and equations	662
Set 60	5.NBT.B.6	Draw division models and explain the strategy	699
Set 19	5.NBT.B.7	Add decimals to hundredths	730
Set 61	5.NBT.B.7	Subtract decimals to hundredths	767
Set 62	5.NBT.B.7	Use place-value models to add and subtract decimals and explain the method	798
Set 20	5.NBT.B.7	Multiply decimals to hundredths	829
Set 63	5.NBT.B.7	Use models and place value to multiply decimals and explain the method	866
Set 21	5.NBT.B.7	Divide decimals to hundredths	897
Set 64	5.NBT.B.7	Use models and place value to divide decimals and explain the method	934
Set 22	5.NBT.B.7	Solve real-world problems with decimal operations	965
Set 23	5.NF.A.1	Add fractions with unlike denominators	996
Set 24	5.NF.A.1	Subtract fractions with unlike denominators	1033
Set 25	5.NF.A.1	Add mixed numbers with unlike denominators	1070
Set 65	5.NF.A.1	Subtract mixed numbers with unlike denominators	1107
Set 26	5.NF.A.2	Solve word problems by adding and subtracting fractions	1138

CONTENTS (continued)

Set 27	5.NF.A.2	Use benchmark fractions to estimate and to check answers	1175
Set 28	5.NF.B.3	Interpret a fraction as division of the numerator by the denominator	1206
Set 29	5.NF.B.3	Solve sharing problems whose answers are fractions or mixed numbers	1243
Set 30	5.NF.B.4.a	Multiply a fraction by a whole number	1274
Set 31	5.NF.B.4.a	Multiply a fraction by a fraction	1311
Set 66	5.NF.B.4.a	Interpret fraction products with tape diagrams and area models	1342
Set 32	5.NF.B.4.b	Find the area of a rectangle with fractional side lengths	1373
Set 67	5.NF.B.4.b	Tile a rectangle with unit-fraction squares and connect the tiling to the area	1410
Set 33	5.NF.B.5.a	Compare the size of a product to the size of one factor without multiplying	1441
Set 34	5.NF.B.5.b	Explain the effect of multiplying by a fraction greater than, less than, or equal to 1	1472
Set 68	5.NF.B.5.b	Explain equivalent fractions as multiplying by a fraction equal to 1	1503
Set 35	5.NF.B.6	Solve real-world problems by multiplying fractions and mixed numbers	1534
Set 36	5.NF.B.6	Solve real-world problems by multiplying mixed numbers	1571
Set 37	5.NF.B.7.a	Divide a unit fraction by a whole number	1608
Set 69	5.NF.B.7.a	Use models to divide a unit fraction by a whole number and check with multiplicati	1645
Set 38	5.NF.B.7.b	Divide a whole number by a unit fraction	1676
Set 70	5.NF.B.7.b	Use models to divide a whole number by a unit fraction and check with multiplicati	1707

CONTENTS (continued)

Set 39	5.NF.B.7.c	Solve word problems that divide unit fractions and whole numbers	1738
Set 40	5.MD.A.1	Convert from a larger unit to a smaller unit	1769
Set 41	5.MD.A.1	Convert from a smaller unit to a larger unit	1806
Set 42	5.MD.A.1	Solve multistep problems that involve converting units	1843
Set 43	5.MD.B.2	Make a line plot of measurements in fractions of a unit	1874
Set 44	5.MD.B.2	Use operations on fractions to solve problems with line plot data	1911
Set 45	5.MD.C.3.a	Understand a unit cube and a cubic unit of volume	1948
Set 46	5.MD.C.3.b	A solid packed with n unit cubes has a volume of n cubic units	1979
Set 47	5.MD.C.4	Measure volume by counting unit cubes	2010
Set 48	5.MD.C.5.a	Relate packing with unit cubes to the formulas $V = l \times w \times h$ and $V = b \times h$	2041
Set 49	5.MD.C.5.b	Apply $V = l \times w \times h$ and $V = b \times h$ to solve real-world problems	2072
Set 50	5.MD.C.5.b	Find a missing dimension when the volume is known	2103
Set 51	5.MD.C.5.c	Find the volume of a figure made of two rectangular prisms	2134
Set 52	5.G.A.1	Understand the coordinate plane: axes, origin, and coordinates	2165
Set 53	5.G.A.2	Graph points in the first quadrant	2196
Set 54	5.G.A.2	Solve real-world problems by graphing points and interpreting coordinates	2227
Set 55	5.G.B.3	Attributes of a category belong to all of its subcategories	2258

CONTENTS (continued)

Set 56	5.G.B.4	Classify quadrilaterals in a hierarchy based on their properties	2295
Set 57	5.G.B.4	Classify triangles by their sides and angles	2326
Set 71	5.G.B.4	Place triangles in a hierarchy of categories and subcategories	2363

Evaluate expressions that use parentheses

MAIN	Grade 5 main question	5.OA.A.1
BACKWARD 1	Do the work in parentheses first	3.OA.D.8
BACKWARD 2	Multiplication and division facts	3.OA.C.7
BACKWARD 3	Adding and subtracting multi-digit numbers	4.NBT.B.4
FORWARD 1	Evaluate expressions with variables	6.EE.A.2.c
FORWARD 2	Evaluate expressions with negative numbers	7.NS.A.3

SET 1 • 5.OA.A.1

MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$$6 \times (8 + 5)$$

Value: _____

Evaluate.

$$(35 - 11) \div 4$$

Value: _____

Evaluate.

$$50 - (6 \times 7)$$

Value: _____

Evaluate.

$$3 + (24 \div 6) \times 5$$

A. 35

B. 135

C. 27

D. 23

Evaluate.

$$(9 + 3) \times (10 - 4)$$

Value: _____

SET 1 • 5.OA.A.1

BACKWARD 1

Do the work in parentheses first • Grade 3 • Operations & Algebraic Thinking •
3.OA.D.8

True or false?

In $4 \times (2 + 5)$, you add $2 + 5$ first.

A. True

B. False

Evaluate.

$$(6 + 4) \times 2$$

Value: _____

Which step comes first in $18 - (3 \times 4)$?

- A.** Subtract $18 - 4$
- B.** Multiply 3×4
- C.** Subtract $18 - 3$
- D.** Add $3 + 4$

Evaluate.

$$20 - (5 + 5)$$

Value: _____

True or false?

$$(10 - 4) + 3 = 10 - 7$$

A. True

B. False

SET 1 • 5.OA.A.1

BACKWARD 2

Multiplication and division facts • Grade 3 • Operations & Algebraic Thinking •
3.OA.C.7

Multiply.

$$7 \times 8$$

Product: _____

Divide.

$$24 \div 6$$

Quotient: _____

True or false?

$$9 \times 7 = 56$$

A. True

B. False

$$48 \div 8 = ?$$

A. 8

B. 5

C. 7

D. 6

Multiply.

$$3 \times 9$$

Product: _____

SET 1 • 5.OA.A.1

BACKWARD 3

Adding and subtracting multi-digit numbers • Grade 4 • Number & Operations
in Base Ten • 4.NBT.B.4

Add.

$$348 + 275$$

Sum: _____

Subtract.

$$502 - 168$$

Difference: _____

True or false?

$$1,250 - 375 = 985$$

A. True

B. False

$$609 + 194 = ?$$

A. 793

B. 703

C. 813

D. 803

Subtract.

$$1,000 - 432$$

Difference: _____

SET 1 • 5.OA.A.1

FORWARD 1

Evaluate expressions with variables • Grade 6 • Expressions & Equations •
6.EE.A.2.c

Evaluate $4(x + 3)$ when $x = 5$.

Value: _____

Evaluate $20 - 2y$ when $y = 6$.

Value: _____

Evaluate $(a + b) \div 3$ when $a = 10$ and $b = 8$.

- A.** 4
- B.** 9
- C.** 54
- D.** 6

Evaluate $3m + 2$ when $m = 7$.

Value: _____

True or false?

When $n = 4$, the value of $5(n - 1)$ is 19.

A. True

B. False

SET 1 • 5.OA.A.1

FORWARD 2

Evaluate expressions with negative numbers • Grade 7 • The Number System •
7.NS.A.3

Find the value.

$$-3 \times (4 - 9)$$

Value: _____

Find the value.

$$(-12 + 4) \div 2$$

Value: _____

Find the value.

$$6 - (2 - 10)$$

A. -2

B. 2

C. 14

D. 18

Find the value.

$$(-5) \times (-2) + (-7)$$

Value: _____

True or false?

$$(-20) \div (3 + 2) = 4$$

A. True

B. False

Evaluate expressions with brackets and braces

MAIN	Grade 5 main question	5.OA.A.1
BACKWARD 1	Do the work in parentheses first	3.OA.D.8
BACKWARD 2	Multiplication and division facts	3.OA.C.7
BACKWARD 3	Adding and subtracting within 100	2.NBT.B.5
FORWARD 1	Evaluate expressions with whole-number exponents	6.EE.A.1
FORWARD 2	Grouping symbols with negative numbers	7.NS.A.3

SET 2 • 5.OA.A.1

MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$$2 \times [5 + (3 \times 4)]$$

Value: _____

Evaluate.

$$[20 - (2 + 6)] \div 3$$

Value: _____

Evaluate.

$$4 \times \{3 + [10 - (2 \times 4)]\}$$

A. 36

B. 20

C. 44

D. 26

Evaluate.

$$60 \div [3 \times (7 - 5)]$$

Value: _____

Evaluate.

$$\{18 - [2 \times (4 + 1)]\} + 9$$

Value: _____

SET 2 • 5.OA.A.1

BACKWARD 1

Do the work in parentheses first • Grade 3 • Operations & Algebraic Thinking •
3.OA.D.8

Evaluate.

$$(12 - 5) \times 3$$

Value: _____

True or false?

In $(8 + 2) \div 5$, you divide $2 \div 5$ first.

A. True

B. False

Evaluate.

$$4 \times (9 - 6)$$

A. 33

B. 10

C. 30

D. 12

Evaluate.

$$30 \div (2 + 4)$$

Value: _____

True or false?

$$(3 + 5) \times 2 = 16$$

A. True

B. False

SET 2 • 5.OA.A.1

BACKWARD 2

Multiplication and division facts • Grade 3 • Operations & Algebraic Thinking •
3.OA.C.7

Multiply.

$$8 \times 6$$

Product: _____

Divide.

$$56 \div 7$$

Quotient: _____

True or false?

$$36 \div 4 = 8$$

A. True

B. False

$$7 \times 9 = ?$$

A. 63

B. 56

C. 72

D. 49

Divide.

$$81 \div 9$$

Quotient: _____

SET 2 • 5.OA.A.1

BACKWARD 3

Adding and subtracting within 100 • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5

Add.

$$46 + 37$$

Sum: _____

Subtract.

$$72 - 28$$

Difference: _____

True or false?

$$55 + 29 = 74$$

A. True

B. False

$$90 - 47 = ?$$

A. 47

B. 43

C. 53

D. 33

Add.

$$38 + 54$$

Sum: _____

SET 2 • 5.OA.A.1

FORWARD 1

Evaluate expressions with whole-number exponents • Grade 6 • Expressions & Equations • 6.EE.A.1

Evaluate.

$$2 \times (3 + 1)^2$$

Value: _____

Evaluate.

$$[10 - 2^3] \times 5$$

Value: _____

Evaluate.

$$4^2 - (6 + 2)$$

- A.** 8
- B.** 2
- C.** 0
- D.** 24

Evaluate.

$$(5^2 - 1) \div 6$$

Value: _____

True or false?

$$3 \times 2^2 = 36$$

A. True

B. False

SET 2 • 5.OA.A.1

FORWARD 2

Grouping symbols with negative numbers • Grade 7 • The Number System •
7.NS.A.3

Find the value.

$$2 \times [-3 + (8 \div (-4))]$$

Value: _____

Find the value.

$$[(-6) + 15] \div (-3)$$

Value: _____

Find the value.

$$-4 \times [5 - (-1)]$$

A. 16

B. -16

C. -24

D. 24

Find the value.

$$\{10 - [2 \times (-3)]\} \div 4$$

Value: _____

True or false?

$$[-8 - (-2)] \times 3 = -30$$

A. True

B. False

Place grouping symbols to make an equation true

MAIN	Grade 5 main question	5.OA.A.1
BACKWARD 1	Do the work in parentheses first	3.OA.D.8
BACKWARD 2	When regrouping keeps the same value	3.OA.B.5
BACKWARD 3	Multiplication and division facts	3.OA.C.7
FORWARD 1	Use parentheses to write equivalent expressions	6.EE.A.3
FORWARD 2	Expand and factor expressions with rational coefficients	7.EE.A.1

SET 3 • 5.OA.A.1

MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Insert parentheses to make the equation true.

$$4 + 6 \times 3 = 30$$

Equation: _____

Insert parentheses to make the equation true.

$$20 - 8 - 2 = 14$$

Equation: _____

Which placement of parentheses makes the equation true?

$$2 \times 5 + 3 \times 4 = 52$$

A. $2 \times (5 + 3) \times 4 = 52$

B. $(2 \times 5 + 3) \times 4 = 52$

C. $2 \times (5 + 3 \times 4) = 52$

D. $2 \times 5 + (3 \times 4) = 52$

Insert parentheses to make the equation true.

$$36 \div 6 - 2 = 9$$

Equation: _____

Insert brackets and parentheses to make the equation true.

$$2 \times 9 - 3 + 1 = 10$$

Equation: _____

SET 3 • 5.OA.A.1

BACKWARD 1

Do the work in parentheses first • Grade 3 • Operations & Algebraic Thinking •
3.OA.D.8

Evaluate.

$$(4 + 6) \times 3$$

Value: _____

Evaluate.

$$4 + (6 \times 3)$$

Value: _____

True or false?

$$(20 - 8) - 2 = 20 - (8 - 2)$$

A. True

B. False

Evaluate.

$$36 \div (6 - 2)$$

A. 4

B. 34

C. 9

D. 7

Evaluate.

$$(36 \div 6) - 2$$

Value: _____

SET 3 • 5.OA.A.1

BACKWARD 2

When regrouping keeps the same value • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$(2 \times 3) \times 4 = 2 \times (3 \times 4)$$

A. True

B. False

True or false?

$$(12 \div 6) \div 2 = 12 \div (6 \div 2)$$

A. True

B. False

Which equation is true?

A. $(5 + 2) + 3 = 5 + (2 + 3)$

B. $(9 - 4) - 1 = 9 - (4 - 1)$

C. $(16 \div 4) \div 2 = 16 \div (4 \div 2)$

D. $(7 - 3) - 2 = 7 - (3 - 2)$

True or false?

$$4 \times (5 \times 2) = (4 \times 5) \times 2$$

A. True

B. False

Fill in the blank.

$$(6 \times 2) \times 5 = 6 \times (2 \times \underline{\quad})$$

Blank: _____

SET 3 • 5.OA.A.1

BACKWARD 3

Multiplication and division facts • Grade 3 • Operations & Algebraic Thinking •
3.OA.C.7

Multiply.

$$9 \times 4$$

Product: _____

Divide.

$$42 \div 6$$

Quotient: _____

True or false?

$$8 \times 7 = 54$$

A. True

B. False

$$64 \div 8 = ?$$

A. 6

B. 7

C. 8

D. 9

Multiply.

$$6 \times 7$$

Product: _____

SET 3 • 5.OA.A.1

FORWARD 1

Use parentheses to write equivalent expressions • Grade 6 • Expressions & Equations • 6.EE.A.3

Expand.

$$3(x + 5)$$

Expression: _____

Expand.

$$4(2y - 1)$$

Expression: _____

Which expression is equivalent to $6(a + 2)$?

- A.** $6a + 2$
- B.** $a + 12$
- C.** $6a + 12$
- D.** $8a$

Factor.

$$5m + 20$$

Expression: _____

True or false?

$$2(x + 7) = 2x + 7$$

A. True

B. False

SET 3 • 5.OA.A.1

FORWARD 2

Expand and factor expressions with rational coefficients • Grade 7 •
Expressions & Equations • 7.EE.A.1

Expand.

$$-2(3x - 4)$$

Expression: _____

Expand.

$$\frac{1}{2}(6y + 10)$$

Expression: _____

Which expression is equivalent to $-5(n + 2)$?

- A.** $-5n + 2$
- B.** $-5n - 10$
- C.** $-5n + 10$
- D.** $5n - 10$

Factor out 3.

$$3x - 12$$

Expression: _____

True or false?

$$0.5(4a - 8) = 2a - 8$$

A. True

B. False

Write numerical expressions from words

MAIN	Grade 5 main question	5.OA.A.2
BACKWARD 1	Products as equal groups	3.OA.A.1
BACKWARD 2	Quotients as equal shares	3.OA.A.2
BACKWARD 3	Sums and differences	2.OA.A.1
FORWARD 1	Write expressions with variables	6.EE.A.2.a
FORWARD 2	Write and solve equations from words	7.EE.B.4.a

SET 4 • 5.OA.A.2

MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2

Write an expression for "add 9 and 6, then multiply by 4."

Expression: _____

Write an expression for "subtract 5 from 30, then divide by 5."

Expression: _____

Which expression shows "triple the sum of 12 and 8"?

A. $12 + 8 \times 3$

B. $3 \times 12 + 8$

C. $3 \times (12 + 8)$

D. $(3 + 12) \times 8$

Write an expression for "divide 40 by 8, then add 7."

Expression: _____

Write an expression for "the product of 6 and the difference of 15 and 9."

Expression: _____

SET 4 • 5.OA.A.2

BACKWARD 1

Products as equal groups • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.1

True or false?

5×7 can mean 5 groups of 7.

A. True

B. False

Which expression shows 4 groups of 6?

A. 46

B. $6 - 4$

C. $4 + 6$

D. 4×6

Write a multiplication expression for 8 groups of 3.

Expression: _____

True or false?

$$3 + 3 + 3 + 3 = 3 \times 3$$

A. True

B. False

What is the product of 7 and 5?

Product: _____

SET 4 • 5.OA.A.2

BACKWARD 2

Quotients as equal shares • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.2

True or false?

$15 \div 3$ can mean 15 shared equally into 3 groups.

A. True

B. False

Which expression shows 20 apples shared equally among 4 bags?

A. 20×4

B. $4 \div 20$

C. $20 \div 4$

D. $20 - 4$

What is the quotient of 18 and 6?

Quotient: _____

Write a division expression for 32 split into groups of 8.

Expression: _____

True or false?

The quotient of 12 and 2 is 24.

A. True

B. False

SET 4 • 5.OA.A.2

BACKWARD 3

Sums and differences • Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1

What is the sum of 14 and 9?

Sum: _____

What is the difference of 20 and 7?

Difference: _____

True or false?

The difference of 15 and 6 is 21.

A. True

B. False

Which word names the answer to an addition problem?

- A.** Difference
- B.** Quotient
- C.** Product
- D.** Sum

Write an expression for "7 more than 25."

Expression: _____

SET 4 • 5.OA.A.2

FORWARD 1

Write expressions with variables • Grade 6 • Expressions & Equations •
6.EE.A.2.a

Write an expression for "the sum of n and 8, times 3."

Expression: _____

Write an expression for "y divided by 5, then minus 2."

Expression: _____

Which expression means "4 less than the product of 2 and x"?

A. $4 - 2x$

B. $2x - 4$

C. $2(x - 4)$

D. $2 + x - 4$

Write an expression for "twice the difference of m and 6."

Expression: _____

True or false?

"9 more than k " can be written as $k + 9$.

A. True

B. False

SET 4 • 5.OA.A.2

FORWARD 2

Write and solve equations from words • Grade 7 • Expressions & Equations •
7.EE.B.4.a

Three times the sum of a number and 4 is 27.

Write and solve an equation to find the number.

Equation and solution: _____

Half of a number, minus 6, is 10.

Write and solve an equation to find the number.

Equation and solution: _____

Which equation represents "2 less than 5 times a number is 33"?

A. $5n + 2 = 33$

B. $5n - 2 = 33$

C. $2 - 5n = 33$

D. $5(n - 2) = 33$

A number divided by 4, plus 3, is 9.

Write and solve an equation to find the number.

Equation and solution: _____

True or false?

The solution of $2(x + 3) = 18$ is $x = 7$.

A. True

B. False

Interpret numerical expressions without evaluating them

MAIN	Grade 5 main question	5.OA.A.2
BACKWARD 1	Multiplicative comparison	4.OA.A.1
BACKWARD 2	Groups of a sum	3.OA.A.1
BACKWARD 3	Equal shares	3.OA.A.2
FORWARD 1	Identify parts of an expression	6.EE.A.2.b
FORWARD 2	Rewrite expressions to show how quantities are related	7.EE.A.2

SET 5 • 5.OA.A.2

MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2

How does $4 \times (563 + 218)$ compare to $563 + 218$?

- A.** It is 4 more.
- B.** It is 4 times as large.
- C.** It is 4 less.
- D.** It is one-fourth as large.

True or false?

$(7,215 - 1,005) \div 5$ is one-fifth as large as $7,215 - 1,005$.

A. True

B. False

Without calculating, how many times as large is $9 \times (48 + 27)$ as $48 + 27$?

Times as large: _____

Which expression is twice as large as 356×12 ?

A. $(356 + 2) \times 12$

B. $356 \times 12 + 2$

C. 356×14

D. $2 \times (356 \times 12)$

True or false?

$6 \times 25 + 4$ is 6 times as large as $25 + 4$.

A. True

B. False

SET 5 • 5.OA.A.2

BACKWARD 1

Multiplicative comparison • Grade 4 • Operations & Algebraic Thinking •
4.OA.A.1

True or false?

35 is 5 times as many as 7.

A. True

B. False

Fill in the blank.

48 is ____ times as many as 6.

Blank: _____

Which equation shows "36 is 4 times as many as 9"?

A. $9 = 4 \times 36$

B. $36 = 4 + 9$

C. $36 - 9 = 4$

D. $36 = 4 \times 9$

True or false?

30 is 3 times as many as 27.

A. True

B. False

Tom has 6 cards. Ana has 3 times as many cards.
How many cards does Ana have?

Cards: _____

SET 5 • 5.OA.A.2

BACKWARD 2

Groups of a sum • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1

True or false?

$3 \times (4 + 1)$ means 3 groups of $(4 + 1)$.

A. True

B. False

Which expression means 6 groups of $(10 + 2)$?

A. $6 \times (10 + 2)$

B. $(6 + 10) \times 2$

C. $6 \times 10 + 2$

D. $6 + (10 + 2)$

How many groups of $(5 + 4)$ are in $7 \times (5 + 4)$?

Groups: _____

True or false?

$$2 \times (8 + 1) = (8 + 1) + 2$$

A. True

B. False

Write 4 groups of 9 as a multiplication expression.

Expression: _____

SET 5 • 5.OA.A.2

BACKWARD 3

Equal shares • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2

True or false?

12 shared equally among 3 people is 4 each.

A. True

B. False

A 20-inch ribbon is cut into 5 equal pieces.
How long is each piece?

Length: _____

Which expression finds one of 4 equal shares of 28?

A. $28 - 4$

B. $28 \div 4$

C. 28×4

D. $4 \div 28$

True or false?

Dividing a number by 2 splits it into 2 equal shares.

A. True

B. False

30 crayons are shared equally among 6 children.
How many crayons does each child get?

Crayons: _____

SET 5 • 5.OA.A.2

FORWARD 1

Identify parts of an expression • Grade 6 • Expressions & Equations •
6.EE.A.2.b

In $5(x + 2)$, what is 5?

- A.** A term of $x + 2$
- B.** A quotient
- C.** A sum
- D.** A factor

How many terms are in $4a + 7b - 1$?

Terms: _____

What is the coefficient of m in $9m + 4$?

Coefficient: _____

True or false?

In $3(y + 6)$, the quantity $(y + 6)$ is a sum of two terms.

A. True

B. False

What is the expression $12 \div (n + 1)$ called?

- A.** A coefficient
- B.** A quotient
- C.** A sum
- D.** A product

SET 5 • 5.OA.A.2

FORWARD 2

Rewrite expressions to show how quantities are related • Grade 7 •
Expressions & Equations • 7.EE.A.2

A price p is increased by 5%.

Which expression shows the new price?

A. $0.05p$

B. $1.05p$

C. $5p$

D. $p + 5$

True or false?

$$x + 0.3x = 0.4x$$

A. True

B. False

Write $n - 0.4n$ as one term.

Expression: _____

Which expression is equivalent to $2(3t)$?

- A.** $5t$
- B.** $2t + 3t$
- C.** $6t$
- D.** $3t + 2$

A sale price is $0.75p$, where p is the original price.
What percent of the original price is taken off?

Percent: _____

Generate two numerical patterns and compare their terms

MAIN	Grade 5 main question	5.OA.B.3
BACKWARD 1	Patterns that follow a rule	4.OA.C.5
BACKWARD 2	Multiplication facts	3.OA.C.7
FORWARD 1	Equations relating dependent and independent variables	6.EE.C.9
FORWARD 2	Equations for proportional relationships	7.RP.A.2.c

SET 6 • 5.OA.B.3

MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3

Pattern A starts at 0 and adds 6.

Pattern B starts at 0 and adds 18.

Write the first 5 terms of each pattern.

Pattern A: _____

Pattern B: _____

Pattern A: 0, 4, 8, 12, 16

Pattern B: 0, 12, 24, 36, 48

How does each term in Pattern B compare to the matching term in Pattern A?

A. 8 more

B. 12 more

C. 4 times as large

D. 3 times as large

Pattern A starts at 0 and adds 5. Pattern B starts at 0 and adds 10.

What is the 6th term of each pattern?

Pattern A: _____

Pattern B: _____

True or false?

Pattern A starts at 0 and adds 2. Pattern B starts at 0 and adds 8. Each term in Pattern B is 4 times the matching term in Pattern A.

- A.** True
- B.** False

Pattern A starts at 0 and adds 7. Pattern B starts at 0 and adds 21.

When Pattern A is 28, what is the matching term in Pattern B?

Term: _____

SET 6 • 5.OA.B.3

BACKWARD 1

Patterns that follow a rule • Grade 4 • Operations & Algebraic Thinking •
4.OA.C.5

The rule is "add 6." The pattern starts at 0.
Write the first five terms.

Terms: _____

True or false?

The rule "add 4," starting at 0, gives 0, 4, 8, 16.

A. True

B. False

What is the rule for the pattern?

0, 9, 18, 27, 36

- A.** Add 9
- B.** Add 18
- C.** Multiply by 9
- D.** Add 8

The rule is "add 11." The pattern starts at 0.
What is the 4th term?

Term: _____

What comes next?

0, 15, 30, 45, ____

Next term: _____

SET 6 • 5.OA.B.3

BACKWARD 2

Multiplication facts • Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7

Multiply.

$$4 \times 7$$

Product: _____

True or false?

$$6 \times 8 = 42$$

A. True

B. False

What number times 3 equals 24?

- A.** 8
- B.** 7
- C.** 21
- D.** 6

Multiply.

$$5 \times 9$$

Product: _____

What number times 6 equals 42?

Number: _____

SET 6 • 5.OA.B.3

FORWARD 1

Equations relating dependent and independent variables • Grade 6 •
Expressions & Equations • 6.EE.C.9

Each box holds 6 cans.

Write an equation for the number of cans c in b boxes.

Equation: _____

In $c = 6b$, which variable is the dependent variable?

- A.** b
- B.** c
- C.** 6
- D.** Neither

Write an equation for y in terms of x .

x	1	2	3	4
y	5	10	15	20

Equation: _____

True or false?

In $y = x + 4$, y is the independent variable.

A. True

B. False

A plant grows 2 cm each week.

Write an equation for its growth g after w weeks.

Equation: _____

SET 6 • 5.OA.B.3

FORWARD 2

Equations for proportional relationships • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c

Pencils cost \$0.35 each.

Write an equation for the cost c of n pencils.

Equation: _____

The table shows a proportional relationship.

Write an equation for y in terms of x .

x	4	6	10
y	10	15	25

Equation: _____

Which equation represents a proportional relationship?

A. $y = 1.5$

B. $y = x + 1.5$

C. $y = 1.5x + 2$

D. $y = 1.5x$

A train travels 150 miles in 2 hours at a constant speed.
Write an equation for the distance d after t hours.

Equation: _____

True or false?

$y = \frac{3}{4}x$ represents a proportional relationship.

A. True

B. False

Explain why two patterns have the relationship they do

MAIN	Grade 5 main question	5.OA.B.3
BACKWARD 1	Notice features of a pattern that the rule does not state	4.OA.C.5
BACKWARD 2	Explain patterns in the multiplication table	3.OA.D.9
FORWARD 1	Use an equation to describe how two quantities change together	6.EE.C.9
FORWARD 2	Identify the constant of proportionality	7.RP.A.2.b

SET 58 • 5.OA.B.3

MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3

Pattern A starts at 0 and adds 4. Pattern B starts at 0 and adds 8.

Each term in Pattern B is twice the matching term in Pattern A.

Explain why.

Explanation: _____

Pattern A starts at 0 and adds 3. Pattern B starts at 0 and adds 15.

Why is each term in Pattern B 5 times the matching term in Pattern A?

- A.** Both start at 0, and B adds 5 times as much as A at every step.
- B.** Pattern B starts 5 higher than Pattern A.
- C.** Pattern B has 5 more terms than Pattern A.
- D.** Each term in B is 12 more than the matching term in A.

True or false?

Pattern A starts at 0 and adds 2. Pattern B starts at 0 and adds 6. Each term in B is 3 times the matching term in A because B adds 3 times as much at every step.

A. True

B. False

True or false?

Pattern A starts at 1 and adds 2. Pattern B starts at 1 and adds 4. Each term in Pattern B is twice the matching term in Pattern A.

A. True

B. False

Pattern A starts at 0 and adds 5. Pattern B starts at 0 and adds 20.

Without listing the terms, how does the 10th term of B compare with the 10th term of A? Explain.

Comparison: _____

Explanation: _____

SET 58 • 5.OA.B.3

BACKWARD 1

Notice features of a pattern that the rule does not state • Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5

True or false?

The rule is "add 2," starting at 1. Every term is odd.

A. True

B. False

The rule is "add 4," starting at 0. Which is true of every term?

- A.** It is a multiple of 4.
- B.** It is odd.
- C.** It ends in 4.
- D.** It is a multiple of 8.

The rule is "add 10," starting at 5.

What do you notice about the ones digit of every term?

Answer: _____

True or false?

The rule is "add 5," starting at 0. The terms alternate between ending in 0 and ending in 5.

- A.** True
- B.** False

The rule is "add 3," starting at 1: 1, 4, 7, 10, 13.

Which describes the terms?

- A.** They alternate odd, even, odd, even.
- B.** They are all odd.
- C.** They are all even.
- D.** They are all multiples of 3.

SET 58 • 5.OA.B.3

BACKWARD 2

Explain patterns in the multiplication table • Grade 3 • Operations & Algebraic Thinking • 3.OA.D.9

True or false?

Every product in the 4s row of the multiplication table is even.

- A. True
- B. False

Why is each product in the 6s row twice the product in the 3s row, in the same column?

- A.** 6 is 2×3 , so $6 \times n$ is $2 \times (3 \times n)$.
- B.** 6 is 3 more than 3.
- C.** The 6s row has twice as many products.
- D.** Products in later rows are always even.

True or false?

Every product in the 5s row ends in 5.

A. True

B. False

The 2s row is 2, 4, 6, 8, 10, ...

What is the same about every product?

Answer: _____

True or false?

In the 9s row, 9×4 is 9 more than 9×3 .

A. True

B. False

SET 58 • 5.OA.B.3

FORWARD 1

Use an equation to describe how two quantities change together • Grade 6 •
Expressions & Equations • 6.EE.C.9

Write an equation for y in terms of x .

Explain how the table shows it.

x	1	2	3	4
y	7	14	21	28

Equation: _____

Explanation: _____

Each time x increases by 1, y increases by 5, and y is 0 when x is 0.

Which equation fits?

A. $y = 5x$

B. $y = 5$

C. $y = x + 5$

D. $x = 5y$

True or false?

In $y = x + 3$, each y -value is 3 times its x -value.

A. True

B. False

A gym charges \$9 for each class.

Write an equation for the cost c of n classes.

Equation: _____

True or false?

For $y = 4x$, when x doubles, y also doubles.

- A. True
- B. False

SET 58 • 5.OA.B.3

FORWARD 2

Identify the constant of proportionality • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

The table shows a proportional relationship.

What is the constant of proportionality?

x	2	5	8
y	6	15	24

Constant: _____

What is the constant of proportionality in $y = 0.6x$?

- A.** 0.6
- B.** 0
- C.** 6
- D.** 1.6

True or false?

In the table x : 4, 6, 10 and y : 2, 3, 5, the constant of proportionality is 2.

- A. True
- B. False

A car travels 50 miles on every 2 gallons of gas.
What is the constant of proportionality, in miles per gallon?

Constant: _____

True or false?

A proportional relationship graphs as a line through $(0, 0)$ and $(1, 7)$. Its constant of proportionality is 7.

A. True

B. False

Form ordered pairs from two patterns and graph them

MAIN	Grade 5 main question	5.OA.B.3
BACKWARD 1	Patterns that follow a rule	4.OA.C.5
BACKWARD 3	Read a scaled bar graph	3.MD.B.3
BACKWARD 4	Whole numbers on a number line	2.MD.B.6
FORWARD 1	Plot pairs of values from a ratio table	6.RP.A.3.a
FORWARD 2	Use a graph to decide whether a relationship is proportional	7.RP.A.2.a

SET 7 • 5.OA.B.3

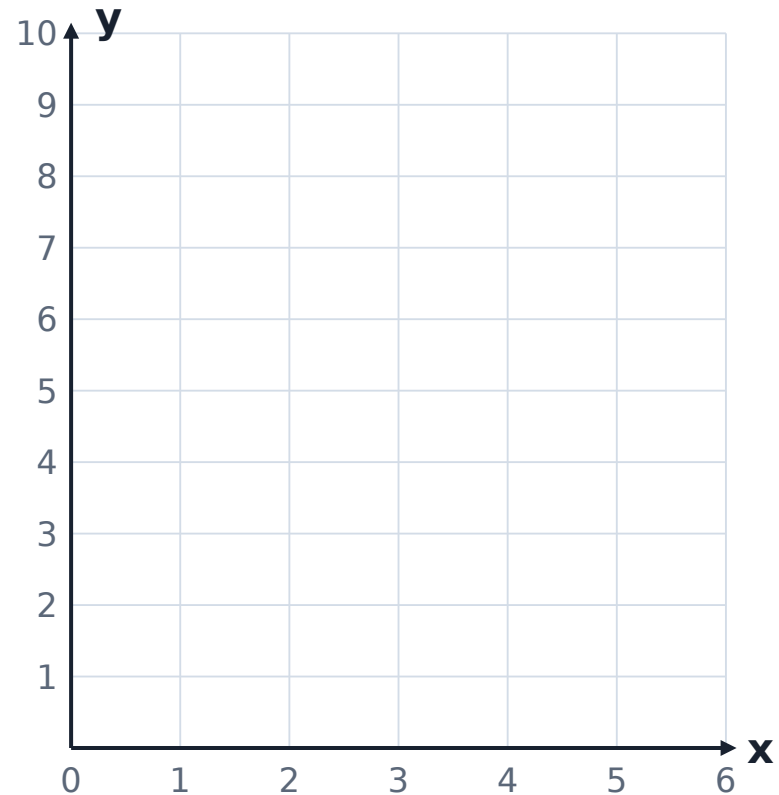
MAIN

Grade 5 main question • Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3

Pattern A (x): start at 0 and add 1.

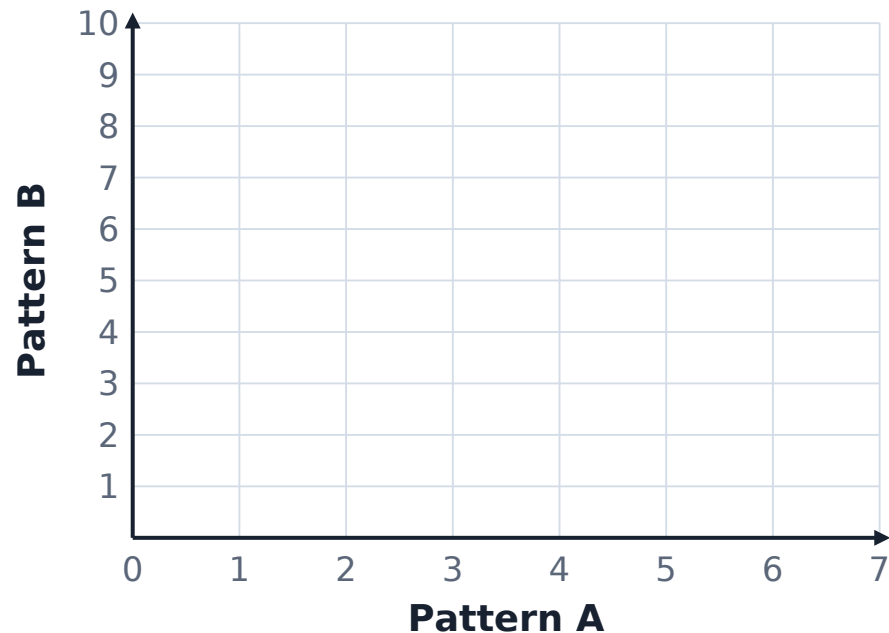
Pattern B (y): start at 0 and add 2.

Graph the ordered pairs for the first 5 terms.



Graph the ordered pairs from the two patterns.

Pattern A	0	2	4	6
Pattern B	0	3	6	9



Pattern A starts at 0 and adds 3. Pattern B starts at 0 and adds 9.

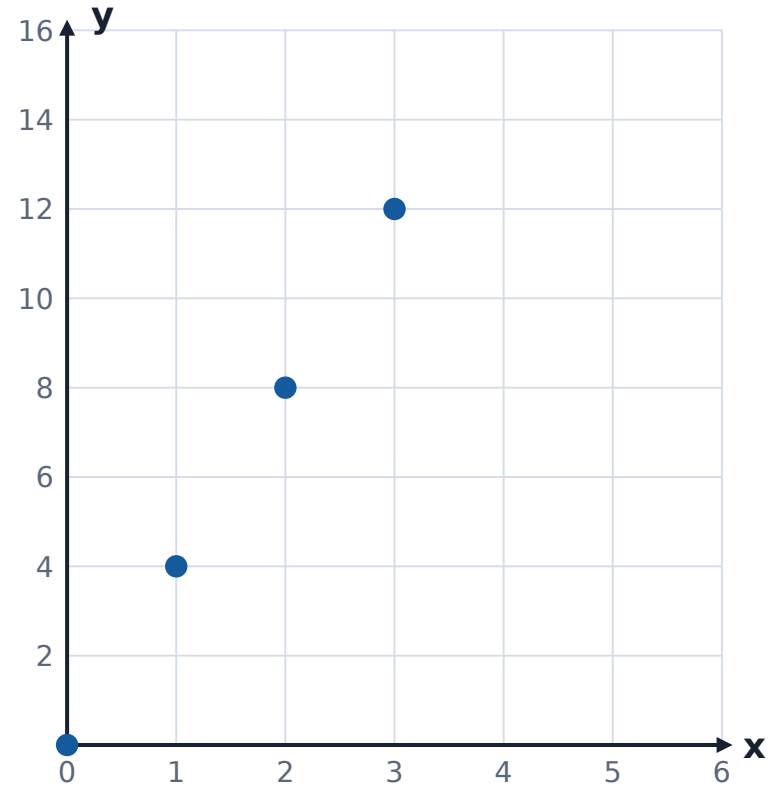
Write the first four ordered pairs (A, B).

Ordered pairs: _____

The points come from two patterns.

Which statement is true?

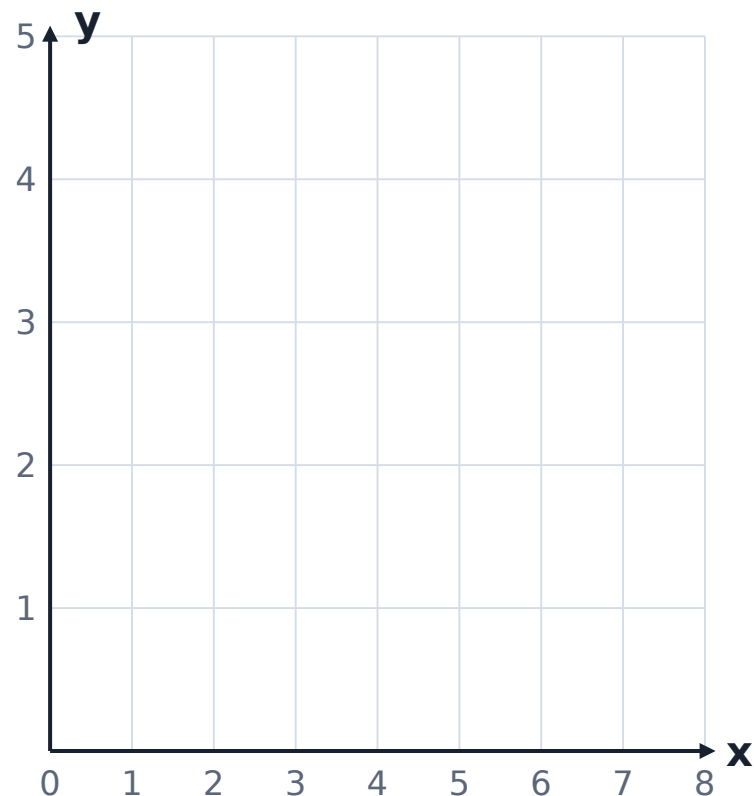
- A.** The y-values increase by 1 each time.
- B.** Each y-value is 4 times its x-value.
- C.** Each x-value is 4 times its y-value.
- D.** Each y-value is 4 more than its x-value.



Pattern A (x): start at 0 and add 2.

Pattern B (y): start at 0 and add 1.

Graph the ordered pairs for the first 4 terms.



SET 7 • 5.OA.B.3

BACKWARD 1

Patterns that follow a rule • Grade 4 • Operations & Algebraic Thinking •
4.OA.C.5

The rule is "add 3." The pattern starts at 0.
What is the 5th term?

Term: _____

True or false?

The pattern 0, 5, 10, 15 follows the rule "add 5."

A. True

B. False

Which pattern follows the rule "add 2," starting at 0?

A. 0, 2, 4, 6

B. 2, 4, 6, 8

C. 0, 1, 2, 3

D. 0, 2, 4, 8

What is the rule for the pattern?

0, 8, 16, 24

Rule: _____

The rule is "add 10." The pattern starts at 0.
What is the 6th term?

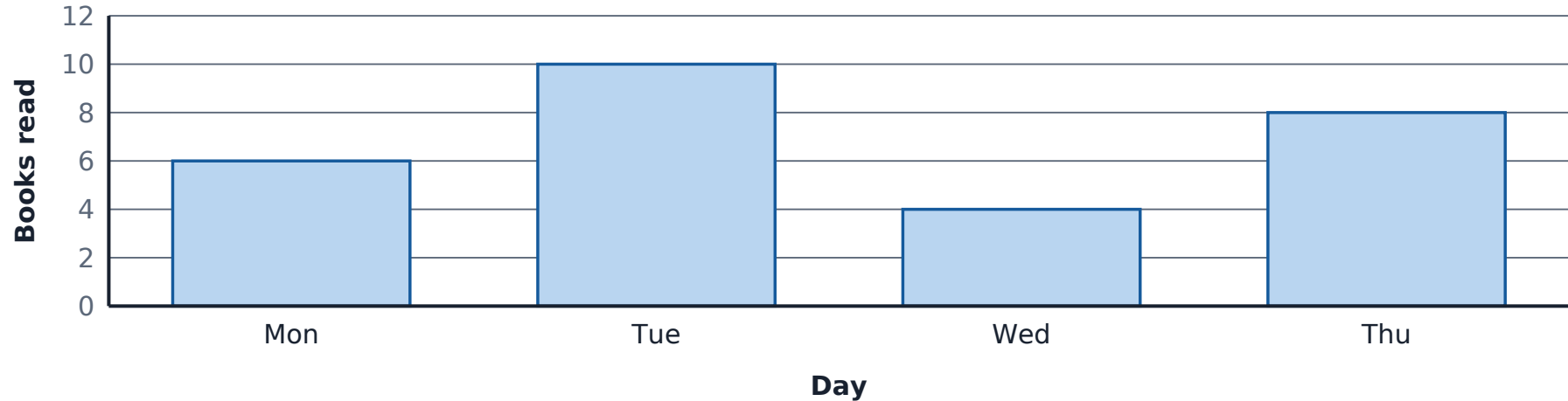
Term: _____

SET 7 • 5.OA.B.3

BACKWARD 3

Read a scaled bar graph • Grade 3 • Measurement & Data • 3.MD.B.3

How many books does each grid line on the scale stand for?



A. 4

B. 1

C. 12

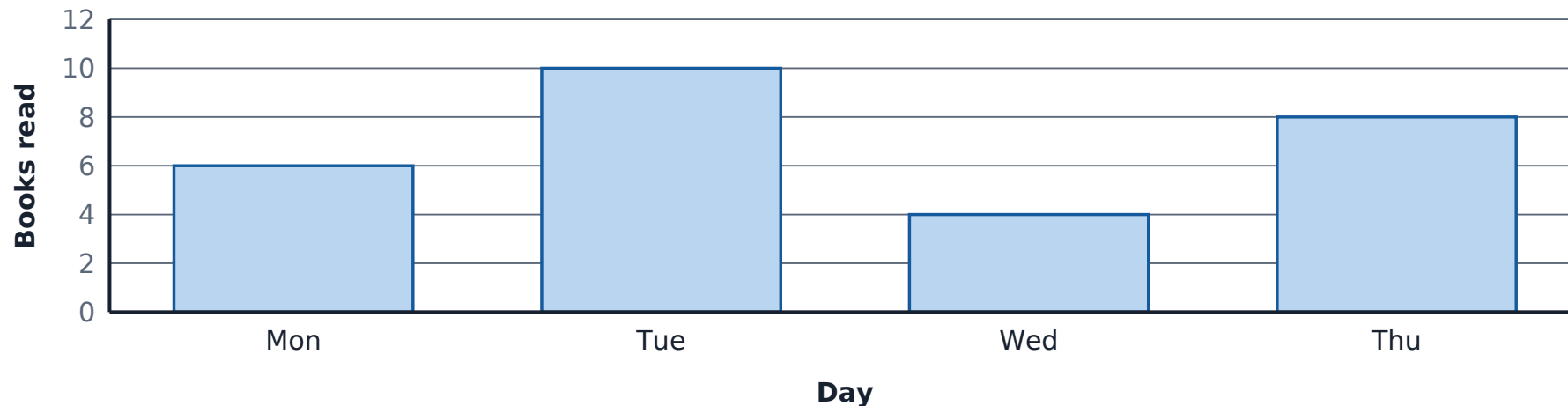
D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q1

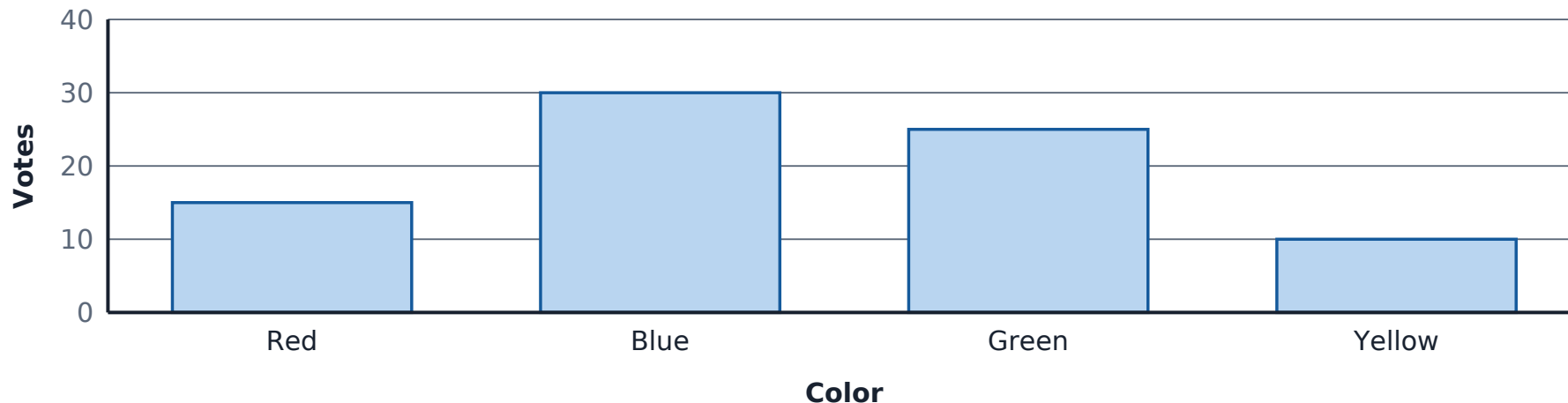
Grade 3 • Measurement & Data • 3.MD.B.3 • Set 7 • BACKWARD 3 • Q1/5

How many books were read on Tuesday?



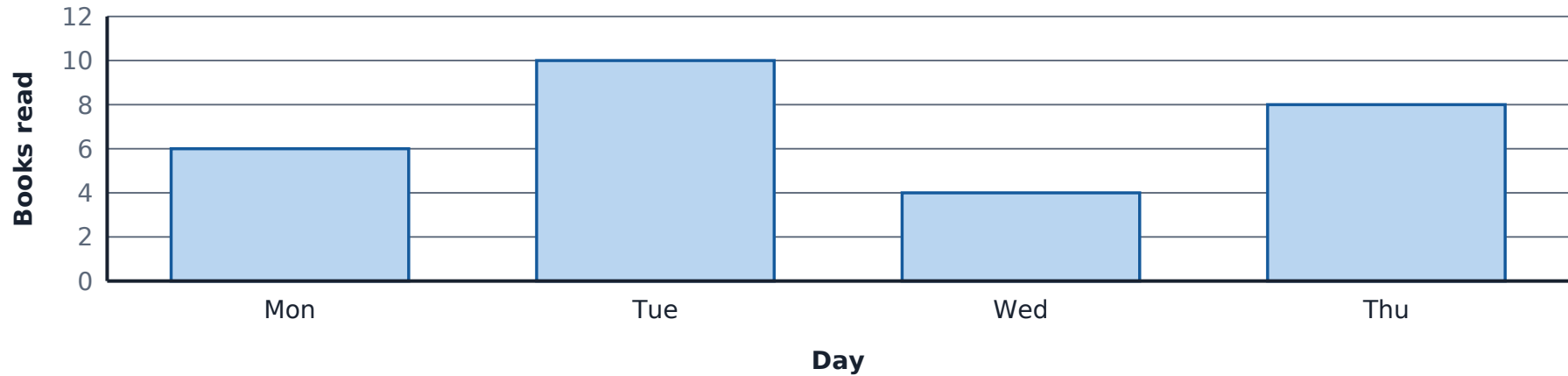
Books: _____

How many votes did Green get?



Votes: _____

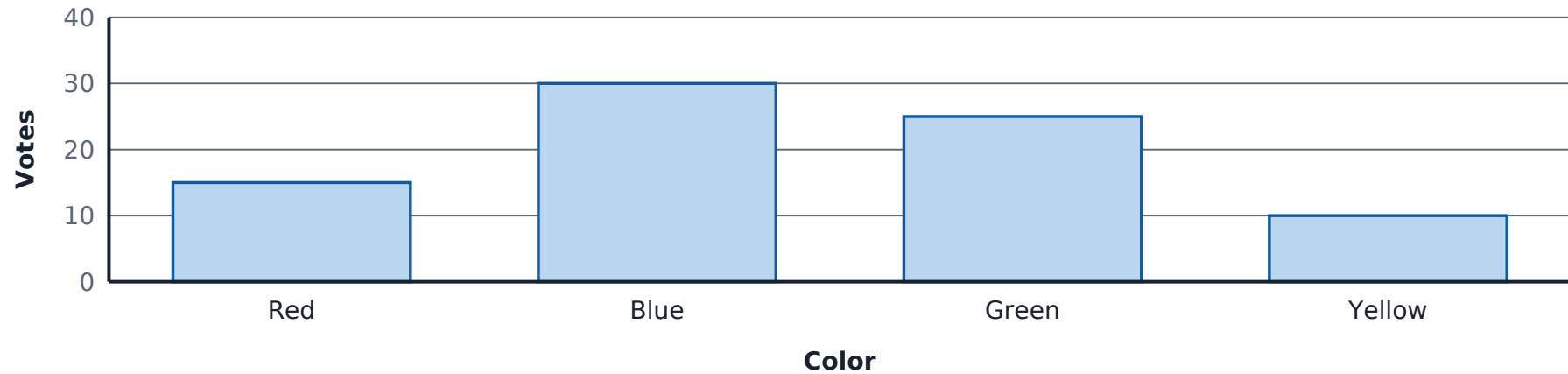
How many more books were read on Tuesday than on Wednesday?



Books: _____

True or false?

Red got 5 more votes than Yellow.



A. True

B. False

SET 7 • 5.OA.B.3

BACKWARD 4

Whole numbers on a number line • Grade 2 • Measurement & Data • 2.MD.B.6

What number is at point A?



A = _____

True or false?

From 0 to 4 on this number line there are 4 equal spaces.



A. True

B. False

The spaces on a number line are equal and each stands for 1.

Which number is 3 spaces to the right of 5?

A. 8

B. 2

C. 53

D. 15

What number is at point B?



B = _____

True or false?

On a number line, 9 is to the left of 6.

A. True

B. False

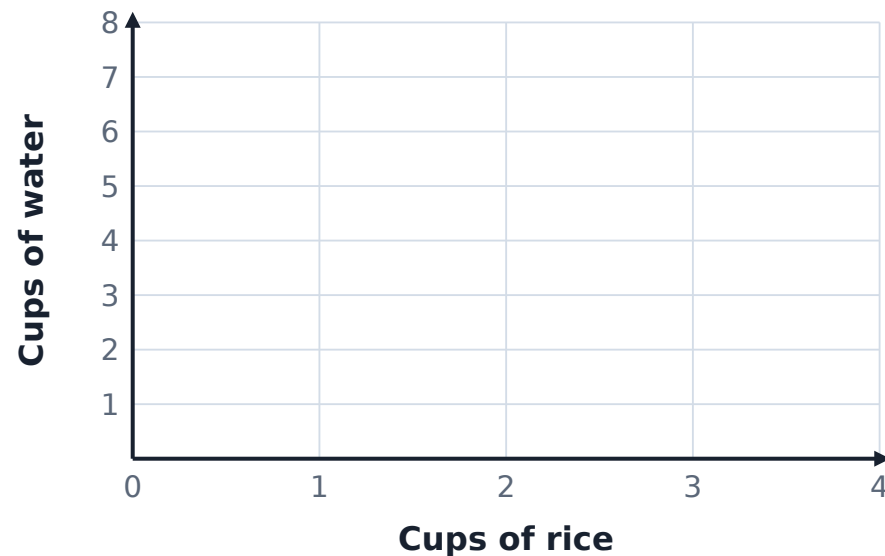
SET 7 • 5.OA.B.3

FORWARD 1

Plot pairs of values from a ratio table • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a

Plot the pairs of values from the ratio table on the coordinate plane.

Cups of rice	1	2	3
Cups of water	2	4	6



The table shows equivalent ratios.

Find the missing value.

Laps	2	4	6	8
Minutes	5	10	?	20

Missing value: _____

Which point comes from the ratio table?

Boxes	1	2	3
Pens	6	12	18

A. (2, 6)

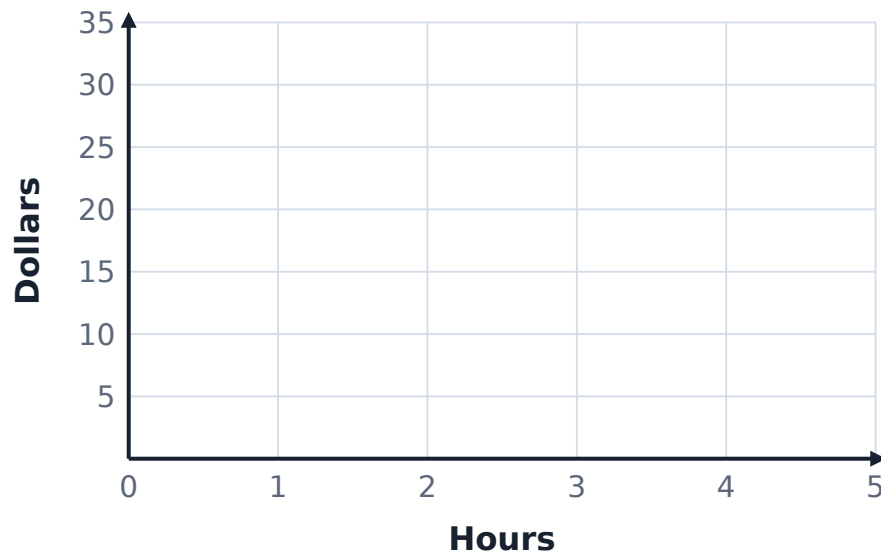
B. (12, 2)

C. (2, 12)

D. (3, 12)

Plot the pairs of values from the ratio table on the coordinate plane.

Hours	1	2	3	4
Dollars	8	16	24	32



True or false?

The points (1, 3), (2, 6), and (3, 9) come from a ratio table with the ratio 1 : 3.

- A. True
- B. False

SET 7 • 5.OA.B.3

FORWARD 2

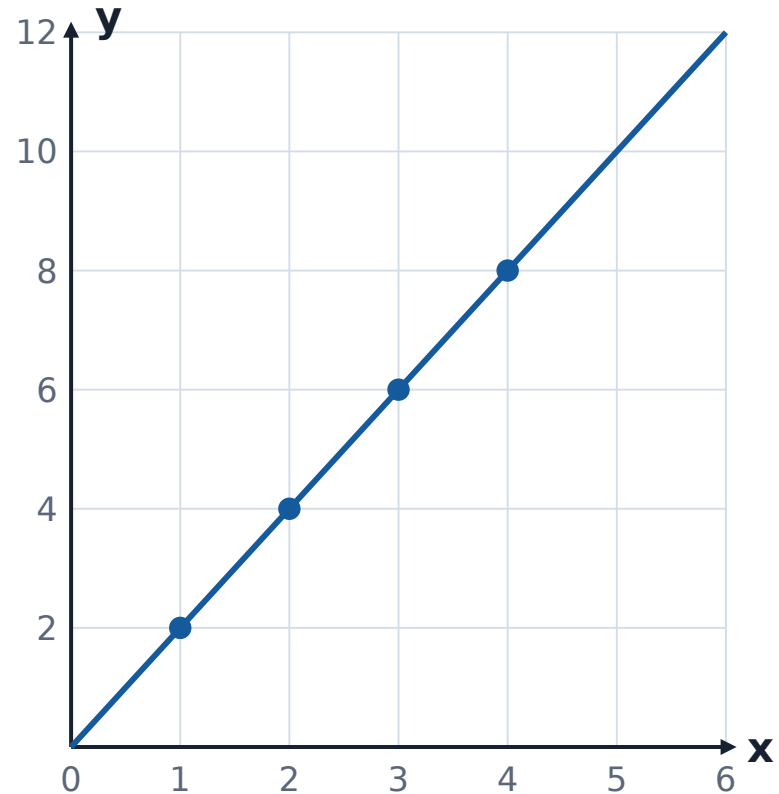
Use a graph to decide whether a relationship is proportional • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

True or false?

The points lie on a straight line through the origin, so the relationship is proportional.

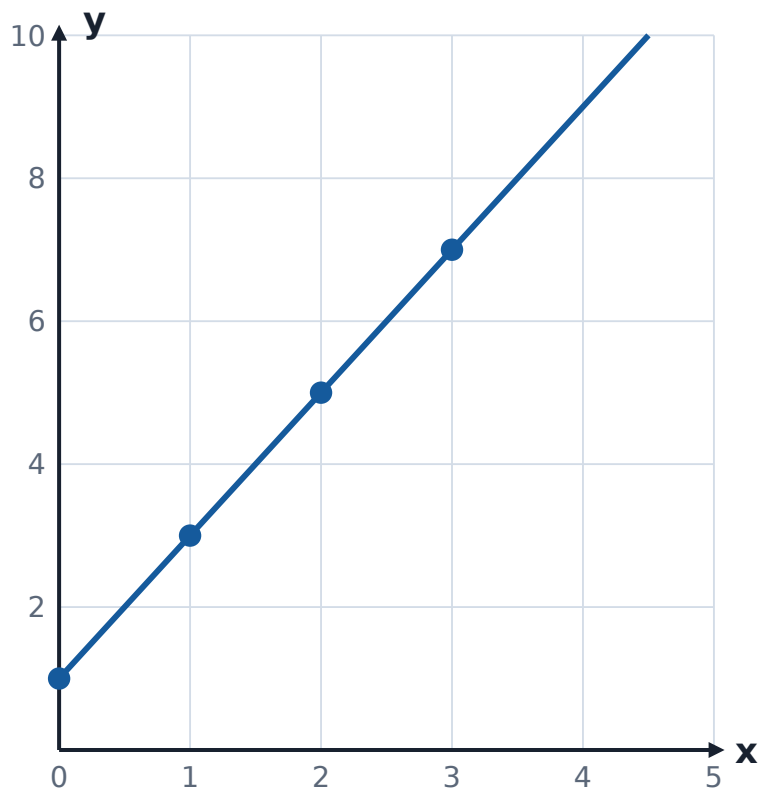
A. True

B. False



Which statement about the graph is true?

- A.** It is not proportional because the line does not pass through $(0, 0)$.
- B.** It is not proportional because the points are not on a line.
- C.** It is proportional because the points are on a line.
- D.** It is proportional because y increases by 2.

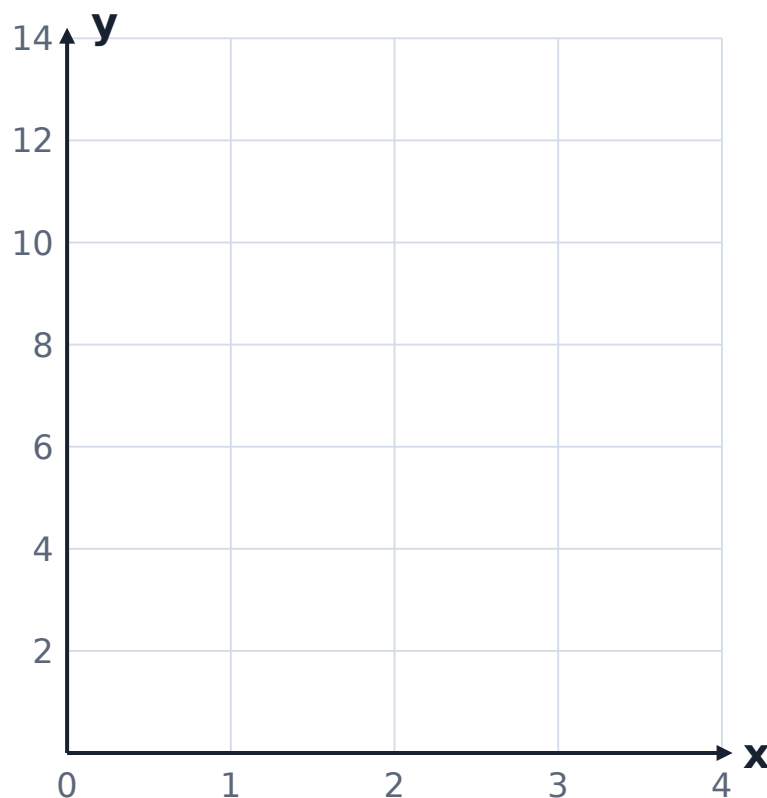


Graph the pairs $(1, 4)$, $(2, 8)$,
and $(3, 12)$.

Is the relationship
proportional? Explain using
your graph.

Answer: _____

Reason: _____



True or false?

The points (1, 2), (2, 5), and (3, 8) show a proportional relationship.

- A. True
- B. False

Which set of points shows a proportional relationship?

- A.** (1, 1), (2, 4), (3, 9)
- B.** (1, 2), (2, 3), (3, 4)
- C.** (0, 3), (1, 6), (2, 9)
- D.** (2, 5), (4, 10), (6, 15)

A digit is worth 10 times as much as the same digit one place to its right

MAIN	Grade 5 main question	5.NBT.A.1
BACKWARD 1	Each place is worth 10 times the place to its right	4.NBT.A.1
BACKWARD 2	Tenths and hundredths as decimals	4.NF.C.6
BACKWARD 3	Tenths as hundredths	4.NF.C.5
FORWARD 1	Compute with multi-digit decimals	6.NS.B.3
FORWARD 2	Real-world problems with rational numbers	7.NS.A.3

SET 8 • 5.NBT.A.1

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.1

In 5,550, how does the value of the 5 in the hundreds place compare to the value of the 5 in the tens place?

- A.** It is 100 times as much.
- B.** It is 10 times as much.
- C.** It is the same.
- D.** It is $\frac{1}{10}$ as much.

In 3.33, the first 3 is in the ones place and the second 3 is in the tenths place.

How many times as much is the first 3 worth as the second 3?

Times as much: _____

True or false?

In 0.77, the 7 in the tenths place is worth 10 times as much as the 7 in the hundredths place.

- A. True
- B. False

Find the value of the 4 in 642.18 and the value of the 4 in 6,421.8.

How many times as much is the second 4 worth?

4 in 642.18: _____

4 in 6,421.8: _____

Times as much: _____

In which number is the 6 worth 10 times as much as the 6 in 3.6?

A. 0.36

B. 3.06

C. 46.1

D. 60.3

SET 8 • 5.NBT.A.1

BACKWARD 1

Each place is worth 10 times the place to its right • Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1

True or false?

In 880, the 8 in the hundreds place is worth 10 times as much as the 8 in the tens place.

A. True

B. False

What is the value of the 7 in 2,740?

Value: _____

Which number is 10 times as much as 60?

- A.** 600
- B.** 6,000
- C.** 70
- D.** 6

How many tens are in 1 hundred?

Tens: _____

True or false?

70 is 10 times as much as 7.

A. True

B. False

SET 8 • 5.NBT.A.1

BACKWARD 2

Tenths and hundredths as decimals • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write $\frac{3}{10}$ as a decimal.

Decimal: _____

Write $\frac{47}{100}$ as a decimal.

Decimal: _____

True or false?

$$0.6 = \frac{6}{100}$$

A. True

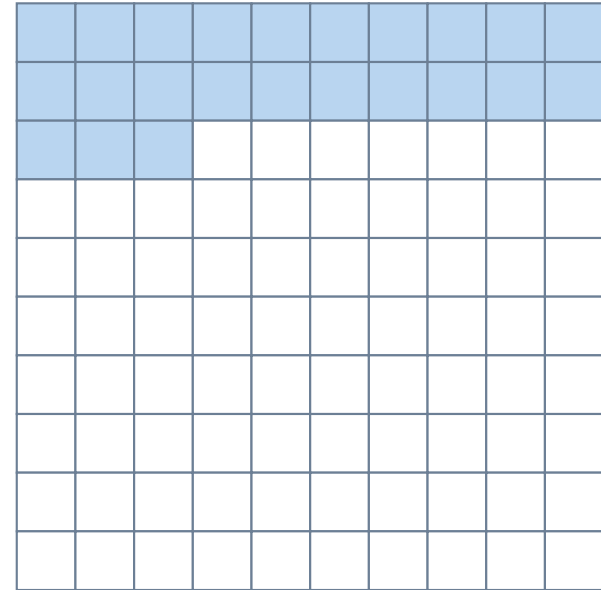
B. False

Which decimal shows 5 hundredths?

- A.** 0.5
- B.** 0.05
- C.** 0.005
- D.** 5.0

What decimal does the shaded part of the grid show?

Decimal: _____



SET 8 • 5.NBT.A.1

BACKWARD 3

Tenths as hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.C.5

True or false?

$$\frac{1}{10} = \frac{10}{100}$$

A. True

B. False

How many hundredths are in 1 tenth?

Hundredths: _____

Which fraction is equal to $\frac{4}{10}$?

A. $\frac{4}{100}$

B. $\frac{400}{100}$

C. $\frac{14}{100}$

D. $\frac{40}{100}$

Find the missing number.

$$\frac{7}{10} = \frac{?}{100}$$

Missing number: _____

True or false?

$$\frac{3}{10} + \frac{5}{100} = \frac{35}{100}$$

A. True

B. False

SET 8 • 5.NBT.A.1

FORWARD 1

Compute with multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Add.

$$27.6 + 4.38$$

Sum: _____

Subtract.

$$50.2 - 7.65$$

Difference: _____

Multiply.

$$1.2 \times 0.4$$

A. 4.8

B. 0.48

C. 0.048

D. 1.6

Divide.

$$7.5 \div 2.5$$

Quotient: _____

True or false?

$$0.3 \times 0.3 = 0.9$$

A. True

B. False

SET 8 • 5.NBT.A.1

FORWARD 2

Real-world problems with rational numbers • Grade 7 • The Number System •
7.NS.A.3

A diver at -4.5 m descends 2.25 m.

What is the diver's new position?

Position: _____

The temperature was 3.5°F . It fell 6.2°F .
What is the new temperature?

Temperature: _____

A stock's value changed by $-\$0.75$ each day for 4 days.
What was the total change?

- A.** $-\$0.30$
- B.** $-\$4.75$
- C.** $\$3.00$
- D.** $-\$3.00$

A bank balance of $-\$12.40$ increases by $\$20.00$.

What is the new balance?

Balance: _____

True or false?

$$-1.5 \times 4 = 6$$

A. True

B. False

A digit is worth $\frac{1}{10}$ of the same digit one place to its left

MAIN	Grade 5 main question	5.NBT.A.1
BACKWARD 1	One place to the right is worth $\frac{1}{10}$ as much	4.NBT.A.1
BACKWARD 2	Unit fractions	3.NF.A.1
BACKWARD 3	Decimals for tenths and hundredths	4.NF.C.6
FORWARD 1	Divide multi-digit decimals	6.NS.B.3
FORWARD 2	Write fractions as decimals using long division	7.NS.A.2.d

SET 9 • 5.NBT.A.1

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.1

True or false?

In 44.4, the 4 in the tenths place is worth $\frac{1}{10}$ of the 4 in the ones place.

- A. True
- B. False

In 2.22, the 2 in the hundredths place is worth what part of the 2 in the tenths place?

A. 10 times as much

B. $\frac{1}{10}$

C. The same amount

D. $\frac{1}{100}$

What is $\frac{1}{10}$ of 50?

Answer: _____

What is $\frac{1}{10}$ of 0.8?

Answer: _____

In which number is the 9 worth $\frac{1}{10}$ of the 9 in 19.4?

- A.** 5.92
- B.** 3.009
- C.** 9.01
- D.** 90.1

SET 9 • 5.NBT.A.1

BACKWARD 1

One place to the right is worth $\frac{1}{10}$ as much • Grade 4 • Number & Operations
in Base Ten • 4.NBT.A.1

True or false?

In 330, the 3 in the tens place is worth $\frac{1}{10}$ of the 3 in the hundreds place.

- A.** True
- B.** False

Divide.

$$400 \div 10$$

Quotient: _____

Which number is $\frac{1}{10}$ of 3,000?

- A.** 3
- B.** 300
- C.** 30,000
- D.** 30

How many hundreds are in 1 thousand?

Hundreds: _____

True or false?

$$60 \div 10 = 600$$

A. True

B. False

SET 9 • 5.NBT.A.1

BACKWARD 2

Unit fractions • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

True or false?

$\frac{1}{8}$ means 1 of 8 equal parts of a whole.

A. True

B. False

A bar is cut into 6 equal parts.

What fraction of the bar is one part?

Fraction: _____

Which fraction is greater?

A. $\frac{1}{8}$

B. $\frac{1}{2}$

True or false?

A whole cut into 4 equal parts has parts that are each $\frac{1}{3}$ of the whole.

- A. True
- B. False

What fraction of the bar is shaded?



Fraction: _____

SET 9 • 5.NBT.A.1

BACKWARD 3

Decimals for tenths and hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write $\frac{9}{100}$ as a decimal.

Decimal: _____

Write 0.8 as a fraction.

Fraction: _____

True or false?

0.4 and 0.04 have the same value.

A. True

B. False

Which decimal is equal to $\frac{6}{10}$?

A. 6.0

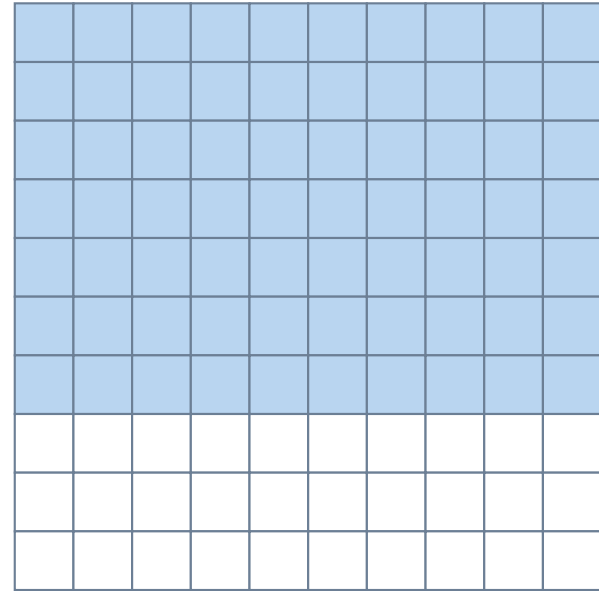
B. 0.06

C. 6.10

D. 0.6

What decimal does the shaded part of the grid show?

Decimal: _____



SET 9 • 5.NBT.A.1

FORWARD 1

Divide multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Divide.

$$4.8 \div 0.4$$

Quotient: _____

Divide.

$$9.36 \div 3$$

Quotient: _____

Divide.

$$0.56 \div 0.7$$

A. 80

B. 0.08

C. 8

D. 0.8

Divide.

$$15 \div 0.25$$

Quotient: _____

True or false?

$$6.3 \div 0.9 = 0.7$$

A. True

B. False

SET 9 • 5.NBT.A.1

FORWARD 2

Write fractions as decimals using long division • Grade 7 • The Number System
• 7.NS.A.2.d

Use long division to write $\frac{7}{8}$ as a decimal. Show your work.

Decimal: _____

Use long division to write $\frac{2}{9}$ as a decimal. Show your work.

Decimal: _____

Which fraction has a terminating decimal?

A. $\frac{5}{6}$

B. $\frac{4}{7}$

C. $\frac{1}{3}$

D. $\frac{3}{20}$

Use long division to write $\frac{9}{20}$ as a decimal. Show your work.

Decimal: _____

True or false?

The decimal form of $\frac{1}{11}$ repeats.

A. True

B. False

Multiply and divide whole numbers by powers of 10; write powers of 10 with exponents

MAIN	Grade 5 main question	5.NBT.A.2
BACKWARD 1	Multiplying and dividing by 10	4.NBT.A.1
BACKWARD 2	Multiplying by multiples of 10	3.NBT.A.3
FORWARD 1	Write and evaluate whole-number exponents	6.EE.A.1
FORWARD 2	Multiply and divide negative numbers by powers of 10 — nearest...	7.NS.A.2.c

SET 10 • 5.NBT.A.2

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.2

Multiply.

$$37 \times 1,000$$

Product: _____

Divide.

$$45,000 \div 100$$

Quotient: _____

Write $10 \times 10 \times 10 \times 10$ as a power of 10.

Power of 10: _____

$$6 \times 10^3 = ?$$

A. 60,000

B. 6,000

C. 18

D. 600

True or false?

$$820 \div 10^2 = 82$$

A. True

B. False

SET 10 • 5.NBT.A.2

BACKWARD 1

Multiplying and dividing by 10 • Grade 4 • Number & Operations in Base Ten •
4.NBT.A.1

Multiply.

$$25 \times 10$$

Product: _____

True or false?

300 is 10 times as much as 30.

A. True

B. False

Which number is 10 times as much as 450?

- A.** 4,500
- B.** 45,000
- C.** 460
- D.** 45

How many tens are in 800?

Tens: _____

Divide.

$$7,000 \div 10$$

Quotient: _____

SET 10 • 5.NBT.A.2

BACKWARD 2

Multiplying by multiples of 10 • Grade 3 • Number & Operations in Base Ten •
3.NBT.A.3

Multiply.

$$4 \times 60$$

Product: _____

Multiply.

$$9 \times 80$$

Product: _____

True or false?

$$7 \times 30 = 2,100$$

A. True

B. False

$$5 \times 90 = ?$$

A. 4,500

B. 450

C. 45

D. 95

Multiply.

$$8 \times 70$$

Product: _____

SET 10 • 5.NBT.A.2

FORWARD 1

Write and evaluate whole-number exponents • Grade 6 • Expressions & Equations • 6.EE.A.1

Evaluate.

$$3^4$$

Value: _____

Write $5 \times 5 \times 5$ using an exponent.

Exponent form: _____

$$2^5 = ?$$

A. 25

B. 10

C. 64

D. 32

Evaluate.

$$10^3 + 2^2$$

Value: _____

True or false?

$$4^3 = 12$$

A. True

B. False

FORWARD 2

Multiply and divide negative numbers by powers of 10 • Grade 7 • The Number System • 7.NS.A.2.c

► No Grade 7 standard directly continues this skill. This branch uses the nearest related Grade 7 standard.

Multiply.

$$-3.5 \times 100$$

Product: _____

Divide.

$$-420 \div 10$$

Quotient: _____

$$-0.06 \times 1,000 = ?$$

- A.** -60
- B.** -0.006
- C.** -6
- D.** 60

Divide.

$$7.2 \div (-10)$$

Quotient: _____

True or false?

$$(-10) \times (-10) \times (-10) = -1,000$$

A. True

B. False

Place the decimal point when multiplying or dividing a decimal by a power of 10

MAIN	Grade 5 main question	5.NBT.A.2
BACKWARD 1	Decimal notation for fractions	4.NF.C.6
BACKWARD 2	Place value when multiplying by 10 or 100	4.NBT.A.1
FORWARD 1	Compute fluently with multi-digit decimals	6.NS.B.3
FORWARD 2	Powers of 10 with negative rational numbers — nearest related (n...	7.NS.A.2.c

SET 11 • 5.NBT.A.2

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.2

Multiply.

$$3.82 \times 100$$

Product: _____

Divide.

$$56.4 \div 10$$

Quotient: _____

Multiply.

$$0.075 \times 10^2$$

Product: _____

Divide.

$$2.9 \div 10^3$$

A. 0.029

B. 0.29

C. 0.0029

D. 2,900

True or false?

When you multiply a number by 1,000, each digit moves 3 places to the left.

A. True

B. False

SET 11 • 5.NBT.A.2

BACKWARD 1

Decimal notation for fractions • Grade 4 • Number & Operations–Fractions •
4.NF.C.6

Write $\frac{35}{100}$ as a decimal.

Decimal: _____

True or false?

$$0.7 = \frac{7}{100}$$

A. True

B. False

Which decimal is $2\frac{4}{10}$?

A. 24

B. 2.4

C. 0.24

D. 2.04

Write 0.09 as a fraction.

Fraction: _____

True or false?

$$1.5 = 1\frac{5}{100}$$

A. True

B. False

SET 11 • 5.NBT.A.2

BACKWARD 2

Place value when multiplying by 10 or 100 • Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1

True or false?

When a whole number is multiplied by 10, each digit moves one place to the left.

A. True

B. False

Multiply.

$$64 \times 100$$

Product: _____

$$3,500 \div 10 = ?$$

- A.** 350
- B.** 3,510
- C.** 35
- D.** 35,000

Divide.

$$9,000 \div 100$$

Quotient: _____

True or false?

$$500 \div 10 = 5$$

A. True

B. False

SET 11 • 5.NBT.A.2

FORWARD 1

Compute fluently with multi-digit decimals • Grade 6 • The Number System •
6.NS.B.3

Multiply.

$$2.4 \times 0.35$$

Product: _____

Divide.

$$1.44 \div 0.12$$

Quotient: _____

Multiply.

$$12.5 \times 0.8$$

A. 100

B. 10

C. 0.1

D. 1

Add.

$$0.875 + 12.4$$

Sum: _____

True or false?

$$4.5 \div 0.05 = 9$$

A. True

B. False

FORWARD 2

Powers of 10 with negative rational numbers • Grade 7 • The Number System •
7.NS.A.2.c

► No Grade 7 standard directly continues this skill. This branch uses the nearest related Grade 7 standard.

Find the value.

$$-0.45 \times 100$$

Value: _____

Find the value.

$$-63.5 \div 1,000$$

Value: _____

Find the value.

$$(-2.7) \times (-10)$$

A. -0.27

B. -27

C. 0.27

D. 27

Find the value.

$$0.8 \times (-100)$$

Value: _____

True or false?

$$-5.2 \div 100 = -0.052$$

A. True

B. False

Explain patterns in the number of zeros and the placement of the decimal point

MAIN	Grade 5 main question	5.NBT.A.2
BACKWARD 1	A digit is worth 10 times as much as in the place to its right	4.NBT.A.1
BACKWARD 2	Tenths as hundredths	4.NF.C.5
FORWARD 1	Explain decimal placement when multiplying and dividing decima...	6.NS.B.3
FORWARD 2	Write fractions as decimals and explain why they terminate or rep...	7.NS.A.2.d

SET 59 • 5.NBT.A.2

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.2

Explain why $46 \times 1,000 = 46,000$ has three more zeros than 46.

Explanation: _____

Why is $5.7 \times 100 = 570$?

- A.** You write two zeros after 5.7.
- B.** Each digit becomes worth 100 times as much, so it moves 2 places to the left.
- C.** Each digit becomes worth $\frac{1}{100}$ as much.
- D.** You move the decimal point 2 places to the left.

True or false?

$38 \div 10 = 3.8$ because dividing by 10 makes each digit worth $\frac{1}{10}$ as much, so each digit moves one place to the right.

A. True

B. False

Mia says $2.4 \times 10 = 2.40$ because "multiplying by 10 adds a zero."

Give the correct product and explain her mistake.

Correct product: _____

Explanation: _____

$$10^3 = 1,000.$$

Explain how the exponent tells you the number of zeros.

Explanation: _____

SET 59 • 5.NBT.A.2

BACKWARD 1

A digit is worth 10 times as much as in the place to its right • Grade 4 •
Number & Operations in Base Ten • 4.NBT.A.1

True or false?

In 770, the 7 in the hundreds place is worth 10 times the 7 in the tens place.

A. True

B. False

Why is $30 \times 10 = 300$?

- A.** The 3 tens become 3 hundreds.
- B.** The 3 moves one place to the right.
- C.** Each digit becomes worth less.
- D.** You add 10 to 30.

How many tens are equal to 6 hundreds?

Tens: _____

True or false?

In 5,505, the 5 in the thousands place is worth 10 times the 5 in the ones place.

- A. True
- B. False

Write the number that is 10 times as much as 408.

Number: _____

SET 59 • 5.NBT.A.2

BACKWARD 2

Tenths as hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.C.5

Write $\frac{3}{10}$ as a fraction with denominator 100.

Fraction: _____

True or false?

$$\frac{6}{10} = \frac{60}{100}$$

A. True

B. False

$$\frac{40}{100} = ?$$

A. $\frac{40}{10}$

B. $\frac{400}{10}$

C. $\frac{4}{10}$

D. $\frac{4}{100}$

Add.

$$\frac{2}{10} + \frac{5}{100}$$

Sum: _____

True or false?

$$\frac{7}{10} = \frac{7}{100}$$

A. True

B. False

SET 59 • 5.NBT.A.2

FORWARD 1

Explain decimal placement when multiplying and dividing decimals • Grade 6 •
The Number System • 6.NS.B.3

Multiply.

$$0.3 \times 0.2$$

Product: _____

Why does $0.4 \times 0.03 = 0.012$ have three decimal places?

- A.** Because $4 \times 3 = 12$.
- B.** Tenths times hundredths gives thousandths.
- C.** Because 0.03 is greater than 0.4.
- D.** Because the product is greater than 1.

Divide.

$$7.2 \div 0.08$$

Quotient: _____

True or false?

$$1.25 \times 0.4 = 0.5$$

A. True

B. False

Explain why $4.8 \div 0.6$ has the same quotient as $48 \div 6$.

Quotient: _____

Explanation: _____

FORWARD 2

Write fractions as decimals and explain why they terminate or repeat • Grade 7

• The Number System • 7.NS.A.2.d

► No Grade 7 standard directly continues this skill. This branch uses the nearest related Grade 7 standard.

Use long division to write $\frac{3}{8}$ as a decimal. Show your work.

Decimal: _____

True or false?

$\frac{3}{40}$ can be written as a decimal that ends.

A. True

B. False

Which decimal equals $\frac{7}{25}$?

A. 0.35

B. 0.28

C. 2.8

D. 0.07

Use long division to write $\frac{5}{12}$ as a decimal. Show your work.

Decimal: _____

True or false?

$$\frac{2}{3} = 0.67 \text{ exactly.}$$

A. True

B. False

Read and write decimals to thousandths using numerals and words

MAIN	Grade 5 main question	5.NBT.A.3.a
BACKWARD 1	Decimals for tenths and hundredths	4.NF.C.6
BACKWARD 2	Reading and writing whole numbers	4.NBT.A.2
BACKWARD 3	Naming tenths and hundredths	4.NF.C.5
FORWARD 1	Find and position rational numbers on a number line	6.NS.C.6.c
FORWARD 2	Write fractions as decimals	7.NS.A.2.d

SET 12 • 5.NBT.A.3.a

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.3.a

Write the number in standard form.

four and twenty-six hundredths

Number: _____

Write 3.508 in words.

Words: _____

Which number is nine and forty-one thousandths?

- A.** 9.410
- B.** 9.41
- C.** 9.0041
- D.** 9.041

Write the number in standard form.

seventeen and three tenths

Number: _____

True or false?

0.072 is read "seventy-two hundredths."

A. True

B. False

SET 12 • 5.NBT.A.3.a

BACKWARD 1

Decimals for tenths and hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write $\frac{6}{100}$ as a decimal.

Decimal: _____

True or false?

$$0.5 = \frac{5}{10}$$

A. True

B. False

Which decimal is equal to $\frac{12}{100}$?

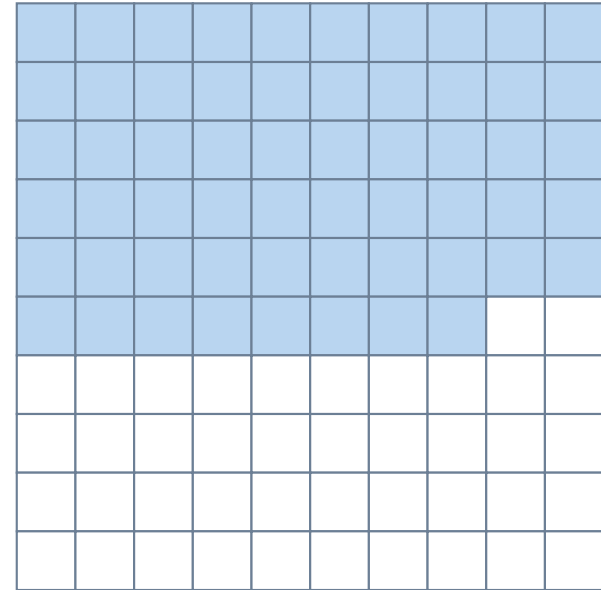
- A.** 1.2
- B.** 12.0
- C.** 0.12
- D.** 0.012

Write $3\frac{7}{10}$ as a decimal.

Decimal: _____

What decimal does the shaded part of the grid show?

Decimal: _____



SET 12 • 5.NBT.A.3.a

BACKWARD 2

Reading and writing whole numbers • Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2

Write the number in standard form.

two thousand, four hundred six

Number: _____

True or false?

5,030 is read "five thousand three hundred."

A. True

B. False

Which number is eighty thousand, nine hundred?

A. 80,900

B. 8,900

C. 89,000

D. 80,090

Write 607 in words.

Words: _____

Write the number in standard form.

twelve thousand, fifty

Number: _____

SET 12 • 5.NBT.A.3.a

BACKWARD 3

Naming tenths and hundredths • Grade 4 • Number & Operations–Fractions •
4.NF.C.5

True or false?

Three tenths is equal to thirty hundredths.

A. True

B. False

How many hundredths are in 4 tenths?

Hundredths: _____

Which fraction is "five hundredths"?

A. $\frac{100}{5}$

B. $\frac{5}{10}$

C. $\frac{50}{100}$

D. $\frac{5}{100}$

Write "nine tenths" as a fraction.

Fraction: _____

True or false?

$$\frac{60}{100} = \frac{6}{10}$$

A. True

B. False

SET 12 • 5.NBT.A.3.a

FORWARD 1

Find and position rational numbers on a number line • Grade 6 • The Number System • 6.NS.C.6.c

What number does point P represent?

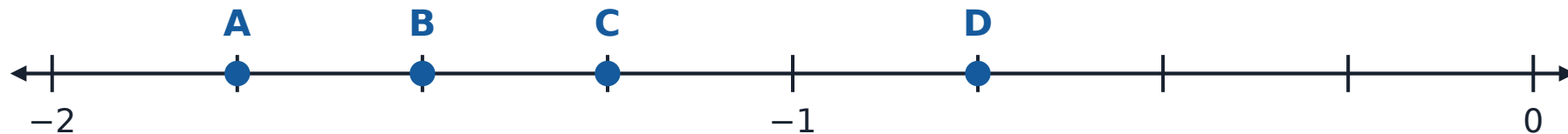


P = _____

Plot and label point Q at 0.45 on the number line.



Which point is at -1.25 ?



A. Point A

B. Point B

C. Point C

D. Point D

What number does point R represent?



R = _____

True or false?

On a number line, -0.6 is to the left of -0.5 .

A. True

B. False

SET 12 • 5.NBT.A.3.a

FORWARD 2

Write fractions as decimals • Grade 7 • The Number System • 7.NS.A.2.d

Use long division to write $\frac{3}{8}$ as a decimal. Show your work.

Decimal: _____

Use long division to write $\frac{5}{12}$ as a decimal. Show your work.

Decimal: _____

$$\frac{11}{20} = ?$$

A. 0.55

B. 5.5

C. 0.2

D. 0.11

Use long division to write $\frac{4}{9}$ as a decimal. Show your work.

Decimal: _____

True or false?

$$\frac{7}{25} = 0.32$$

A. True

B. False

Write decimals to thousandths in expanded form

MAIN	Grade 5 main question	5.NBT.A.3.a
BACKWARD 1	Expanded form of whole numbers	4.NBT.A.2
BACKWARD 2	Decimals and fractions with denominators 10 and 100	4.NF.C.6
FORWARD 1	Write expanded form with exponents	6.EE.A.1
FORWARD 2	Decimal expansions of fractions — nearest related (no direct Gra...	7.NS.A.2.d

SET 13 • 5.NBT.A.3.a

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.3.a

Write 352.68 in expanded form.

Expanded form: _____

Write the number in standard form.

$$4 \times 10 + 7 \times 1 + 3 \times \frac{1}{10} + 9 \times \frac{1}{1000}$$

Number: _____

Which is the expanded form of 6.045?

A. $6 + 45$

B. $6 \times 10 + 4 \times \frac{1}{100} + 5 \times \frac{1}{1000}$

C. $6 \times 1 + 4 \times \frac{1}{100} + 5 \times \frac{1}{1000}$

D. $6 \times 1 + 4 \times \frac{1}{10} + 5 \times \frac{1}{100}$

Write the number in standard form.

$$8 \times 100 + 2 \times \frac{1}{10} + 6 \times \frac{1}{100}$$

Number: _____

True or false?

$$0.908 = 9 \times \frac{1}{10} + 8 \times \frac{1}{1000}$$

A. True

B. False

SET 13 • 5.NBT.A.3.a

BACKWARD 1

Expanded form of whole numbers • Grade 4 • Number & Operations in Base Ten • 4.NBT.A.2

Write 4,371 in expanded form.

Expanded form: _____

True or false?

$$6,052 = 6,000 + 500 + 2$$

A. True

B. False

Which number is $7,000 + 400 + 9$?

- A.** 7,409
- B.** 7,490
- C.** 74,009
- D.** 7,049

Write $3 \times 1,000 + 8 \times 100 + 5 \times 1$ in standard form.

Number: _____

True or false?

$$900 + 20 + 4 = 9,024$$

A. True

B. False

SET 13 • 5.NBT.A.3.a

BACKWARD 2

Decimals and fractions with denominators 10 and 100 • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write 0.6 as a fraction.

Fraction: _____

True or false?

$$\frac{4}{100} = 0.4$$

A. True

B. False

Which fraction is equal to 0.25?

A. $\frac{1}{25}$

B. $\frac{25}{100}$

C. $\frac{25}{10}$

D. $\frac{2}{5}$

Write $2 + \frac{3}{10} + \frac{7}{100}$ as a decimal.

Decimal: _____

True or false?

$$\frac{9}{10} = 0.09$$

A. True

B. False

SET 13 • 5.NBT.A.3.a

FORWARD 1

Write expanded form with exponents • Grade 6 • Expressions & Equations •
6.EE.A.1

Write $5,000 + 300 + 20 + 7$ using powers of 10.

Expression: _____

Evaluate.

$$4 \times 10^3 + 6 \times 10^1$$

Value: _____

Which number is equal to 10^4 ?

- A.** 40
- B.** 100,000
- C.** 1,000
- D.** 10,000

Write $600 + 90 + 1$ using powers of 10.

Expression: _____

True or false?

$$2 \times 10^2 + 5 = 250$$

A. True

B. False

FORWARD 2

Decimal expansions of fractions • Grade 7 • The Number System • 7.NS.A.2.d

► No Grade 7 standard directly continues this skill. This branch uses the nearest related Grade 7 standard.

Write $\frac{3}{4}$ as a decimal.

Decimal: _____

Use long division to write $\frac{1}{6}$ as a decimal. Show your work.

Decimal: _____

Which fraction is equal to 0.666... (the 6 repeats)?

A. $\frac{3}{5}$

B. $\frac{6}{10}$

C. $\frac{2}{3}$

D. $\frac{1}{6}$

Use long division to write $\frac{5}{8}$ as a decimal. Show your work.

Decimal: _____

True or false?

$$\frac{2}{5} = 0.4$$

A. True

B. False

Compare decimals to thousandths

MAIN	Grade 5 main question	5.NBT.A.3.b
BACKWARD 1	Comparing decimals to hundredths	4.NF.C.7
BACKWARD 2	The symbols $>$, $<$, and $=$	1.NBT.B.3
FORWARD 1	Order rational numbers	6.NS.C.7.b
FORWARD 2	Compare fractions and decimals using decimal expansions	7.NS.A.2.d

SET 14 • 5.NBT.A.3.b

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.3.b

Write $>$, $<$, or $=$ to compare.

0.47 0.405

Symbol: _____

Write $>$, $<$, or $=$ to compare.

3.06 3.060

Symbol: _____

Which number is greatest?

- A.** 5.2
- B.** 5.19
- C.** 5.093
- D.** 5.199

Order from least to greatest.

2.31, 2.301, 2.13, 2.3

Order: _____

True or false?

$$0.8 < 0.79$$

A. True

B. False

SET 14 • 5.NBT.A.3.b

BACKWARD 1

Comparing decimals to hundredths • Grade 4 • Number & Operations–Fractions
• 4.NF.C.7

Write $>$, $<$, or $=$ to compare.

0.6 0.58

Symbol: _____

True or false?

$$0.3 = 0.03$$

A. True

B. False

Which number is less?

A. 0.09

B. 0.1

Write $>$, $<$, or $=$ to compare.

0.75 0.8

Symbol: _____

True or false?

$$0.42 > 0.5$$

A. True

B. False

SET 14 • 5.NBT.A.3.b

BACKWARD 2

The symbols $>$, $<$, and $=$ • Grade 1 • Number & Operations in Base Ten •
1.NBT.B.3

Which symbol means "is less than"?

A. $<$

B. $>$

C. $=$

True or false?

$$9 > 6$$

A. True

B. False

Write $>$, $<$, or $=$ to compare.

52 25

Symbol: _____

True or false?

The symbol $>$ means "is less than."

A. True

B. False

Which statement is true?

A. $14 < 41$

B. $14 = 41$

C. $14 > 41$

SET 14 • 5.NBT.A.3.b

FORWARD 1

Order rational numbers • Grade 6 • The Number System • 6.NS.C.7.b

Write $>$, $<$, or $=$ to compare.

-0.4 ____ -0.45

Symbol: _____

Which number is least?

- A.** -2
- B.** -2.5
- C.** 0.5
- D.** -2.05

Order from least to greatest.

-1.2 , 0.3 , -1.25 , 0

Order: _____

True or false?

$$-3.7 > -3.6$$

A. True

B. False

A temperature of -2.5°C is colder than a temperature of -1.8°C .

Write an inequality to show this.

Inequality: _____

SET 14 • 5.NBT.A.3.b

FORWARD 2

Compare fractions and decimals using decimal expansions • Grade 7 • The Number System • 7.NS.A.2.d

Write $\frac{5}{8}$ as a decimal.

Is $\frac{5}{8}$ greater than or less than 0.6?

Decimal: _____

Answer: _____

Which number is greatest?

A. $\frac{4}{5}$

B. 0.7

C. $\frac{3}{4}$

D. 0.78

Order from least to greatest.

$$\frac{2}{3}, 0.6, \frac{5}{8}$$

Order: _____

True or false?

$$\frac{1}{3} > 0.33$$

A. True

B. False

Write $>$, $<$, or $=$ to compare.

$$\frac{7}{20} \text{ ____ } 0.4$$

Symbol: _____

Round decimals to any place

MAIN

Grade 5 main question

5.NBT.A.4

BACKWARD 1

Rounding whole numbers

4.NBT.A.3

BACKWARD 2

Decimals on a number line

4.NF.C.6

FORWARD 1

Estimate to check computations with decimals

6.NS.B.3

FORWARD 2

Use estimation to judge whether an answer is reasonable

7.EE.B.3

SET 15 • 5.NBT.A.4

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.4

Round 6.483 to the nearest tenth.

Rounded: _____

Round 12.749 to the nearest hundredth.

Rounded: _____

Round 0.865 to the nearest whole number.

A. 0.9

B. 1

C. 0

D. 0.87

Round 39.96 to the nearest tenth.

Rounded: _____

True or false?

4.351 rounded to the nearest hundredth is 4.36.

A. True

B. False

SET 15 • 5.NBT.A.4

BACKWARD 1

Rounding whole numbers • Grade 4 • Number & Operations in Base Ten •
4.NBT.A.3

Round 4,682 to the nearest hundred.

Rounded: _____

True or false?

350 rounded to the nearest hundred is 300.

A. True

B. False

Round 7,249 to the nearest thousand.

- A.** 7,200
- B.** 7,000
- C.** 7,250
- D.** 8,000

Round 85 to the nearest ten.

Rounded: _____

True or false?

1,349 rounded to the nearest hundred is 1,400.

- A. True
- B. False

SET 15 • 5.NBT.A.4

BACKWARD 2

Decimals on a number line • Grade 4 • Number & Operations–Fractions •
4.NF.C.6

What decimal is at point A?



A = _____

True or false?

0.7 is closer to 1 than to 0.

A. True

B. False

Is 0.43 closer to 0.4 or to 0.5?

A. 0.4

B. 0.5

What decimal is at point B?



B = _____

True or false?

2.5 is halfway between 2 and 3.

A. True

B. False

SET 15 • 5.NBT.A.4

FORWARD 1

Estimate to check computations with decimals • Grade 6 • The Number System
• 6.NS.B.3

Round each number to the nearest whole number. Then add to estimate.

$$18.7 + 6.25$$

Estimate: _____

Which is the best estimate of 4.9×6.1 ?

A. 36

B. 30

C. 24

D. 11

Estimate $31.8 \div 7.9$ by rounding each number to the nearest whole number.

Estimate: _____

True or false?

29.6 – 10.2 is about 40.

A. True

B. False

Kim says $3.9 \times 2.1 = 81.9$.

Use an estimate to decide whether her answer is reasonable. Write yes or no.

Answer: _____

SET 15 • 5.NBT.A.4

FORWARD 2

Use estimation to judge whether an answer is reasonable • Grade 7 •
Expressions & Equations • 7.EE.B.3

A \$24.75 shirt is 20% off.

Round the price to the nearest dollar. Then use the rounded price to estimate the sale price.

Rounded price: _____

Estimated sale price: _____

A worker earns \$19.85 per hour for 9.75 hours.
Which is the best estimate of her pay?

- A.** \$20
- B.** \$2,000
- C.** \$200
- D.** \$100

True or false?

A 6.2% tax on \$49.50 is about \$3.

A. True

B. False

Nina says $\frac{3}{4} \times 39.8$ is about 40.

Round 39.8 to the nearest whole number and estimate $\frac{3}{4} \times 39.8$. Is her estimate reasonable?

Your estimate: _____

Reasonable? _____

A board is $8\frac{1}{4}$ feet long. It is cut into pieces that are $1\frac{7}{8}$ feet long.

Estimate how many whole pieces can be cut.

Estimate: _____

Multiply multi-digit whole numbers using the standard algorithm

MAIN	Grade 5 main question	5.NBT.B.5
BACKWARD 1	Multiplying by a one-digit or two-digit number	4.NBT.B.5
BACKWARD 2	Multiplication facts	3.OA.C.7
BACKWARD 3	Adding partial products	4.NBT.B.4
FORWARD 1	Multiply multi-digit decimals	6.NS.B.3
FORWARD 2	Multiply integers	7.NS.A.2.c

SET 16 • 5.NBT.B.5

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.5

Use the standard algorithm to multiply. Show your work.

$$246 \times 37$$

Product: _____

Use the standard algorithm to multiply. Show your work.

$$1,385 \times 24$$

Product: _____

This is the standard algorithm for 508×63 with one partial product missing.
What is the missing partial product?

$$\begin{array}{r} 508 \\ \times 63 \\ \hline 1,524 \\ + \quad ? \\ \hline 32,004 \end{array}$$

A. 1,524

B. 31,904

C. 30,480

D. 3,048

Use the standard algorithm to multiply. Show your work.

$$79 \times 86$$

Product: _____

A theater has 48 rows with 125 seats in each row.

Use the standard algorithm to find how many seats are in the theater. Show your work.

Seats: _____

SET 16 • 5.NBT.B.5

BACKWARD 1

Multiplying by a one-digit or two-digit number • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5

Multiply.

$$346 \times 7$$

Product: _____

Multiply.

$$23 \times 45$$

Product: _____

True or false?

$$1,203 \times 4 = 4,082$$

A. True

B. False

$$62 \times 18 = ?$$

A. 1,126

B. 496

C. 1,116

D. 1,006

Multiply.

$$58 \times 30$$

Product: _____

SET 16 • 5.NBT.B.5

BACKWARD 2

Multiplication facts • Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7

Multiply.

$$7 \times 6$$

Product: _____

Multiply.

$$8 \times 9$$

Product: _____

True or false?

$$6 \times 9 = 56$$

A. True

B. False

$$4 \times 8 = ?$$

A. 24

B. 36

C. 32

D. 28

Multiply.

$$9 \times 9$$

Product: _____

SET 16 • 5.NBT.B.5

BACKWARD 3

Adding partial products • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.4

Add.

$$7,380 + 1,722$$

Sum: _____

Add.

$$27,700 + 5,540$$

Sum: _____

True or false?

$$30,480 + 1,524 = 31,904$$

A. True

B. False

$$6,400 + 394 = ?$$

A. 6,784

B. 6,794

C. 7,794

D. 6,694

Add.

$$4,800 + 1,200$$

Sum: _____

SET 16 • 5.NBT.B.5

FORWARD 1

Multiply multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Multiply.

$$2.46 \times 3.7$$

Product: _____

Multiply.

$$13.85 \times 2.4$$

Product: _____

Multiply.

$$5.08 \times 6.3$$

A. 31.904

B. 320.04

C. 3.2004

D. 32.004

Multiply.

$$0.79 \times 8.6$$

Product: _____

True or false?

$$1.25 \times 4.8 = 60$$

A. True

B. False

SET 16 • 5.NBT.B.5

FORWARD 2

Multiply integers • Grade 7 • The Number System • 7.NS.A.2.c

Find the product.

$$(-24) \times 15$$

Product: _____

Find the product.

$$(-12) \times (-35)$$

Product: _____

Find the product.

$$18 \times (-21)$$

- A.** 378
- B.** -378
- C.** -39
- D.** -388

Find the product.

$$(-3) \times (-4) \times (-25)$$

Product: _____

True or false?

$$(-6)(-7) = -42$$

A. True

B. False

Divide with up to four-digit dividends and two-digit divisors

MAIN	Grade 5 main question	5.NBT.B.6
BACKWARD 1	Dividing by a one-digit number	4.NBT.B.6
BACKWARD 2	Multiplying to check a quotient	4.NBT.B.5
BACKWARD 3	Division facts	3.OA.C.7
FORWARD 1	Divide multi-digit numbers using the standard algorithm	6.NS.B.2
FORWARD 2	Divide integers	7.NS.A.2.b

SET 17 • 5.NBT.B.6

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.6

Divide.

$$2,016 \div 24$$

Quotient: _____

Divide.

$$7,830 \div 45$$

Quotient: _____

Divide.

$$952 \div 17$$

A. 54

B. 46

C. 56

D. 57

Divide.

$$3,724 \div 38$$

Quotient: _____

Divide. Write the quotient and the remainder.

$$1,000 \div 32$$

Quotient: _____

Remainder: _____

SET 17 • 5.NBT.B.6

BACKWARD 1

Dividing by a one-digit number • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Divide.

$$852 \div 6$$

Quotient: _____

Divide.

$$1,935 \div 5$$

Quotient: _____

True or false?

$$728 \div 8 = 901$$

A. True

B. False

$$504 \div 7 = ?$$

A. 82

B. 71

C. 62

D. 72

Divide. Write the quotient and the remainder.

$$95 \div 4$$

Quotient and remainder: _____

SET 17 • 5.NBT.B.6

BACKWARD 2

Multiplying to check a quotient • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.5

True or false?

To check $2,016 \div 24 = 84$, you can multiply 24×84 .

- A. True
- B. False

Multiply to check $2,016 \div 24 = 84$.

$$24 \times 84$$

Product: _____

Which product checks $952 \div 17 = 56$?

- A.** $17 + 56$
- B.** 17×56
- C.** $56 - 17$
- D.** 952×17

Multiply.

$$38 \times 98$$

Product: _____

True or false?

$$32 \times 31 + 8 = 1,000$$

A. True

B. False

SET 17 • 5.NBT.B.6

BACKWARD 3

Division facts • Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7

Divide.

$$72 \div 8$$

Quotient: _____

Divide.

$$54 \div 9$$

Quotient: _____

True or false?

$$63 \div 7 = 8$$

A. True

B. False

$$40 \div 5 = ?$$

A. 6

B. 7

C. 8

D. 9

Divide.

$$35 \div 7$$

Quotient: _____

SET 17 • 5.NBT.B.6

FORWARD 1

Divide multi-digit numbers using the standard algorithm • Grade 6 • The Number System • 6.NS.B.2

Use the standard algorithm to divide. Show your work.

$$6,432 \div 48$$

Quotient: _____

Use the standard algorithm to divide. Show your work.

$$25,875 \div 69$$

Quotient: _____

Divide.

$$14,040 \div 52$$

- A.** 207
- B.** 2,700
- C.** 270
- D.** 27

Use the standard algorithm to divide. Show your work.

$$9,594 \div 41$$

Quotient: _____

True or false?

$$11,700 \div 45 = 2,600$$

A. True

B. False

SET 17 • 5.NBT.B.6

FORWARD 2

Divide integers • Grade 7 • The Number System • 7.NS.A.2.b

Find the quotient.

$$-2,016 \div 24$$

Quotient: _____

Find the quotient.

$$(-952) \div (-17)$$

Quotient: _____

Find the quotient.

$$780 \div (-15)$$

- A.** -52
- B.** -53
- C.** -765
- D.** 52

Find the quotient.

$$-(144 \div 12)$$

Quotient: _____

True or false?

$$(-90) \div 18 = 90 \div (-18)$$

A. True

B. False

Explain division using area models and equations

MAIN	Grade 5 main question	5.NBT.B.6
BACKWARD 1	Area models for dividing by a one-digit number	4.NBT.B.6
BACKWARD 2	Area as length times width	3.MD.C.7.b
BACKWARD 3	Splitting a rectangle to find its area	3.MD.C.7.c
FORWARD 1	Use the distributive property with the greatest common factor	6.NS.B.4
FORWARD 2	Factor and expand linear expressions	7.EE.A.1

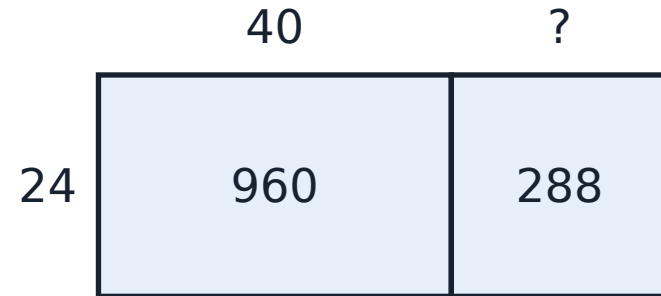
SET 18 • 5.NBT.B.6

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.6

The area model shows
 $1,248 \div 24$.

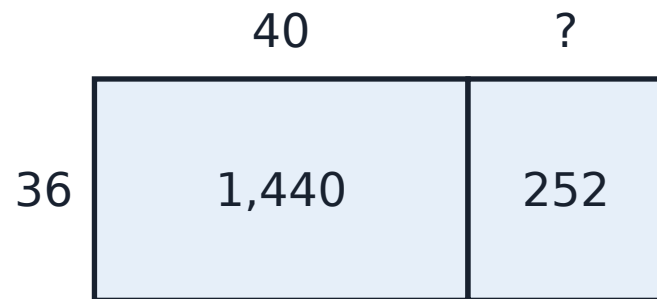
What is the quotient?



Quotient: _____

The area model shows
 $1,692 \div 36$.

What is the quotient?



Quotient: _____

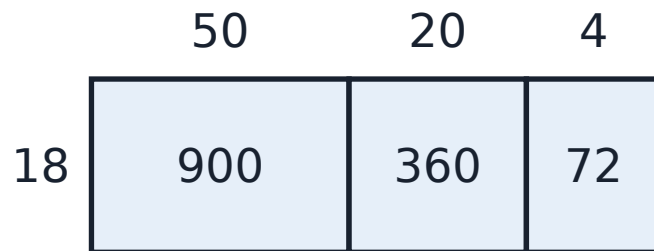
Which equation matches the area model?

A. $1,332 \div 74 = 20$

B. $1,332 \div 18 = 74$

C. $18 \times 50 = 1,332$

D. $900 \div 18 = 74$



Use an area model or equations to divide. Show your model or equations.

$$2,184 \div 28$$

Quotient: _____

True or false?

$$1,248 \div 24 = (960 \div 24) + (288 \div 24)$$

A. True

B. False

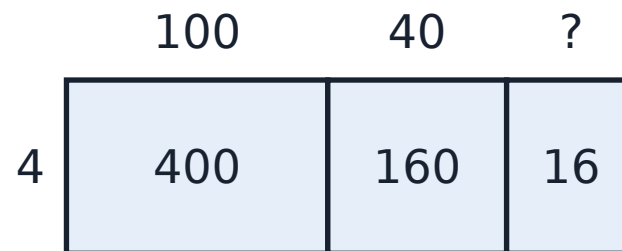
SET 18 • 5.NBT.B.6

BACKWARD 1

Area models for dividing by a one-digit number • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

The area model shows
 $576 \div 4$.

What is the quotient?



Quotient: _____

True or false?

$$468 \div 3 = (300 \div 3) + (150 \div 3) + (18 \div 3)$$

A. True

B. False

Which shows $245 \div 5$ split into easier parts?

- A.** $(200 \div 5) + (45 \div 5)$
- B.** $(245 \div 2) + (245 \div 3)$
- C.** $200 \div (5 + 45)$
- D.** $(200 \div 5) \times (45 \div 5)$

Divide.

$$735 \div 7$$

Quotient: _____

A rectangle has an area of 312 square units and a width of 6 units.

What is its length?

Length: _____

SET 18 • 5.NBT.B.6

BACKWARD 2

Area as length times width • Grade 3 • Measurement & Data • 3.MD.C.7.b

A rectangle is 9 units long and 7 units wide.
What is its area?

Area: _____

True or false?

A rectangle with an area of 40 square units and a side of 5 units has another side of 8 units.

A. True

B. False

Which equation gives the area of a 6-by-8 rectangle?

A. $8 - 6 = 2$

B. $6 + 8 = 14$

C. $6 \times 8 = 48$

D. $2 \times (6 + 8) = 28$

A rectangle has an area of 36 square units. One side is 4 units long.

How long is the other side?

Length: _____

Find the area of the rectangle.

Area: _____



SET 18 • 5.NBT.B.6

BACKWARD 3

Splitting a rectangle to find its area • Grade 3 • Measurement & Data •
3.MD.C.7.c

True or false?

The area of a rectangle that is 4 by $(10 + 3)$ equals $4 \times 10 + 4 \times 3$.

A. True

B. False

A 5-by-12 rectangle is split into a 5-by-10 rectangle and a 5-by-2 rectangle.

What is the total area?

Area: _____

A 7-by-13 rectangle is split into a 7-by-10 part and a 7-by-3 part.

Which expression gives its area?

A. $7 \times 10 + 7 \times 3$

B. $7 \times 10 + 3$

C. $7 + 10 + 3$

D. $7 \times 10 \times 3$

Fill in the blank.

$$6 \times 14 = 6 \times 10 + 6 \times \underline{\quad}$$

Blank: _____

True or false?

$$8 \times 12 = 8 \times 10 + 2$$

A. True

B. False

SET 18 • 5.NBT.B.6

FORWARD 1

Use the distributive property with the greatest common factor • Grade 6 • The Number System • 6.NS.B.4

Use the greatest common factor to write $36 + 48$ as a product.

Expression: _____

Find the greatest common factor of 32 and 56.

GCF: _____

Which expression is equal to $45 + 60$ and uses the greatest common factor?

- A.** $15(3 + 6)$
- B.** $15(3 + 4)$
- C.** $3(15 + 20)$
- D.** $5(9 + 12)$

Complete the equation.

$$24 + 40 = 8(\underline{\quad} + \underline{\quad})$$

First blank: _____

Second blank: _____

True or false?

$$18 + 27 = 9(2 + 3)$$

A. True

B. False

SET 18 • 5.NBT.B.6

FORWARD 2

Factor and expand linear expressions • Grade 7 • Expressions & Equations •
7.EE.A.1

Factor using the greatest common factor.

$$12x + 30$$

Expression: _____

Expand.

$$7(4y - 3)$$

Expression: _____

Which expression is equivalent to $18a - 24$?

- A.** $3(6a - 4)$
- B.** $6(3a - 24)$
- C.** $18(a - 6)$
- D.** $6(3a - 4)$

Factor out -4 .

$$-8n - 20$$

Expression: _____

True or false?

$$4(2.5x + 3) = 10x + 3$$

A. True

B. False

Draw division models and explain the strategy

MAIN	Grade 5 main question	5.NBT.B.6
BACKWARD 1	Area models for dividing by a one-digit number	4.NBT.B.6
BACKWARD 2	Division as an unknown-factor problem	3.OA.B.6
FORWARD 1	Connect the standard division algorithm to partial quotients	6.NS.B.2
FORWARD 2	Solve real-world problems by dividing rational numbers	7.NS.A.3

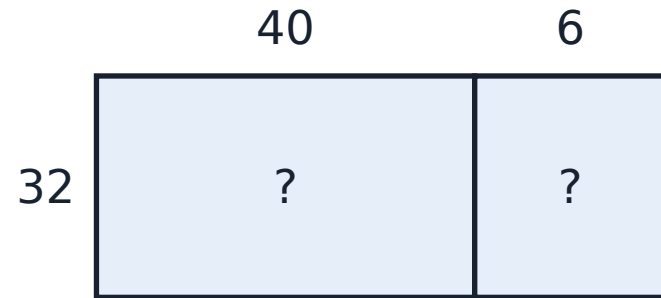
SET 60 • 5.NBT.B.6

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.6

Complete the area model for $1,472 \div 32$ by writing the area of each part.

Then write the quotient.



Quotient: _____

Draw an area model to divide $1,615 \div 19$.

Explain how your model shows the quotient.

Quotient: _____

Explanation: _____

Lena finds $1,380 \div 23$ by thinking $23 \times 60 = 1,380$.

Which statement explains her strategy?

- A.** She multiplied 1,380 by 23 to get 60.
- B.** She divided 23 by 1,380 to get 60.
- C.** Division finds the missing factor:
 $23 \times ? = 1,380$, so the quotient is 60.
- D.** She subtracted 23 from 1,380 to get 60.

Explain how these equations show $2,345 \div 35$.

$$35 \times 60 = 2,100$$

$$35 \times 7 = 245$$

$$2,100 + 245 = 2,345$$

Quotient: _____

Explanation: _____

True or false?

An area model for $1,512 \div 36$ can use parts with areas 1,440 and 72, so the quotient is $40 + 2 = 42$.

- A.** True
- B.** False

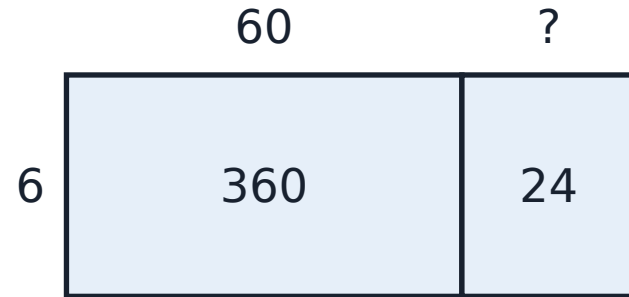
SET 60 • 5.NBT.B.6

BACKWARD 1

Area models for dividing by a one-digit number • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

The area model shows
 $384 \div 6$.

What is the missing length?



Length: _____

True or false?

An area model for $528 \div 4$ can be split into $400 \div 4$, $120 \div 4$, and $8 \div 4$.

- A. True
- B. False

An area model for $945 \div 9$ has parts with areas 900 and 45.

What are the lengths of the parts?

A. 100 and 5

B. 90 and 5

C. 10 and 5

D. 100 and 45

Divide.

$$672 \div 8$$

Quotient: _____

True or false?

In an area model for $945 \div 9$, the side length 9 is the quotient.

- A. True
- B. False

SET 60 • 5.NBT.B.6

BACKWARD 2

Division as an unknown-factor problem • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6

Find the unknown factor.

$$6 \times ? = 42$$

Factor: _____

True or false?

$32 \div 4 = 8$ because $4 \times 8 = 32$.

A. True

B. False

Which multiplication fact helps you find $56 \div 7$?

A. $8 + 7 = 15$

B. $7 \times 7 = 49$

C. $56 \times 7 = 392$

D. $7 \times 8 = 56$

Find the unknown factor.

$$? \times 9 = 81$$

Factor: _____

True or false?

To find $27 \div 3$, you can ask "27 times what number is 3?"

- A. True
- B. False

SET 60 • 5.NBT.B.6

FORWARD 1

Connect the standard division algorithm to partial quotients • Grade 6 • The Number System • 6.NS.B.2

Use the standard algorithm to divide. Show your work.

$$8,640 \div 36$$

Quotient: _____

In the standard algorithm for $5,712 \div 28$, the first step uses $28 \times 200 = 5,600$.

What does the 200 stand for?

- A.** Part of the quotient
- B.** The dividend
- C.** The remainder
- D.** Part of the divisor

True or false?

The steps of the standard algorithm for $1,836 \div 17$ match an area model with parts 17×100 and 17×8 .

A. True

B. False

Use the standard algorithm to divide. Show your work.

$$13,248 \div 46$$

Quotient: _____

True or false?

$$3,825 \div 45 = 95$$

A. True

B. False

SET 60 • 5.NBT.B.6

FORWARD 2

Solve real-world problems by dividing rational numbers • Grade 7 • The Number System • 7.NS.A.3

A diver goes down 126 feet in 18 equal stages.

Write the change in depth for each stage as a signed number.

Change: _____

A business lost \$4,320 over 24 months, the same amount each month.

Which number shows the change each month?

A. $-\$108$

B. $\$180$

C. $-\$1,800$

D. $-\$180$

The temperature fell 22.5°F in 9 hours at a steady rate.
Write the change per hour as a signed number.

Change: _____

True or false?

A change of $-\$1,512$ shared equally over 36 weeks is a change of $-\$42$ each week.

A. True

B. False

A tank loses $\frac{3}{4}$ gallon every hour.

After how many hours is the change in the amount of water -6 gallons?

Hours: _____

Add decimals to hundredths

MAIN

Grade 5 main question

5.NBT.B.7

BACKWARD 1 Adding multi-digit whole numbers

4.NBT.B.4

BACKWARD 2 Adding tenths and hundredths

4.NF.C.5

BACKWARD 3 Writing decimals with the same number of places

4.NF.C.6

FORWARD 1 Add multi-digit decimals

6.NS.B.3

FORWARD 2 Add rational numbers

7.NS.A.1.d

SET 19 • 5.NBT.B.7

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Add.

$$23.45 + 8.7$$

Sum: _____

Add.

$$15.3 + 6.48$$

Sum: _____

Add.

$$4.06 + 12.9$$

- A.** 16.15
- B.** 16.069
- C.** 5.35
- D.** 16.96

Add.

$$9.65 + 0.8$$

Sum: _____

Maya had \$18.50. She earned \$6.75 more.
How much money does she have now?

Money: _____

SET 19 • 5.NBT.B.7

BACKWARD 1

Adding multi-digit whole numbers • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4

Add.

$$2,345 + 870$$

Sum: _____

Add.

$$1,530 + 648$$

Sum: _____

True or false?

$$406 + 1,290 = 1,596$$

A. True

B. False

$$1,265 + 735 = ?$$

- A.** 2,100
- B.** 2,000
- C.** 1,900
- D.** 1,990

Add.

$$1,850 + 675$$

Sum: _____

SET 19 • 5.NBT.B.7

BACKWARD 2

Adding tenths and hundredths • Grade 4 • Number & Operations–Fractions •
4.NF.C.5

Add.

$$\frac{3}{10} + \frac{45}{100}$$

Sum: _____

True or false?

$$\frac{6}{10} + \frac{2}{100} = \frac{8}{100}$$

A. True

B. False

$$\frac{7}{10} + \frac{15}{100} = ?$$

A. $\frac{85}{100}$

B. $\frac{22}{100}$

C. $\frac{22}{110}$

D. $\frac{85}{10}$

Write $\frac{4}{10}$ as hundredths.

Fraction: _____

True or false?

$$\frac{5}{10} + \frac{5}{100} = \frac{10}{100}$$

A. True

B. False

SET 19 • 5.NBT.B.7

BACKWARD 3

Writing decimals with the same number of places • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

True or false?

$$8.7 = 8.70$$

A. True

B. False

Write 15.3 with two decimal places.

Decimal: _____

Which decimal is equal to 20?

- A.** 20.00
- B.** 0.20
- C.** 2.00
- D.** 200.0

True or false?

$$6.4 = 6.04$$

A. True

B. False

Write 12.9 with a digit in the hundredths place, without changing its value.

Decimal: _____

SET 19 • 5.NBT.B.7

FORWARD 1

Add multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Add.

$$128.47 + 9.605$$

Sum: _____

Add.

$$54.2 + 18.376$$

Sum: _____

Add.

$$0.952 + 3.48 + 11.6$$

A. 16.132

B. 4.53

C. 15.032

D. 16.032

Add.

$$99.6 + 0.485$$

Sum: _____

True or false?

$$7.3 + 2.145 = 2.875$$

A. True

B. False

SET 19 • 5.NBT.B.7

FORWARD 2

Add rational numbers • Grade 7 • The Number System • 7.NS.A.1.d

Find the value.

$$-12.5 + 4.75$$

Value: _____

Find the value.

$$3.2 + (-8.45)$$

Value: _____

Find the value.

$$-6.3 + (-2.85)$$

- A.** 9.15
- B.** -9.15
- C.** -3.45
- D.** 3.45

Find the value.

$$-4.6 + 10.25$$

Value: _____

True or false?

$$-1.5 + 1.5 = 0$$

A. True

B. False

Subtract decimals to hundredths

MAIN

Grade 5 main question

5.NBT.B.7

BACKWARD 1

Subtracting multi-digit whole numbers

4.NBT.B.4

BACKWARD 2

Subtracting mixed numbers in tenths and hundredths

4.NF.B.3.c

FORWARD 1

Subtract multi-digit decimals

6.NS.B.3

FORWARD 2

Subtract rational numbers

7.NS.A.1.d

SET 61 • 5.NBT.B.7

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Subtract.

$$15.3 - 6.48$$

Difference: _____

Subtract.

$$20 - 7.35$$

Difference: _____

Maya had \$18.50. She spent \$6.75.

How much money does she have left?

Money left: _____

Subtract.

$$9.04 - 3.7$$

A. 6.34

B. 5.34

C. 8.67

D. 5.44

True or false?

$$6.5 - 2.75 = 3.75$$

A. True

B. False

SET 61 • 5.NBT.B.7

BACKWARD 1

Subtracting multi-digit whole numbers • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4

Subtract.

$$1,530 - 648$$

Difference: _____

$$2,000 - 735 = ?$$

A. 2,735

B. 1,275

C. 1,265

D. 1,365

Subtract.

$$1,850 - 675$$

Difference: _____

True or false?

$$5,000 - 1,234 = 3,766$$

A. True

B. False

Subtract.

$$4,206 - 1,519$$

Difference: _____

SET 61 • 5.NBT.B.7

BACKWARD 2

Subtracting mixed numbers in tenths and hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.c

Subtract.

$$5\frac{3}{10} - 2\frac{6}{10}$$

Difference: _____

True or false?

$$4 - \frac{35}{100} = 3\frac{65}{100}$$

A. True

B. False

$$6\frac{2}{10} - 1\frac{9}{10} = ?$$

A. $5\frac{7}{10}$

B. $4\frac{7}{10}$

C. $4\frac{3}{10}$

D. $5\frac{3}{10}$

Subtract.

$$3\frac{40}{100} - 1\frac{25}{100}$$

Difference: _____

True or false?

$$2\frac{1}{10} - \frac{5}{10} = 2\frac{4}{10}$$

A. True

B. False

SET 61 • 5.NBT.B.7

FORWARD 1

Subtract multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Subtract.

$$54.2 - 18.376$$

Difference: _____

Subtract.

$$100 - 0.485$$

Difference: _____

True or false?

$$7.3 - 2.145 = 5.245$$

A. True

B. False

Subtract.

$$12.06 - 4.875$$

Difference: _____

Subtract.

$$30 - 0.009$$

A. 29.991

B. 30.009

C. 29.001

D. 29.91

SET 61 • 5.NBT.B.7

FORWARD 2

Subtract rational numbers • Grade 7 • The Number System • 7.NS.A.1.d

Find the value.

$$3.2 - 8.45$$

Value: _____

Find the value.

$$-4.6 - (-10.25)$$

Value: _____

Find the value.

$$-2.75 - 4.5$$

Value: _____

Find the value.

$$1.2 - (-3.65)$$

- A.** 2.45
- B.** -4.85
- C.** -2.45
- D.** 4.85

True or false?

$$-8.1 - (-8.1) = 0$$

A. True

B. False

Use place-value models to add and subtract decimals and explain the method

MAIN	Grade 5 main question	5.NBT.B.7
BACKWARD 1	Add tenths and hundredths	4.NF.C.5
BACKWARD 2	Rename a unit as ten of the next smaller unit	4.NBT.A.1
FORWARD 1	Add and subtract multi-digit decimals with the standard algorithm	6.NS.B.3
FORWARD 2	Add and subtract rational numbers	7.NS.A.1.d

SET 62 • 5.NBT.B.7

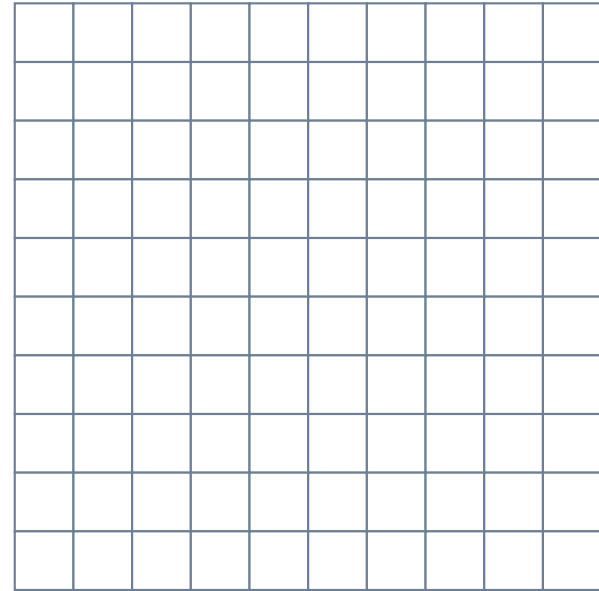
MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Shade the grid to show
 $0.4 + 0.25$.

Then write the sum.

Sum: _____



Explain why you line up the decimal points to add $3.6 + 2.45$.

Sum: _____

Explanation: _____

Which describes a place-value model for $1.3 - 0.45$?

- A.** Add 4 tenths and 5 hundredths to 1 one and 3 tenths.
- B.** Start with 1 one and 3 tenths, trade 1 tenth for 10 hundredths, then take away 4 tenths and 5 hundredths.
- C.** Take away 1 one and 3 tenths from 4 tenths and 5 hundredths.
- D.** Take away 4 ones and 5 tenths from 1 one and 3 tenths.

True or false?

To find $0.7 - 0.25$, you can rename 0.7 as 70 hundredths and subtract 25 hundredths to get 45 hundredths.

- A.** True
- B.** False

Raj added $4.2 + 1.35$ and got 1.77.

Find the correct sum and explain his mistake.

Correct sum: _____

Explanation: _____

SET 62 • 5.NBT.B.7

BACKWARD 1

Add tenths and hundredths • Grade 4 • Number & Operations–Fractions •
4.NF.C.5

Add.

$$\frac{4}{10} + \frac{25}{100}$$

Sum: _____

True or false?

$$\frac{3}{10} = \frac{30}{100}$$

A. True

B. False

$$\frac{8}{10} + \frac{7}{100} = ?$$

A. $\frac{87}{100}$

B. $\frac{15}{100}$

C. $\frac{15}{110}$

D. $\frac{87}{10}$

Write $\frac{6}{10}$ as hundredths.

Fraction: _____

True or false?

$$\frac{2}{10} + \frac{9}{100} = \frac{11}{100}$$

A. True

B. False

SET 62 • 5.NBT.B.7

BACKWARD 2

Rename a unit as ten of the next smaller unit • Grade 4 • Number & Operations
in Base Ten • 4.NBT.A.1

How many tens are equal to 1 hundred?

Tens: _____

True or false?

3 hundreds 4 tens can be renamed as 2 hundreds 14 tens.

A. True

B. False

Which shows 52 renamed so that you can take away 8 ones?

- A.** 6 tens 2 ones
- B.** 4 tens 12 ones
- C.** 4 tens 2 ones
- D.** 5 tens 12 ones

Rename 4 thousands as hundreds.

Hundreds: _____

True or false?

7 tens 3 ones = 6 tens 3 ones

A. True

B. False

SET 62 • 5.NBT.B.7

FORWARD 1

Add and subtract multi-digit decimals with the standard algorithm • Grade 6 •
The Number System • 6.NS.B.3

Add.

$$36.075 + 8.9$$

Sum: _____

Subtract.

$$41.2 - 7.865$$

Difference: _____

Add.

$$0.68 + 4.795$$

- A.** 4.863
- B.** 11.475
- C.** 5.375
- D.** 5.475

Subtract.

$$9 - 2.468$$

Difference: _____

True or false?

$$12.5 + 0.375 = 12.875$$

A. True

B. False

SET 62 • 5.NBT.B.7

FORWARD 2

Add and subtract rational numbers • Grade 7 • The Number System •
7.NS.A.1.d

Find the value.

$$-3.25 + 7.6$$

Value: _____

Find the value.

$$0.8 - 2.35$$

Value: _____

Find the value.

$$-1.75 + (-0.6)$$

- A.** 2.35
- B.** 1.15
- C.** -2.35
- D.** -1.15

True or false?

$$5.5 - 7.25 = -1.75$$

A. True

B. False

Find the value.

$$-9.4 - (-2.65)$$

Value: _____

Multiply decimals to hundredths

MAIN	Grade 5 main question	5.NBT.B.7
BACKWARD 1	Multiplying whole numbers	4.NBT.B.5
BACKWARD 2	Decimals as tenths and hundredths	4.NF.C.6
BACKWARD 4	Multiply a whole number by tenths or hundredths	4.NF.B.4.b
FORWARD 1	Multiply multi-digit decimals	6.NS.B.3
FORWARD 2	Multiply rational numbers	7.NS.A.2.c

SET 20 • 5.NBT.B.7

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Multiply.

$$3.4 \times 6$$

Product: _____

Multiply.

$$0.7 \times 0.5$$

Product: _____

Multiply.

$$2.5 \times 1.2$$

A. 3

B. 30

C. 3.7

D. 0.3

Multiply.

$$0.06 \times 8$$

Product: _____

A pen costs \$1.25.

How much do 7 pens cost?

Cost: _____

SET 20 • 5.NBT.B.7

BACKWARD 1

Multiplying whole numbers • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.5

Multiply.

$$34 \times 6$$

Product: _____

Multiply.

$$25 \times 12$$

Product: _____

True or false?

$$125 \times 7 = 785$$

A. True

B. False

$$18 \times 4 = ?$$

A. 72

B. 82

C. 48

D. 64

Multiply.

$$106 \times 8$$

Product: _____

SET 20 • 5.NBT.B.7

BACKWARD 2

Decimals as tenths and hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write 35 hundredths as a decimal.

Decimal: _____

True or false?

0.4 is 4 tenths.

A. True

B. False

Which decimal is 204 tenths?

- A.** 204
- B.** 20.4
- C.** 2.04
- D.** 0.204

How many hundredths are in 0.48?

Hundredths: _____

True or false?

0.06 is 6 tenths.

A. True

B. False

SET 20 • 5.NBT.B.7

BACKWARD 4

Multiply a whole number by tenths or hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.B.4.b

Multiply.

$$4 \times \frac{3}{10}$$

Product: _____

True or false?

$$6 \times \frac{5}{100} = \frac{30}{100}$$

A. True

B. False

$$3 \times \frac{7}{10} = ?$$

A. $\frac{7}{30}$

B. $\frac{21}{10}$

C. $\frac{10}{10}$

D. $\frac{21}{30}$

A cup holds $\frac{3}{10}$ liter.

How many liters do 5 cups hold? Write a fraction.

Liters: _____

True or false?

$$8 \times \frac{2}{100} = \frac{16}{800}$$

A. True

B. False

SET 20 • 5.NBT.B.7

FORWARD 1

Multiply multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Multiply.

$$3.45 \times 0.6$$

Product: _____

Multiply.

$$12.08 \times 2.5$$

Product: _____

Multiply.

$$0.125 \times 0.4$$

- A.** 0.5
- B.** 0.05
- C.** 5
- D.** 0.005

Multiply.

$$7.2 \times 1.15$$

Product: _____

True or false?

$$0.03 \times 0.03 = 0.09$$

A. True

B. False

SET 20 • 5.NBT.B.7

FORWARD 2

Multiply rational numbers • Grade 7 • The Number System • 7.NS.A.2.c

Find the product.

$$-3.4 \times 6$$

Product: _____

Find the product.

$$(-0.7) \times (-0.5)$$

Product: _____

Find the product.

$$2.5 \times (-1.2)$$

A. -0.3

B. -30

C. 3

D. -3

Find the product.

$$(-0.06) \times 8$$

Product: _____

True or false?

$$(-1.25) \times (-4) = -5$$

A. True

B. False

Use models and place value to multiply decimals and explain the method

MAIN	Grade 5 main question	5.NBT.B.7
BACKWARD 1	Multiply a fraction in tenths or hundredths by a whole number	4.NF.B.4.b
BACKWARD 2	Multiplication as equal groups	3.OA.A.1
FORWARD 1	Multiply multi-digit decimals	6.NS.B.3
FORWARD 2	Multiply rational numbers	7.NS.A.2.c

SET 63 • 5.NBT.B.7

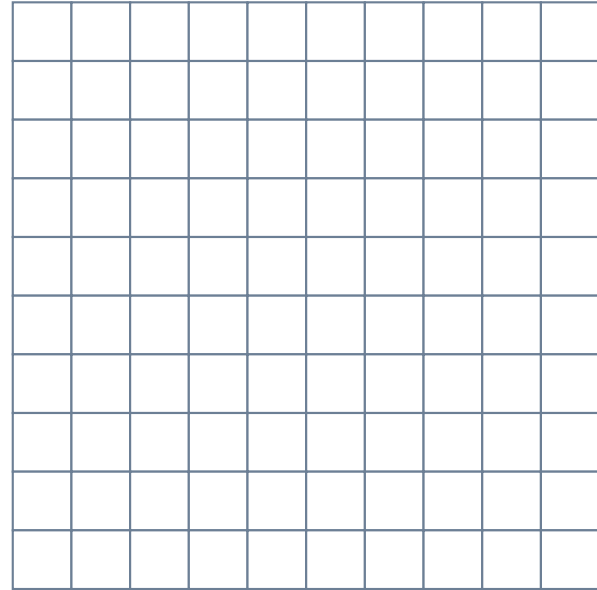
MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Shade the grid to show
 3×0.15 .

Then write the product.

Product: _____



On a hundredths grid, 0.3×0.4 is where 3 tenths (3 columns) and 4 tenths (4 rows) overlap.

Explain why the product is 0.12.

Explanation: _____

Which explains why $4 \times 0.25 = 1$?

- A.** 4 groups of 25 hundredths is 100 hundredths, which is 1 whole.
- B.** $4 + 0.25$ is about 4, so the product is about 1.
- C.** 0.25 has two decimal places, so the product is 0.01.
- D.** $4 \times 25 = 100$, so the product is 100.

True or false?

$2 \times 0.35 = 0.70$ because 2 groups of 35 hundredths is 70 hundredths.

A. True

B. False

Lia says $0.5 \times 0.6 = 3.0$.

Give the correct product and use place value to explain her mistake.

Correct product: _____

Explanation: _____

SET 63 • 5.NBT.B.7

BACKWARD 1

Multiply a fraction in tenths or hundredths by a whole number • Grade 4 •
Number & Operations–Fractions • 4.NF.B.4.b

Multiply.

$$3 \times \frac{15}{100}$$

Product: _____

True or false?

$$2 \times \frac{35}{100} = \frac{70}{100}$$

A. True

B. False

$$4 \times \frac{6}{10} = ?$$

A. $\frac{6}{40}$

B. $\frac{24}{40}$

C. $\frac{24}{10}$

D. $\frac{10}{10}$

Write $7 \times \frac{2}{10}$ as one fraction.

Fraction: _____

True or false?

$$6 \times \frac{4}{100} = \frac{24}{600}$$

A. True

B. False

SET 63 • 5.NBT.B.7

BACKWARD 2

Multiplication as equal groups • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.1

There are 3 groups with 15 counters in each group.
How many counters are there in all?

Counters: _____

True or false?

4×25 can mean 4 groups of 25.

A. True

B. False

Which can 2×35 mean?

- A.** 2 more than 35
- B.** 2 groups of 3 and 5
- C.** 35 take away 2
- D.** 2 groups of 35

Write a multiplication equation for 6 groups of 4.

Equation: _____

True or false?

3×8 means 3 more than 8.

A. True

B. False

SET 63 • 5.NBT.B.7

FORWARD 1

Multiply multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Multiply.

$$0.24 \times 0.5$$

Product: _____

Multiply.

$$3.6 \times 0.15$$

Product: _____

Multiply.

$$1.6 \times 0.45$$

A. 0.072

B. 0.72

C. 2.05

D. 7.2

Multiply.

$$0.08 \times 0.7$$

Product: _____

True or false?

$$2.5 \times 0.04 = 1$$

A. True

B. False

SET 63 • 5.NBT.B.7

FORWARD 2

Multiply rational numbers • Grade 7 • The Number System • 7.NS.A.2.c

Find the product.

$$(-0.3) \times 0.4$$

Product: _____

Find the product.

$$(-2.5) \times (-0.6)$$

Product: _____

Find the product.

$$3 \times (-0.15)$$

A. -0.45

B. 0.45

C. -0.045

D. -4.5

True or false?

$$(-0.2) \times (-0.2) \times (-0.2) = -0.008$$

A. True

B. False

Find the product.

$$-1.2 \times 0.05$$

Product: _____

Divide decimals to hundredths

MAIN	Grade 5 main question	5.NBT.B.7
BACKWARD 1	Dividing whole numbers	4.NBT.B.6
BACKWARD 2	Division as an unknown-factor problem	3.OA.B.6
BACKWARD 4	Rename decimals as tenths or hundredths	4.NF.C.6
FORWARD 1	Divide multi-digit decimals	6.NS.B.3
FORWARD 2	Divide rational numbers	7.NS.A.2.b

SET 21 • 5.NBT.B.7

MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Divide.

$$8.4 \div 4$$

Quotient: _____

Divide.

$$3.6 \div 0.6$$

Quotient: _____

Divide.

$$0.45 \div 0.05$$

A. 0.09

B. 0.9

C. 90

D. 9

Divide.

$$7.5 \div 0.3$$

Quotient: _____

A rope is 4.8 meters long. It is cut into pieces that are each 0.6 meter long.

How many pieces are there?

Pieces: _____

SET 21 • 5.NBT.B.7

BACKWARD 1

Dividing whole numbers • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Divide.

$$84 \div 4$$

Quotient: _____

Divide.

$$360 \div 6$$

Quotient: _____

True or false?

$$450 \div 5 = 9$$

A. True

B. False

$$750 \div 3 = ?$$

- A.** 25
- B.** 250
- C.** 2,500
- D.** 260

Divide.

$$48 \div 6$$

Quotient: _____

SET 21 • 5.NBT.B.7

BACKWARD 2

Division as an unknown-factor problem • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6

Find $36 \div 6$ by thinking $6 \times \square = 36$.

Quotient: _____

True or false?

$7 \times 9 = 63$, so $63 \div 7 = 9$.

A. True

B. False

Which fact helps you find $56 \div 8$?

A. $56 \times 8 = 448$

B. $8 \times 7 = 56$

C. $8 + 48 = 56$

D. $8 - 7 = 1$

What number goes in the box?

$$\square \times 5 = 45$$

Number: _____

True or false?

$24 \div 3 = 6$ because $3 \times 6 = 24$.

A. True

B. False

SET 21 • 5.NBT.B.7

BACKWARD 4

Rename decimals as tenths or hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

How many tenths are in 3.6?

Tenths: _____

True or false?

0.45 is 45 hundredths.

A. True

B. False

How many hundredths are in 0.05?

A. 500

B. 0.5

C. 5

D. 50

How many tenths are in 0.6?

Tenths: _____

True or false?

7.5 is 75 hundredths.

A. True

B. False

SET 21 • 5.NBT.B.7

FORWARD 1

Divide multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Divide.

$$13.65 \div 0.35$$

Quotient: _____

Divide.

$$9.072 \div 2.4$$

Quotient: _____

Divide.

$$0.081 \div 0.09$$

A. 90

B. 0.09

C. 0.9

D. 9

Divide.

$$52.5 \div 1.25$$

Quotient: _____

True or false?

$$1.8 \div 0.006 = 3$$

A. True

B. False

SET 21 • 5.NBT.B.7

FORWARD 2

Divide rational numbers • Grade 7 • The Number System • 7.NS.A.2.b

Find the quotient.

$$-8.4 \div 4$$

Quotient: _____

Find the quotient.

$$(-3.6) \div (-0.6)$$

Quotient: _____

Find the quotient.

$$0.45 \div (-0.05)$$

A. -9

B. -0.9

C. 0.9

D. 9

Find the quotient.

$$-7.5 \div 0.3$$

Quotient: _____

True or false?

$$-(4.8 \div 0.6) = (-4.8) \div 0.6$$

A. True

B. False

Use models and place value to divide decimals and explain the method

MAIN	Grade 5 main question	5.NBT.B.7
BACKWARD 1	Division as the number of equal groups	3.OA.A.2
BACKWARD 2	Decimals as tenths and hundredths	4.NF.C.6
FORWARD 1	Divide multi-digit decimals	6.NS.B.3
FORWARD 2	Divide rational numbers	7.NS.A.2.b

SET 64 • 5.NBT.B.7

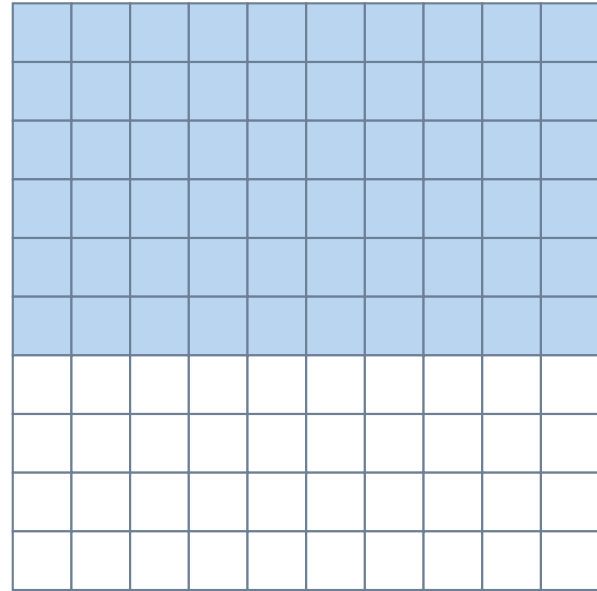
MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

The grid shows 0.6 shaded.

Circle groups of 0.15 to find $0.6 \div 0.15$. Then write the quotient.

Quotient: _____



Which explains why $1.8 \div 0.3 = 6$?

- A.** 0.3 has one decimal place, so the quotient is 0.6.
- B.** There are 6 groups of 1.8 in 0.3.
- C.** $1.8 - 0.3 = 6$
- D.** There are 6 groups of 3 tenths in 18 tenths.

Explain how to find $2.4 \div 4$ by thinking of 2.4 as 24 tenths.

Quotient: _____

Explanation: _____

True or false?

$4.5 \div 0.9 = 5$ because there are 5 groups of 9 tenths in 45 tenths.

A. True

B. False

Sam says $0.8 \div 0.2 = 0.4$.

Give the correct quotient and use a model or place value to explain why he is wrong.

Correct quotient: _____

Explanation: _____

SET 64 • 5.NBT.B.7

BACKWARD 1

Division as the number of equal groups • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2

18 counters are put into groups of 3.

How many groups are there?

Groups: _____

True or false?

$24 \div 4$ can mean "how many groups of 4 are in 24?"

- A. True
- B. False

Which question does $35 \div 5$ answer?

- A.** What is 35 take away 5?
- B.** What is 35 more than 5?
- C.** How many groups of 5 are in 35?
- D.** How many are in 5 groups of 35?

There are 28 apples. Each bag holds 7 apples.

How many bags are needed?

Bags: _____

True or false?

$12 \div 3$ means 12 groups of 3.

A. True

B. False

SET 64 • 5.NBT.B.7

BACKWARD 2

Decimals as tenths and hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

How many tenths are in 1.8?

Tenths: _____

True or false?

0.15 is 15 hundredths.

A. True

B. False

How many hundredths are in 0.6?

- A.** 60
- B.** 0.6
- C.** 600
- D.** 6

Write 45 tenths as a decimal.

Decimal: _____

True or false?

2.4 is 24 hundredths.

A. True

B. False

SET 64 • 5.NBT.B.7

FORWARD 1

Divide multi-digit decimals • Grade 6 • The Number System • 6.NS.B.3

Divide.

$$3.75 \div 0.25$$

Quotient: _____

Divide.

$$0.096 \div 0.12$$

Quotient: _____

Divide.

$$8.1 \div 0.09$$

A. 9

B. 90

C. 0.9

D. 900

Divide.

$$14.4 \div 1.2$$

Quotient: _____

True or false?

$$0.5 \div 0.02 = 2.5$$

A. True

B. False

SET 64 • 5.NBT.B.7

FORWARD 2

Divide rational numbers • Grade 7 • The Number System • 7.NS.A.2.b

Find the quotient.

$$-1.8 \div 0.3$$

Quotient: _____

Find the quotient.

$$(-0.6) \div (-0.15)$$

Quotient: _____

Find the quotient.

$$2.4 \div (-4)$$

A. 0.6

B. -6

C. -0.06

D. -0.6

True or false?

$$(-4.5) \div (-0.9) = -5$$

A. True

B. False

Find the quotient.

$$-0.08 \div 0.2$$

Quotient: _____

Solve real-world problems with decimal operations

MAIN	Grade 5 main question	5.NBT.B.7
BACKWARD 1	Word problems with money and measurement	4.MD.A.2
BACKWARD 2	Hundredths on a grid	4.NF.C.6
FORWARD 1	Real-world problems with multi-digit decimals	6.NS.B.3
FORWARD 2	Real-world problems with rational numbers	7.NS.A.3

SET 22 • 5.NBT.B.7

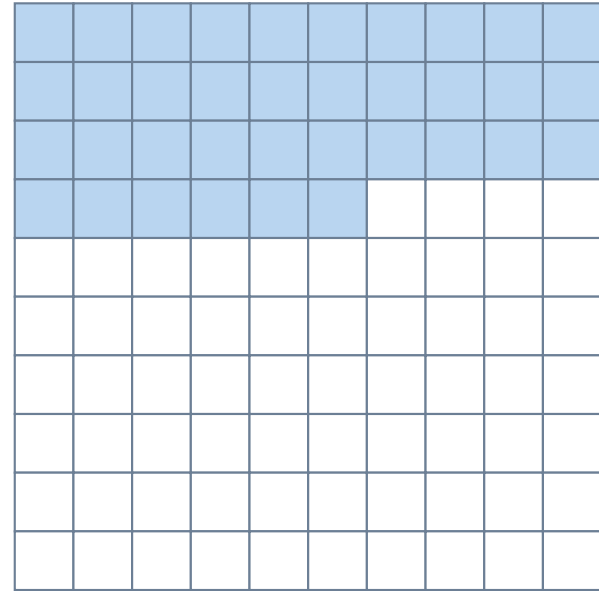
MAIN

Grade 5 main question • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

The grid shows 0.36 shaded.

How much more must be shaded to show 0.5?

Amount: _____



Juice costs \$2.35 a bottle. Ben buys 4 bottles and pays with a \$20 bill.

How much change does he get?

Change: _____

A 2.5-pound bag of nuts is shared equally among 5 friends.

How many pounds does each friend get?

- A.** 0.05 pound
- B.** 0.5 pound
- C.** 2 pounds
- D.** 12.5 pounds

A runner ran 3.75 km on Monday and 4.6 km on Tuesday.

How much farther did she run on Tuesday?

Distance: _____

Tiles cost \$0.45 each.

How much do 12 tiles cost?

Cost: _____

SET 22 • 5.NBT.B.7

BACKWARD 1

Word problems with money and measurement • Grade 4 • Measurement & Data • 4.MD.A.2

Pens cost 75 cents each.

How much do 4 pens cost, in dollars?

Cost: _____

A board is 120 cm long. It is cut into 4 equal pieces.
How long is each piece?

Length: _____

True or false?

Two quarters and three dimes are worth 65 cents.

A. True

B. False

Ann has \$5. She spends \$2.25.

How much money is left?

A. \$7.25

B. \$3.25

C. \$2.75

D. \$2.25

A jug holds 2 liters.

How many milliliters is that?

Milliliters: _____

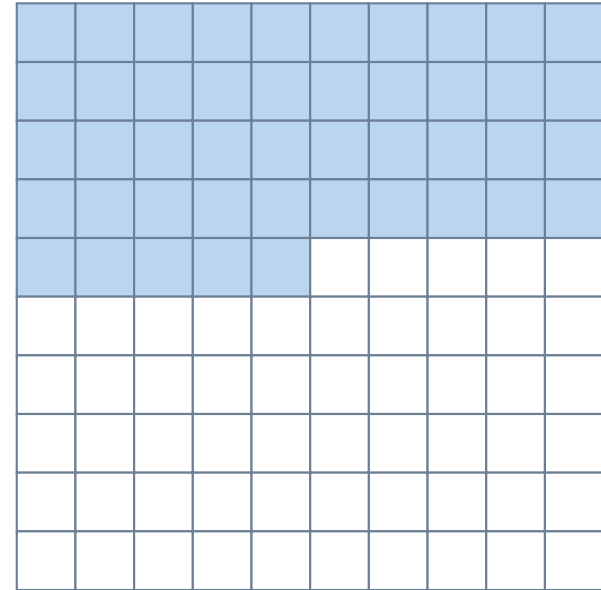
SET 22 • 5.NBT.B.7

BACKWARD 2

Hundredths on a grid • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

What decimal does the shaded part of the grid show?

Decimal: _____

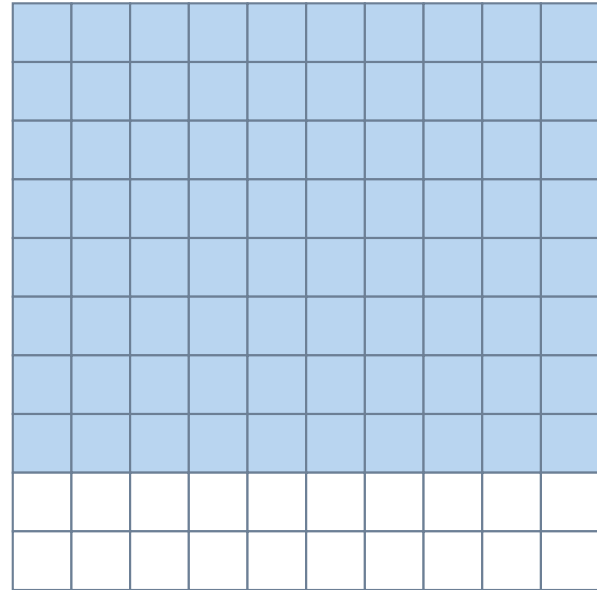


True or false?

The shaded part of the grid shows 0.8.

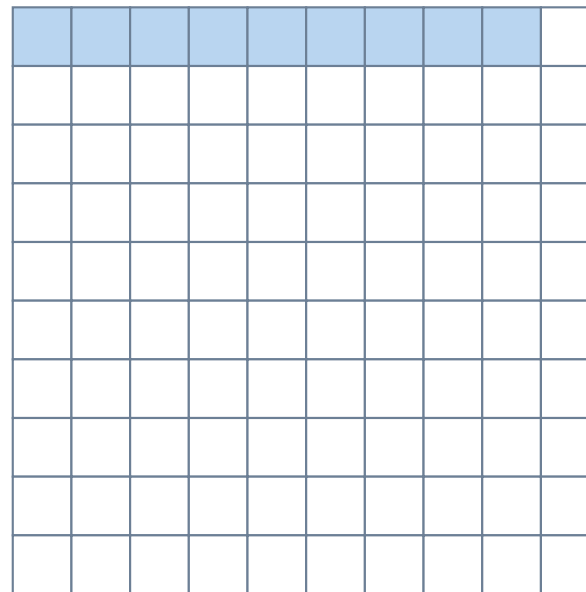
A. True

B. False



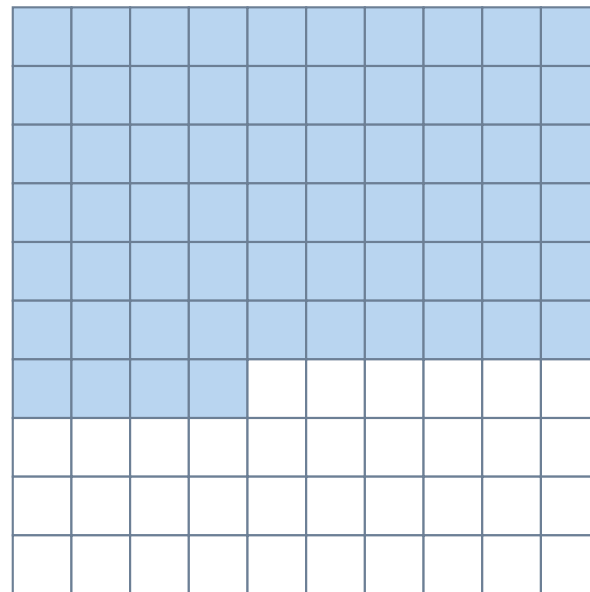
Which decimal does the shaded part of the grid show?

- A.** 0.9
- B.** 9
- C.** 0.009
- D.** 0.09



How many more squares
must be shaded to fill the
whole grid?

Squares: _____



True or false?

0.5 of a hundredths grid is 5 squares.

A. True

B. False

SET 22 • 5.NBT.B.7

FORWARD 1

Real-world problems with multi-digit decimals • Grade 6 • The Number System
• 6.NS.B.3

Gas costs \$3.85 per gallon.

How much do 12 gallons cost?

Cost: _____

A board is 6.75 m long. It is cut into pieces that are 0.45 m long.

How many pieces are there?

Pieces: _____

Rice costs \$1.20 per pound.

How much do 2.5 pounds cost?

- A.** \$3.00
- B.** \$30.00
- C.** \$2.70
- D.** \$3.70

Lee buys items that cost \$12.49, \$3.05, and \$7.60.

What is the total cost?

Total: _____

True or false?

A 2.4-pound bag split into 0.3-pound portions makes 8 portions.

- A. True
- B. False

SET 22 • 5.NBT.B.7

FORWARD 2

Real-world problems with rational numbers • Grade 7 • The Number System •
7.NS.A.3

A submarine at -45.5 m rises 12.75 m and then dives 20 m.

What is its final position?

Position: _____

A store's profit was $-\$125.40$ in May and $\$310.15$ in June.

What was the total profit for the two months?

Total: _____

The temperature was 4°F . It dropped 1.5°F each hour for 6 hours.

What was the final temperature?

A. -5°F

B. 13°F

C. 5°F

D. -9°F

An account changes by $-\$2.40$ each week for 5 weeks.
What is the total change?

Total change: _____

True or false?

$$-3.6 \div 0.4 = -0.9$$

A. True

B. False

Add fractions with unlike denominators

MAIN	Grade 5 main question	5.NF.A.1
BACKWARD 1	Equivalent fractions	4.NF.A.1
BACKWARD 2	Adding fractions with like denominators	4.NF.B.3.a
BACKWARD 3	Common multiples	4.OA.B.4
FORWARD 1	Find the least common multiple	6.NS.B.4
FORWARD 2	Add rational numbers	7.NS.A.1.d

SET 23 • 5.NF.A.1

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Add.

$$\frac{2}{3} + \frac{1}{4}$$

Sum: _____

Add.

$$\frac{3}{5} + \frac{1}{10}$$

Sum: _____

Add.

$$\frac{5}{6} + \frac{3}{8}$$

A. $1\frac{5}{24}$

B. $1\frac{1}{24}$

C. $\frac{8}{24}$

D. $\frac{8}{14}$

Add.

$$\frac{1}{2} + \frac{2}{7}$$

Sum: _____

Add.

$$\frac{3}{4} + \frac{5}{12} + \frac{1}{6}$$

Sum: _____

SET 23 • 5.NF.A.1

BACKWARD 1

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Find the missing number.

$$\frac{2}{3} = \frac{?}{12}$$

Missing number: _____

True or false?

$$\frac{1}{4} = \frac{4}{12}$$

A. True

B. False

Which fraction is equivalent to $\frac{3}{5}$?

A. $\frac{6}{5}$

B. $\frac{5}{3}$

C. $\frac{3}{10}$

D. $\frac{6}{10}$

Find the missing number.

$$\frac{5}{6} = \frac{20}{?}$$

Missing number: _____

True or false?

$$\frac{2}{7} = \frac{4}{21}$$

A. True

B. False

SET 23 • 5.NF.A.1

BACKWARD 2

Adding fractions with like denominators • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.a

Add.

$$\frac{3}{8} + \frac{2}{8}$$

Sum: _____

True or false?

$$\frac{4}{12} + \frac{5}{12} = \frac{9}{24}$$

A. True

B. False

$$\frac{7}{10} + \frac{6}{10} = ?$$

A. $1\frac{6}{10}$

B. $1\frac{3}{10}$

C. $\frac{13}{20}$

D. $\frac{1}{10}$

Add.

$$\frac{1}{6} + \frac{4}{6}$$

Sum: _____

True or false?

$$\frac{2}{5} + \frac{3}{5} = 1$$

A. True

B. False

SET 23 • 5.NF.A.1

BACKWARD 3

Common multiples • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List the first five positive multiples of 4.

Multiples: _____

True or false?

10 is a multiple of both 3 and 4.

A. True

B. False

Which number is a multiple of both 6 and 8?

A. 16

B. 30

C. 24

D. 14

Name a positive number that is a multiple of both 5 and 10.

Number: _____

True or false?

18 is a multiple of both 4 and 6.

A. True

B. False

SET 23 • 5.NF.A.1

FORWARD 1

Find the least common multiple • Grade 6 • The Number System • 6.NS.B.4

Find the least common multiple of 6 and 10.

LCM: _____

Find the least common multiple of 8 and 12.

LCM: _____

What is the least common multiple of 4 and 9?

- A.** 72
- B.** 13
- C.** 18
- D.** 36

Find the least common multiple of 5 and 15.

LCM: _____

True or false?

The least common multiple of 6 and 9 is 54.

A. True

B. False

SET 23 • 5.NF.A.1

FORWARD 2

Add rational numbers • Grade 7 • The Number System • 7.NS.A.1.d

Find the value.

$$-\frac{2}{3} + \frac{1}{4}$$

Value: _____

Find the value.

$$\frac{3}{5} + \left(-\frac{7}{10}\right)$$

Value: _____

Find the value.

$$-\frac{1}{2} + \left(-\frac{3}{8}\right)$$

A. $-\frac{7}{8}$

B. $-\frac{4}{10}$

C. $\frac{1}{8}$

D. $-\frac{1}{8}$

Find the value.

$$-1\frac{1}{3} + \frac{5}{6}$$

Value: _____

True or false?

$$-\frac{3}{4} + \frac{3}{4} = 0$$

A. True

B. False

Subtract fractions with unlike denominators

MAIN	Grade 5 main question	5.NF.A.1
BACKWARD 1	Equivalent fractions	4.NF.A.1
BACKWARD 2	Subtracting fractions with like denominators	4.NF.B.3.a
BACKWARD 3	Common multiples	4.OA.B.4
FORWARD 1	Solve equations of the form $x + p = q$ with fractions	6.EE.B.7
FORWARD 2	Subtract rational numbers	7.NS.A.1.c

SET 24 • 5.NF.A.1

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Subtract.

$$\frac{5}{6} - \frac{1}{4}$$

Difference: _____

Subtract.

$$\frac{7}{8} - \frac{1}{2}$$

Difference: _____

Subtract.

$$\frac{4}{5} - \frac{2}{3}$$

A. $\frac{2}{2}$

B. $\frac{1}{5}$

C. $\frac{6}{15}$

D. $\frac{2}{15}$

Subtract.

$$\frac{9}{10} - \frac{2}{5}$$

Difference: _____

Subtract.

$$1 - \frac{3}{7}$$

Difference: _____

SET 24 • 5.NF.A.1

BACKWARD 1

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Find the missing number.

$$\frac{1}{4} = \frac{?}{12}$$

Missing number: _____

True or false?

$$\frac{1}{2} = \frac{4}{8}$$

A. True

B. False

Which fraction is equivalent to $\frac{2}{3}$?

A. $\frac{10}{15}$

B. $\frac{4}{9}$

C. $\frac{3}{4}$

D. $\frac{2}{6}$

Find the missing number.

$$\frac{4}{5} = \frac{?}{15}$$

Missing number: _____

True or false?

$$\frac{3}{4} = \frac{9}{16}$$

A. True

B. False

SET 24 • 5.NF.A.1

BACKWARD 2

Subtracting fractions with like denominators • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.a

Subtract.

$$\frac{7}{9} - \frac{2}{9}$$

Difference: _____

True or false?

$$\frac{5}{8} - \frac{3}{8} = \frac{2}{8}$$

A. True

B. False

$$\frac{11}{12} - \frac{4}{12} = ?$$

A. $\frac{7}{24}$

B. $\frac{7}{12}$

C. $\frac{15}{12}$

D. $\frac{4}{12}$

Subtract.

$$1 - \frac{3}{5}$$

Difference: _____

True or false?

$$\frac{9}{10} - \frac{4}{10} = \frac{5}{20}$$

A. True

B. False

SET 24 • 5.NF.A.1

BACKWARD 3

Common multiples • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List the first four positive multiples of 6.

Multiples: _____

True or false?

15 is a multiple of both 3 and 5.

A. True

B. False

Which number is a multiple of both 4 and 5?

A. 15

B. 20

C. 9

D. 25

Name a positive number that is a multiple of both 3 and 8.

Number: _____

True or false?

16 is a multiple of both 4 and 6.

A. True

B. False

SET 24 • 5.NF.A.1

FORWARD 1

Solve equations of the form $x + p = q$ with fractions • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

$$x + \frac{1}{3} = \frac{5}{6}$$

x = _____

Solve for y.

$$y - \frac{2}{5} = \frac{3}{10}$$

$$y = \underline{\hspace{10cm}}$$

Solve for n.

$$\frac{3}{4} + n = 1\frac{1}{2}$$

A. $1\frac{1}{4}$

B. $\frac{3}{4}$

C. $2\frac{1}{4}$

D. $\frac{1}{4}$

Solve for m.

$$m + \frac{5}{8} = 2$$

$$m = \underline{\hspace{10cm}}$$

True or false?

The solution of $x - \frac{1}{6} = \frac{2}{3}$ is $x = \frac{5}{6}$.

A. True

B. False

SET 24 • 5.NF.A.1

FORWARD 2

Subtract rational numbers • Grade 7 • The Number System • 7.NS.A.1.c

Find the value.

$$\frac{1}{4} - \frac{5}{6}$$

Value: _____

Find the value.

$$-\frac{2}{3} - \frac{1}{6}$$

Value: _____

Find the value.

$$\frac{1}{2} - \left(-\frac{3}{4}\right)$$

A. $-\frac{1}{4}$

B. $\frac{1}{4}$

C. $1\frac{1}{4}$

D. $-1\frac{1}{4}$

Find the value.

$$-\frac{3}{5} - (-\frac{1}{2})$$

Value: _____

True or false?

$$\frac{2}{9} - \frac{5}{9} = \frac{1}{3}$$

A. True

B. False

Add mixed numbers with unlike denominators

MAIN	Grade 5 main question	5.NF.A.1
BACKWARD 1	Adding mixed numbers with like denominators	4.NF.B.3.c
BACKWARD 2	Mixed numbers and fractions greater than 1	4.NF.B.3.b
BACKWARD 3	Equivalent fractions	4.NF.A.1
FORWARD 1	Solve equations of the form $x + p = q$ with mixed numbers	6.EE.B.7
FORWARD 2	Solve equations of the form $px + q = r$ with fractions	7.EE.B.4.a

SET 25 • 5.NF.A.1

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Add.

$$2\frac{1}{3} + 1\frac{1}{2}$$

Sum: _____

Add.

$$4\frac{3}{4} + 2\frac{1}{6}$$

Sum: _____

Add.

$$1\frac{2}{5} + 3\frac{1}{2}$$

A. $4\frac{3}{7}$

B. $4\frac{3}{10}$

C. $5\frac{1}{10}$

D. $4\frac{9}{10}$

Add.

$$3\frac{5}{8} + 2\frac{3}{4}$$

Sum: _____

Add.

$$5\frac{2}{3} + 1\frac{5}{9}$$

Sum: _____

SET 25 • 5.NF.A.1

BACKWARD 1

Adding mixed numbers with like denominators • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.c

Add.

$$1\frac{2}{5} + 2\frac{1}{5}$$

Sum: _____

Add.

$$4\frac{1}{6} + 1\frac{4}{6}$$

Sum: _____

True or false?

$$2\frac{3}{4} + 1\frac{3}{4} = 3\frac{6}{8}$$

A. True

B. False

$$2\frac{2}{3} + 1\frac{2}{3} = ?$$

A. $3\frac{1}{3}$

B. $3\frac{2}{3}$

C. $4\frac{1}{3}$

D. $4\frac{2}{3}$

Add.

$$3\frac{7}{8} + \frac{3}{8}$$

Sum: _____

SET 25 • 5.NF.A.1

BACKWARD 2

Mixed numbers and fractions greater than 1 • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.b

Write $2\frac{3}{4}$ as a fraction.

Fraction: _____

Write $\frac{17}{5}$ as a mixed number.

Mixed number: _____

True or false?

$$1\frac{1}{3} = \frac{4}{3}$$

A. True

B. False

Which fraction is equal to $3\frac{1}{2}$?

A. $\frac{31}{2}$

B. $\frac{7}{2}$

C. $\frac{5}{2}$

D. $\frac{6}{2}$

True or false?

$$5\frac{2}{3} = 4\frac{5}{3}$$

A. True

B. False

SET 25 • 5.NF.A.1

BACKWARD 3

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Find the missing number.

$$\frac{3}{4} = \frac{?}{8}$$

Missing number: _____

True or false?

$$\frac{1}{6} = \frac{2}{12}$$

A. True

B. False

Which fraction is equivalent to $\frac{1}{2}$?

A. $\frac{5}{10}$

B. $\frac{2}{3}$

C. $\frac{2}{2}$

D. $\frac{1}{4}$

Find the missing number.

$$\frac{2}{5} = \frac{?}{10}$$

Missing number: _____

True or false?

$$\frac{5}{6} = \frac{10}{18}$$

A. True

B. False

SET 25 • 5.NF.A.1

FORWARD 1

Solve equations of the form $x + p = q$ with mixed numbers • Grade 6 •
Expressions & Equations • 6.EE.B.7

Solve for x.

$$x + 2\frac{1}{2} = 6\frac{1}{4}$$

x = _____

Solve for y.

$$y - 1\frac{2}{3} = 3\frac{1}{6}$$

y = _____

Solve for n.

$$n + 4\frac{3}{8} = 7$$

A. $3\frac{5}{8}$

B. $2\frac{3}{8}$

C. $11\frac{3}{8}$

D. $2\frac{5}{8}$

A board was $8\frac{1}{2}$ feet long. After a piece was cut off, it was $5\frac{3}{4}$ feet long.

Write and solve an equation to find the length p of the piece.

Equation and solution: _____

True or false?

The solution of $x - 1\frac{1}{4} = 2\frac{1}{2}$ is $x = 1\frac{1}{4}$.

A. True

B. False

SET 25 • 5.NF.A.1

FORWARD 2

Solve equations of the form $px + q = r$ with fractions • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Solve for x.

$$2x + \frac{1}{2} = 3\frac{1}{2}$$

x = _____

Solve for x.

$$\frac{1}{3}x - 1\frac{1}{4} = 2\frac{3}{4}$$

x = _____

Solve for x.

$$4x - 2\frac{1}{3} = 1$$

A. $1\frac{1}{12}$

B. $\frac{1}{6}$

C. $\frac{5}{6}$

D. $\frac{1}{3}$

Jo bought 3 bags of flour that weigh the same and one $1\frac{1}{2}$ -pound bag of sugar. The total weight was $9\frac{3}{4}$ pounds.

How much does each bag of flour weigh?

Weight: _____

True or false?

The solution of $\frac{1}{2}(x + 3) = 4$ is $x = 5$.

A. True

B. False

Subtract mixed numbers with unlike denominators

MAIN	Grade 5 main question	5.NF.A.1
BACKWARD 1	Subtracting mixed numbers with like denominators	4.NF.B.3.c
BACKWARD 2	Rename a mixed number by decomposing a whole	4.NF.B.3.b
FORWARD 1	Solve equations of the form $x + p = q$ with mixed numbers	6.EE.B.7
FORWARD 2	Subtract rational numbers	7.NS.A.1.c

SET 65 • 5.NF.A.1

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Subtract.

$$5\frac{3}{4} - 2\frac{1}{6}$$

Difference: _____

Subtract.

$$4\frac{1}{5} - 1\frac{1}{2}$$

A. $3\frac{3}{10}$

B. $2\frac{7}{10}$

C. $3\frac{7}{10}$

D. $2\frac{3}{10}$

Subtract.

$$6 - 2\frac{2}{3}$$

Difference: _____

Subtract.

$$7\frac{1}{4} - 3\frac{5}{6}$$

Difference: _____

True or false?

$$3\frac{1}{2} - 1\frac{2}{3} = 1\frac{5}{6}$$

A. True

B. False

SET 65 • 5.NF.A.1

BACKWARD 1

Subtracting mixed numbers with like denominators • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.c

Subtract.

$$4\frac{5}{6} - 1\frac{4}{6}$$

Difference: _____

$$5\frac{1}{3} - 2\frac{2}{3} = ?$$

A. $3\frac{1}{3}$

B. $3\frac{2}{3}$

C. $2\frac{2}{3}$

D. $2\frac{1}{3}$

Subtract.

$$6\frac{1}{4} - 3\frac{3}{4}$$

Difference: _____

True or false?

$$7\frac{2}{5} - 4\frac{4}{5} = 3\frac{2}{5}$$

A. True

B. False

Subtract.

$$5 - 1\frac{3}{8}$$

Difference: _____

SET 65 • 5.NF.A.1

BACKWARD 2

Rename a mixed number by decomposing a whole • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.b

Fill in the blank.

$$3\frac{1}{4} = 2\frac{?}{4}$$

Blank: _____

True or false?

$$5\frac{2}{3} = 4\frac{5}{3}$$

A. True

B. False

Which is equal to $4\frac{1}{8}$?

A. $4\frac{9}{8}$

B. $5\frac{1}{8}$

C. $3\frac{1}{8}$

D. $3\frac{9}{8}$

Fill in the blank.

$$6 = 5\frac{?}{5}$$

Blank: _____

True or false?

$$2\frac{3}{10} = 1\frac{3}{10}$$

A. True

B. False

SET 65 • 5.NF.A.1

FORWARD 1

Solve equations of the form $x + p = q$ with mixed numbers • Grade 6 •
Expressions & Equations • 6.EE.B.7

Solve for x.

$$x + 3\frac{2}{3} = 7\frac{1}{4}$$

x = _____

Solve for y.

$$9\frac{1}{2} - y = 4\frac{5}{6}$$

y = _____

Solve for n.

$$n + 1\frac{5}{8} = 4\frac{1}{4}$$

A. $2\frac{5}{8}$

B. $2\frac{3}{8}$

C. $3\frac{5}{8}$

D. $5\frac{7}{8}$

A tank had $10\frac{1}{4}$ gallons of water. After some water was used, $6\frac{2}{3}$ gallons were left.

Write and solve an equation to find the amount w used.

Equation and solution: _____

True or false?

The solution of $x + 2\frac{3}{4} = 5$ is $x = 2\frac{1}{4}$.

A. True

B. False

SET 65 • 5.NF.A.1

FORWARD 2

Subtract rational numbers • Grade 7 • The Number System • 7.NS.A.1.c

Find the value.

$$2\frac{1}{4} - 5\frac{1}{2}$$

Value: _____

Find the value.

$$-1\frac{2}{3} - 2\frac{1}{6}$$

Value: _____

Find the value.

$$3\frac{1}{2} - (-1\frac{3}{4})$$

A. $1\frac{3}{4}$

B. $-1\frac{3}{4}$

C. $5\frac{1}{4}$

D. $-5\frac{1}{4}$

Find the value.

$$-4\frac{1}{3} - (-1\frac{5}{6})$$

Value: _____

True or false?

$$1\frac{1}{4} - 3 = -1\frac{3}{4}$$

A. True

B. False

Solve word problems by adding and subtracting fractions

MAIN	Grade 5 main question	5.NF.A.2
BACKWARD 1	Word problems with like denominators	4.NF.B.3.d
BACKWARD 2	Choosing addition or subtraction	2.OA.A.1
BACKWARD 4	Equivalent fractions	4.NF.A.1
FORWARD 1	Write and solve equations of the form $x + p = q$	6.EE.B.7
FORWARD 2	Write and solve equations of the form $px + q = r$	7.EE.B.4.a

SET 26 • 5.NF.A.2

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.A.2

Mia walked $\frac{3}{4}$ mile to school and then $\frac{2}{3}$ mile to the park.

How far did she walk in all?

Distance: _____

A jug had $\frac{7}{8}$ gallon of milk. Ben drank $\frac{1}{3}$ gallon.

How much milk is left?

Milk left: _____

Ana used $2\frac{1}{2}$ cups of flour and $1\frac{3}{4}$ cups of sugar.

How many cups did she use in all?

A. $4\frac{1}{4}$ cups

B. $3\frac{3}{4}$ cups

C. $4\frac{3}{4}$ cups

D. $3\frac{1}{4}$ cups

A ribbon is $5\frac{1}{6}$ feet long. Lea cuts off $2\frac{1}{2}$ feet.

How long is the ribbon now?

Length: _____

Tom read $\frac{1}{4}$ of a book on Monday and $\frac{3}{8}$ of the book on Tuesday.

What fraction of the book does he still have to read?

Fraction: _____

SET 26 • 5.NF.A.2

BACKWARD 1

Word problems with like denominators • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.d

Kim ate $\frac{3}{8}$ of a pizza and Joe ate $\frac{2}{8}$ of the pizza.

What fraction of the pizza did they eat in all?

Fraction: _____

A rope is $\frac{9}{10}$ meter long. $\frac{4}{10}$ meter is cut off.

How much rope is left?

Rope left: _____

True or false?

$\frac{2}{6}$ of a garden is corn and $\frac{3}{6}$ is beans. Corn and beans cover $\frac{5}{6}$ of the garden.

A. True

B. False

Lu had $\frac{7}{12}$ of a cake. She gave away $\frac{5}{12}$ of the cake.

How much of the cake is left?

A. $\frac{12}{12}$

B. $\frac{2}{12}$

C. $\frac{35}{12}$

D. $\frac{2}{24}$

A tank is $\frac{3}{5}$ full. Water is added to fill another $\frac{1}{5}$ of the tank.

How full is the tank now?

Fraction: _____

SET 26 • 5.NF.A.2

BACKWARD 2

Choosing addition or subtraction • Grade 2 • Operations & Algebraic Thinking •
2.OA.A.1

A class has 14 boys and 13 girls.

Which operation finds the number of students in all?

- A.** Subtraction
- B.** Division
- C.** Addition
- D.** Multiplication

True or false?

To find how much is left after some is used, you subtract.

A. True

B. False

Sam had 42 marbles. He lost 15 marbles.
How many marbles are left?

Marbles: _____

A shelf holds 35 books. 18 books are on it now.

Which equation finds how many more books fit?

A. $35 - 18 = \square$

B. $18 - 35 = \square$

C. $35 \times 18 = \square$

D. $35 + 18 = \square$

Liz has 26 stickers. She gets 17 more.

How many stickers does she have now?

Stickers: _____

SET 26 • 5.NF.A.2

BACKWARD 4

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Find the missing number.

$$\frac{3}{4} = \frac{?}{12}$$

Missing number: _____

True or false?

$$\frac{2}{3} = \frac{8}{12}$$

A. True

B. False

Which fraction is equivalent to $\frac{5}{6}$?

A. $\frac{15}{16}$

B. $\frac{10}{12}$

C. $\frac{5}{12}$

D. $\frac{6}{5}$

Rewrite $\frac{1}{2}$ and $\frac{1}{3}$ as fractions with the denominator 6.

$$\frac{1}{2} = \underline{\hspace{10cm}}$$

$$\frac{1}{3} = \underline{\hspace{10cm}}$$

True or false?

$$\frac{3}{8} = \frac{6}{24}$$

A. True

B. False

SET 26 • 5.NF.A.2

FORWARD 1

Write and solve equations of the form $x + p = q$ • Grade 6 • Expressions & Equations • 6.EE.B.7

After using $\frac{2}{3}$ cup of oil, Ray has $\frac{3}{4}$ cup left.

Write and solve an equation to find how much oil c he started with.

Equation and solution: _____

A plant grew $\frac{5}{8}$ inch and is now $4\frac{1}{4}$ inches tall.

Write and solve an equation to find its starting height h .

Equation and solution: _____

$\frac{3}{10}$ inch of rain fell on day 1. In all, $\frac{9}{10}$ inch fell on days 1 and 2.

Which equation finds the rain r on day 2?

A. $\frac{3}{10}r = \frac{9}{10}$

B. $r + \frac{9}{10} = \frac{3}{10}$

C. $\frac{3}{10} + r = \frac{9}{10}$

D. $r - \frac{3}{10} = \frac{9}{10}$

Solve for x.

$$x + \frac{2}{9} = \frac{2}{3}$$

x = _____

True or false?

If $n - \frac{1}{4} = \frac{5}{8}$, then $n = \frac{7}{8}$.

A. True

B. False

SET 26 • 5.NF.A.2

FORWARD 2

Write and solve equations of the form $px + q = r$ • Grade 7 • Expressions & Equations • 7.EE.B.4.a

A recipe uses $\frac{3}{4}$ cup of sugar per batch, plus $\frac{1}{2}$ cup for frosting. Ali used $2\frac{3}{4}$ cups of sugar in all.

How many batches did he make?

Batches: _____

Solve for x.

$$\frac{2}{3}x + \frac{1}{6} = \frac{5}{6}$$

x = _____

A 6-foot board is cut into 4 equal pieces, with $\frac{1}{2}$ foot left over.

How long is each piece?

A. $1\frac{1}{2}$ feet

B. $1\frac{5}{8}$ feet

C. $1\frac{3}{8}$ feet

D. $5\frac{1}{2}$ feet

Solve for x.

$$5x - \frac{3}{4} = 4\frac{1}{4}$$

x = _____

True or false?

The solution of $2x + \frac{1}{3} = 1$ is $x = \frac{2}{3}$.

A. True

B. False

Use benchmark fractions to estimate and to check answers

MAIN	Grade 5 main question	5.NF.A.2
BACKWARD 1	Comparing fractions to $\frac{1}{2}$ and 1	4.NF.A.2
BACKWARD 2	Fractions on a number line	3.NF.A.2
FORWARD 1	Reason about the size of a quotient of fractions	6.NS.A.1
FORWARD 2	Use estimation to judge whether an answer is reasonable	7.EE.B.3

SET 27 • 5.NF.A.2

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.A.2

True or false?

Ella says $\frac{3}{4} + \frac{1}{6} = \frac{4}{10}$. Her answer must be wrong, because $\frac{4}{10}$ is less than $\frac{3}{4}$, one of the fractions she added.

A. True

B. False

Which is the best estimate of $\frac{7}{8} + \frac{5}{12}$?

- A.** About $\frac{1}{2}$
- B.** About $2\frac{1}{2}$
- C.** About $1\frac{1}{2}$
- D.** About 3

Is $\frac{3}{5} + \frac{4}{9}$ more or less than 1?

Use benchmark fractions to explain.

Answer: _____

Explanation: _____

Jen ran $\frac{9}{10}$ mile and walked $\frac{1}{8}$ mile.

Estimate the total distance to the nearest whole mile.

Estimate: _____

Leo says $\frac{5}{6} - \frac{1}{3} = \frac{4}{3}$.

Which statement explains why his answer is not reasonable?

- A.** The answer is greater than $\frac{5}{6}$, but subtracting should give less than $\frac{5}{6}$.
- B.** The denominators are different.
- C.** $\frac{4}{3}$ is less than 1.
- D.** The answer should be 0.

SET 27 • 5.NF.A.2

BACKWARD 1

Comparing fractions to $\frac{1}{2}$ and 1 • Grade 4 • Number & Operations–Fractions •
4.NF.A.2

True or false?

$\frac{5}{8}$ is less than $\frac{1}{2}$.

A. True

B. False

Which fraction is greater than $\frac{1}{2}$?

A. $\frac{1}{3}$

B. $\frac{3}{8}$

C. $\frac{2}{5}$

D. $\frac{4}{7}$

Is $\frac{5}{9}$ closest to 0, $\frac{1}{2}$, or 1?

Answer: _____

True or false?

$\frac{6}{10}$ is less than $\frac{1}{2}$.

A. True

B. False

Is $\frac{11}{12}$ closer to $\frac{1}{2}$ or to 1?

Answer: _____

SET 27 • 5.NF.A.2

BACKWARD 2

Fractions on a number line • Grade 3 • Number & Operations–Fractions •
3.NF.A.2

What fraction is at point A?



A = _____

True or false?

On a number line, $\frac{1}{2}$ is halfway between 0 and 1.

A. True

B. False

Which point is at $\frac{1}{3}$?



A. Point A

B. Point B

C. Point C

D. Point D

What fraction is at point B?



B = _____

True or false?

$\frac{7}{8}$ is closer to 1 than to $\frac{1}{2}$.

A. True

B. False

SET 27 • 5.NF.A.2

FORWARD 1

Reason about the size of a quotient of fractions • Grade 6 • The Number System • 6.NS.A.1

True or false?

$\frac{3}{4} \div \frac{1}{8}$ is greater than 1.

A. True

B. False

Without dividing, which quotient is less than 1?

A. $\frac{3}{4} \div \frac{1}{2}$

B. $\frac{1}{3} \div \frac{1}{2}$

C. $2 \div \frac{1}{2}$

D. $\frac{5}{6} \div \frac{1}{3}$

Divide.

$$\frac{5}{6} \div \frac{5}{12}$$

Quotient: _____

Estimate $2\frac{7}{8} \div \frac{1}{2}$ by first rounding $2\frac{7}{8}$ to the nearest whole number.

Estimate: _____

True or false?

$\frac{1}{2} \div 4$ is greater than $\frac{1}{2}$.

A. True

B. False

SET 27 • 5.NF.A.2

FORWARD 2

Use estimation to judge whether an answer is reasonable • Grade 7 •
Expressions & Equations • 7.EE.B.3

A board $9\frac{7}{8}$ feet long is cut into 3 equal pieces.

Round the length of the board to the nearest whole foot.
Then use it to estimate the length of each piece.

Rounded length: _____

Estimate: _____

Ann earns \$11.85 per hour and works $7\frac{3}{4}$ hours.

Which is the best estimate of her pay?

- A.** \$96
- B.** \$84
- C.** \$120
- D.** \$77

True or false?

$\frac{3}{4}$ of \$79.95 is about \$60.

A. True

B. False

Kai says $4\frac{5}{6} + 3\frac{1}{8}$ is about 9.

Round each mixed number to the nearest whole number and estimate the sum. Is the exact sum more or less than 9?

Your estimate: _____

More or less than 9? _____

A recipe needs $2\frac{1}{3}$ cups of flour per batch.

Round $2\frac{1}{3}$ to the nearest whole number. Then estimate the flour needed for 4 batches.

Rounded amount: _____

Estimate: _____

Interpret a fraction as division of the numerator by the denominator

MAIN	Grade 5 main question	5.NF.B.3
BACKWARD 1	Unit fractions	3.NF.A.1
BACKWARD 2	Division with remainders	4.NBT.B.6
BACKWARD 3	Mixed numbers and fractions greater than 1	4.NF.B.3.b
FORWARD 1	Unit rates written as fractions	6.RP.A.2
FORWARD 2	Write fractions as decimals using division	7.NS.A.2.d

SET 28 • 5.NF.B.3

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.B.3

Write $3 \div 5$ as a fraction.

Fraction: _____

4 friends share 3 pizzas equally.

How much pizza does each friend get?

Each friend gets: _____

Which division expression is equal to $\frac{7}{4}$?

A. $7 \div 4$

B. $7 - 4$

C. $4 \div 7$

D. 7×4

Write $\frac{13}{6}$ as a division expression. Then write the quotient as a mixed number.

Division: _____

Mixed number: _____

True or false?

When 2 pounds of rice are shared equally among 3 bags, each bag gets $\frac{2}{3}$ pound.

- A. True
- B. False

SET 28 • 5.NF.B.3

BACKWARD 1

Unit fractions • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

True or false?

$\frac{1}{4}$ is one of 4 equal parts of a whole.

A. True

B. False

A sandwich is cut into 3 equal parts.

What fraction of the sandwich is one part?

Fraction: _____

What fraction of the bar is shaded?



A. $\frac{8}{3}$

B. $\frac{3}{8}$

C. $\frac{5}{8}$

D. $\frac{3}{5}$

How many $\frac{1}{6}$ s make $\frac{5}{6}$?

Answer: _____

True or false?

$\frac{3}{8}$ is 3 parts of size $\frac{1}{8}$.

A. True

B. False

SET 28 • 5.NF.B.3

BACKWARD 2

Division with remainders • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Divide. Write the quotient and the remainder.

$$17 \div 5$$

Quotient and remainder: _____

True or false?

$$23 \div 4 = 6 \text{ R } 1$$

A. True

B. False

$$29 \div 6 = ?$$

A. 5 R 5

B. 5 R 1

C. 4 R 3

D. 4 R 5

Divide. Write the quotient and the remainder.

$$50 \div 8$$

Quotient and remainder: _____

True or false?

$$15 \div 2 = 7 \text{ R } 2$$

A. True

B. False

SET 28 • 5.NF.B.3

BACKWARD 3

Mixed numbers and fractions greater than 1 • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.b

Write $\frac{9}{4}$ as a mixed number.

Mixed number: _____

Write $1\frac{2}{3}$ as a fraction.

Fraction: _____

True or false?

$$\frac{11}{5} = 1\frac{1}{5}$$

A. True

B. False

Which mixed number is equal to $\frac{7}{3}$?

A. $2\frac{2}{3}$

B. $3\frac{1}{3}$

C. $2\frac{1}{3}$

D. $1\frac{1}{3}$

Write $\frac{15}{4}$ as a mixed number.

Mixed number: _____

SET 28 • 5.NF.B.3

FORWARD 1

Unit rates written as fractions • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.2

A recipe uses 3 cups of flour for every 4 cups of water.
How many cups of flour are used per cup of water?

Cups of flour: _____

8 apples cost \$6.

What is the cost per apple?

Cost per apple: _____

A car goes 150 miles on 4 gallons of gas.

What is the rate in miles per gallon?

A. $\frac{4}{150}$

B. 600

C. 154

D. 37.5

There are 5 cups of juice for every 2 pitchers.

How many cups of juice are there per pitcher?

Cups per pitcher: _____

True or false?

The ratio 3 : 8 has the unit rate $\frac{3}{8}$.

A. True

B. False

SET 28 • 5.NF.B.3

FORWARD 2

Write fractions as decimals using division • Grade 7 • The Number System •
7.NS.A.2.d

Write $5 \div 8$ as a decimal.

Decimal: _____

Use long division to write $\frac{7}{12}$ as a decimal. Show your work.

Decimal: _____

Which fraction has a decimal form that repeats?

A. $\frac{7}{20}$

B. $\frac{5}{6}$

C. $\frac{9}{25}$

D. $\frac{3}{8}$

Write $11 \div 4$ as a decimal.

Decimal: _____

True or false?

$$\frac{1}{3} = 0.3$$

A. True

B. False

Solve sharing problems whose answers are fractions or mixed numbers

MAIN	Grade 5 main question	5.NF.B.3
BACKWARD 1	Division word problems	3.OA.A.3
BACKWARD 2	Whole numbers as fractions	3.NF.A.3.c
FORWARD 1	Solve unit rate problems	6.RP.A.3.b
FORWARD 2	Unit rates with fractional quantities	7.RP.A.1

SET 29 • 5.NF.B.3

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.B.3

6 friends share 10 granola bars equally.

How many bars does each friend get? Write a mixed number.

Bars: _____

A 50-pound bag of rice is shared equally among 8 families.

How many pounds does each family get?

Pounds: _____

9 yards of ribbon are cut into 4 equal pieces.

How long is each piece?

A. $4\frac{1}{2}$ yards

B. $\frac{4}{9}$ yard

C. $2\frac{1}{9}$ yards

D. $2\frac{1}{4}$ yards

7 liters of juice are poured equally into 3 jugs.

How many liters are in each jug? Between which two whole numbers is your answer?

Liters: _____

Between: _____

True or false?

When 5 cookies are shared equally by 2 children, each child gets $2\frac{1}{2}$ cookies.

A. True

B. False

SET 29 • 5.NF.B.3

BACKWARD 1

Division word problems • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3

24 stickers are shared equally by 6 friends.

How many stickers does each friend get?

Stickers: _____

35 chairs are set out in 5 equal rows.

How many chairs are in each row?

A. 30

B. 6

C. 40

D. 7

True or false?

If 4 boxes hold 28 toys, with the same number in each box, each box holds 7 toys.

- A. True
- B. False

A baker puts 45 muffins into bags of 9.
How many bags does she fill?

Bags: _____

18 pencils are shared equally by 3 students.

How many pencils does each student get?

Pencils: _____

SET 29 • 5.NF.B.3

BACKWARD 2

Whole numbers as fractions • Grade 3 • Number & Operations–Fractions •
3.NF.A.3.c

True or false?

$$\frac{6}{3} = 3$$

A. True

B. False

Write 4 as a fraction with a denominator of 1.

Fraction: _____

Which fraction is equal to 3?

A. $\frac{3}{3}$

B. $\frac{2}{6}$

C. $\frac{3}{2}$

D. $\frac{6}{2}$

How many halves are in 5 wholes?

Halves: _____

True or false?

$$\frac{8}{8} = 8$$

A. True

B. False

SET 29 • 5.NF.B.3

FORWARD 1

Solve unit rate problems • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.b

3 pounds of grapes cost \$7.

What is the cost per pound?

Cost per pound: _____

A car travels 100 miles in 3 hours at a constant speed.
At this rate, how far does it travel in 6 hours?

Distance: _____

8 cans of soup cost \$6.

At this rate, how much do 12 cans cost?

A. \$16

B. \$8

C. \$9

D. \$4

A runner runs 5 miles in 40 minutes at a constant pace.
At this pace, how long does she take to run 8 miles?

Time: _____

True or false?

If 4 pens cost \$3, then 10 pens cost \$7.00.

- A. True
- B. False

SET 29 • 5.NF.B.3

FORWARD 2

Unit rates with fractional quantities • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

A snail crawls $\frac{2}{3}$ foot in $\frac{1}{4}$ hour.

What is its speed in feet per hour?

Speed: _____

A recipe uses $\frac{3}{4}$ cup of sugar for every $\frac{1}{2}$ cup of milk.

How many cups of sugar are used per cup of milk?

Cups of sugar: _____

Jo walks $\frac{1}{2}$ mile in $\frac{1}{6}$ hour.

What is her speed?

A. 2 miles per hour

B. 3 miles per hour

C. $\frac{1}{3}$ mile per hour

D. $\frac{1}{12}$ mile per hour

A pump fills $\frac{5}{8}$ of a tank in $\frac{1}{2}$ hour.

How much of the tank does it fill per hour?

Tanks per hour: _____

True or false?

$\frac{1}{2}$ cup of juice for every $\frac{1}{4}$ cup of water is a rate of 2 cups of juice per cup of water.

- A. True
- B. False

Multiply a fraction by a whole number

MAIN	Grade 5 main question	5.NF.B.4.a
BACKWARD 1	A fraction times a whole number as repeated unit fractions	4.NF.B.4.b
BACKWARD 2	Parts of a whole	3.NF.A.1
BACKWARD 3	Splitting a number into equal parts	3.OA.A.2
FORWARD 1	Find a percent of a quantity	6.RP.A.3.c
FORWARD 2	Multistep percent problems	7.RP.A.3

SET 30 • 5.NF.B.4.a

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.4.a

Multiply.

$$\frac{3}{4} \times 12$$

Product: _____

Multiply.

$$5 \times \frac{2}{3}$$

Product: _____

What is $\frac{2}{5}$ of 20?

A. 10

B. 8

C. 4

D. 50

Multiply.

$$\frac{5}{8} \times 6$$

Product: _____

True or false?

$\frac{1}{3} \times 15$ means 15 split into 3 equal parts, with 1 of the parts taken.

- A. True
- B. False

SET 30 • 5.NF.B.4.a

BACKWARD 1

A fraction times a whole number as repeated unit fractions • Grade 4 • Number & Operations–Fractions • 4.NF.B.4.b

Write $\frac{4}{5}$ as a whole number times $\frac{1}{5}$.

Expression: _____

True or false?

$$3 \times \frac{2}{7} = \frac{6}{21}$$

A. True

B. False

$$3 \times \frac{3}{8} = ?$$

A. $3\frac{3}{8}$

B. $\frac{6}{8}$

C. $\frac{9}{8}$

D. $\frac{9}{24}$

Multiply.

$$4 \times \frac{2}{9}$$

Product: _____

True or false?

$$2 \times \frac{3}{5} = \frac{6}{10}$$

A. True

B. False

SET 30 • 5.NF.B.4.a

BACKWARD 2

Parts of a whole • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

True or false?

$\frac{1}{4}$ of a bar means 1 of 4 equal parts.

A. True

B. False

What fraction of the bar is shaded?



Fraction: _____

What fraction is 2 parts of size $\frac{1}{3}$?

A. $\frac{2}{3}$

B. $\frac{1}{6}$

C. $\frac{1}{3}$

D. $\frac{3}{2}$

How many $\frac{1}{8}$ s are in $\frac{5}{8}$?

Answer: _____

True or false?

$\frac{3}{4}$ means 3 parts of size $\frac{1}{4}$.

A. True

B. False

SET 30 • 5.NF.B.4.a

BACKWARD 3

Splitting a number into equal parts • Grade 3 • Operations & Algebraic Thinking
• 3.OA.A.2

12 is split into 4 equal parts.

How much is in each part?

Answer: _____

True or false?

20 split into 5 equal parts gives 4 in each part.

A. True

B. False

Which expression shows 18 split into 3 equal parts?

A. $3 \div 18$

B. $18 - 3$

C. $18 \div 3$

D. 18×3

Divide.

$$24 \div 8$$

Quotient: _____

30 is split into 6 equal parts.

How much is in each part?

Answer: _____

SET 30 • 5.NF.B.4.a

FORWARD 1

Find a percent of a quantity • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.c

What is 25% of 48?

Answer: _____

What is 60% of 35?

Answer: _____

A \$40 game is 15% off.

How many dollars are taken off the price?

- A.** \$6
- B.** \$15
- C.** \$4
- D.** \$34

A class has 30 students. 70% of them ride the bus.
How many students ride the bus?

Students: _____

True or false?

50% of 18 is 36.

A. True

B. False

SET 30 • 5.NF.B.4.a

FORWARD 2

Multistep percent problems • Grade 7 • Ratios & Proportional Relationships •
7.RP.A.3

A \$60 coat is 30% off.

What is the sale price?

Sale price: _____

A meal costs \$25. A 20% tip is added.

What is the total cost?

Total: _____

A \$80 bike has a 6% sales tax added.

What is the total cost?

- A.** \$86.00
- B.** \$4.80
- C.** \$85.60
- D.** \$84.80

A town of 500 people grew by 12%.

What is the new population?

Population: _____

True or false?

A 25% markup on a \$40 item makes the price \$50.

A. True

B. False

Multiply a fraction by a fraction

MAIN	Grade 5 main question	5.NF.B.4.a
BACKWARD 1	A whole number times a fraction	4.NF.B.4.b
BACKWARD 2	Equivalent fractions and simplest form	4.NF.A.1
FORWARD 1	Divide fractions by fractions	6.NS.A.1
FORWARD 2	Multiply rational numbers	7.NS.A.2.c

SET 31 • 5.NF.B.4.a

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.4.a

Multiply.

$$\frac{3}{4} \times \frac{2}{7}$$

Product: _____

Multiply.

$$\frac{3}{4} \times \frac{1}{6}$$

Product: _____

Multiply.

$$\frac{5}{6} \times \frac{3}{10}$$

A. $\frac{15}{6}$

B. $\frac{1}{2}$

C. $\frac{1}{4}$

D. $\frac{8}{16}$

What is $\frac{1}{2}$ of $\frac{3}{8}$?

Answer: _____

True or false?

$$\frac{2}{5} \times \frac{2}{3} = \frac{4}{15}$$

A. True

B. False

SET 31 • 5.NF.B.4.a

BACKWARD 1

A whole number times a fraction • Grade 4 • Number & Operations–Fractions •
4.NF.B.4.b

Multiply.

$$2 \times \frac{3}{10}$$

Product: _____

True or false?

$$5 \times \frac{1}{4} = \frac{5}{20}$$

A. True

B. False

$$4 \times \frac{2}{3} = ?$$

A. $\frac{8}{12}$

B. $\frac{8}{3}$

C. $\frac{6}{3}$

D. $\frac{2}{12}$

Multiply.

$$3 \times \frac{5}{12}$$

Product: _____

True or false?

$$6 \times \frac{1}{6} = 1$$

A. True

B. False

SET 31 • 5.NF.B.4.a

BACKWARD 2

Equivalent fractions and simplest form • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Write $\frac{6}{8}$ as an equivalent fraction with a denominator of 4.

Fraction: _____

True or false?

$$\frac{10}{15} = \frac{3}{4}$$

A. True

B. False

Which fraction is equivalent to $\frac{4}{12}$?

A. $\frac{4}{3}$

B. $\frac{1}{4}$

C. $\frac{2}{4}$

D. $\frac{1}{3}$

Find the missing number.

$$\frac{9}{12} = \frac{?}{4}$$

Missing number: _____

True or false?

$$\frac{5}{20} = \frac{1}{5}$$

A. True

B. False

SET 31 • 5.NF.B.4.a

FORWARD 1

Divide fractions by fractions • Grade 6 • The Number System • 6.NS.A.1

Divide.

$$\frac{2}{3} \div \frac{4}{5}$$

Quotient: _____

Divide.

$$\frac{3}{4} \div \frac{1}{6}$$

Quotient: _____

Divide.

$$\frac{5}{8} \div \frac{1}{4}$$

A. $\frac{5}{32}$

B. $\frac{1}{2}$

C. $2\frac{1}{2}$

D. $1\frac{1}{4}$

How many $\frac{3}{8}$ -cup servings are in $\frac{3}{4}$ cup?

Servings: _____

True or false?

$$\frac{1}{2} \div \frac{1}{4} = \frac{1}{8}$$

A. True

B. False

SET 31 • 5.NF.B.4.a

FORWARD 2

Multiply rational numbers • Grade 7 • The Number System • 7.NS.A.2.c

Find the product.

$$-\frac{2}{3} \times \frac{3}{4}$$

Product: _____

Find the product.

$$\left(-\frac{4}{5}\right) \times \left(-\frac{5}{6}\right)$$

Product: _____

Find the product.

$$\frac{3}{7} \times \left(-\frac{7}{9}\right)$$

A. $-\frac{3}{7}$

B. $\frac{1}{3}$

C. $-\frac{1}{3}$

D. $\frac{10}{16}$

Find the product.

$$\left(-\frac{1}{2}\right) \times \left(-\frac{1}{2}\right) \times \left(-\frac{1}{2}\right)$$

Product: _____

True or false?

$$\left(-\frac{2}{5}\right) \times \frac{5}{2} = 1$$

A. True

B. False

Interpret fraction products with tape diagrams and area models

MAIN	Grade 5 main question	5.NF.B.4.a
BACKWARD 1	Fractions of a whole in a model	3.NF.A.1
BACKWARD 2	Split a whole number into equal parts	3.OA.A.2
FORWARD 1	Use visual models to divide fractions by fractions	6.NS.A.1
FORWARD 2	Multiply rational numbers	7.NS.A.2.c

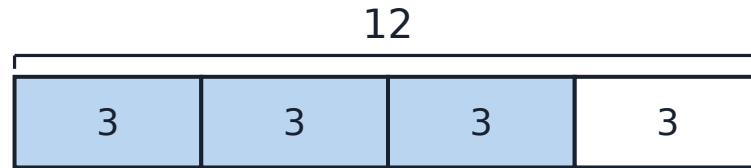
SET 66 • 5.NF.B.4.a

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.4.a

The tape diagram shows 12 split into 4 equal parts, with 3 parts shaded.

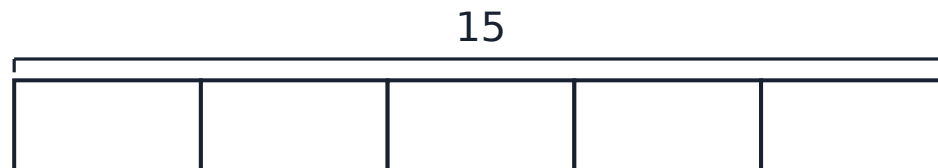
Which product does the shaded part show?



- A.** $\frac{3}{4} \times 12$ **B.** $\frac{4}{3} \times 12$ **C.** $\frac{1}{4} \times 3$ **D.** 3×12

Shade the tape diagram to show $\frac{2}{5} \times 15$.

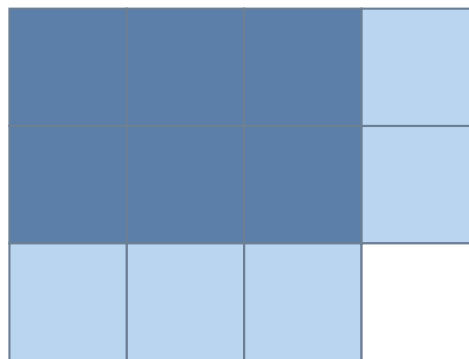
Then write the product.



Product: _____

In the model, the darker part shows a product of two fractions.

Which equation does the model show?



- A.** $\frac{2}{3} + \frac{3}{4} = \frac{5}{7}$ **B.** $\frac{2}{3} \times \frac{3}{4} = \frac{5}{7}$
- C.** $\frac{1}{3} \times \frac{1}{4} = \frac{1}{12}$ **D.** $\frac{2}{3} \times \frac{3}{4} = \frac{6}{12}$

Shade 3 of the 5 columns.
Then shade 1 of the 2 rows.

What product of fractions
does the part shaded twice
show, and what is it?

Product: _____

Value: _____

True or false?

In a model of $\frac{3}{4} \times 8$, 8 is split into 4 equal parts and 3 of the parts are taken, so the product is 6.

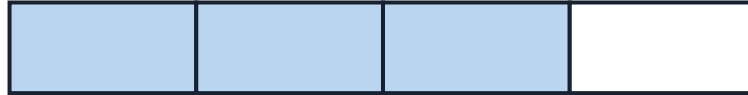
- A.** True
- B.** False

SET 66 • 5.NF.B.4.a

BACKWARD 1

Fractions of a whole in a model • Grade 3 • Number & Operations–Fractions •
3.NF.A.1

What fraction of the bar is shaded?



Fraction: _____

True or false?

$\frac{2}{3}$ of a bar means 2 of 3 equal parts.

A. True

B. False

A bar is split into 6 equal parts and 4 parts are shaded.
What fraction is shaded?

A. $\frac{6}{4}$

B. $\frac{2}{6}$

C. $\frac{4}{6}$

D. $\frac{4}{10}$

How many $\frac{1}{3}$ s make $\frac{2}{3}$?

Answer: _____

True or false?

A bar split into 8 equal parts with 3 parts shaded shows $\frac{3}{5}$.

- A.** True
- B.** False

SET 66 • 5.NF.B.4.a

BACKWARD 2

Split a whole number into equal parts • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2

15 is split into 5 equal parts.

How much is in each part?

Answer: _____

True or false?

21 split into 3 equal parts gives 7 in each part.

A. True

B. False

Which expression shows 32 split into 4 equal parts?

A. $32 \div 4$

B. $32 - 4$

C. $4 \div 32$

D. 32×4

The tape diagram shows 16 split into 8 equal parts.
What number goes in each part?



Answer: _____

True or false?

36 split into 6 equal parts gives 5 in each part.

A. True

B. False

SET 66 • 5.NF.B.4.a

FORWARD 1

Use visual models to divide fractions by fractions • Grade 6 • The Number System • 6.NS.A.1

How many $\frac{1}{6}$ s are in $\frac{2}{3}$?

Answer: _____

True or false?

$\frac{3}{4} \div \frac{1}{8} = 6$ because there are 6 eighths in $\frac{3}{4}$.

A. True

B. False

A model shows 2 wholes, each cut into thirds.

How many $\frac{1}{3}$ s are in 2?

A. $\frac{2}{3}$

B. $\frac{1}{6}$

C. 5

D. 6

Divide.

$$\frac{5}{6} \div \frac{1}{12}$$

Quotient: _____

True or false?

$$\frac{2}{5} \div \frac{1}{10} = \frac{2}{50}$$

A. True

B. False

SET 66 • 5.NF.B.4.a

FORWARD 2

Multiply rational numbers • Grade 7 • The Number System • 7.NS.A.2.c

Find the product.

$$\frac{3}{4} \times (-8)$$

Product: _____

True or false?

$$-\frac{2}{3} \times \frac{3}{4} = -\frac{1}{2}$$

A. True

B. False

Find the product.

$$\left(-\frac{1}{2}\right) \times \left(-\frac{3}{5}\right)$$

A. $-\frac{3}{10}$

B. $\frac{3}{10}$

C. $-\frac{4}{7}$

D. $\frac{4}{7}$

Find the product.

$$-\frac{2}{5} \times 15$$

Product: _____

True or false?

$$(-1) \times \left(-\frac{5}{6}\right) = -\frac{5}{6}$$

A. True

B. False

Find the area of a rectangle with fractional side lengths

MAIN

Grade 5 main question

5.NF.B.4.b

BACKWARD 1

Area by tiling with unit squares

3.MD.C.7.a

BACKWARD 2

The area formula for rectangles

4.MD.A.3

BACKWARD 4

Multiply a whole number by a fraction

4.NF.B.4.b

FORWARD 1

Area of triangles and other polygons with fractional measures

6.G.A.1

FORWARD 2

Area of composite figures

7.G.B.6

SET 32 • 5.NF.B.4.b

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.4.b

Find the area of the rectangle.

Area: _____



A rectangle is $2\frac{1}{2}$ inches long and $\frac{4}{5}$ inch wide.

What is its area?

Area: _____

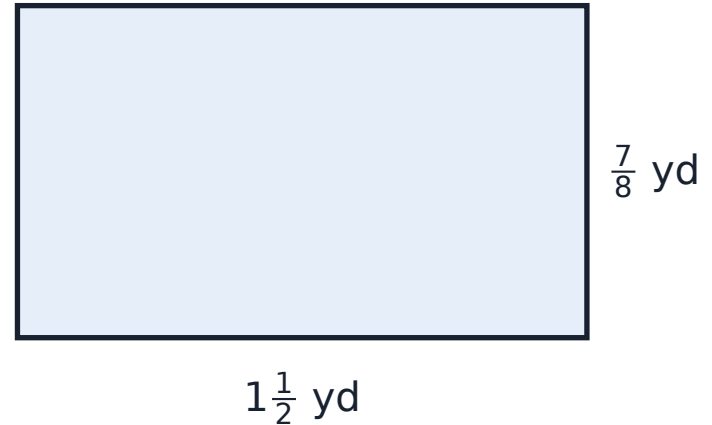
A rectangle that is $\frac{2}{3}$ unit by $\frac{3}{4}$ unit is tiled with small rectangles that are each $\frac{1}{3}$ unit by $\frac{1}{4}$ unit.

What is the area of the large rectangle?

- A.** $\frac{6}{12}$ square unit
- B.** $\frac{6}{7}$ square unit
- C.** $\frac{5}{7}$ square unit
- D.** $1\frac{5}{12}$ square units

Find the area of the rectangle.

Area: _____



True or false?

A rectangle that is $\frac{1}{2}$ foot by $\frac{1}{2}$ foot has an area of 1 square foot.

- A. True
- B. False

SET 32 • 5.NF.B.4.b

BACKWARD 1

Area by tiling with unit squares • Grade 3 • Measurement & Data • 3.MD.C.7.a

A rectangle is covered by 4 rows of 6 unit squares, with no gaps.

What is its area?

Area: _____

True or false?

Tiling a 3-by-5 rectangle takes 8 unit squares.

A. True

B. False

How many unit squares cover a 7-by-2 rectangle?

A. 72

B. 9

C. 14

D. 18

A rectangle has 5 rows of unit squares with 8 squares in each row.

What is its area?

Area: _____

True or false?

The number of unit squares that cover a rectangle equals its length times its width.

- A. True
- B. False

SET 32 • 5.NF.B.4.b

BACKWARD 2

The area formula for rectangles • Grade 4 • Measurement & Data • 4.MD.A.3

A rectangle is 12 m long and 5 m wide.

What is its area?

Area: _____

A rectangle has an area of 54 square feet and a length of 9 feet.

What is its width?

Width: _____

True or false?

The formula $A = l \times w$ gives the area of a rectangle.

A. True

B. False

A rectangle is 15 cm long and 4 cm wide.

What is its area?

- A.** 60 square cm
- B.** 45 square cm
- C.** 38 square cm
- D.** 19 square cm

Find the area of the rectangle.

Area: _____



SET 32 • 5.NF.B.4.b

BACKWARD 4

Multiply a whole number by a fraction • Grade 4 • Number & Operations–Fractions • 4.NF.B.4.b

A rectangle is 3 units long and $\frac{2}{5}$ unit wide.

Write its area as a fraction.

Area: _____

True or false?

$$4 \times \frac{3}{8} = \frac{12}{8}$$

A. True

B. False

$$5 \times \frac{2}{3} = ?$$

A. $\frac{10}{3}$

B. $\frac{10}{15}$

C. $\frac{2}{15}$

D. $\frac{7}{3}$

Multiply.

$$6 \times \frac{3}{4}$$

Product: _____

True or false?

$$2 \times \frac{5}{6} = \frac{10}{12}$$

A. True

B. False

SET 32 • 5.NF.B.4.b

FORWARD 1

Area of triangles and other polygons with fractional measures • Grade 6 •
Geometry • 6.G.A.1

A triangle has a base of $4\frac{1}{2}$ inches and a height of 2 inches.

What is its area?

Area: _____

A parallelogram has a base of $3\frac{1}{4}$ feet and a height of 4 feet.

What is its area?

Area: _____

A triangle has a base of $\frac{3}{4}$ m and a height of $\frac{2}{3}$ m.

What is its area?

A. $\frac{1}{2}$ square meter

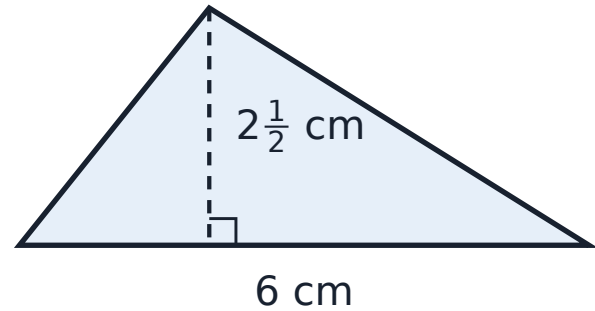
B. $\frac{5}{7}$ square meter

C. $\frac{1}{4}$ square meter

D. $\frac{6}{7}$ square meter

Find the area of the triangle.

Area: _____



True or false?

A trapezoid with bases of $2\frac{1}{2}$ m and $3\frac{1}{2}$ m and a height of 2 m has an area of 6 square meters.

- A. True
- B. False

SET 32 • 5.NF.B.4.b

FORWARD 2

Area of composite figures • Grade 7 • Geometry • 7.G.B.6

A figure is made of a rectangle that is $6\frac{1}{2}$ ft by 4 ft and a triangle with a base of 4 ft and a height of 3 ft.

What is the total area?

Area: _____

A rectangular garden is 10 m by $7\frac{1}{2}$ m. A square patio with $2\frac{1}{2}$ -m sides is inside the garden.

What is the area of the garden that is not patio?

Area: _____

A $\frac{1}{2}$ -by-2 rectangle is cut out of a 5-by-4 rectangle.

What is the area of the part that is left?

- A.** 21 square units
- B.** 18 square units
- C.** 19 square units
- D.** 20 square units

A kite is made of two triangles. Each triangle has a base of 8 in. and a height of $3\frac{1}{2}$ in.

What is the area of the kite?

Area: _____

True or false?

A figure made of two $\frac{1}{2}$ -by-3 rectangles that do not overlap has an area of 6 square units.

- A. True
- B. False

Tile a rectangle with unit-fraction squares and connect the tiling to the area

MAIN	Grade 5 main question	5.NF.B.4.b
BACKWARD 1	Measure area by counting unit squares	3.MD.C.6
BACKWARD 2	Unit fractions of a length	3.NF.A.1
FORWARD 1	Pack a prism with unit-fraction cubes to find its volume	6.G.A.2
FORWARD 2	Area of composite figures with fractional measures	7.G.B.6

SET 67 • 5.NF.B.4.b

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.4.b

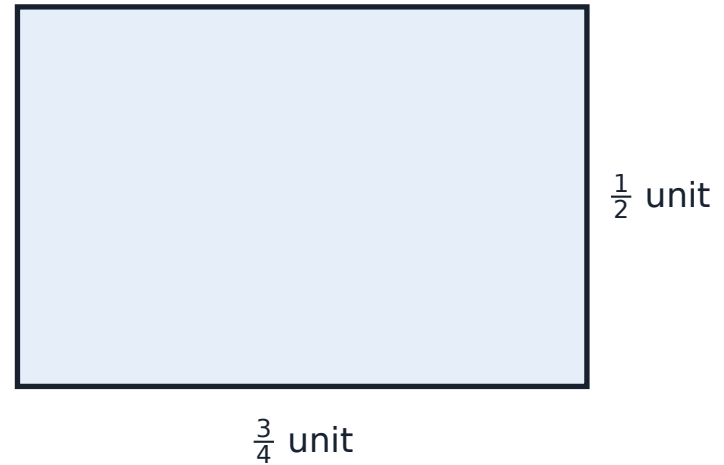
The rectangle is $\frac{3}{4}$ unit long and $\frac{1}{2}$ unit wide.

Draw lines to tile it with squares that are $\frac{1}{4}$ unit on each side.

How many squares fit, and what is the area?

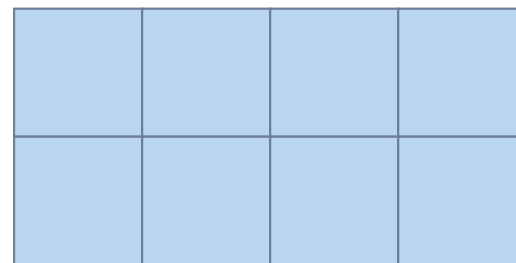
Squares: _____

Area: _____



The rectangle is tiled with squares that are $\frac{1}{3}$ unit on each side.

What is its area? Explain how the tiling shows it.



Area: _____

Explanation: _____

How many squares with $\frac{1}{6}$ -unit sides tile a rectangle that is $\frac{5}{6}$ unit by $\frac{1}{2}$ unit?

- A. 5
- B. 30
- C. 8
- D. 15

True or false?

Tiling a square with $\frac{2}{3}$ -unit sides takes 4 squares with $\frac{1}{3}$ -unit sides, so its area is $\frac{4}{9}$ square unit.

A. True

B. False

A rectangle is tiled with 3 rows of 4 squares. Each square has $\frac{1}{4}$ -unit sides.

What are the rectangle's side lengths and its area? Show that multiplying the side lengths gives the same area.

Side lengths: _____

Area: _____

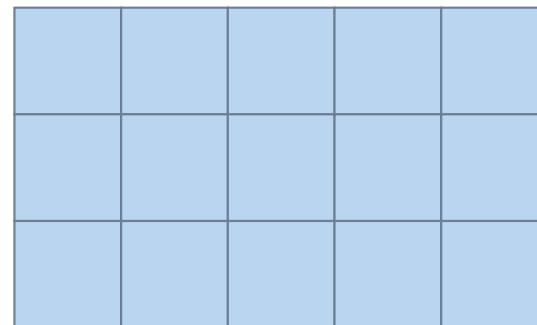
SET 67 • 5.NF.B.4.b

BACKWARD 1

Measure area by counting unit squares • Grade 3 • Measurement & Data •
3.MD.C.6

Each small square is 1 square unit.

What is the area of the rectangle?



Area: _____

True or false?

A shape covered by 9 unit squares with no gaps or overlaps has an area of 9 square units.

- A.** True
- B.** False

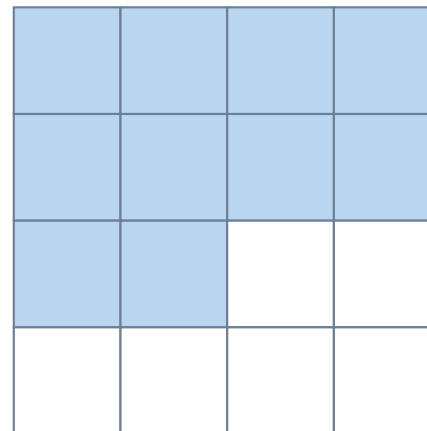
A rectangle is covered by 2 rows of 6 unit squares.

What is its area?

- A.** 8 square units
- B.** 6 square units
- C.** 12 square units
- D.** 16 square units

Each small square is 1 square inch.

What is the area of the shaded part?



Area: _____

True or false?

If the unit squares overlap, counting them still gives the exact area.

- A. True
- B. False

SET 67 • 5.NF.B.4.b

BACKWARD 2

Unit fractions of a length • Grade 3 • Number & Operations–Fractions •
3.NF.A.1

A 1-unit side is split into 3 equal parts.

How long is each part?

Length: _____

True or false?

Four lengths of $\frac{1}{4}$ unit make 1 unit.

A. True

B. False

How many lengths of $\frac{1}{6}$ unit make $\frac{5}{6}$ unit?

A. 11

B. 5

C. 6

D. 1

A side is made of 3 lengths of $\frac{1}{4}$ unit.

How long is the side?

Length: _____

True or false?

Two lengths of $\frac{1}{8}$ unit make $\frac{2}{16}$ unit.

A. True

B. False

SET 67 • 5.NF.B.4.b

FORWARD 1

Pack a prism with unit-fraction cubes to find its volume • Grade 6 • Geometry •
6.G.A.2

A prism is 1 unit by $\frac{1}{2}$ unit by $\frac{1}{2}$ unit.

How many cubes with edges $\frac{1}{2}$ unit long pack it?

Cubes: _____

True or false?

A cube with edges $\frac{1}{2}$ unit long has a volume of $\frac{1}{8}$ cubic unit.

A. True

B. False

A box is packed with 24 cubes that each have edges $\frac{1}{2}$ unit long.

What is its volume?

A. 12 cubic units

B. 24 cubic units

C. 3 cubic units

D. 6 cubic units

Find the volume of a prism that is $\frac{3}{2}$ units by 1 unit by $\frac{1}{2}$ unit.

Volume: _____

True or false?

Packing a 1-by-1-by-1 cube with cubes that have edges $\frac{1}{3}$ unit long takes 9 cubes.

A. True

B. False

SET 67 • 5.NF.B.4.b

FORWARD 2

Area of composite figures with fractional measures • Grade 7 • Geometry •
7.G.B.6

A $\frac{1}{4}$ -m by $\frac{1}{3}$ -m rectangle is cut out of a $\frac{3}{4}$ -m by $\frac{2}{3}$ -m rectangle.

What area is left?

Area: _____

True or false?

Two rectangles that do not overlap, $\frac{1}{2}$ ft by 3 ft and $\frac{1}{2}$ ft by 1 ft, have a total area of 2 square feet.

A. True

B. False

A square with sides of $\frac{5}{2}$ units has a square with sides of $\frac{1}{2}$ unit removed.

What is the area of the figure?

A. 5 square units

B. 6 square units

C. 2 square units

D. $6\frac{1}{4}$ square units

A rug is $2\frac{1}{2}$ ft by 4 ft. A square corner with sides of $1\frac{1}{2}$ ft is cut off.

What is the area of the rug that is left?

Area: _____

True or false?

A triangle with a base of $\frac{3}{4}$ unit and a height of $\frac{2}{3}$ unit has an area of $\frac{1}{2}$ square unit.

A. True

B. False

Compare the size of a product to the size of one factor without multiplying

MAIN	Grade 5 main question	5.NF.B.5.a
BACKWARD 1	Multiplicative comparison	4.OA.A.2
BACKWARD 2	Comparing fractions to 1	4.NF.A.2
FORWARD 1	Scale quantities in ratio tables	6.RP.A.3.a
FORWARD 2	Use a scale to find lengths in scale drawings	7.G.A.1

SET 33 • 5.NF.B.5.a

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.5.a

Without multiplying, how does $\frac{3}{4} \times 20$ compare to 20?

- A.** It is less than 20.
- B.** It is equal to 20.
- C.** It is 4 times 20.
- D.** It is greater than 20.

True or false?

$5 \times 3\frac{1}{2}$ is greater than 5.

A. True

B. False

Without multiplying, is $\frac{2}{3} \times 15$ greater than, less than, or equal to 15?

Answer: _____

Which product is greater than 8?

A. $\frac{9}{8} \times 8$

B. $\frac{1}{2} \times 8$

C. $\frac{7}{8} \times 8$

D. $\frac{8}{8} \times 8$

A photo is 6 inches wide. It is resized to $\frac{5}{4}$ of its width.

Is the new width greater than or less than 6 inches?

Answer: _____

SET 33 • 5.NF.B.5.a

BACKWARD 1

Multiplicative comparison • Grade 4 • Operations & Algebraic Thinking •
4.OA.A.2

True or false?

3×12 is 3 more than 12.

A. True

B. False

A dog weighs 4 times as much as a 6-pound cat.
How much does the dog weigh?

Weight: _____

Which number is 5 times as much as 7?

A. 57

B. 12

C. 2

D. 35

True or false?

2×9 is less than 9.

A. True

B. False

Fill in the blank.

42 is ____ times as much as 7.

Blank: _____

SET 33 • 5.NF.B.5.a

BACKWARD 2

Comparing fractions to 1 • Grade 4 • Number & Operations–Fractions •
4.NF.A.2

True or false?

$\frac{4}{5}$ is greater than 1.

A. True

B. False

Which fraction is less than 1?

A. $\frac{6}{7}$

B. $\frac{9}{9}$

C. $\frac{7}{6}$

D. $\frac{3}{2}$

Write $>$, $<$, or $=$ to compare.

$$\frac{8}{8} \text{ ____ } 1$$

Symbol: _____

True or false?

$$\frac{3}{5} > 1$$

A. True

B. False

Is $\frac{11}{10}$ greater than, less than, or equal to 1?

Answer: _____

SET 33 • 5.NF.B.5.a

FORWARD 1

Scale quantities in ratio tables • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.a

A recipe uses 2 cups of rice for every 3 cups of water.
How many cups of water are needed for 8 cups of rice?

Cups of water: _____

A ratio table starts with 4 : 10.

Which ratio could also be in the table?

- A.** 4 : 5
- B.** 2 : 5
- C.** 8 : 15
- D.** 10 : 4

The table shows equivalent ratios.

Find the missing value.

Red	3	6	?
Blue	5	10	25

Missing value: _____

True or false?

Multiplying both numbers in $6 : 9$ by $\frac{1}{3}$ gives $2 : 3$, an equivalent ratio.

- A.** True
- B.** False

A map shows 5 km for every 2 cm.

How many kilometers do 7 cm show?

Kilometers: _____

SET 33 • 5.NF.B.5.a

FORWARD 2

Use a scale to find lengths in scale drawings • Grade 7 • Geometry • 7.G.A.1

A scale drawing uses 1 inch = 4 feet. A wall is $3\frac{1}{2}$ inches long in the drawing.

How long is the real wall?

Length: _____

A room is 15 feet long.

How long is it on a drawing with a scale of 1 inch = 3 feet?

Length: _____

A drawing is reduced to $\frac{1}{2}$ of its size.

A 10-cm side becomes how long?

- A.** 2 cm
- B.** 20 cm
- C.** 10 cm
- D.** 5 cm

A drawing that is 4 cm by 6 cm is enlarged by a scale factor of $\frac{3}{2}$.

What are the new dimensions?

Dimensions: _____

True or false?

Using a scale factor less than 1 makes a drawing smaller.

- A. True
- B. False

Explain the effect of multiplying by a fraction greater than, less than, or equal to 1

MAIN	Grade 5 main question	5.NF.B.5.b
BACKWARD 2	Fractions equal to 1	3.NF.A.3.c
BACKWARD 3	Multiply a whole number by a fraction	4.NF.B.4.b
FORWARD 1	Percents greater than and less than 100%	6.RP.A.3.c
FORWARD 2	Use a multiplier for percent increase and decrease	7.RP.A.3

SET 34 • 5.NF.B.5.b

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.5.b

True or false?

Multiplying 12 by $\frac{7}{6}$ gives a product greater than 12.

- A.** True
- B.** False

Which statement explains why $\frac{4}{5} \times 30$ is less than 30?

- A.** $\frac{4}{5}$ is less than 1.
- B.** $\frac{4}{5}$ is greater than 1.
- C.** 30 is greater than 1.
- D.** The product is a fraction.

Explain why multiplying 24 by $\frac{5}{8}$ gives a product less than 24.

Explanation: _____

Which expression has a value equal to 9?

A. $\frac{6}{5} \times 9$

B. $\frac{5}{6} \times 9$

C. $\frac{1}{5} \times 9$

D. $\frac{5}{5} \times 9$

Without multiplying, order the products from least to greatest.

$$\frac{3}{4} \times 10, \frac{4}{3} \times 10, \frac{4}{4} \times 10$$

Order: _____

SET 34 • 5.NF.B.5.b

BACKWARD 2

Fractions equal to 1 • Grade 3 • Number & Operations–Fractions • 3.NF.A.3.c

True or false?

$$\frac{4}{4} = 4$$

A. True

B. False

Write 1 as a fraction with a denominator of 6.

Fraction: _____

Which fraction is equal to 1?

A. $\frac{0}{8}$

B. $\frac{8}{1}$

C. $\frac{1}{8}$

D. $\frac{8}{8}$

True or false?

$\frac{6}{6}$ is greater than 1.

A. True

B. False

How many eighths make 1 whole?

Eighths: _____

SET 34 • 5.NF.B.5.b

BACKWARD 3

Multiply a whole number by a fraction • Grade 4 • Number & Operations–Fractions • 4.NF.B.4.b

True or false?

$$3 \times \frac{2}{3} = 2$$

A. True

B. False

Multiply.

$$4 \times \frac{3}{4}$$

Product: _____

$$6 \times \frac{3}{2} = ?$$

A. 4

B. $\frac{9}{2}$

C. $\frac{18}{12}$

D. 9

Multiply.

$$5 \times \frac{2}{5}$$

Product: _____

True or false?

$$2 \times \frac{7}{8} = \frac{14}{16}$$

A. True

B. False

SET 34 • 5.NF.B.5.b

FORWARD 1

Percents greater than and less than 100% • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

True or false?

150% of 40 is less than 40.

A. True

B. False

What is 120% of 50?

Answer: _____

Which value is less than 80?

- A.** 200% of 80
- B.** 125% of 80
- C.** 75% of 80
- D.** 100% of 80

What is 100% of 37?

Answer: _____

What is 250% of 12?

Answer: _____

SET 34 • 5.NF.B.5.b

FORWARD 2

Use a multiplier for percent increase and decrease • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

A \$60 price increases by 15%.

Which expression gives the new price?

A. 1.15×60

B. $60 + 15$

C. 0.85×60

D. 0.15×60

A \$90 jacket is 20% off.

Multiply by a decimal to find the sale price.

Sale price: _____

True or false?

Multiplying a price by 0.7 decreases it by 30%.

A. True

B. False

A population of 2,000 grows by 5%.

What is the new population?

Population: _____

Which multiplier shows a 40% decrease?

A. 0.6

B. 0.4

C. 1.4

D. 1.04

Explain equivalent fractions as multiplying by a fraction equal to 1

MAIN	Grade 5 main question	5.NF.B.5.b
BACKWARD 1	Equivalent fractions	4.NF.A.1
BACKWARD 2	Equivalent fractions with a model	3.NF.A.3.b
FORWARD 1	Find equivalent ratios	6.RP.A.3.a
FORWARD 2	Test for a proportional relationship with equivalent ratios	7.RP.A.2.a

SET 68 • 5.NF.B.5.b

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.5.b

Use the fact that $\frac{4}{4} = 1$ to explain why $\frac{2}{3} = \frac{8}{12}$.

Explanation: _____

Which fraction equal to 1 turns $\frac{3}{5}$ into $\frac{9}{15}$?

A. $\frac{3}{3}$

B. $\frac{15}{15}$

C. $\frac{9}{9}$

D. $\frac{5}{5}$

True or false?

$$\frac{5}{6} \times \frac{2}{2} = \frac{10}{12}, \text{ and } \frac{10}{12} = \frac{5}{6} \text{ because } \frac{2}{2} = 1.$$

A. True

B. False

What fraction equal to 1 can you multiply $\frac{7}{10}$ by to get $\frac{28}{40}$?

Fraction: _____

True or false?

$\frac{3}{4} \times \frac{2}{3} = \frac{6}{12}$ shows that $\frac{3}{4} = \frac{6}{12}$.

A. True

B. False

SET 68 • 5.NF.B.5.b

BACKWARD 1

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

True or false?

To write a fraction equal to $\frac{3}{5}$, you can multiply its numerator and denominator by the same nonzero whole number.

A. True

B. False

Find the missing number.

$$\frac{3}{5} = \frac{?}{20}$$

Missing number: _____

Which fraction is equivalent to $\frac{7}{8}$?

A. $\frac{7}{16}$

B. $\frac{14}{24}$

C. $\frac{21}{24}$

D. $\frac{8}{7}$

Find the missing number.

$$\frac{1}{3} = \frac{4}{?}$$

Missing number: _____

True or false?

$$\frac{2}{9} = \frac{4}{27}$$

A. True

B. False

SET 68 • 5.NF.B.5.b

BACKWARD 2

Equivalent fractions with a model • Grade 3 • Number & Operations–Fractions •
3.NF.A.3.b

True or false?

$$\frac{1}{2} = \frac{2}{4}$$

A. True

B. False

Find the missing number.

$$\frac{2}{3} = \frac{?}{6}$$

Missing number: _____

Which fraction is equal to $\frac{3}{4}$?

A. $\frac{6}{8}$

B. $\frac{3}{8}$

C. $\frac{4}{3}$

D. $\frac{4}{6}$

The bar shows $\frac{1}{3}$ shaded. Each part is cut in half.

How many sixths are shaded?



Sixths: _____

True or false?

$$\frac{2}{4} = \frac{2}{8}$$

A. True

B. False

SET 68 • 5.NF.B.5.b

FORWARD 1

Find equivalent ratios • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.a

Write a ratio equivalent to 3 : 5 by multiplying both numbers by 4.

Ratio: _____

True or false?

6 : 8 and 9 : 12 are equivalent ratios.

A. True

B. False

Which ratio is equivalent to 2 : 7?

- A.** 4 : 9
- B.** 7 : 2
- C.** 2 : 14
- D.** 6 : 21

Find the missing value.

$$5 : 6 = 20 : ?$$

Missing value: _____

True or false?

4 : 10 and 6 : 12 are equivalent ratios.

A. True

B. False

SET 68 • 5.NF.B.5.b

FORWARD 2

Test for a proportional relationship with equivalent ratios • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

True or false?

The table shows a proportional relationship.

x	2	4	6
y	5	10	15

A. True

B. False

Which table shows a proportional relationship?

- A.** x: 1, 2, 3 and y: 4, 6, 8
- B.** x: 1, 2, 3 and y: 4, 8, 12
- C.** x: 1, 2, 3 and y: 3, 5, 7
- D.** x: 1, 2, 3 and y: 5, 6, 7

Are 3 : 4 and 12 : 16 equivalent ratios? Explain using a fraction equal to 1.

Answer: _____

Explanation: _____

True or false?

3 : 5 and 9 : 20 are equivalent ratios.

A. True

B. False

The ratios $2 : 9$ and $8 : y$ are equivalent.

Find y .

$y =$ _____

Solve real-world problems by multiplying fractions and mixed numbers

MAIN	Grade 5 main question	5.NF.B.6
BACKWARD 1	Word problems: a whole number times a fraction	4.NF.B.4.c
BACKWARD 2	Mixed numbers as fractions	4.NF.B.3.b
BACKWARD 4	Word problems that add fractions with like denominators	4.NF.B.3.d
FORWARD 1	Word problems with division of fractions	6.NS.A.1
FORWARD 2	Real-world problems with rational numbers	7.NS.A.3

SET 35 • 5.NF.B.6

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.B.6

A recipe uses $\frac{3}{4}$ cup of oats per batch.

How many cups of oats are in $2\frac{1}{2}$ batches?

Cups: _____

A garden is 6 m long. $\frac{2}{3}$ of its length is planted with beans.

How many meters are planted with beans?

Meters: _____

Sam walks $\frac{5}{8}$ mile each day.

How far does he walk in 4 days?

A. $\frac{5}{32}$ mile

B. $\frac{9}{8}$ miles

C. $2\frac{1}{2}$ miles

D. $4\frac{5}{8}$ miles

A board is $3\frac{1}{3}$ feet long. Jo uses $\frac{3}{5}$ of the board.

How many feet does she use?

Feet: _____

A pitcher holds $1\frac{1}{2}$ liters. It is $\frac{2}{3}$ full.

How many liters are in the pitcher?

Liters: _____

SET 35 • 5.NF.B.6

BACKWARD 1

Word problems: a whole number times a fraction • Grade 4 • Number & Operations–Fractions • 4.NF.B.4.c

Each serving is $\frac{2}{3}$ cup.

How many cups are in 5 servings?

Cups: _____

True or false?

4 children each drink $\frac{3}{8}$ liter of milk. They drink $1\frac{1}{2}$ liters in all.

- A. True
- B. False

Each piece of ribbon is $\frac{5}{6}$ yard long.

How long are 3 pieces in all?

A. $\frac{5}{18}$ yard

B. $2\frac{1}{2}$ yards

C. $3\frac{5}{6}$ yards

D. $\frac{15}{18}$ yard

A bug crawls $\frac{3}{10}$ meter each minute.

How far does it crawl in 7 minutes?

Distance: _____

Each bag holds $\frac{1}{4}$ pound of nuts.

How many pounds are in 6 bags?

Pounds: _____

SET 35 • 5.NF.B.6

BACKWARD 2

Mixed numbers as fractions • Grade 4 • Number & Operations–Fractions •
4.NF.B.3.b

Write $2\frac{1}{2}$ as a fraction.

Fraction: _____

Write $3\frac{1}{3}$ as a fraction.

Fraction: _____

True or false?

$$1\frac{1}{2} = \frac{11}{2}$$

A. True

B. False

Which fraction is equal to $4\frac{2}{5}$?

A. $\frac{22}{5}$

B. $\frac{42}{5}$

C. $\frac{11}{5}$

D. $\frac{24}{5}$

Write $\frac{15}{8}$ as a mixed number.

Mixed number: _____

SET 35 • 5.NF.B.6

BACKWARD 4

Word problems that add fractions with like denominators • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.d

A recipe uses $\frac{3}{4}$ cup of oats. Ann makes the recipe twice.

How many cups of oats does she use? Write a fraction.

Cups: _____

True or false?

$$\frac{2}{5} + \frac{2}{5} + \frac{2}{5} = \frac{6}{5}$$

A. True

B. False

Ty walks $\frac{3}{8}$ mile to school and $\frac{3}{8}$ mile home.

How far does he walk in all?

A. $\frac{9}{8}$ mile

B. $\frac{6}{8}$ mile

C. $\frac{3}{8}$ mile

D. $\frac{6}{16}$ mile

A plant grew $\frac{1}{6}$ inch on Monday and $\frac{4}{6}$ inch on Tuesday.

How much did it grow in all?

Growth: _____

True or false?

$$\frac{5}{12} + \frac{5}{12} = \frac{10}{24}$$

A. True

B. False

SET 35 • 5.NF.B.6

FORWARD 1

Word problems with division of fractions • Grade 6 • The Number System •
6.NS.A.1

How many $\frac{3}{4}$ -cup servings are in 6 cups of soup?

Servings: _____

A $\frac{7}{8}$ -mile trail is split into $\frac{1}{8}$ -mile sections.

How many sections are there?

Sections: _____

A rectangle has an area of $\frac{3}{4}$ square foot and a width of $\frac{1}{2}$ foot.

What is its length?

A. $\frac{3}{8}$ foot

B. $1\frac{1}{4}$ feet

C. $1\frac{1}{2}$ feet

D. $\frac{2}{3}$ foot

Ann has $2\frac{1}{2}$ pounds of clay. Each pot uses $\frac{5}{8}$ pound.

How many pots can she make?

Pots: _____

True or false?

$$\frac{2}{3} \div \frac{1}{6} = \frac{1}{9}$$

A. True

B. False

SET 35 • 5.NF.B.6

FORWARD 2

Real-world problems with rational numbers • Grade 7 • The Number System •
7.NS.A.3

A diver descends $\frac{3}{4}$ foot per second for $6\frac{1}{2}$ seconds.

What is the diver's change in depth?

Change: _____

The temperature was 2°F . It dropped $\frac{1}{2}^{\circ}\text{F}$ each hour for 9 hours.

What was the final temperature?

Temperature: _____

A stock loses $\frac{3}{8}$ dollar each day for 4 days.

What is the total change in value?

A. $-4\frac{3}{8}$ dollars

B. $-\frac{3}{32}$ dollar

C. $-1\frac{1}{2}$ dollars

D. $1\frac{1}{2}$ dollars

A tank contains $12\frac{1}{2}$ gallons of water. $\frac{2}{5}$ of the water drains out.

How many gallons drain out?

Gallons: _____

True or false?

$$-\frac{2}{3} \times 4\frac{1}{2} = -3$$

A. True

B. False

Solve real-world problems by multiplying mixed numbers

MAIN

Grade 5 main question

5.NF.B.6

BACKWARD 1

Mixed numbers and fractions greater than 1

4.NF.B.3.b

BACKWARD 3

Breaking apart a factor to multiply

3.OA.B.5

BACKWARD 4

Measurement word problems with fractions

4.MD.A.2

FORWARD 1

Volume of prisms with fractional edge lengths

6.G.A.2

FORWARD 2

Surface area of prisms with fractional edge lengths

7.G.B.6

SET 36 • 5.NF.B.6

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions • 5.NF.B.6

A bag of flour weighs $2\frac{1}{2}$ pounds.

How much do 3 bags weigh?

Weight: _____

A garden bed is $1\frac{1}{3}$ yards long and $1\frac{1}{2}$ yards wide.

What is its area?

Area: _____

A hiker walks $2\frac{1}{4}$ miles each hour.

How far does she walk in $1\frac{2}{3}$ hours?

A. $3\frac{1}{6}$ miles

B. $4\frac{1}{2}$ miles

C. $2\frac{2}{12}$ miles

D. $3\frac{3}{4}$ miles

A poster is $3\frac{1}{5}$ feet long. A copy is made $2\frac{1}{2}$ times as long.

How long is the copy?

Length: _____

True or false?

A recipe uses $1\frac{1}{2}$ cups of milk. Making $1\frac{1}{2}$ times the recipe uses $1\frac{1}{4}$ cups of milk.

A. True

B. False

SET 36 • 5.NF.B.6

BACKWARD 1

Mixed numbers and fractions greater than 1 • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.b

Write $1\frac{3}{4}$ as a fraction.

Fraction: _____

Write $\frac{13}{3}$ as a mixed number.

Mixed number: _____

True or false?

$$2\frac{2}{3} = \frac{22}{3}$$

A. True

B. False

Which mixed number is equal to $\frac{9}{2}$?

A. $2\frac{1}{9}$

B. $4\frac{1}{9}$

C. $9\frac{1}{2}$

D. $4\frac{1}{2}$

Write $5\frac{1}{2}$ as a fraction.

Fraction: _____

SET 36 • 5.NF.B.6

BACKWARD 3

Breaking apart a factor to multiply • Grade 3 • Operations & Algebraic Thinking
• 3.OA.B.5

True or false?

$$4 \times (5 + 2) = 4 \times 5 + 4 \times 2$$

A. True

B. False

Fill in the blank.

$$6 \times 12 = 6 \times 10 + 6 \times \underline{\quad}$$

Blank: _____

Which expression is equal to 3×15 ?

A. $3 \times 10 + 5$

B. $3 + 10 \times 5$

C. $3 \times 10 \times 5$

D. $3 \times 10 + 3 \times 5$

True or false?

$$7 \times 13 = 7 \times 10 + 3$$

A. True

B. False

Fill in the blank.

$$8 \times (4 + 1) = 8 \times 4 + 8 \times \underline{\hspace{1cm}}$$

Blank: _____

SET 36 • 5.NF.B.6

BACKWARD 4

Measurement word problems with fractions • Grade 4 • Measurement & Data •
4.MD.A.2

A ribbon is $\frac{3}{4}$ yard long.

How long are 4 ribbons in all?

Length: _____

True or false?

Each lap is $\frac{1}{2}$ mile. 6 laps is 3 miles.

A. True

B. False

A jug holds 2 liters.

How many milliliters are in 3 jugs?

- A.** 6,000 mL
- B.** 5,000 mL
- C.** 2,003 mL
- D.** 600 mL

Each box weighs $1\frac{1}{4}$ pounds.

How much do 2 boxes weigh?

Weight: _____

True or false?

3 pieces of wood, each $\frac{2}{5}$ meter long, make 1 meter in all.

A. True

B. False

SET 36 • 5.NF.B.6

FORWARD 1

Volume of prisms with fractional edge lengths • Grade 6 • Geometry • 6.G.A.2

A box is $2\frac{1}{2}$ ft by 2 ft by $1\frac{1}{2}$ ft.

What is its volume?

Volume: _____

A cube has edges that are $2\frac{1}{2}$ inches long.

What is its volume?

Volume: _____

A prism has a base area of $3\frac{1}{3}$ square cm and a height of $1\frac{1}{2}$ cm.

What is its volume?

A. $3\frac{1}{2}$ cubic cm

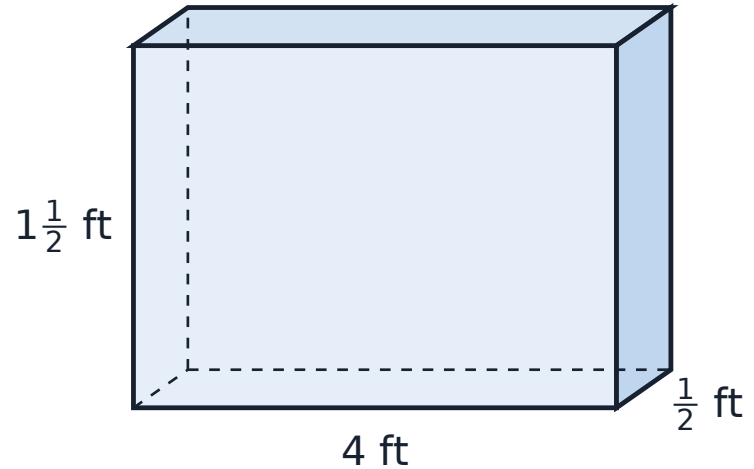
B. 10 cubic cm

C. $4\frac{5}{6}$ cubic cm

D. 5 cubic cm

Find the volume of the prism.

Volume: _____



True or false?

A box that is $\frac{1}{2}$ m by $\frac{1}{2}$ m by $\frac{1}{2}$ m has a volume of $\frac{1}{8}$ cubic meter.

A. True

B. False

SET 36 • 5.NF.B.6

FORWARD 2

Surface area of prisms with fractional edge lengths • Grade 7 • Geometry •
7.G.B.6

A rectangular prism is 4 cm by 3 cm by $2\frac{1}{2}$ cm.

What is its surface area?

Surface area: _____

A cube has edges that are $2\frac{1}{2}$ inches long.

What is its surface area?

Surface area: _____

A box is 5 in. by 4 in. by $1\frac{1}{2}$ in.

What is its surface area?

- A.** 30 square inches
- B.** 40 square inches
- C.** 67 square inches
- D.** $33\frac{1}{2}$ square inches

A cube has edges that are $1\frac{1}{2}$ m long.

What is its surface area?

Surface area: _____

True or false?

A prism that is 2 ft by $1\frac{1}{2}$ ft by 1 ft has a surface area of 13 square feet.

- A. True
- B. False

Divide a unit fraction by a whole number

MAIN	Grade 5 main question	5.NF.B.7.a
BACKWARD 1	Unit fractions get smaller as the number of parts grows	3.NF.A.1
BACKWARD 2	Sharing equally	3.OA.A.2
BACKWARD 4	Equivalent fractions	4.NF.A.1
FORWARD 1	Divide fractions	6.NS.A.1
FORWARD 2	Divide rational numbers	7.NS.A.2.b

SET 37 • 5.NF.B.7.a

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.7.a

Divide.

$$\frac{1}{7} \div 2$$

Quotient: _____

Divide.

$$\frac{1}{2} \div 5$$

Quotient: _____

$$\frac{1}{4} \div 2 = ?$$

A. $\frac{1}{2}$

B. $\frac{2}{4}$

C. $\frac{1}{8}$

D. 8

$\frac{1}{5}$ of a pan of brownies is shared equally by 3 friends.

What fraction of the whole pan does each friend get?

Fraction: _____

True or false?

$$\frac{1}{6} \div 3 = \frac{1}{2}$$

A. True

B. False

SET 37 • 5.NF.B.7.a

BACKWARD 1

Unit fractions get smaller as the number of parts grows • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

True or false?

When a whole is cut into more equal parts, each part is smaller.

A. True

B. False

Which unit fraction is the smallest?

A. $\frac{1}{6}$

B. $\frac{1}{8}$

C. $\frac{1}{2}$

D. $\frac{1}{3}$

A bar is cut into 6 equal parts.

What fraction of the bar is one part?

Fraction: _____

True or false?

$$\frac{1}{3} < \frac{1}{4}$$

A. True

B. False

What fraction of the bar is shaded?



Fraction: _____

SET 37 • 5.NF.B.7.a

BACKWARD 2

Sharing equally • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2

15 is shared equally among 5 groups.

How much is in each group?

Answer: _____

True or false?

Sharing 8 equally between 2 groups gives 4 in each group.

A. True

B. False

Which expression shows 21 shared equally among 7 people?

A. $21 \div 7$

B. $21 - 7$

C. $7 \div 21$

D. 21×7

Divide.

$$16 \div 4$$

Quotient: _____

36 is shared equally among 6 people.

How much does each person get?

Answer: _____

SET 37 • 5.NF.B.7.a

BACKWARD 4

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Find the missing number.

$$\frac{1}{3} = \frac{?}{12}$$

Missing number: _____

True or false?

$$\frac{1}{2} = \frac{5}{10}$$

A. True

B. False

Which fraction is equal to $\frac{1}{4}$?

A. $\frac{4}{1}$

B. $\frac{1}{8}$

C. $\frac{2}{4}$

D. $\frac{2}{8}$

Find the missing number.

$$\frac{1}{5} = \frac{3}{?}$$

Missing number: _____

True or false?

$$\frac{1}{6} = \frac{3}{12}$$

A. True

B. False

SET 37 • 5.NF.B.7.a

FORWARD 1

Divide fractions • Grade 6 • The Number System • 6.NS.A.1

Divide.

$$\frac{3}{4} \div 3$$

Quotient: _____

Divide.

$$\frac{2}{5} \div 4$$

Quotient: _____

Divide.

$$\frac{5}{6} \div \frac{1}{3}$$

A. $\frac{5}{18}$

B. $2\frac{1}{2}$

C. $\frac{6}{9}$

D. $\frac{1}{2}$

$\frac{3}{4}$ pound of trail mix is shared equally among 6 bags.

How much trail mix is in each bag?

Pounds: _____

True or false?

$$\frac{2}{3} \div 2 = 1\frac{1}{3}$$

A. True

B. False

SET 37 • 5.NF.B.7.a

FORWARD 2

Divide rational numbers • Grade 7 • The Number System • 7.NS.A.2.b

Find the quotient.

$$-\frac{1}{3} \div 4$$

Quotient: _____

Find the quotient.

$$\frac{3}{5} \div \left(-\frac{3}{10}\right)$$

Quotient: _____

Find the quotient.

$$\left(-\frac{1}{2}\right) \div \left(-\frac{1}{4}\right)$$

A. $\frac{1}{8}$

B. $-\frac{1}{8}$

C. -2

D. 2

Find the quotient.

$$-\frac{4}{9} \div \frac{2}{3}$$

Quotient: _____

True or false?

$$-\left(\frac{1}{2} \div 5\right) = \left(-\frac{1}{2}\right) \div 5$$

A. True

B. False

Use models to divide a unit fraction by a whole number and check with multiplication

MAIN	Grade 5 main question	5.NF.B.7.a
BACKWARD 1	Unit fractions	3.NF.A.1
BACKWARD 2	A fraction as a multiple of a unit fraction	4.NF.B.4.a
FORWARD 1	Divide fractions and check with multiplication	6.NS.A.1
FORWARD 2	Divide rational numbers	7.NS.A.2.b

SET 69 • 5.NF.B.7.a

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.7.a

The bar shows $\frac{1}{3}$ shaded.

Split every third into 2 equal parts to find $\frac{1}{3} \div 2$. Then write the quotient.



Quotient: _____

Explain how you can use multiplication to check that

$$\frac{1}{4} \div 3 = \frac{1}{12}.$$

Explanation: _____

Which equation checks $\frac{1}{5} \div 4 = \frac{1}{20}$?

A. $\frac{1}{20} \div 4 = \frac{1}{80}$

B. $\frac{1}{5} + 4 = 4\frac{1}{5}$

C. $\frac{1}{20} \times 4 = \frac{1}{5}$

D. $\frac{1}{5} \times 4 = \frac{4}{5}$

True or false?

$\frac{1}{2} \div 3 = \frac{1}{6}$ because 3 groups of $\frac{1}{6}$ make $\frac{1}{2}$.

A. True

B. False

4 friends share $\frac{1}{2}$ of a pizza equally.

How much of the whole pizza does each friend get?
Describe a model that shows it.

Share: _____

Model: _____

SET 69 • 5.NF.B.7.a

BACKWARD 1

Unit fractions • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

A bar is cut into 8 equal parts.

What fraction of the bar is one part?

Fraction: _____

True or false?

2 parts of size $\frac{1}{6}$ make $\frac{2}{6}$.

A. True

B. False

What fraction of the bar is shaded?



A. $\frac{5}{6}$

B. $\frac{1}{5}$

C. $\frac{1}{6}$

D. $\frac{6}{1}$

A whole is cut into 3 equal parts.

What fraction of the whole is one part?

Fraction: _____

True or false?

$\frac{1}{4}$ is one of 3 equal parts of a whole.

A. True

B. False

SET 69 • 5.NF.B.7.a

BACKWARD 2

A fraction as a multiple of a unit fraction • Grade 4 • Number & Operations–Fractions • 4.NF.B.4.a

Write $\frac{3}{12}$ as a whole number times $\frac{1}{12}$.

Expression: _____

True or false?

$$4 \times \frac{1}{8} = \frac{4}{8}$$

A. True

B. False

Which fraction equals $6 \times \frac{1}{10}$?

A. $\frac{1}{60}$

B. $\frac{6}{10}$

C. $\frac{10}{6}$

D. $\frac{6}{60}$

Multiply.

$$2 \times \frac{1}{6}$$

Product: _____

True or false?

$$3 \times \frac{1}{9} = \frac{3}{27}$$

A. True

B. False

SET 69 • 5.NF.B.7.a

FORWARD 1

Divide fractions and check with multiplication • Grade 6 • The Number System
• 6.NS.A.1

Divide.

$$\frac{2}{3} \div 4$$

Quotient: _____

True or false?

$$\frac{3}{4} \div \frac{1}{4} = 3 \text{ because } 3 \times \frac{1}{4} = \frac{3}{4}.$$

A. True

B. False

Which equation checks $\frac{2}{3} \div \frac{3}{4} = \frac{8}{9}$?

A. $\frac{8}{9} \times \frac{3}{4} = \frac{2}{3}$

B. $\frac{3}{4} \div \frac{8}{9} = \frac{27}{32}$

C. $\frac{8}{9} \times \frac{2}{3} = \frac{16}{27}$

D. $\frac{2}{3} \times \frac{3}{4} = \frac{1}{2}$

Divide.

$$\frac{4}{5} \div 2$$

Quotient: _____

True or false?

$$\frac{1}{2} \div \frac{1}{3} = \frac{1}{6}$$

A. True

B. False

SET 69 • 5.NF.B.7.a

FORWARD 2

Divide rational numbers • Grade 7 • The Number System • 7.NS.A.2.b

Find the quotient.

$$-\frac{1}{4} \div 2$$

Quotient: _____

True or false?

$$-\frac{1}{3} \div 5 = -\frac{1}{15}$$

A. True

B. False

Find the quotient.

$$\frac{1}{6} \div (-3)$$

A. $\frac{1}{18}$

B. $-\frac{1}{2}$

C. -18

D. $-\frac{1}{18}$

Find the quotient.

$$\left(-\frac{1}{2}\right) \div (-4)$$

Quotient: _____

True or false?

$$\left(-\frac{1}{5}\right) \div 2 = \frac{1}{10}$$

A. True

B. False

Divide a whole number by a unit fraction

MAIN

Grade 5 main question

5.NF.B.7.b

BACKWARD 1

Division as "how many groups?"

3.OA.A.2

BACKWARD 2

Unit fractions that make a whole

3.NF.A.1

FORWARD 1

Divide whole numbers by fractions

6.NS.A.1

FORWARD 2

Unit rates from quantities measured in fractions

7.RP.A.1

SET 38 • 5.NF.B.7.b

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.7.b

Divide.

$$6 \div \frac{1}{5}$$

Quotient: _____

Divide.

$$3 \div \frac{1}{6}$$

Quotient: _____

How many $\frac{1}{3}$ s are in 7?

A. 10

B. 21

C. $\frac{7}{3}$

D. $\frac{1}{21}$

Divide.

$$9 \div \frac{1}{4}$$

Quotient: _____

True or false?

$$5 \div \frac{1}{2} = 2\frac{1}{2}$$

A. True

B. False

SET 38 • 5.NF.B.7.b

BACKWARD 1

Division as "how many groups?" • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.2

How many groups of 3 are in 18?

Groups: _____

True or false?

$20 \div 5$ can mean "How many 5s are in 20?"

A. True

B. False

Pam puts 24 eggs into cartons of 6.

Which equation finds the number of cartons?

A. $6 \div 24 = \square$

B. $24 - 6 = \square$

C. $24 \times 6 = \square$

D. $24 \div 6 = \square$

How many 4s are in 32?

Answer: _____

A 30-inch ribbon is cut into 5-inch pieces.
How many pieces are there?

Pieces: _____

SET 38 • 5.NF.B.7.b

BACKWARD 2

Unit fractions that make a whole • Grade 3 • Number & Operations–Fractions •
3.NF.A.1

How many $\frac{1}{4}$ s make 1 whole?

Answer: _____

True or false?

There are 3 thirds in 2 wholes.

A. True

B. False

How many $\frac{1}{6}$ s make 2 wholes?

- A.** 2
- B.** 8
- C.** 12
- D.** 6

How many $\frac{1}{8}$ s make 1 whole?

Answer: _____

True or false?

There are 2 halves in 2 wholes.

A. True

B. False

SET 38 • 5.NF.B.7.b

FORWARD 1

Divide whole numbers by fractions • Grade 6 • The Number System • 6.NS.A.1

Divide.

$$6 \div \frac{3}{4}$$

Quotient: _____

Divide.

$$4 \div \frac{2}{5}$$

Quotient: _____

How many $\frac{2}{3}$ -cup servings are in 8 cups?

A. 6

B. $\frac{1}{12}$

C. $\frac{16}{3}$

D. 12

Divide.

$$5 \div \frac{5}{6}$$

Quotient: _____

True or false?

$$3 \div \frac{3}{8} = \frac{9}{8}$$

A. True

B. False

SET 38 • 5.NF.B.7.b

FORWARD 2

Unit rates from quantities measured in fractions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

A car uses $\frac{1}{4}$ gallon of gas to go 6 miles.

How many miles does it go per gallon?

Miles per gallon: _____

A machine makes 5 parts every $\frac{1}{3}$ minute.

How many parts does it make per minute?

Parts per minute: _____

Lee runs 2 miles in $\frac{1}{6}$ hour.

What is his speed?

- A.** $\frac{1}{12}$ mile per hour
- B.** 3 miles per hour
- C.** 8 miles per hour
- D.** 12 miles per hour

A hose fills 10 liters in $\frac{1}{2}$ minute.

How many liters does it fill per minute?

Liters per minute: _____

True or false?

Walking 3 miles in $\frac{3}{4}$ hour is a speed of 4 miles per hour.

- A. True
- B. False

Use models to divide a whole number by a unit fraction and check with multiplication

MAIN	Grade 5 main question	5.NF.B.7.b
BACKWARD 1	Write whole numbers as sums of unit fractions	4.NF.B.3.b
BACKWARD 2	A fraction as a multiple of a unit fraction	4.NF.B.4.a
FORWARD 1	Divide whole numbers by fractions and check with multiplication	6.NS.A.1
FORWARD 2	Divide rational numbers	7.NS.A.2.b

SET 70 • 5.NF.B.7.b

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.7.b

Each bar is 1 whole.

Split each whole into fourths to find $2 \div \frac{1}{4}$. Then write the quotient.

1 whole



1 whole



Quotient: _____

Explain how to use multiplication to check that $5 \div \frac{1}{3} = 15$.

Explanation: _____

Which equation checks $6 \div \frac{1}{2} = 12$?

A. $12 \div \frac{1}{2} = 24$

B. $6 \times \frac{1}{2} = 3$

C. $12 \times \frac{1}{2} = 6$

D. $6 \times 12 = 72$

True or false?

$3 \div \frac{1}{5} = 15$ because each of the 3 wholes has 5 fifths.

A. True

B. False

Write a story problem for $4 \div \frac{1}{2}$ and solve it. Describe a model that shows the answer.

Story and answer: _____

Model: _____

SET 70 • 5.NF.B.7.b

BACKWARD 1

Write whole numbers as sums of unit fractions • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.b

Write 1 as a sum of thirds.

Sum: _____

True or false?

$$2 = \frac{5}{5} + \frac{5}{5}$$

A. True

B. False

Which sum equals 1?

A. $\frac{1}{4} + \frac{1}{4} + \frac{1}{4}$

B. $\frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4}$

C. $\frac{1}{4} + \frac{1}{4}$

D. $\frac{1}{2} + \frac{1}{4}$

Write 2 as a fraction with a denominator of 6.

Fraction: _____

True or false?

$$3 = \frac{3}{3}$$

A. True

B. False

SET 70 • 5.NF.B.7.b

BACKWARD 2

A fraction as a multiple of a unit fraction • Grade 4 • Number & Operations–Fractions • 4.NF.B.4.a

Write $\frac{8}{4}$ as a whole number times $\frac{1}{4}$.

Expression: _____

True or false?

$$15 \times \frac{1}{3} = 5$$

A. True

B. False

What is $10 \times \frac{1}{5}$?

A. 2

B. $\frac{1}{50}$

C. 50

D. $\frac{10}{50}$

Multiply.

$$6 \times \frac{1}{2}$$

Product: _____

True or false?

$$12 \times \frac{1}{4} = 48$$

A. True

B. False

SET 70 • 5.NF.B.7.b

FORWARD 1

Divide whole numbers by fractions and check with multiplication • Grade 6 •
The Number System • 6.NS.A.1

Divide.

$$3 \div \frac{3}{5}$$

Quotient: _____

True or false?

$2 \div \frac{2}{3} = 3$ because $3 \times \frac{2}{3} = 2$.

A. True

B. False

Which equation checks $4 \div \frac{2}{7} = 14$?

A. $4 \times 14 = 56$

B. $4 \times \frac{2}{7} = \frac{8}{7}$

C. $14 \div \frac{2}{7} = 49$

D. $14 \times \frac{2}{7} = 4$

How many $\frac{3}{4}$ -foot pieces can be cut from a 9-foot board?

Pieces: _____

True or false?

$$6 \div \frac{3}{4} = \frac{9}{2}$$

A. True

B. False

SET 70 • 5.NF.B.7.b

FORWARD 2

Divide rational numbers • Grade 7 • The Number System • 7.NS.A.2.b

Find the quotient.

$$-3 \div \frac{1}{4}$$

Quotient: _____

True or false?

$$4 \div \left(-\frac{1}{2}\right) = -8$$

A. True

B. False

Find the quotient.

$$(-5) \div \left(-\frac{1}{3}\right)$$

A. 15

B. $\frac{5}{3}$

C. -15

D. $-\frac{5}{3}$

Find the quotient.

$$-2 \div \frac{1}{6}$$

Quotient: _____

True or false?

$$(-6) \div \frac{1}{2} = -3$$

A. True

B. False

Solve word problems that divide unit fractions and whole numbers

MAIN	Grade 5 main question	5.NF.B.7.c
BACKWARD 1	Division word problems	3.OA.A.3
BACKWARD 3	Word problems: a whole number times a unit fraction	4.NF.B.4.c
FORWARD 1	Word problems that divide fractions by fractions	6.NS.A.1
FORWARD 2	Real-world problems that divide rational numbers	7.NS.A.3

SET 39 • 5.NF.B.7.c

MAIN

Grade 5 main question • Grade 5 • Number & Operations–Fractions •
5.NF.B.7.c

How many $\frac{1}{4}$ -cup scoops of birdseed are in 3 cups?

Scoops: _____

4 students share $\frac{1}{3}$ liter of paint equally.

How much paint does each student get?

Liters: _____

A 5-meter rope is cut into pieces that are $\frac{1}{4}$ meter long.

How many pieces are there?

A. 20

B. $\frac{1}{20}$

C. $\frac{5}{4}$

D. 9

$\frac{1}{8}$ of a pizza is shared equally by 2 people.

What fraction of the whole pizza does each person get?

Fraction: _____

True or false?

A 6-mile trail is split into $\frac{1}{2}$ -mile sections. There are 12 sections.

- A.** True
- B.** False

SET 39 • 5.NF.B.7.c

BACKWARD 1

Division word problems • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3

Kim has 28 beads. Each bracelet uses 7 beads.
How many bracelets can she make?

Bracelets: _____

True or false?

If 32 students sit 4 at each table, there are 6 tables.

- A.** True
- B.** False

Raj shares 40 cards equally among 5 friends.

How many cards does each friend get?

- A.** 8
- B.** 200
- C.** 35
- D.** 45

A 27-foot fence is split into 9 equal sections.
How long is each section?

Length: _____

48 eggs are packed 6 to a carton.

How many cartons are filled?

Cartons: _____

SET 39 • 5.NF.B.7.c

BACKWARD 3

Word problems: a whole number times a unit fraction • Grade 4 • Number & Operations–Fractions • 4.NF.B.4.c

Each scoop holds $\frac{1}{4}$ cup.

How many cups are in 8 scoops?

Cups: _____

True or false?

5 servings of $\frac{1}{3}$ pound each weigh $\frac{5}{3}$ pounds in all.

A. True

B. False

Each piece of ribbon is $\frac{1}{2}$ meter long.

How long are 6 pieces in all?

A. $\frac{1}{12}$ meter

B. $6\frac{1}{2}$ meters

C. 12 meters

D. 3 meters

Each lap is $\frac{1}{8}$ mile.

How far are 16 laps?

Distance: _____

True or false?

4 cups that each hold $\frac{1}{3}$ liter hold $\frac{4}{12}$ liter in all.

A. True

B. False

SET 39 • 5.NF.B.7.c

FORWARD 1

Word problems that divide fractions by fractions • Grade 6 • The Number System • 6.NS.A.1

How many $\frac{3}{8}$ -pound bags can be filled from 3 pounds of rice?

Bags: _____

$\frac{2}{3}$ of a cake is shared equally by 4 people.

What fraction of the whole cake does each person get?

Fraction: _____

A path is $\frac{9}{10}$ mile long. It is split into sections that are $\frac{3}{10}$ mile long.

How many sections are there?

A. $\frac{27}{100}$

B. 6

C. $\frac{1}{3}$

D. 3

A rectangle has an area of 2 square yards and a length of $2\frac{1}{2}$ yards.

What is its width?

Width: _____

True or false?

$\frac{3}{4}$ cup of oil fills exactly 3 bottles that hold $\frac{1}{4}$ cup each.

A. True

B. False

SET 39 • 5.NF.B.7.c

FORWARD 2

Real-world problems that divide rational numbers • Grade 7 • The Number System • 7.NS.A.3

A diver descends 18 feet in $\frac{1}{2}$ minute at a constant rate.

What is the change in depth per minute?

Change per minute: _____

The temperature fell 5°F in $\frac{1}{4}$ hour.

At this rate, what is the change in temperature per hour?

Change per hour: _____

A bank account changes by $-\$45$ in $\frac{1}{3}$ month at a steady rate.

What is the change per month?

A. $\$135$

B. $-\$45$

C. $-\$15$

D. $-\$135$

Find the value.

$$-6 \div \frac{1}{3}$$

Value: _____

True or false?

$$-2 \div \frac{1}{4} = -\frac{1}{2}$$

A. True

B. False

Convert from a larger unit to a smaller unit

MAIN	Grade 5 main question	5.MD.A.1
BACKWARD 1	Relative sizes of measurement units	4.MD.A.1
BACKWARD 2	Multiplying whole numbers	4.NBT.B.5
BACKWARD 4	Multiply whole numbers by 10, 100, and 1,000	4.NBT.A.1
FORWARD 1	Use ratio reasoning to convert measurement units	6.RP.A.3.d
FORWARD 2	Unit rates with fractional quantities and different units	7.RP.A.1

SET 40 • 5.MD.A.1

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.A.1

How many inches are in $4\frac{1}{2}$ feet?

Inches: _____

How many ounces are in 3 pounds?

Ounces: _____

A track is 2.5 kilometers long.

How many meters is that?

- A.** 250 m
- B.** 25,000 m
- C.** 2,500 m
- D.** 25 m

Convert.

6 yards = ____ feet

Feet: _____

A jug holds 3 gallons.

How many quarts is that?

Quarts: _____

SET 40 • 5.MD.A.1

BACKWARD 1

Relative sizes of measurement units • Grade 4 • Measurement & Data •
4.MD.A.1

How many inches are in 1 foot?

Inches: _____

True or false?

1 pound = 16 ounces

A. True

B. False

How many quarts are in 1 gallon?

- A.** 2
- B.** 16
- C.** 8
- D.** 4

How many meters are in 1 kilometer?

Meters: _____

True or false?

1 yard = 12 inches

A. True

B. False

SET 40 • 5.MD.A.1

BACKWARD 2

Multiplying whole numbers • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.5

Multiply.

$$16 \times 3$$

Product: _____

Multiply.

$$1,000 \times 7$$

Product: _____

True or false?

$$3 \times 36 = 98$$

A. True

B. False

$$12 \times 9 = ?$$

- A.** 96
- B.** 118
- C.** 21
- D.** 108

Multiply.

$$60 \times 5$$

Product: _____

SET 40 • 5.MD.A.1

BACKWARD 4

Multiply whole numbers by 10, 100, and 1,000 • Grade 4 • Number & Operations in Base Ten • 4.NBT.A.1

Multiply.

$$25 \times 100$$

Product: _____

True or false?

$$7 \times 1,000 = 700$$

A. True

B. False

$$42 \times 10 = ?$$

- A.** 4.2
- B.** 420
- C.** 4,200
- D.** 42

Multiply.

$$16 \times 1,000$$

Product: _____

True or false?

3×100 is 10 times as much as 3×10 .

A. True

B. False

SET 40 • 5.MD.A.1

FORWARD 1

Use ratio reasoning to convert measurement units • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d

There are 2.54 centimeters in 1 inch.

How many centimeters are in 5 inches?

Centimeters: _____

A car travels 3 miles every 4 minutes.

At this rate, how many miles does it travel in 1 hour?

Miles: _____

There are 3 feet in 1 yard.

How many yards are in 27 feet?

- A.** 81
- B.** 30
- C.** 9
- D.** 24

1 kilogram is about 2.2 pounds.

About how many pounds are in 15 kilograms?

Pounds: _____

True or false?

If 1 cup = 8 fluid ounces, then 5 cups = 13 fluid ounces.

- A. True
- B. False

SET 40 • 5.MD.A.1

FORWARD 2

Unit rates with fractional quantities and different units • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

A snail moves 6 feet in $\frac{1}{2}$ hour.

What is its speed in feet per minute?

Speed: _____

A cyclist rides $\frac{3}{4}$ mile in 3 minutes.

What is the speed in miles per hour?

Speed: _____

A faucet drips $\frac{1}{2}$ cup every 5 minutes.

How many cups does it drip per hour?

A. $2\frac{1}{2}$

B. $\frac{1}{10}$

C. 6

D. 10

A plant grows $\frac{1}{4}$ inch per week.

How many inches does it grow in 52 weeks?

Inches: _____

True or false?

Running $1\frac{1}{2}$ miles in 15 minutes is a speed of 6 miles per hour.

- A. True
- B. False

Convert from a smaller unit to a larger unit

MAIN	Grade 5 main question	5.MD.A.1
BACKWARD 1	Relative sizes of measurement units	4.MD.A.1
BACKWARD 2	Dividing by a one-digit number	4.NBT.B.6
BACKWARD 4	Write tenths and hundredths as decimals	4.NF.C.6
FORWARD 1	Use ratio reasoning to convert measurement units	6.RP.A.3.d
FORWARD 2	Equations for unit conversions	7.RP.A.2.c

SET 41 • 5.MD.A.1

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.A.1

Convert.

30 inches = ____ feet

Feet: _____

Convert.

750 milliliters = ____ liters

Liters: _____

A rope is 45 feet long.

How many yards is that?

A. 15

B. 5

C. 135

D. 42

Convert.

40 ounces = ____ pounds

Pounds: _____

Convert.

1,350 grams = ____ kilograms

Kilograms: _____

SET 41 • 5.MD.A.1

BACKWARD 1

Relative sizes of measurement units • Grade 4 • Measurement & Data •
4.MD.A.1

How many milliliters are in 1 liter?

Milliliters: _____

True or false?

1 kilogram = 1,000 grams

A. True

B. False

How many feet are in 1 yard?

A. 10

B. 3

C. 36

D. 12

How many cups are in 1 pint?

Cups: _____

True or false?

1 foot = 10 inches

A. True

B. False

SET 41 • 5.MD.A.1

BACKWARD 2

Dividing by a one-digit number • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Divide.

$$36 \div 3$$

Quotient: _____

Divide.

$$64 \div 4$$

Quotient: _____

True or false?

$$84 \div 7 = 14$$

A. True

B. False

$$90 \div 3 = ?$$

- A.** 33
- B.** 300
- C.** 27
- D.** 30

Divide.

$$5,000 \div 5$$

Quotient: _____

SET 41 • 5.MD.A.1

BACKWARD 4

Write tenths and hundredths as decimals • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write $\frac{75}{100}$ as a decimal.

Decimal: _____

True or false?

$$\frac{5}{10} = 0.5$$

A. True

B. False

Which decimal is equal to $\frac{45}{100}$?

- A.** 45
- B.** 0.045
- C.** 0.45
- D.** 4.5

Write $2\frac{6}{10}$ as a decimal.

Decimal: _____

True or false?

$$\frac{8}{10} = 0.08$$

A. True

B. False

SET 41 • 5.MD.A.1

FORWARD 1

Use ratio reasoning to convert measurement units • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d

There are 12 inches in 1 foot.

How many feet are in 102 inches?

Feet: _____

There are 1,000 meters in 1 kilometer.

How many kilometers are in 4,200 meters?

Kilometers: _____

A recipe needs 10 cups of milk. There are 4 cups in 1 quart.

How many quarts of milk are needed?

A. 40

B. 6

C. $2\frac{1}{2}$

D. 14

A trail is 5 kilometers long. 1 kilometer is about 0.62 mile.

About how many miles long is the trail?

Miles: _____

True or false?

There are 60 minutes in 1 hour, so 150 minutes = $1\frac{1}{2}$ hours.

A. True

B. False

SET 41 • 5.MD.A.1

FORWARD 2

Equations for unit conversions • Grade 7 • Ratios & Proportional Relationships •
7.RP.A.2.c

There are 2.54 centimeters in 1 inch.

Write an equation for the number of centimeters c in i inches.

Equation: _____

Write an equation for the number of feet f in y yards.

Equation: _____

Which equation converts k kilograms to g grams?

A. $k = 1,000g$

B. $g = 1,000k$

C. $g = k \div 1,000$

D. $g = k + 1,000$

Write an equation for the number of pounds p in z ounces.

Equation: _____

True or false?

The equation $m = 60h$ gives the number of minutes m in h hours.

- A. True
- B. False

Solve multistep problems that involve converting units

MAIN	Grade 5 main question	5.MD.A.1
BACKWARD 1	Word problems with measurement	4.MD.A.2
BACKWARD 2	Relative sizes of measurement units	4.MD.A.1
FORWARD 1	Convert units to compare rates	6.RP.A.3.d
FORWARD 2	Multistep ratio problems	7.RP.A.3

SET 42 • 5.MD.A.1

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.A.1

Ana has a 3-yard ribbon. She cuts off 4 feet.
How many feet of ribbon are left?

Feet: _____

A recipe needs 2 pounds of flour. Jo has 20 ounces.
How many more ounces does she need?

Ounces: _____

A runner ran 1.2 km and then 850 m.

How many meters did she run in all?

- A.** 1,050 m
- B.** 2,050 m
- C.** 9,700 m
- D.** 862 m

A jug holds 2 liters of water. Six glasses of 250 milliliters each are poured from it.

How many milliliters are left in the jug?

Milliliters: _____

A board is 8 feet long. It is cut into pieces that are 16 inches long.

How many pieces are there?

Pieces: _____

SET 42 • 5.MD.A.1

BACKWARD 1

Word problems with measurement • Grade 4 • Measurement & Data •
4.MD.A.2

A movie starts at 2:15 and lasts 1 hour 40 minutes.
What time does it end?

Time: _____

True or false?

3 quarters and 2 dimes make 85 cents.

A. True

B. False

Jake has 2 meters of string. He uses 75 cm.

How many centimeters are left?

A. 1.25 cm

B. 75 cm

C. 125 cm

D. 275 cm

A bag holds 3 kilograms of apples.

How many grams is that?

Grams: _____

Liz runs 400 meters four times.

How many meters does she run in all?

Meters: _____

SET 42 • 5.MD.A.1

BACKWARD 2

Relative sizes of measurement units • Grade 4 • Measurement & Data •
4.MD.A.1

How many ounces are in 2 pounds?

Ounces: _____

True or false?

1 gallon = 4 quarts

A. True

B. False

How many centimeters are in 1 meter?

- A.** 1,000
- B.** 100
- C.** 10
- D.** 12

How many feet are in 3 yards?

Feet: _____

True or false?

1 liter = 100 milliliters

A. True

B. False

SET 42 • 5.MD.A.1

FORWARD 1

Convert units to compare rates • Grade 6 • Ratios & Proportional Relationships
• 6.RP.A.3.d

A car uses 2 gallons of gas to drive 64 miles. There are 4 quarts in 1 gallon.

How many miles does the car drive per quart?

Miles: _____

Mo walks 3 kilometers in 40 minutes.

At this rate, how many meters does he walk per minute?

Meters per minute: _____

Fabric costs \$6 per yard.

How much do 12 feet of fabric cost?

A. \$72

B. \$18

C. \$24

D. \$2

A 24-ounce bag of rice costs \$3. There are 16 ounces in 1 pound.

What is the cost per pound?

Cost per pound: _____

True or false?

60 miles per hour is the same speed as 1 mile per minute.

- A. True
- B. False

SET 42 • 5.MD.A.1

FORWARD 2

Multistep ratio problems • Grade 7 • Ratios & Proportional Relationships •
7.RP.A.3

A 2-liter bottle of soda costs \$1.80. A 500-milliliter can costs \$0.60.

Which is the better buy per liter?

Better buy: _____

A map scale is 1 cm : 2 km. Two towns are 7.5 cm apart on the map. A car drives 60 km per hour.

How many minutes does the drive take?

Minutes: _____

One gallon of paint covers 350 square feet.

How many gallons are needed to cover 1,400 square feet?

A. 1,050

B. 3

C. 4

D. 5

A recipe for 4 people uses 600 grams of pasta.
How many kilograms are needed for 10 people?

Kilograms: _____

True or false?

At 5 miles per hour, a 2.5-mile walk takes 50 minutes.

A. True

B. False

Make a line plot of measurements in fractions of a unit

MAIN	Grade 5 main question	5.MD.B.2
BACKWARD 1	Reading a line plot with fractions	4.MD.B.4
BACKWARD 2	Fractions on a number line	3.NF.A.2
BACKWARD 3	Line plots of whole-number measurements	2.MD.D.9
FORWARD 1	Display numerical data in a dot plot	6.SP.B.4
FORWARD 2	Compare the difference in centers with the variability	7.SP.B.3

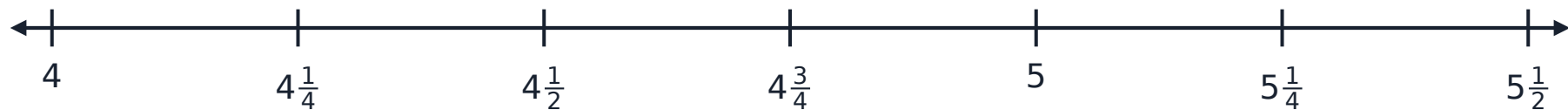
SET 43 • 5.MD.B.2

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.B.2

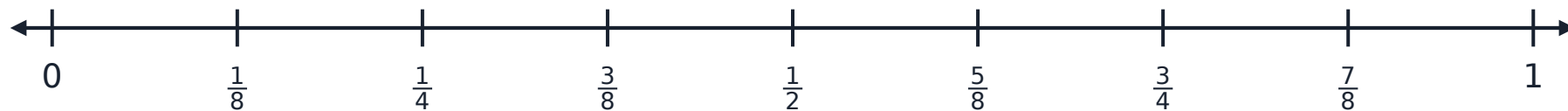
Pencil lengths (inches): $4\frac{1}{2}$, 5, $4\frac{1}{2}$, $5\frac{1}{4}$, $4\frac{3}{4}$, 5, $4\frac{1}{2}$

Make a line plot of the data.



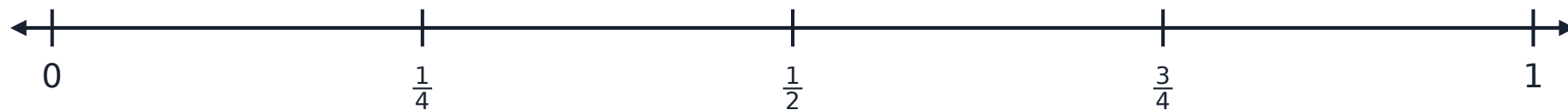
Rainfall (inches): $\frac{1}{4}$, $\frac{1}{2}$, $\frac{1}{8}$, $\frac{1}{4}$, $\frac{3}{8}$, $\frac{1}{4}$, $\frac{1}{2}$

Make a line plot of the data.



Apple weights (pounds): $\frac{3}{4}$, $\frac{1}{2}$, $\frac{3}{4}$, 1 , $\frac{1}{2}$, $\frac{3}{4}$, $\frac{1}{4}$

Make a line plot of the data.



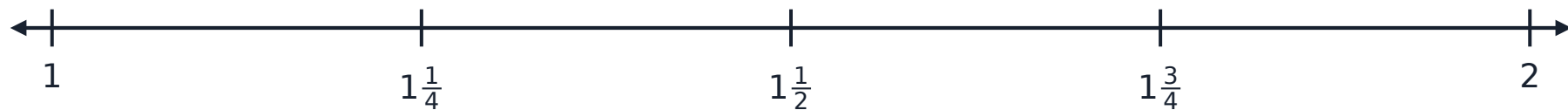
Lengths (inches): $\frac{1}{8}$, $\frac{3}{8}$, $\frac{3}{8}$, $\frac{5}{8}$, $\frac{3}{8}$

In a line plot, how many X's are drawn at $\frac{3}{8}$?

X's: _____

Water in each bottle (liters): $1\frac{1}{2}$, 2, $1\frac{1}{4}$, $1\frac{1}{2}$, $1\frac{3}{4}$, 2, $1\frac{1}{2}$

Make a line plot of the data.



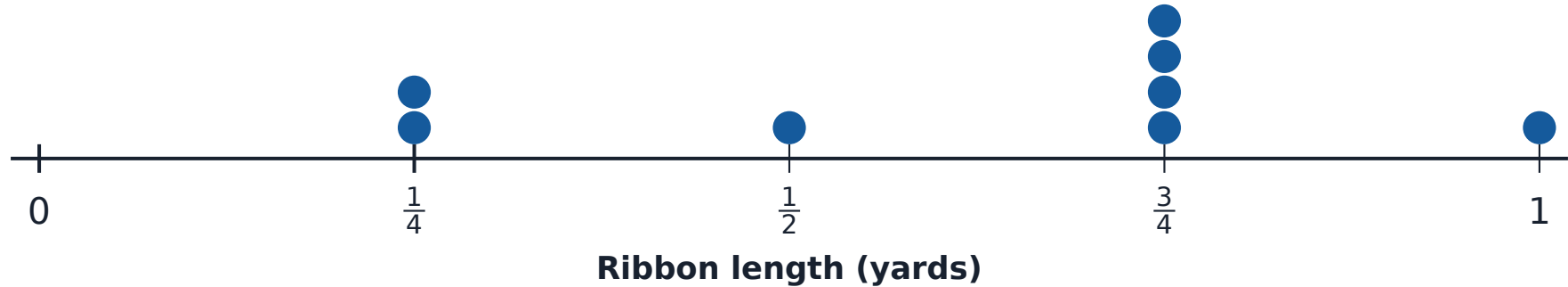
SET 43 • 5.MD.B.2

BACKWARD 1

Reading a line plot with fractions • Grade 4 • Measurement & Data • 4.MD.B.4

The line plot shows ribbon lengths.

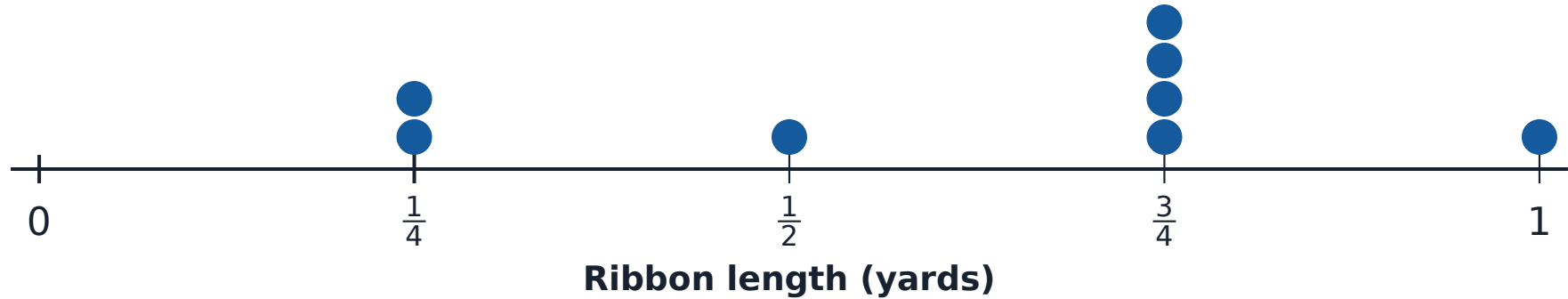
How many ribbons are $\frac{3}{4}$ yard long?



Ribbons: _____

True or false?

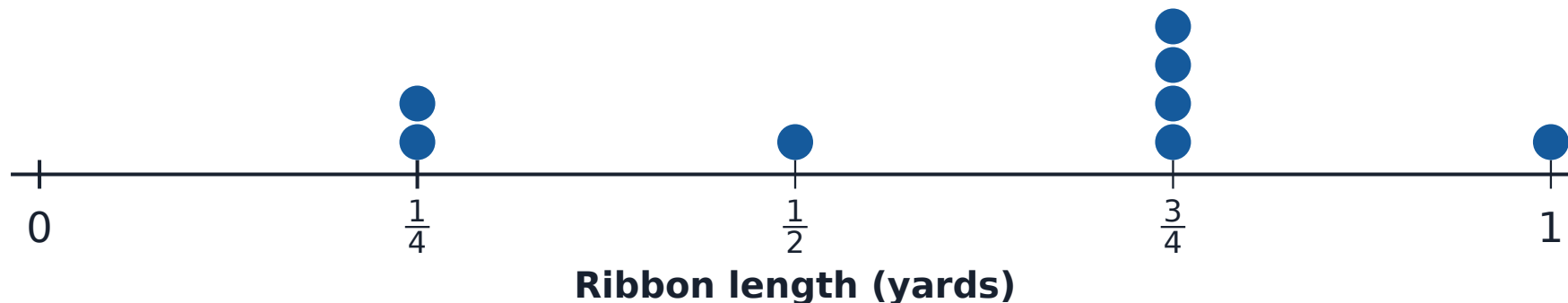
Two ribbons are $\frac{1}{4}$ yard long.



A. True

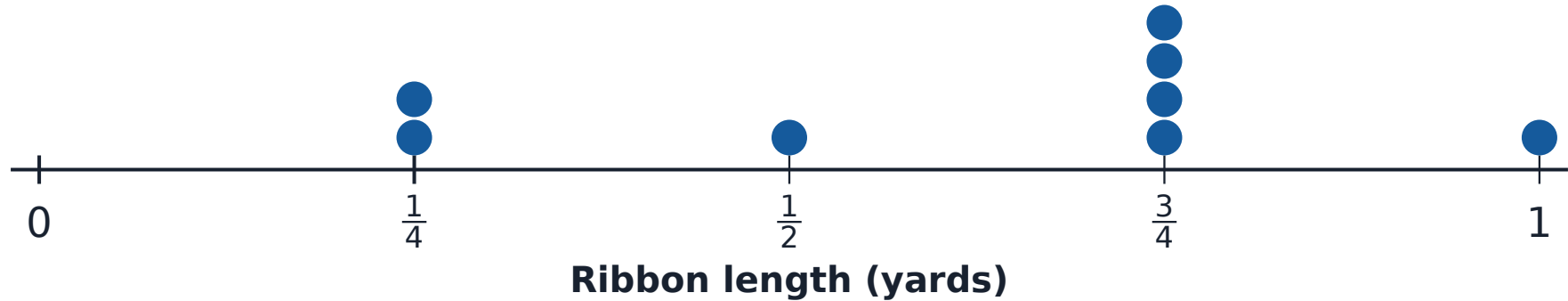
B. False

Which ribbon length is the most common?



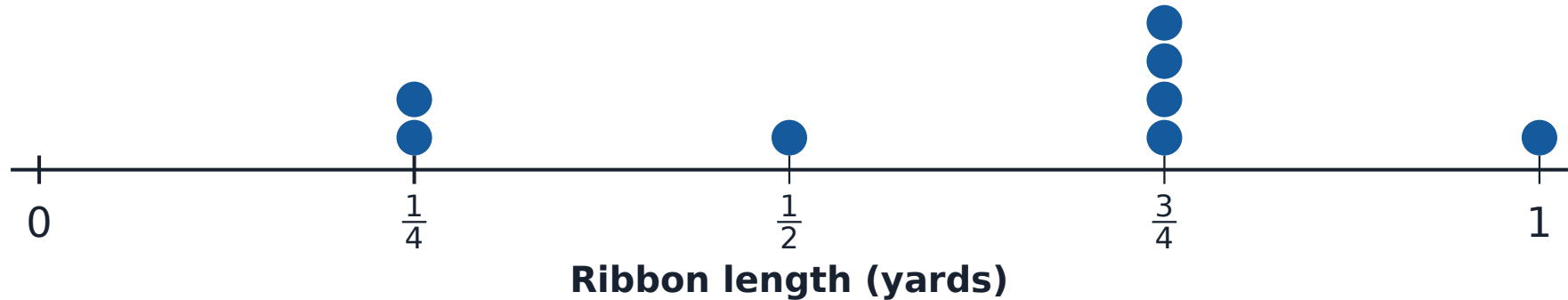
- A.** 1 yard **B.** $\frac{1}{2}$ yard **C.** $\frac{1}{4}$ yard **D.** $\frac{3}{4}$ yard

How many ribbons were measured?



Ribbons: _____

What is the difference between the longest and the shortest ribbon?



Difference: _____

SET 43 • 5.MD.B.2

BACKWARD 2

Fractions on a number line • Grade 3 • Number & Operations–Fractions •
3.NF.A.2

What fraction is at point P?



P = _____

True or false?

$\frac{2}{4}$ and $\frac{1}{2}$ are at the same point on a number line.

A. True

B. False

Into how many equal parts is the space from 0 to 1 divided?



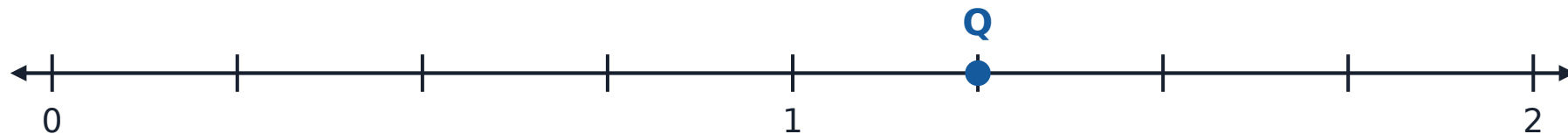
A. 5

B. 7

C. 6

D. 1

What number is at point Q?



Q = _____

True or false?

On a number line from 0 to 1 marked in eighths, $\frac{6}{8}$ is one mark to the left of 1.

- A. True
- B. False

SET 43 • 5.MD.B.2

BACKWARD 3

Line plots of whole-number measurements • Grade 2 • Measurement & Data •
2.MD.D.9

Lengths (inches): 5, 6, 6, 8, 6

In a line plot, how many X's are drawn at 6?

X's: _____

True or false?

A line plot shows each measurement as a mark above a number line.

- A.** True
- B.** False

Which number line would you use for a line plot of 3, 4, 4, 7?

- A.** A number line from 0 to 8
- B.** A number line from 0 to 3
- C.** A number line from 5 to 6
- D.** A number line from 10 to 20

Lengths (cm): 2, 4, 4, 5

Make a line plot of the data.



Lengths (feet): 9, 10, 10, 10, 12

Which length gets the most X's?

Length: _____

SET 43 • 5.MD.B.2

FORWARD 1

Display numerical data in a dot plot • Grade 6 • Statistics & Probability •
6.SP.B.4

Hours of sleep: 8, 9, 9, 10, 8, 9, 7

Make a dot plot of the data.



Number of siblings: 0, 1, 1, 2, 3, 1, 0, 2

Make a dot plot of the data.



Which number line fits a dot plot of 12, 15, 15, 18, 20?

- A.** A number line from 0 to 5
- B.** A number line from 20 to 40
- C.** A number line from 10 to 20
- D.** A number line from 15 to 18

Data: 4, 6, 6, 6, 7, 9

In a dot plot, how many dots go at 6?

Dots: _____

True or false?

A dot plot shows every value in a data set.

- A.** True
- B.** False

SET 43 • 5.MD.B.2

FORWARD 2

Compare the difference in centers with the variability • Grade 7 • Statistics & Probability • 7.SP.B.3

Plot A has a mean of 5 inches and Plot B has a mean of 8 inches. Each plot has a mean absolute deviation (MAD) of 1 inch.

The difference in the means is how many times the MAD?



Plot A: plant heights (inches)



Plot B: plant heights (inches)

Times the MAD: _____

True or false?

The difference in the means (3 inches) is 3 times the MAD, so the two distributions overlap only a little.



Plot A: plant heights (inches)



Plot B: plant heights (inches)

A. True

B. False

Which statement compares the two plots correctly?



Plot A: plant heights (inches)



Plot B: plant heights (inches)

- A.** The centers differ by less than 1 MAD, so the plots are about the same.
- B.** Plot A has much more variation than Plot B.
- C.** The plots cannot be compared because they show different plants.
- D.** The centers differ by 3 times the MAD, so Plot B plants are noticeably taller.

Two teams have mean heights of 150 cm and 156 cm.
Each team has a MAD of 3 cm.

The difference in the means is how many times the
MAD?

Times the MAD: _____

True or false?

Two data sets have means of 20 and 21, and each has a MAD of 4. The difference in the means is large compared with the variability.

- A.** True
- B.** False

Use operations on fractions to solve problems with line plot data

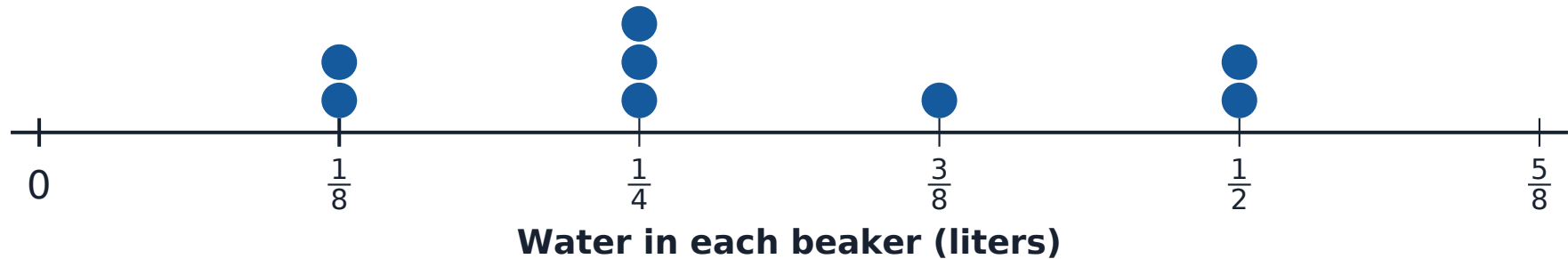
MAIN	Grade 5 main question	5.MD.B.2
BACKWARD 1	Solving problems from a line plot with like denominators	4.MD.B.4
BACKWARD 4	Rename halves and fourths as eighths	4.NF.A.1
BACKWARD 5	Add fractions with like denominators	4.NF.B.3.a
FORWARD 1	Find the mean of a data set	6.SP.B.5.c
FORWARD 2	Compare two populations using means	7.SP.B.4

SET 44 • 5.MD.B.2

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.B.2

The line plot shows the water in some beakers.
What is the total amount of water?



Total: _____

If all of the water were shared equally among the beakers, how much would each beaker hold?



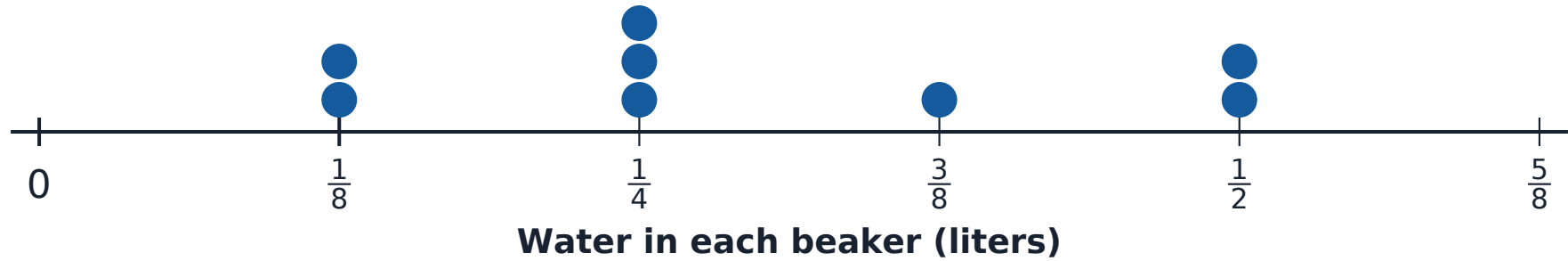
Liters: _____

How much more water is in the fullest beaker than in the least full beaker?



- A.** $\frac{1}{2}$ liter **B.** $\frac{1}{8}$ liter **C.** $\frac{5}{8}$ liter **D.** $\frac{3}{8}$ liter

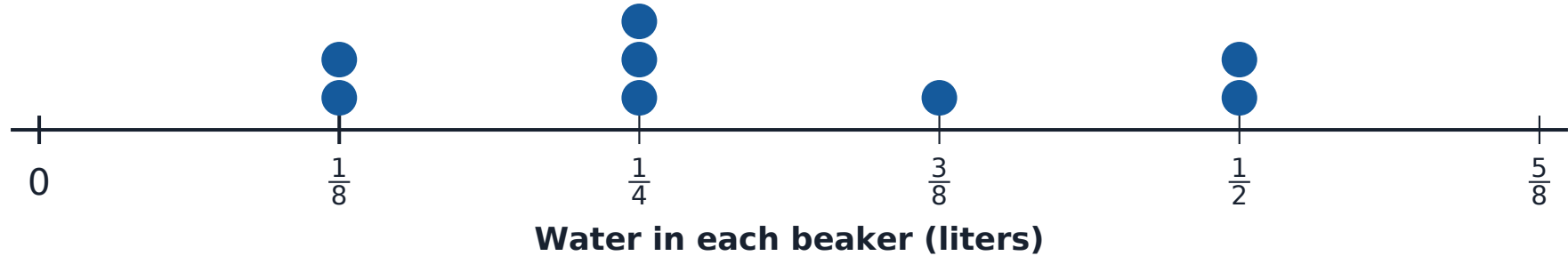
What is the total amount of water in the beakers that hold $\frac{1}{4}$ liter each?



Total: _____

True or false?

The beakers that hold $\frac{1}{2}$ liter each contain 1 liter in all.



A. True

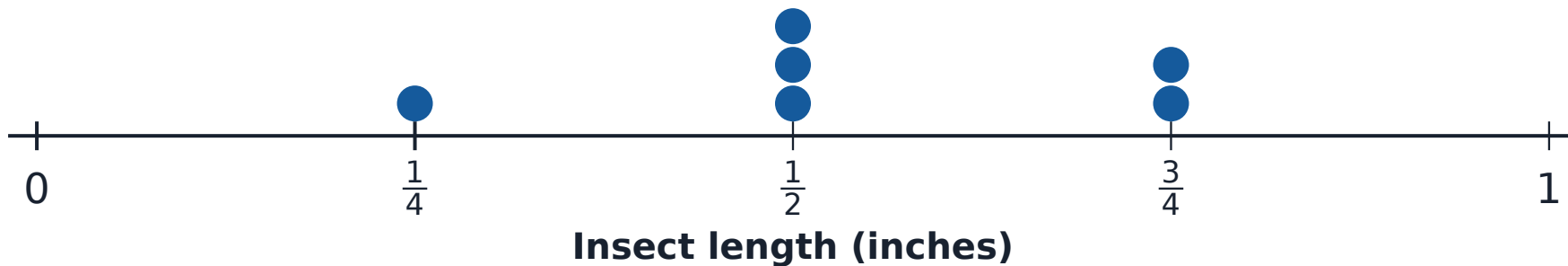
B. False

SET 44 • 5.MD.B.2

BACKWARD 1

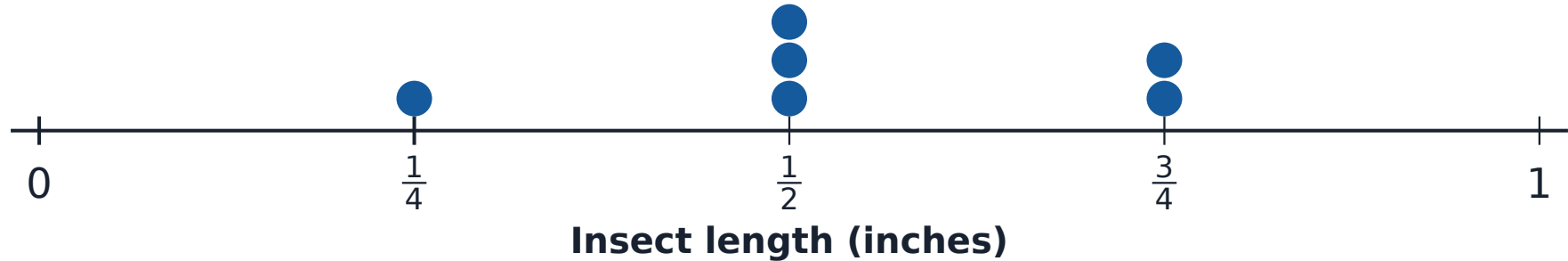
Solving problems from a line plot with like denominators • Grade 4 •
Measurement & Data • 4.MD.B.4

How many insects are $\frac{1}{2}$ inch long?



Insects: _____

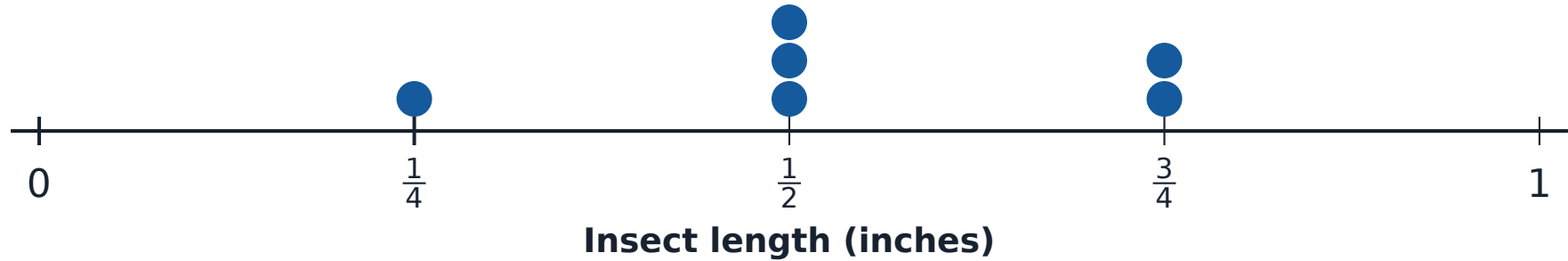
What is the difference between the longest and the shortest insect?



Difference: _____

True or false?

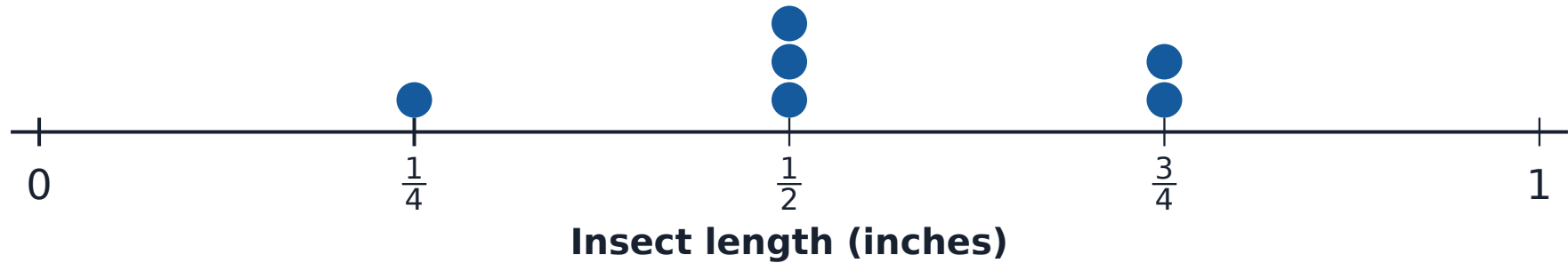
The total length of the $\frac{3}{4}$ -inch insects is $1\frac{1}{2}$ inches.



A. True

B. False

How many insects were measured?



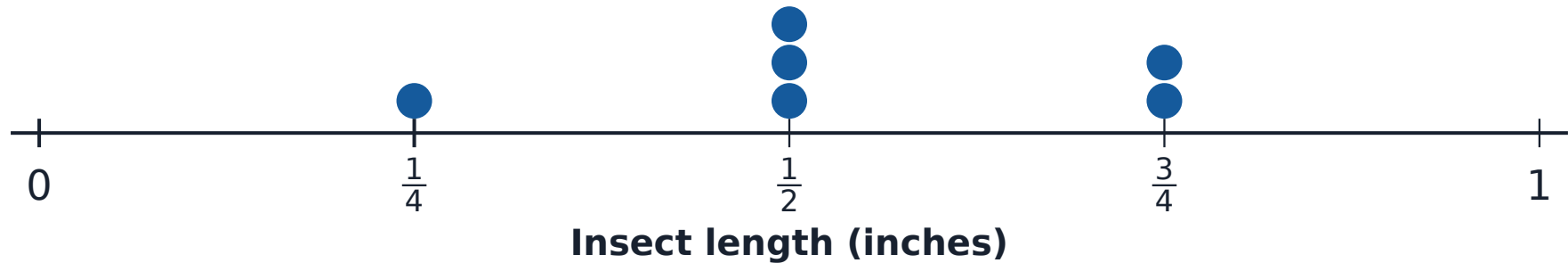
A. 4

B. 3

C. 5

D. 6

What is the total length of the $\frac{1}{2}$ -inch insects?



Total: _____

SET 44 • 5.MD.B.2

BACKWARD 4

Rename halves and fourths as eighths • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Write $\frac{1}{4}$ as eighths.

Fraction: _____

True or false?

$$\frac{1}{2} = \frac{4}{8}$$

A. True

B. False

Which fraction is equal to $\frac{3}{4}$?

A. $\frac{6}{8}$

B. $\frac{3}{8}$

C. $\frac{7}{8}$

D. $\frac{4}{8}$

Find the missing number.

$$\frac{1}{2} = \frac{?}{8}$$

Missing number: _____

True or false?

$$\frac{3}{8} = \frac{3}{4}$$

A. True

B. False

SET 44 • 5.MD.B.2

BACKWARD 5

Add fractions with like denominators • Grade 4 • Number & Operations–Fractions • 4.NF.B.3.a

Add.

$$\frac{1}{8} + \frac{2}{8}$$

Sum: _____

True or false?

$$\frac{3}{8} + \frac{3}{8} = \frac{6}{16}$$

A. True

B. False

$$\frac{2}{8} + \frac{4}{8} + \frac{1}{8} = ?$$

A. $\frac{6}{8}$

B. $\frac{7}{8}$

C. $\frac{7}{24}$

D. 1

Add.

$$\frac{5}{8} + \frac{3}{8}$$

Sum: _____

True or false?

$$\frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{3}{8}$$

A. True

B. False

SET 44 • 5.MD.B.2

FORWARD 1

Find the mean of a data set • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the mean.

3, 5, 6, 8, 8

Mean: _____

Find the mean.

12, 15, 21

Mean: _____

Four friends have 2, 4, 5, and 9 marbles. They share the marbles equally.

How many marbles does each friend get?

A. 4.5

B. 5

C. 20

D. 4

The dot plot shows the goals scored in 6 games.
What is the mean number of goals?



Mean: _____

True or false?

The mean of 4, 4, and 10 is 4.

A. True

B. False

SET 44 • 5.MD.B.2

FORWARD 2

Compare two populations using means • Grade 7 • Statistics & Probability •
7.SP.B.4

Random samples: Sample A has a mean of 15 minutes.
Sample B has a mean of 22 minutes.

How much greater is the mean of Sample B?

Difference: _____

Two random samples of plant heights have means of 18 cm and 24 cm. Both samples have similar variability.

Which conclusion is best?

- A.** The populations are the same.
- B.** Plants in the second population tend to be taller.
- C.** Plants in the first population tend to be taller.
- D.** No conclusion is possible.

True or false?

Comparing the means of random samples is one way to compare two populations.

- A.** True
- B.** False

In random samples, Class X sleeps a mean of 7.5 hours and Class Y sleeps a mean of 8.25 hours.

Which class probably sleeps more?

Class: _____

A random sample of 30 students from School A spends a mean of 40 minutes on homework. A random sample of 30 students from School B spends a mean of 50 minutes. The samples have similar variability.

What can you infer about all the students at the two schools?

Inference: _____

Understand a unit cube and a cubic unit of volume

MAIN	Grade 5 main question	5.MD.C.3.a
BACKWARD 1	Unit squares and square units	3.MD.C.5.a
BACKWARD 2	Faces, edges, and vertices of a cube	2.G.A.1
FORWARD 1	Volume of cubes with fractional edge lengths	6.G.A.2
FORWARD 2	Volume of right prisms	7.G.B.6

SET 45 • 5.MD.C.3.a

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.3.a

True or false?

A cube with edges that are 1 unit long is called a unit cube.

- A.** True
- B.** False

What is the volume of a unit cube?

- A.** 1 square unit
- B.** 6 square units
- C.** 1 cubic unit
- D.** 3 cubic units

What is the volume of a cube with edges that are 1 cm long?

Volume: _____

Which object has a volume closest to 1 cubic inch?

- A.** A shoebox
- B.** A sugar cube
- C.** A bathtub
- D.** A school bus

True or false?

Volume measures the amount of space inside a solid figure.

A. True

B. False

SET 45 • 5.MD.C.3.a

BACKWARD 1

Unit squares and square units • Grade 3 • Measurement & Data • 3.MD.C.5.a

True or false?

A square with sides that are 1 unit long is called a unit square.

A. True

B. False

What is the area of a unit square?

- A.** 1 square unit
- B.** 1 cubic unit
- C.** 2 square units
- D.** 4 square units

How many unit squares cover a 3-by-3 square?

Unit squares: _____

True or false?

Area is measured in cubic units.

A. True

B. False

A square has sides that are 1 inch long.

What is its area?

Area: _____

SET 45 • 5.MD.C.3.a

BACKWARD 2

Faces, edges, and vertices of a cube • Grade 2 • Geometry • 2.G.A.1

How many faces does a cube have?

Faces: _____

True or false?

All the faces of a cube are squares.

A. True

B. False

How many edges does a cube have?

- A.** 6
- B.** 12
- C.** 4
- D.** 8

How many vertices (corners) does a cube have?

Vertices: _____

True or false?

All the edges of a cube are the same length.

A. True

B. False

SET 45 • 5.MD.C.3.a

FORWARD 1

Volume of cubes with fractional edge lengths • Grade 6 • Geometry • 6.G.A.2

What is the volume of a cube with edges that are $\frac{1}{2}$ inch long?

Volume: _____

How many cubes with $\frac{1}{2}$ -inch edges fit in a cube with 1-inch edges?

Cubes: _____

A cube has edges that are $\frac{1}{3}$ foot long.

What is its volume?

A. $\frac{1}{3}$ cubic foot

B. $\frac{1}{9}$ cubic foot

C. $\frac{1}{27}$ cubic foot

D. 1 cubic foot

A box is packed with 16 cubes that each have $\frac{1}{2}$ -cm edges.

What is the volume of the box?

Volume: _____

True or false?

Four cubes with $\frac{1}{2}$ -unit edges have a total volume of 1 cubic unit.

A. True

B. False

SET 45 • 5.MD.C.3.a

FORWARD 2

Volume of right prisms • Grade 7 • Geometry • 7.G.B.6

A prism has a base area of 24 square cm and a height of 5 cm.

What is its volume?

Volume: _____

A cube has edges that are 3 m long.

What is its volume?

Volume: _____

A triangular prism has a triangle base with an area of 6 square inches. The prism is 10 inches long.

What is its volume?

A. 60 cubic inches

B. 30 cubic inches

C. 16 cubic inches

D. 600 cubic inches

A box is 2.5 cm by 4 cm by 6 cm.

What is its volume?

Volume: _____

True or false?

Volume is measured in cubic units.

A. True

B. False

A solid packed with n unit cubes has a volume of n cubic units

MAIN

Grade 5 main question

5.MD.C.3.b

BACKWARD 1

Measuring area by counting unit squares

3.MD.C.6

BACKWARD 2

Arrays and repeated addition

2.OA.C.4

FORWARD 1

Pack prisms with fractional unit cubes

6.G.A.2

FORWARD 2

Solve real-world volume problems

7.G.B.6

SET 46 • 5.MD.C.3.b

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.3.b

A box is filled with 30 unit cubes with no gaps or overlaps.

What is its volume?

Volume: _____

True or false?

If unit cubes fill a box but leave gaps, the number of cubes is less than the volume of the box.

- A. True
- B. False

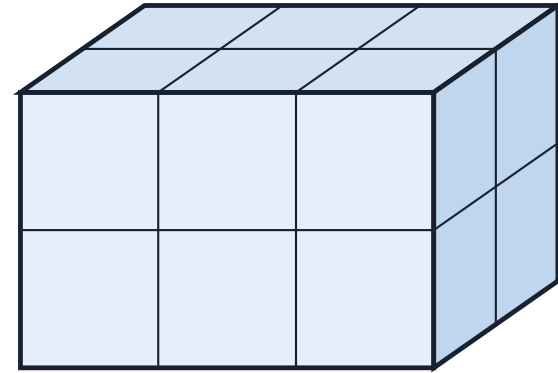
A solid is built from 18 unit cubes with no gaps or overlaps.

What is its volume?

- A.** 18 square units
- B.** 18 cubic units
- C.** 54 cubic units
- D.** 6 cubic units

Each cube is 1 cubic unit.

What is the volume of the solid?



Volume: _____

A tower is made of 9 cubes stacked with no gaps. Each cube is 1 cubic inch.

What is the volume of the tower?

Volume: _____

SET 46 • 5.MD.C.3.b

BACKWARD 1

Measuring area by counting unit squares • Grade 3 • Measurement & Data •
3.MD.C.6

A shape is covered by 12 unit squares with no gaps or overlaps.

What is its area?

Area: _____

True or false?

Overlapping squares give a wrong count for area.

A. True

B. False

A rectangle is covered by 20 unit squares.

What is its area?

- A.** 10 square units
- B.** 40 square units
- C.** 20 cubic units
- D.** 20 square units

Each square is 1 square cm. A rectangle is covered by 15 of these squares.

What is its area?

Area: _____

True or false?

Area can be found by counting the unit squares that cover a shape.

- A.** True
- B.** False

SET 46 • 5.MD.C.3.b

BACKWARD 2

Arrays and repeated addition • Grade 2 • Operations & Algebraic Thinking •
2.OA.C.4

An array has 4 rows with 5 in each row.

How many are there in all?

Total: _____

True or false?

3 rows of 5 make 8.

A. True

B. False

Which equation matches an array with 2 rows of 5?

A. $2 + 5 = 7$

B. $5 + 5 = 10$

C. $5 - 2 = 3$

D. $2 + 2 = 4$

An array has 5 rows with 3 in each row.

How many are there in all?

Total: _____

Write an addition equation for 3 rows of 4.

Equation: _____

SET 46 • 5.MD.C.3.b

FORWARD 1

Pack prisms with fractional unit cubes • Grade 6 • Geometry • 6.G.A.2

A box is packed with cubes that have $\frac{1}{2}$ -inch edges. It is 4 cubes long, 2 cubes wide, and 2 cubes tall.

What is the volume of the box?

Volume: _____

How many cubes with $\frac{1}{4}$ -inch edges fill a cube with 1-inch edges?

Cubes: _____

A prism is packed with 27 cubes that each have $\frac{1}{3}$ -foot edges.

What is its volume?

- A.** 27 cubic feet
- B.** 9 cubic feet
- C.** 1 cubic foot
- D.** 3 cubic feet

A box is $1\frac{1}{2}$ in. by 1 in. by $\frac{1}{2}$ in.

How many cubes with $\frac{1}{2}$ -inch edges fill it?

Cubes: _____

True or false?

40 cubes with $\frac{1}{2}$ -unit edges have a total volume of 20 cubic units.

- A. True
- B. False

SET 46 • 5.MD.C.3.b

FORWARD 2

Solve real-world volume problems • Grade 7 • Geometry • 7.G.B.6

A fish tank is 30 cm by 20 cm by 25 cm.

What is its volume?

Volume: _____

A prism has a volume of 96 cubic inches and a base area of 12 square inches.

What is its height?

Height: _____

A triangular prism has a triangle base with a base of 4 m and a height of 3 m. The prism is 7 m long.

What is its volume?

A. 84 cubic meters

B. 42 cubic meters

C. 21 cubic meters

D. 14 cubic meters

A storage box is 1.5 m by 1 m by 0.8 m.

What is its volume?

Volume: _____

True or false?

Two prisms with the same base area and the same height have the same volume.

A. True

B. False

Measure volume by counting unit cubes

MAIN	Grade 5 main question	5.MD.C.4
BACKWARD 1	Counting the squares in one layer	3.MD.C.7.a
BACKWARD 2	Equal groups	3.OA.A.1
FORWARD 1	Volume with fractional edge lengths	6.G.A.2
FORWARD 2	Volume of right prisms	7.G.B.6

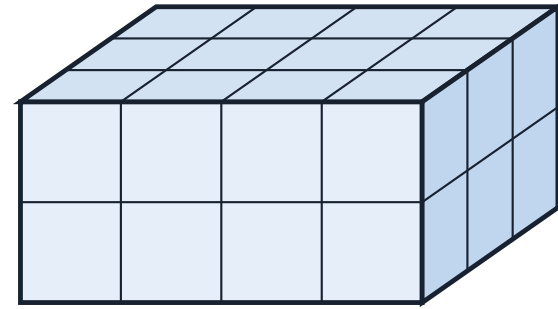
SET 47 • 5.MD.C.4

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.4

Each cube is 1 cubic cm.

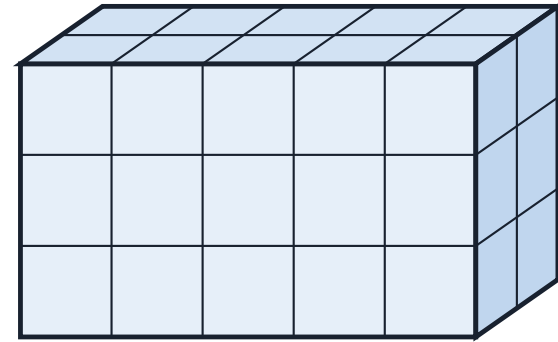
What is the volume of the prism?



Volume: _____

Each cube is 1 cubic inch.

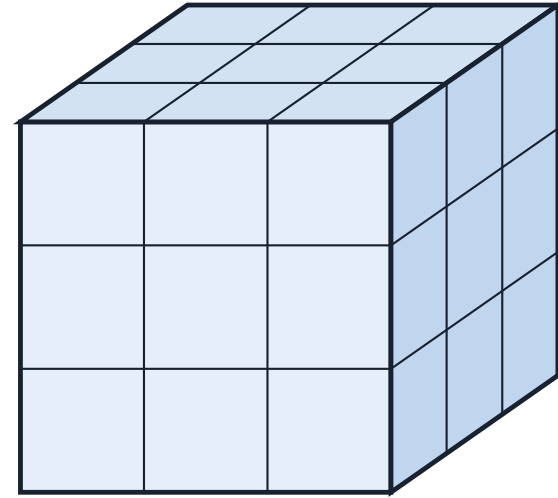
What is the volume of the prism?



Volume: _____

Each cube is 1 cubic foot.

What is the volume of the prism?



- A.** 18 cubic feet **B.** 54 cubic feet
- C.** 27 cubic feet **D.** 9 cubic feet

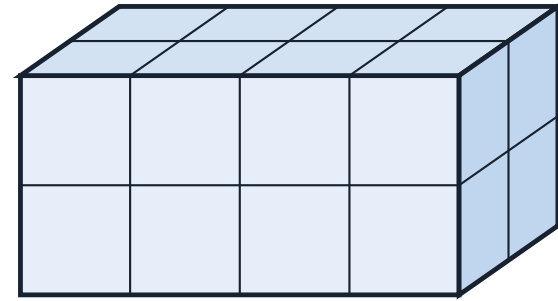
A box is filled with number cubes. The bottom layer has 6 rows of 4 cubes, and there are 3 layers.

What is the volume of the box, measured in number cubes?

Volume: _____

True or false?

Each cube is 1 cubic cm.
The volume of the prism is
16 cubic cm.



A. True

B. False

SET 47 • 5.MD.C.4

BACKWARD 1

Counting the squares in one layer • Grade 3 • Measurement & Data •
3.MD.C.7.a

The bottom layer of a box has 5 rows of 4 cubes.
How many cubes are in the layer?

Cubes: _____

True or false?

A layer with 3 rows of 6 cubes has 18 cubes.

A. True

B. False

How many unit squares cover a 4-by-6 rectangle?

- A.** 10
- B.** 46
- C.** 24
- D.** 20

A rectangle is tiled with 7 rows of 3 unit squares.
How many squares are there?

Squares: _____

True or false?

The number of unit squares that tile a rectangle is its length times its width.

- A. True
- B. False

SET 47 • 5.MD.C.4

BACKWARD 2

Equal groups • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1

There are 3 layers with 12 cubes in each layer.
How many cubes are there in all?

Cubes: _____

True or false?

4 layers of 10 cubes is 14 cubes.

A. True

B. False

Which expression shows 5 layers of 8 cubes?

A. $8 - 5$

B. 5×8

C. $5 + 8$

D. 58

Multiply.

$$6 \times 9$$

Product: _____

There are 2 layers with 15 cubes in each layer.
How many cubes are there in all?

Cubes: _____

SET 47 • 5.MD.C.4

FORWARD 1

Volume with fractional edge lengths • Grade 6 • Geometry • 6.G.A.2

A prism is 3 cubes long, 2 cubes wide, and 2 cubes tall.
Each cube has $\frac{1}{2}$ -inch edges.

What is the volume of the prism?

Volume: _____

A box is $2\frac{1}{2}$ cm by 2 cm by 1 cm.

What is its volume?

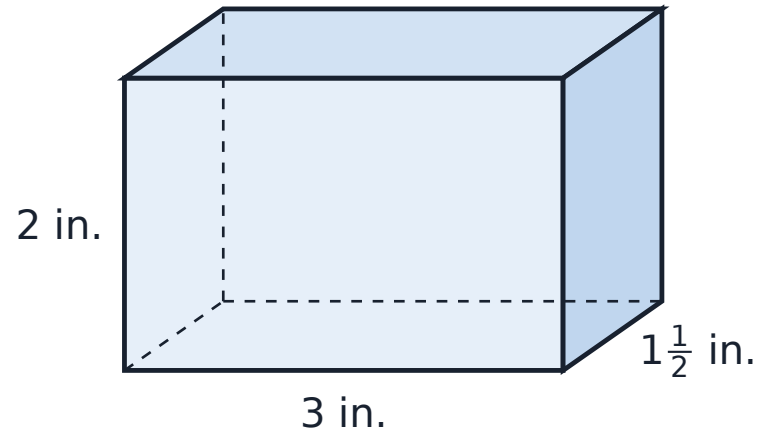
Volume: _____

How many cubes with $\frac{1}{2}$ -foot edges fill a box that is 2 ft by 1 ft by 1 ft?

- A.** 4
- B.** 8
- C.** 16
- D.** 2

Find the volume of the prism.

Volume: _____



True or false?

A cube with edges of $1\frac{1}{2}$ feet has a volume of $2\frac{1}{4}$ cubic feet.

- A.** True
- B.** False

SET 47 • 5.MD.C.4

FORWARD 2

Volume of right prisms • Grade 7 • Geometry • 7.G.B.6

A prism has a rectangular base that is 5 units by 4 units.
It is 6 units tall.

What is its volume?

Volume: _____

A box has a volume of 120 cubic cm. Its base is 6 cm by 5 cm.

What is its height?

Height: _____

A prism has a base area of 15 square feet and a height of 4 feet.

What is its volume?

- A.** 60 cubic feet
- B.** 30 cubic feet
- C.** 120 cubic feet
- D.** 19 cubic feet

A triangular prism has a base area of 9 square cm and a length of 12 cm.

What is its volume?

Volume: _____

True or false?

A box that is 10 in. by 2 in. by 3 in. has a volume of 15 cubic inches.

A. True

B. False

Relate packing with unit cubes to the formulas $V = l \times w \times h$ and $V = b \times h$

MAIN	Grade 5 main question	5.MD.C.5.a
BACKWARD 1	Area by tiling	3.MD.C.7.a
BACKWARD 2	Multiplying three numbers in any grouping	3.OA.B.5
FORWARD 1	Apply the volume formulas with fractional edge lengths	6.G.A.2
FORWARD 2	Volume of right prisms using $V = b \times h$	7.G.B.6

SET 48 • 5.MD.C.5.a

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.5.a

A prism is packed with unit cubes. It is 5 cubes long, 3 cubes wide, and 4 layers tall.

What is its volume?

Volume: _____

The bottom layer of a prism has 12 unit cubes. There are 5 layers.

Which expression gives the volume?

A. $12 \times 12 \times 5$

B. 12×5

C. 5×5

D. $12 + 5$

True or false?

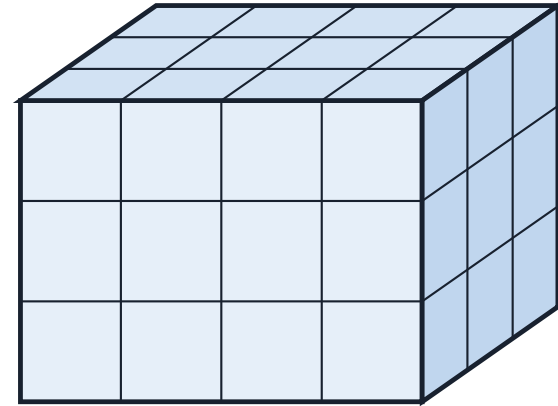
For a prism that is 4 by 3 by 2, $(4 \times 3) \times 2$ and $4 \times (3 \times 2)$ give the same volume.

A. True

B. False

Each cube is 1 cubic unit.

Find the volume by
multiplying the area of the
base by the height.



Volume: _____

A prism has a base area of 18 square units and a height of 4 units.

What is its volume?

Volume: _____

SET 48 • 5.MD.C.5.a

BACKWARD 1

Area by tiling • Grade 3 • Measurement & Data • 3.MD.C.7.a

A rectangle is tiled with unit squares in 3 rows of 4.
What is its area?

Area: _____

True or false?

The area of a 5-by-2 rectangle is 10 square units.

- A. True
- B. False

Which expression gives the area of a 6-by-3 rectangle?

A. $6 - 3$

B. $6 + 3$

C. 6×3

D. $2 \times (6 + 3)$

How many unit squares tile a 9-by-2 rectangle?

Squares: _____

True or false?

Tiling a rectangle and multiplying its side lengths give the same area.

A. True

B. False

SET 48 • 5.MD.C.5.a

BACKWARD 2

Multiplying three numbers in any grouping • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$(2 \times 3) \times 4 = 2 \times 3 + 4$$

A. True

B. False

Fill in the blank.

$$(5 \times 2) \times 3 = 5 \times (2 \times \underline{\quad})$$

Blank: _____

Which expression is equal to $4 \times 5 \times 2$?

A. 9×2

B. $20 + 2$

C. $4 + 5 + 2$

D. 4×10

Multiply.

$$3 \times 2 \times 5$$

Product: _____

True or false?

$$6 \times (2 \times 3) = 6 + 6$$

A. True

B. False

SET 48 • 5.MD.C.5.a

FORWARD 1

Apply the volume formulas with fractional edge lengths • Grade 6 • Geometry •
6.G.A.2

Use $V = l \times w \times h$ to find V when $l = 3\frac{1}{2}$ in., $w = 2$ in.,
and $h = 1\frac{1}{2}$ in.

Volume: _____

Use $V = b \times h$ to find the volume of a prism with $b = 6\frac{1}{4}$ square cm and $h = 4$ cm.

Volume: _____

A prism is packed with cubes that have edges $\frac{1}{2}$ unit long: 6 cubes long, 4 cubes wide, and 2 cubes tall.

Which shows its volume?

A. $6 + 4 + 2 = 12$ cubic units

B. $3 \times 2 \times 1 = 6$ cubic units

C. $6 \times 4 \times 2 = 48$ cubic units

D. $\frac{1}{2} \times 48 = 24$ cubic units

A cube has edges that are $\frac{3}{4}$ foot long.

What is its volume?

Volume: _____

True or false?

For a rectangular prism, $V = l \times w \times h$ and $V = b \times h$ give the same volume.

- A. True
- B. False

SET 48 • 5.MD.C.5.a

FORWARD 2

Volume of right prisms using $V = b \times h$ • Grade 7 • Geometry • 7.G.B.6

A triangular prism has a triangle base with a base of 6 cm and a height of 4 cm. The prism is 10 cm long.

What is its volume?

Volume: _____

A prism has a trapezoid base with an area of 14 square inches. The prism is 5 inches tall.

What is its volume?

Volume: _____

A prism has a volume of 84 cubic meters and a base area of 12 square meters.

What is its height?

A. 1,008 m

B. 7 m

C. 72 m

D. 96 m

A rectangular prism is 8 by 5 by 2.5.

What is its volume?

Volume: _____

True or false?

$V = b \times h$ works for triangular prisms as well as rectangular prisms.

A. True

B. False

Apply $V = l \times w \times h$ and $V = b \times h$ to solve real-world problems

MAIN	Grade 5 main question	5.MD.C.5.b
BACKWARD 1	Multiplying multi-digit numbers	4.NBT.B.5
BACKWARD 2	The area of the base	4.MD.A.3
FORWARD 1	Real-world volume with fractional edge lengths	6.G.A.2
FORWARD 2	Real-world volume problems with right prisms	7.G.B.6

SET 49 • 5.MD.C.5.b

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.5.b

A fish tank is 24 inches long, 12 inches wide, and 15 inches tall.

What is its volume?

Volume: _____

A sandbox has a base area of 30 square feet. It is filled with sand to a depth of 2 feet.

How much sand does it hold?

Volume: _____

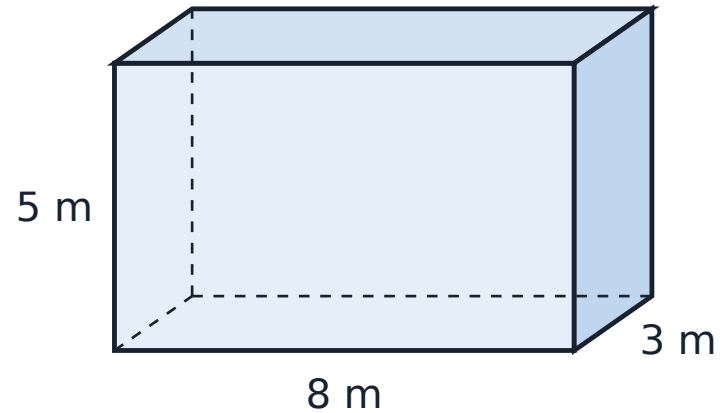
A box is 9 cm long, 4 cm wide, and 5 cm tall.

What is its volume?

- A.** 90 cubic cm
- B.** 36 cubic cm
- C.** 180 cubic cm
- D.** 18 cubic cm

Find the volume of the prism.

Volume: _____



A cube-shaped box has edges that are 7 inches long.
What is its volume?

Volume: _____

SET 49 • 5.MD.C.5.b

BACKWARD 1

Multiplying multi-digit numbers • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.5

Multiply.

$$24 \times 12$$

Product: _____

Multiply.

$$288 \times 5$$

Product: _____

True or false?

$$36 \times 5 = 150$$

A. True

B. False

$$49 \times 7 = ?$$

- A.** 56
- B.** 343
- C.** 283
- D.** 353

Multiply.

$$15 \times 24$$

Product: _____

SET 49 • 5.MD.C.5.b

BACKWARD 2

The area of the base • Grade 4 • Measurement & Data • 4.MD.A.3

A rectangle is 24 in. by 12 in.

What is its area?

Area: _____

True or false?

A 9-by-4 rectangle has an area of 36 square units.

A. True

B. False

The base of a box is 8 m by 3 m.

What is the area of the base?

- A.** 11 square meters
- B.** 24 square meters
- C.** 22 square meters
- D.** 48 square meters

A square has sides that are 7 inches long.
What is its area?

Area: _____

A rectangle has an area of 30 square feet and a length of 6 feet.

What is its width?

Width: _____

SET 49 • 5.MD.C.5.b

FORWARD 1

Real-world volume with fractional edge lengths • Grade 6 • Geometry • 6.G.A.2

A box is $1\frac{1}{2}$ ft by 1 ft by $\frac{3}{4}$ ft.

What is its volume?

Volume: _____

A prism has a base area of $12\frac{1}{2}$ square inches and a height of 3 inches.

What is its volume?

Volume: _____

A cube has edges that are $2\frac{1}{2}$ cm long.

What is its volume?

A. $6\frac{1}{4}$ cubic cm

B. 15 cubic cm

C. $7\frac{1}{2}$ cubic cm

D. $15\frac{5}{8}$ cubic cm

A drawer is 2 ft by $1\frac{1}{2}$ ft by $\frac{1}{2}$ ft.

What is its volume?

Volume: _____

True or false?

A box that is 3 by $\frac{1}{2}$ by 4 has a volume of 12 cubic units.

A. True

B. False

SET 49 • 5.MD.C.5.b

FORWARD 2

Real-world volume problems with right prisms • Grade 7 • Geometry • 7.G.B.6

A tent is a triangular prism. Its triangle has a base of 2 m and a height of 1.5 m. The tent is 3 m long.

What is its volume?

Volume: _____

A pool is 10 m long, 5 m wide, and 2 m deep. It is filled to $\frac{3}{4}$ of its depth.

How much water is in the pool?

Volume: _____

A box has a volume of 360 cubic inches. Its base is 12 in. by 6 in.

What is its height?

A. 3 inches

B. 30 inches

C. 5 inches

D. 60 inches

A rectangular prism is 4.5 cm by 2 cm by 3 cm.
What is its volume?

Volume: _____

True or false?

A triangular prism with a base area of 8 square feet and a length of 5 feet holds 40 cubic feet.

- A. True
- B. False

Find a missing dimension when the volume is known

MAIN	Grade 5 main question	5.MD.C.5.b
BACKWARD 1	Dividing by a one-digit number	4.NBT.B.6
BACKWARD 2	Finding an unknown factor	3.OA.A.4
FORWARD 1	Solve one-step equations of the form $px = q$	6.EE.B.7
FORWARD 2	Solve two-step equations	7.EE.B.4.a

SET 50 • 5.MD.C.5.b

MAIN

Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.5.b

A box has a volume of 96 cubic inches. It is 6 inches long and 4 inches wide.

How tall is it?

Height: _____

A prism has a volume of 150 cubic cm and a height of 6 cm.

What is the area of its base?

Base area: _____

A tank holds 240 cubic feet. Its base is 8 feet by 5 feet.

How deep is the tank?

- A.** 6 feet
- B.** 30 feet
- C.** 40 feet
- D.** 227 feet

A rectangular prism has a volume of 84 cubic meters. It is 7 m long and 4 m tall.

How wide is it?

Width: _____

True or false?

A box with a volume of 60 cubic units and a base area of 12 square units is 5 units tall.

A. True

B. False

SET 50 • 5.MD.C.5.b

BACKWARD 1

Dividing by a one-digit number • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Divide.

$$96 \div 4$$

Quotient: _____

Divide.

$$150 \div 6$$

Quotient: _____

True or false?

$$240 \div 8 = 3$$

A. True

B. False

$$84 \div 7 = ?$$

A. 12

B. 77

C. 11

D. 13

Divide.

$$60 \div 5$$

Quotient: _____

SET 50 • 5.MD.C.5.b

BACKWARD 2

Finding an unknown factor • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.4

What number goes in the box?

$$6 \times \square = 48$$

Number: _____

True or false?

If $\square \times 5 = 35$, then $\square = 7$.

A. True

B. False

What number goes in the box?

$$4 \times \square = 36$$

- A.** 9
- B.** 8
- C.** 32
- D.** 40

What number goes in the box?

$$\square \times 7 = 63$$

Number: _____

True or false?

If $8 \times \square = 56$, then $\square = 6$.

A. True

B. False

SET 50 • 5.MD.C.5.b

FORWARD 1

Solve one-step equations of the form $px = q$ • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for h.

$$24h = 96$$

$$h = \underline{\hspace{10cm}}$$

Solve for b.

$$6b = 150$$

$$b = \underline{\hspace{10cm}}$$

Solve for w.

$$28w = 84$$

A. 56

B. 2,352

C. 112

D. 3

A prism has a volume of 52.5 cubic cm and a base area of 15 square cm.

Write and solve an equation to find its height h .

Equation and solution: _____

True or false?

The solution of $\frac{3}{4}x = 12$ is $x = 9$.

A. True

B. False

SET 50 • 5.MD.C.5.b

FORWARD 2

Solve two-step equations • Grade 7 • Expressions & Equations • 7.EE.B.4.a

A box is 4 in. wide, 5 in. tall, and $(x + 2)$ in. long. Its volume is 120 cubic inches.

Find x .

$x =$ _____

Solve for x.

$$6(x + 3) = 54$$

x = _____

Solve for h.

$$12h + 8 = 68$$

A. 5

B. 6.3

C. 60

D. 4

Two boxes have a total volume of 100 cubic feet. One box holds 40 cubic feet. The other has a base area of 12 square feet.

Write and solve an equation to find the height h of the other box.

Equation and solution: _____

True or false?

The solution of $3(x - 2) = 21$ is $x = 9$.

A. True

B. False

Find the volume of a figure made of two rectangular prisms

MAIN	Grade 5 main question	5.MD.C.5.c
BACKWARD 1	Area of figures made of rectangles	3.MD.C.7.d
BACKWARD 3	The area formula for rectangles	4.MD.A.3
FORWARD 1	Volume of composite figures with fractional edge lengths	6.G.A.2
FORWARD 2	Volume of composite solids	7.G.B.6

SET 51 • 5.MD.C.5.c

MAIN

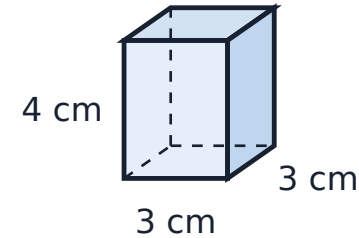
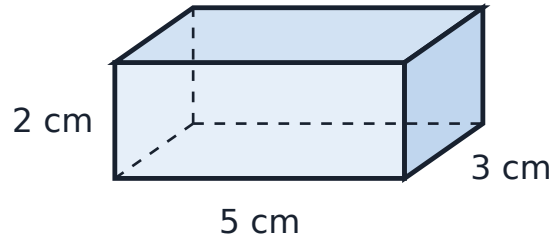
Grade 5 main question • Grade 5 • Measurement & Data • 5.MD.C.5.c

A figure is made of two rectangular prisms that do not overlap. One is 6 ft by 4 ft by 3 ft. The other is 2 ft by 4 ft by 5 ft.

What is the total volume?

Volume: _____

A figure is made of the two prisms shown, which do not overlap.
What is the total volume?



Volume: _____

An L-shaped step is made of a 4-by-2-by-1 prism with a 2-by-2-by-1 prism on top.

What is its volume?

- A.** 12 cubic units
- B.** 8 cubic units
- C.** 10 cubic units
- D.** 16 cubic units

A building is a 10 m by 8 m by 6 m prism with a 4 m by 8 m by 3 m prism attached to one side.

What is the total volume?

Volume: _____

True or false?

The volume of a figure made of two prisms that do not overlap is the sum of their volumes.

- A. True
- B. False

SET 51 • 5.MD.C.5.c

BACKWARD 1

Area of figures made of rectangles • Grade 3 • Measurement & Data •
3.MD.C.7.d

A figure is split into two rectangles that do not overlap.
Their areas are 15 square units and 8 square units.

What is the area of the figure?

Area: _____

True or false?

The area of a figure made of two non-overlapping parts is the sum of the areas of the parts.

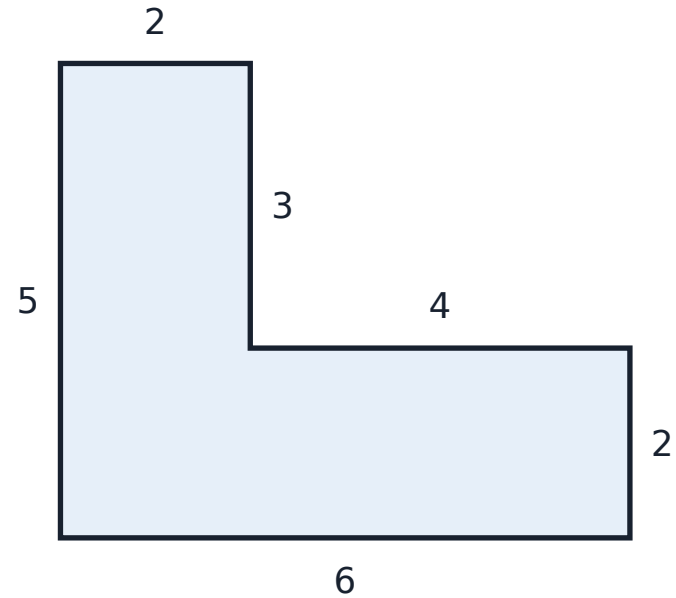
A. True

B. False

The figure is split into a 6-by-2 rectangle along the bottom and a rectangle above it.

What is the area of the 6-by-2 rectangle?

- A.** 12 square units
- B.** 16 square units
- C.** 6 square units
- D.** 8 square units



A figure is made of two squares that do not overlap.
Their areas are 9 square units and 25 square units.

To find the area of the figure, what do you do with the two areas?

- A.** Divide them
- B.** Multiply them
- C.** Subtract them
- D.** Add them

A square with an area of 4 square units is cut out of a rectangle with an area of 48 square units.

What is the area of the part that is left?

Area: _____

SET 51 • 5.MD.C.5.c

BACKWARD 3

The area formula for rectangles • Grade 4 • Measurement & Data • 4.MD.A.3

A rectangle is 6 units by 4 units.

What is its area?

Area: _____

True or false?

A rectangle with an area of 32 square units and a length of 8 units has a width of 4 units.

A. True

B. False

A rectangle is 10 m by 8 m.

What is its area?

- A.** 36 square meters
- B.** 800 square meters
- C.** 80 square meters
- D.** 18 square meters

A rectangle has an area of 45 square feet and a width of 5 feet.

What is its length?

Length: _____

True or false?

A 4-by-8 rectangle has an area of 24 square units.

A. True

B. False

SET 51 • 5.MD.C.5.c

FORWARD 1

Volume of composite figures with fractional edge lengths • Grade 6 • Geometry
• 6.G.A.2

A figure is made of two prisms that do not overlap: $2\frac{1}{2}$ ft by 2 ft by 1 ft, and 1 ft by 2 ft by $\frac{1}{2}$ ft.

What is the total volume?

Volume: _____

Two prisms are joined. One has a volume of $4\frac{1}{2}$ cubic inches. The other is 3 in. by 1 in. by $\frac{1}{2}$ in.

What is the total volume?

Volume: _____

A step is made of a 3-by-1-by- $\frac{1}{2}$ prism and a 1-by-1-by- $\frac{1}{2}$ prism that do not overlap.

What is its volume?

A. 2 cubic units

B. $\frac{1}{2}$ cubic unit

C. 4 cubic units

D. $1\frac{1}{2}$ cubic units

A block that is 1 in. by 1 in. by $2\frac{1}{2}$ in. is cut out of a box that is 4 in. by 3 in. by $2\frac{1}{2}$ in.

What volume is left?

Volume: _____

True or false?

A figure made of two cubes with $\frac{1}{2}$ -foot edges has a volume of 1 cubic foot.

- A. True
- B. False

SET 51 • 5.MD.C.5.c

FORWARD 2

Volume of composite solids • Grade 7 • Geometry • 7.G.B.6

A figure is a 6-by-4-by-3 rectangular prism with a triangular prism on top. The triangle has a base of 4 and a height of 2, and the triangular prism is 6 long.

What is the total volume?

Volume: _____

A concrete step is made of two prisms: 1.5 m by 0.5 m by 0.2 m and 1.5 m by 0.25 m by 0.4 m.

What is the total volume?

Volume: _____

A 10-by-10-by-10 cube has a 2-by-2-by-10 hole cut all the way through it.

What volume remains?

- A.** 960 cubic units
- B.** 800 cubic units
- C.** 1,000 cubic units
- D.** 40 cubic units

A shed is a 3 m by 4 m by 2.5 m prism with a roof shaped like a triangular prism. The triangle has a base of 3 m and a height of 1 m, and the roof is 4 m long.

What is the total volume?

Volume: _____

True or false?

The volume of a composite solid can be found by adding the volumes of its parts.

A. True

B. False

Understand the coordinate plane: axes, origin, and coordinates

MAIN	Grade 5 main question	5.G.A.1
BACKWARD 1	Whole numbers on a number line	2.MD.B.6
BACKWARD 2	Perpendicular lines	4.G.A.1
FORWARD 1	Name the coordinates of points in all four quadrants	6.NS.C.6.c
FORWARD 2	Interpret a point on the graph of a proportional relationship	7.RP.A.2.d

SET 52 • 5.G.A.1

MAIN

Grade 5 main question • Grade 5 • Geometry • 5.G.A.1

What is the point where the x-axis and the y-axis meet called?

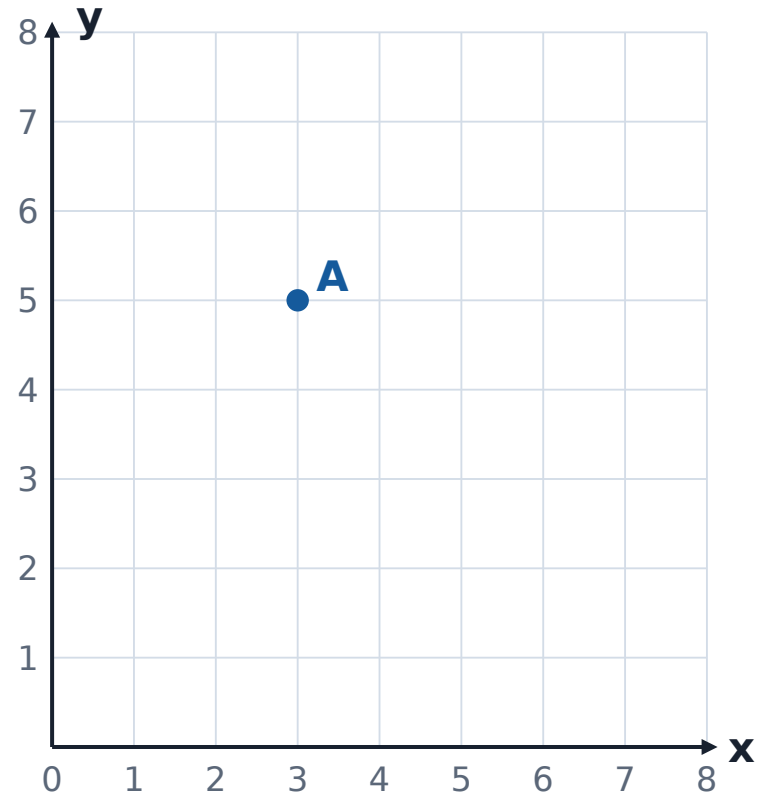
- A.** The y-axis
- B.** The origin
- C.** The x-coordinate
- D.** A quadrant

In the ordered pair $(7, 2)$, what does the 7 tell you?

Answer: _____

What are the coordinates of point A?

A = _____



True or false?

To reach $(0, 6)$, you start at the origin and move 6 units to the right along the x-axis.

A. True

B. False

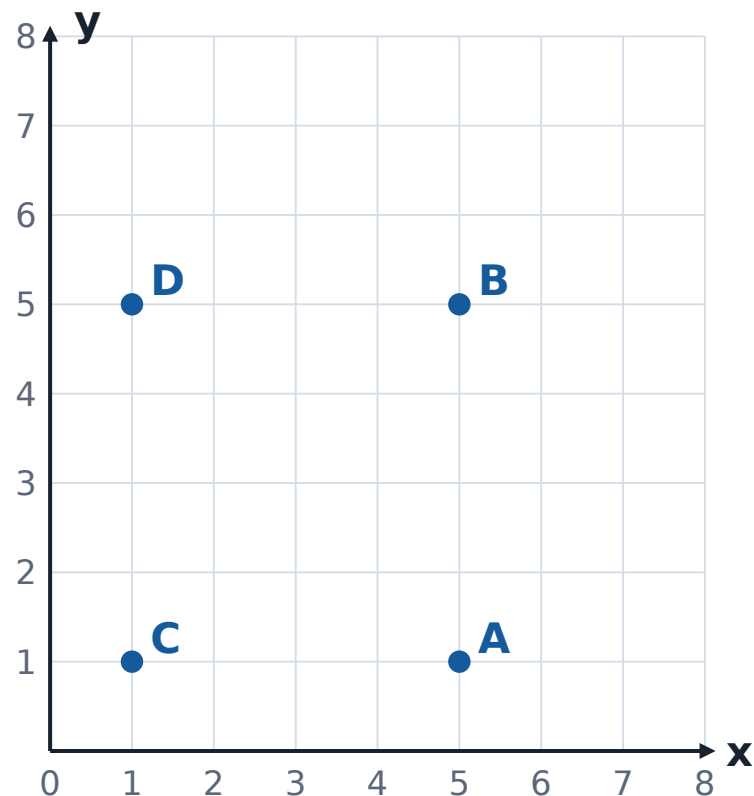
Which point is at (5, 1)?

A. Point A

B. Point B

C. Point C

D. Point D



SET 52 • 5.G.A.1

BACKWARD 1

Whole numbers on a number line • Grade 2 • Measurement & Data • 2.MD.B.6

What number is at point P?



P = _____

True or false?

On a number line, numbers increase from left to right.

- A.** True
- B.** False

Which number is 4 units to the right of 0?

- A.** 4
- B.** 8
- C.** 40
- D.** 0

How many units is it from 0 to 9 on a number line?

Units: _____

True or false?

On a number line, 3 is farther from 0 than 7 is.

A. True

B. False

SET 52 • 5.G.A.1

BACKWARD 2

Perpendicular lines • Grade 4 • Geometry • 4.G.A.1

True or false?

Perpendicular lines meet to form right angles.

A. True

B. False

The x-axis and the y-axis are ____.

- A.** perpendicular
- B.** parallel
- C.** curved
- D.** the same line

How many degrees are in a right angle?

Degrees: _____

True or false?

Parallel lines never meet.

A. True

B. False

Which letter is made of two perpendicular line segments?

A. S

B. L

C. O

D. C

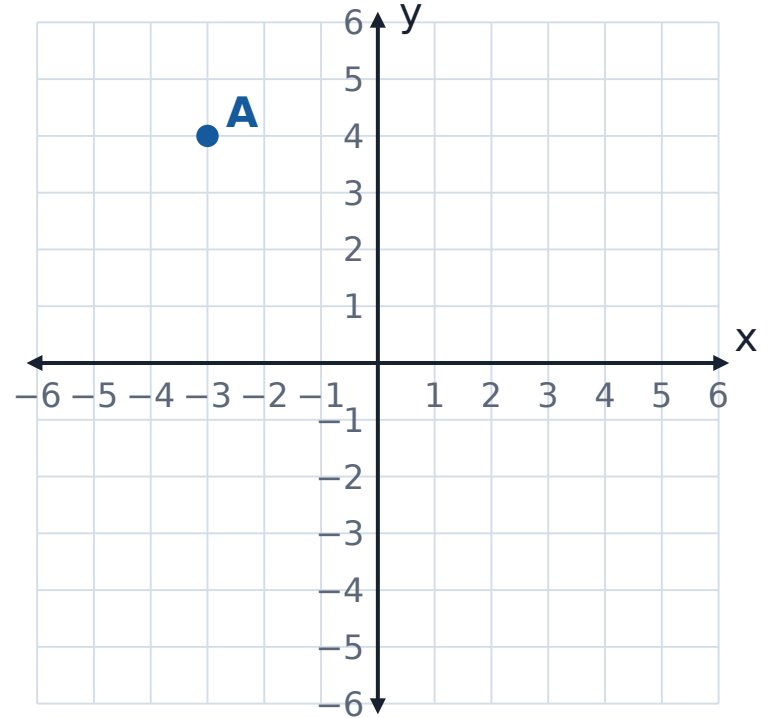
SET 52 • 5.G.A.1

FORWARD 1

Name the coordinates of points in all four quadrants • Grade 6 • The Number System • 6.NS.C.6.c

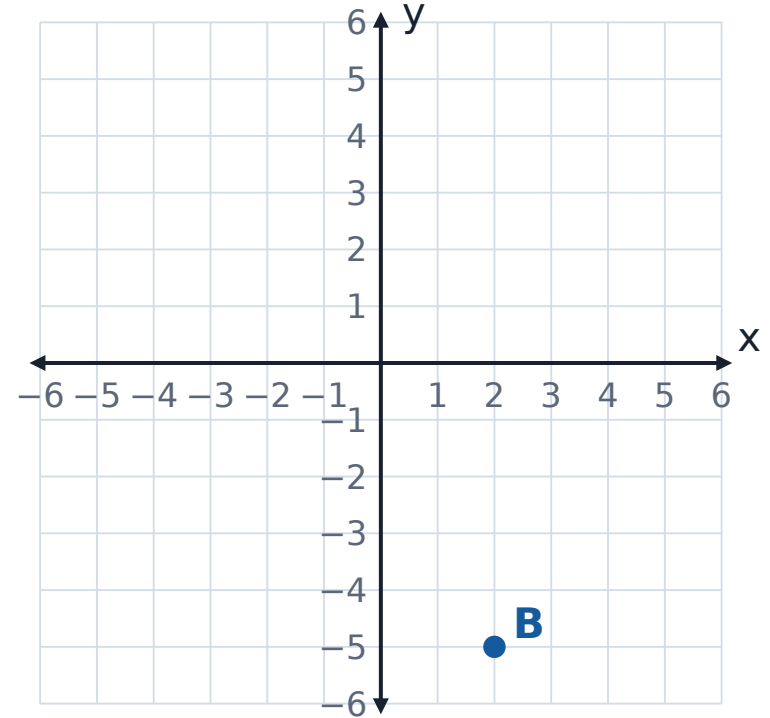
What are the coordinates of point A?

A = _____



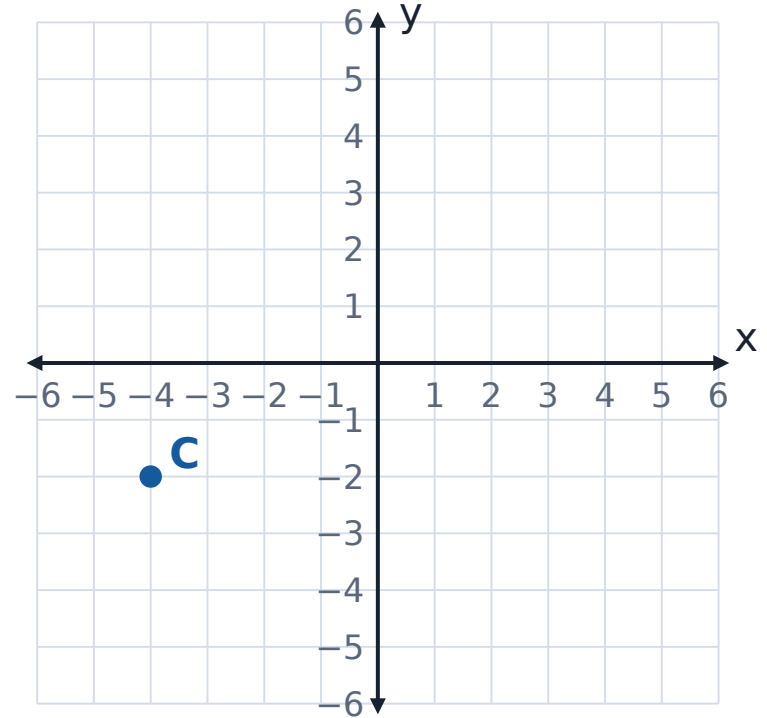
What are the coordinates of point B?

B = _____



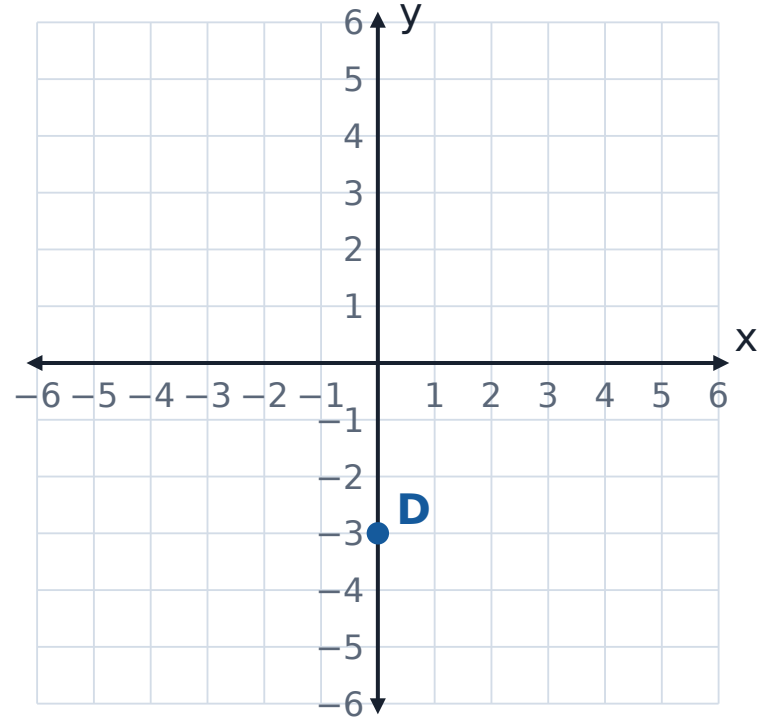
Which ordered pair names point C?

- A.** $(-4, -2)$
- B.** $(-2, -4)$
- C.** $(4, 2)$
- D.** $(-4, 2)$



What are the coordinates of point D?

D = _____

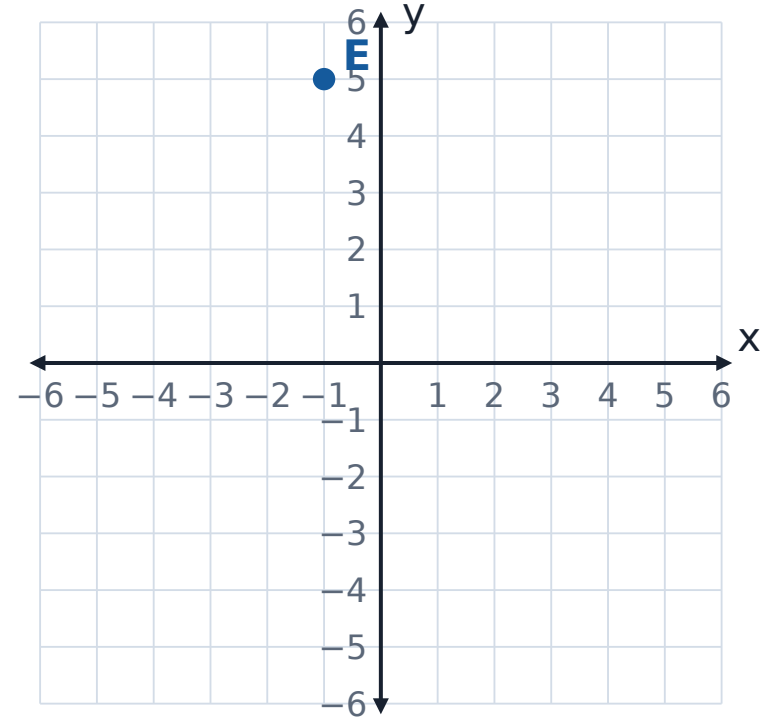


True or false?

Point E is located at $(5, -1)$.

A. True

B. False



SET 52 • 5.G.A.1

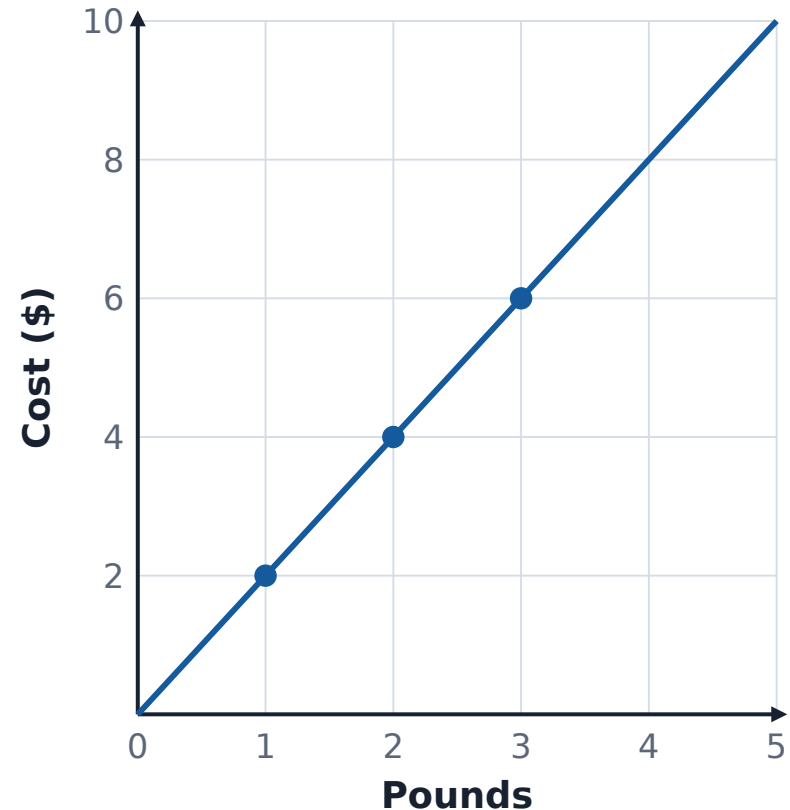
FORWARD 2

Interpret a point on the graph of a proportional relationship • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d

The graph shows the cost of pears.

What does the point (3, 6) represent?

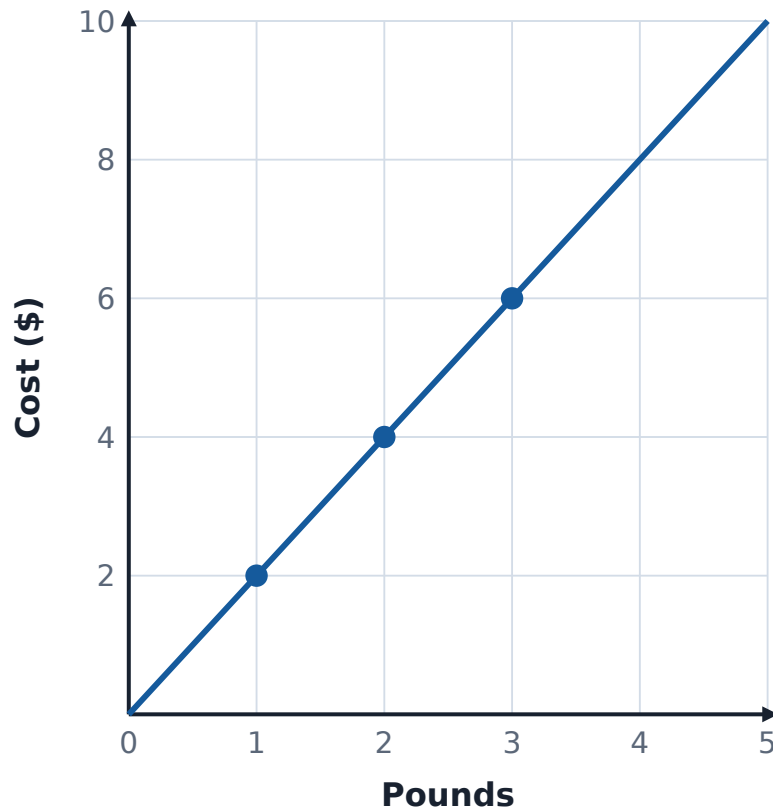
Meaning: _____



The graph shows the cost of pears.

What does the point (1, 2) represent?

- A.** The cost starts at \$2.
- B.** 2 pounds cost \$1.
- C.** \$1 buys 2 pounds.
- D.** 1 pound costs \$2.



True or false?

On the graph of a proportional relationship between pounds and cost, the point $(0, 0)$ means 0 pounds cost \$0.

- A. True
- B. False

The graph of a proportional relationship passes through (1, r) and (4, 20).

What is r ?

$r =$ _____

A graph shows a proportional relationship between the time x , in hours, that a bus drives and the distance y , in miles. It passes through $(2, 90)$.

What does this point mean?

- A.** The speed is 2 miles per hour.
- B.** In 90 hours, the distance is 2 miles.
- C.** The speed is 90 miles per hour.
- D.** In 2 hours, the distance is 90 miles.

Graph points in the first quadrant

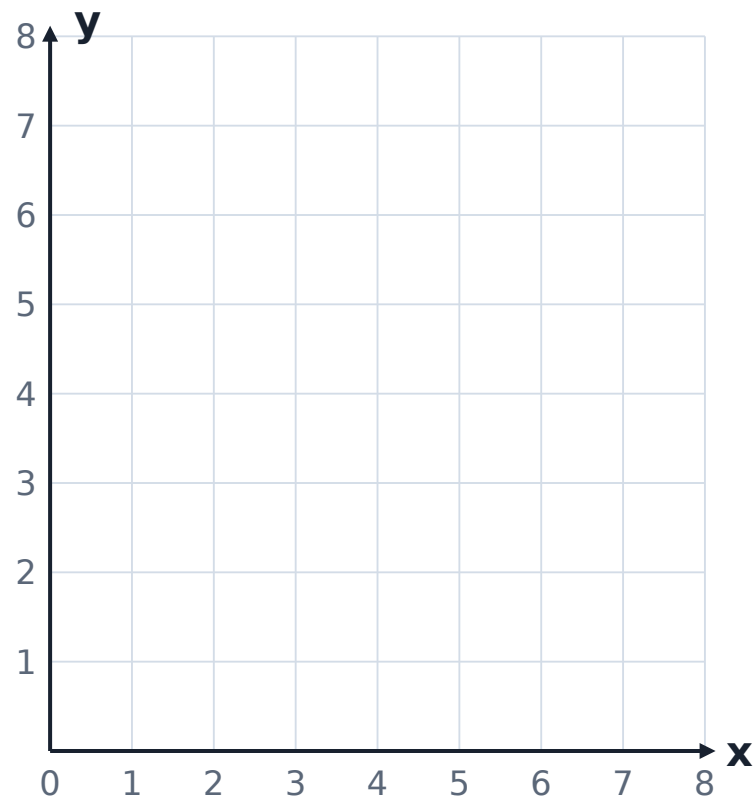
MAIN	Grade 5 main question	5.G.A.2
BACKWARD 2	Read a scaled bar graph	3.MD.B.3
BACKWARD 3	Fractions and whole numbers on a number line	3.NF.A.2
FORWARD 1	Reflect points across the axes	6.NS.C.6.b
FORWARD 2	Graph pairs to decide whether a relationship is proportional	7.RP.A.2.a

SET 53 • 5.G.A.2

MAIN

Grade 5 main question • Grade 5 • Geometry • 5.G.A.2

Plot and label the points $A(2, 7)$, $B(6, 3)$, and $C(0, 5)$.

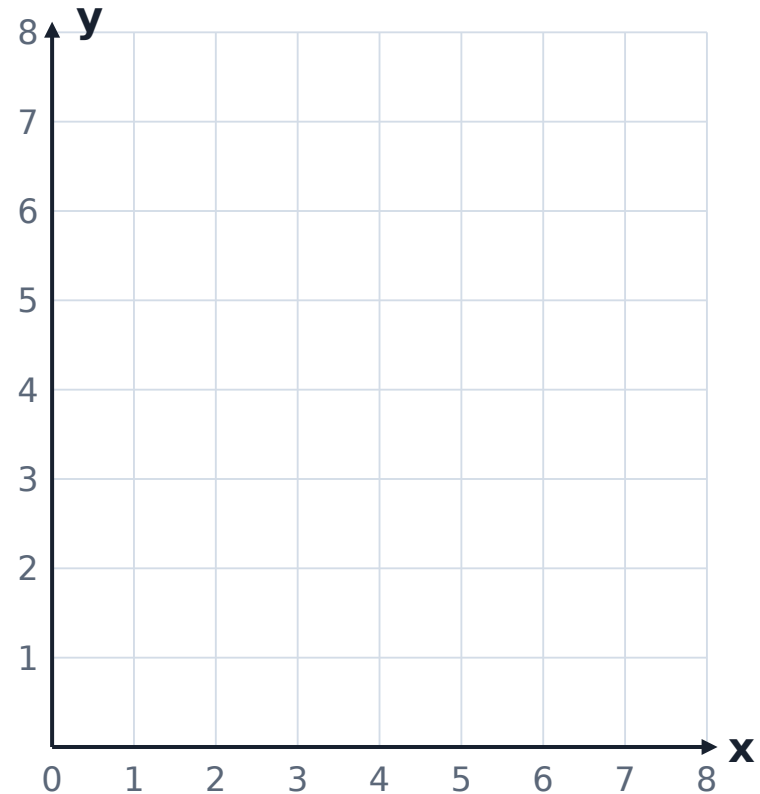


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

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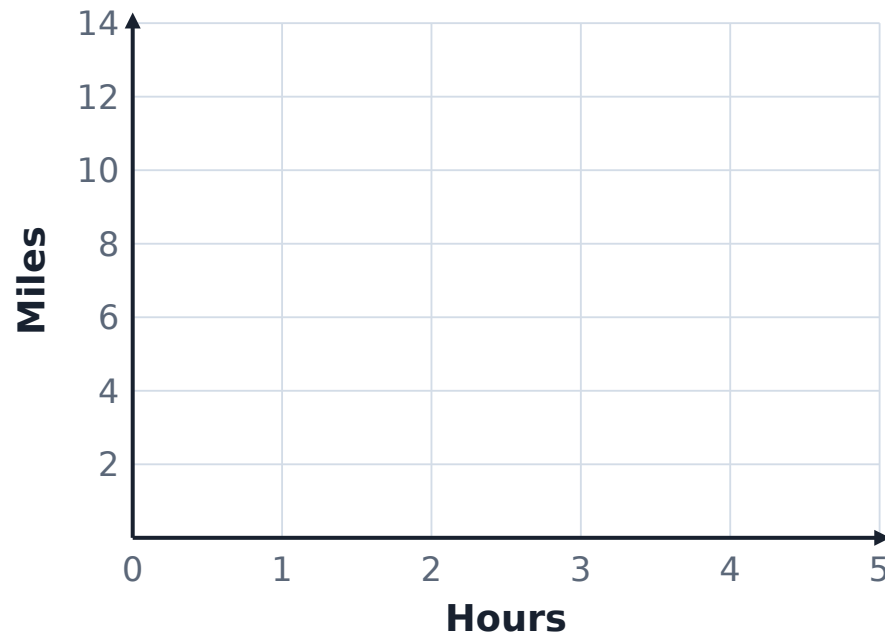
Grade 5 • Geometry • 5.G.A.2 • Set 53 • MAIN • Q1/5

Plot and label the points D(4, 4), E(7, 0), and F(1, 6).

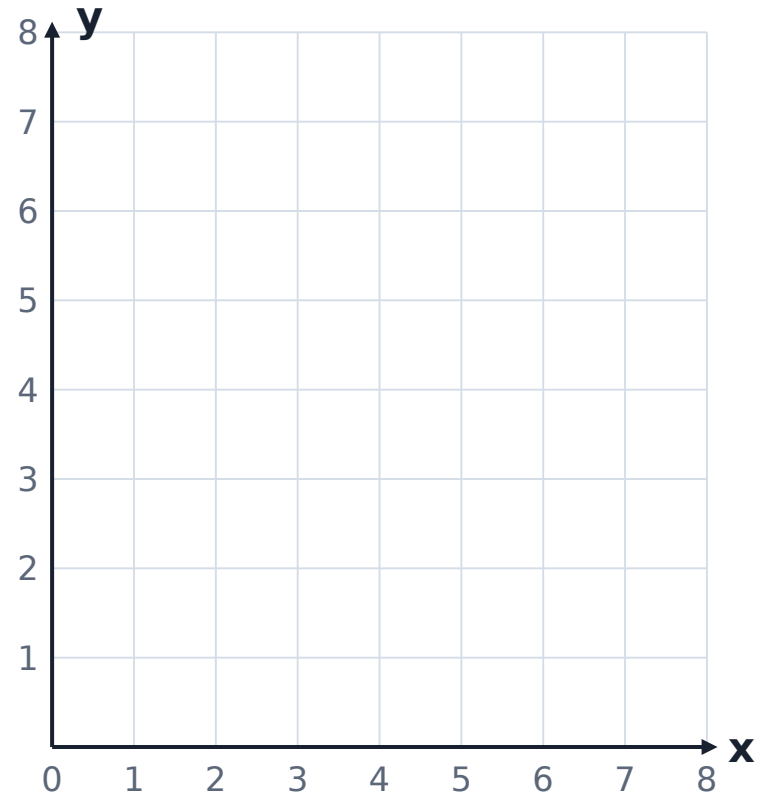


Plot the points from the table.

Hours	1	2	3	4
Miles	3	6	9	12



Plot and label the points $G(3, 2\frac{1}{2})$ and $H(5\frac{1}{2}, 4)$.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

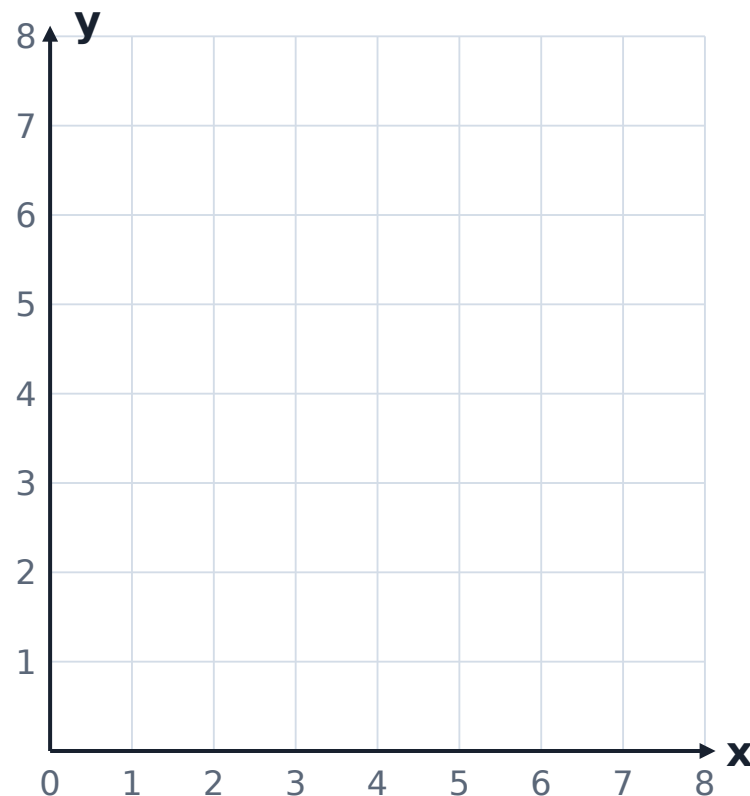
ID S53-M-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 53 • MAIN • Q4/5

Plot $P(2, 3)$ and $Q(2, 7)$.

How far apart are the two points?

Distance: _____

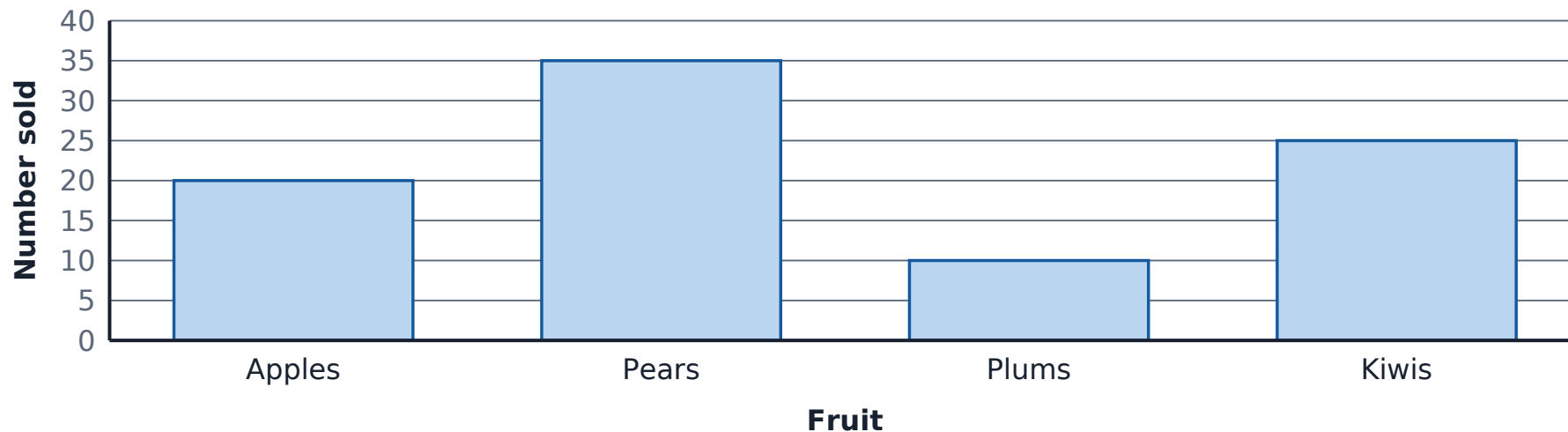


SET 53 • 5.G.A.2

BACKWARD 2

Read a scaled bar graph • Grade 3 • Measurement & Data • 3.MD.B.3

What does each grid line on the scale stand for?



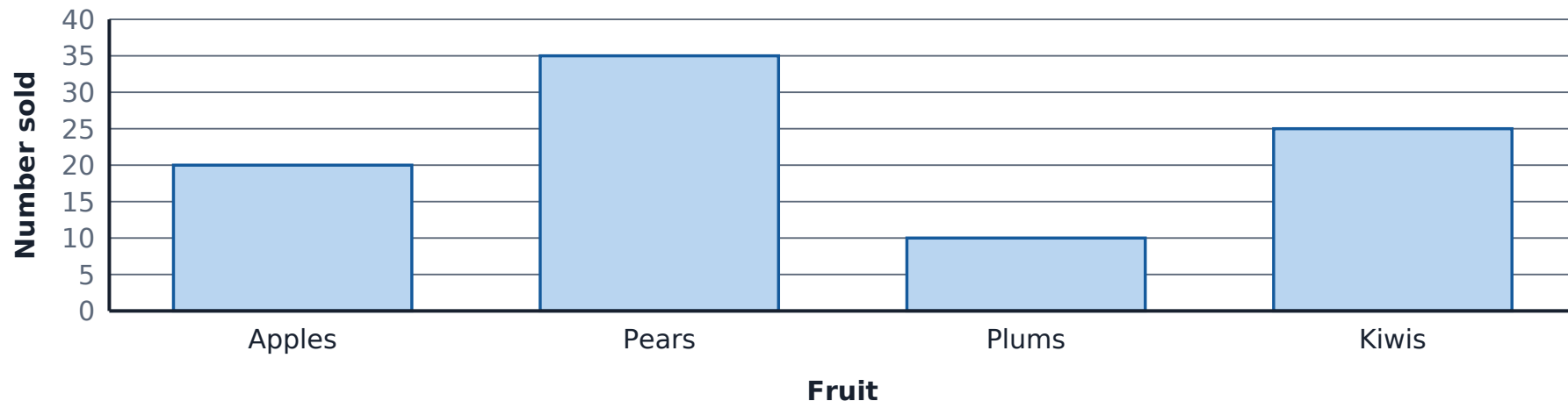
A. 10

B. 20

C. 5

D. 1

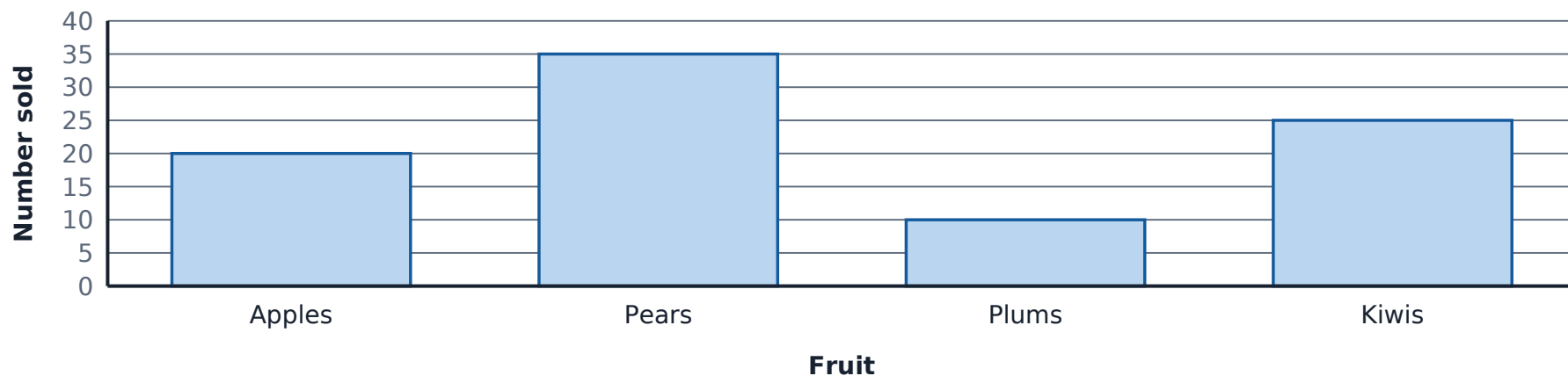
How many pears were sold?



Pears: _____

True or false?

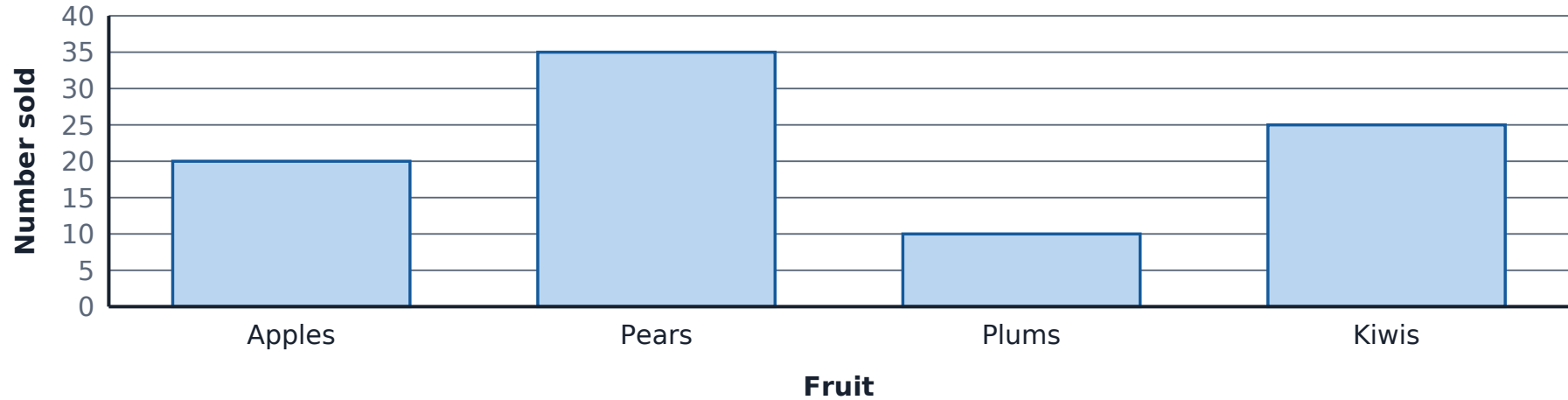
More kiwis than apples were sold.



A. True

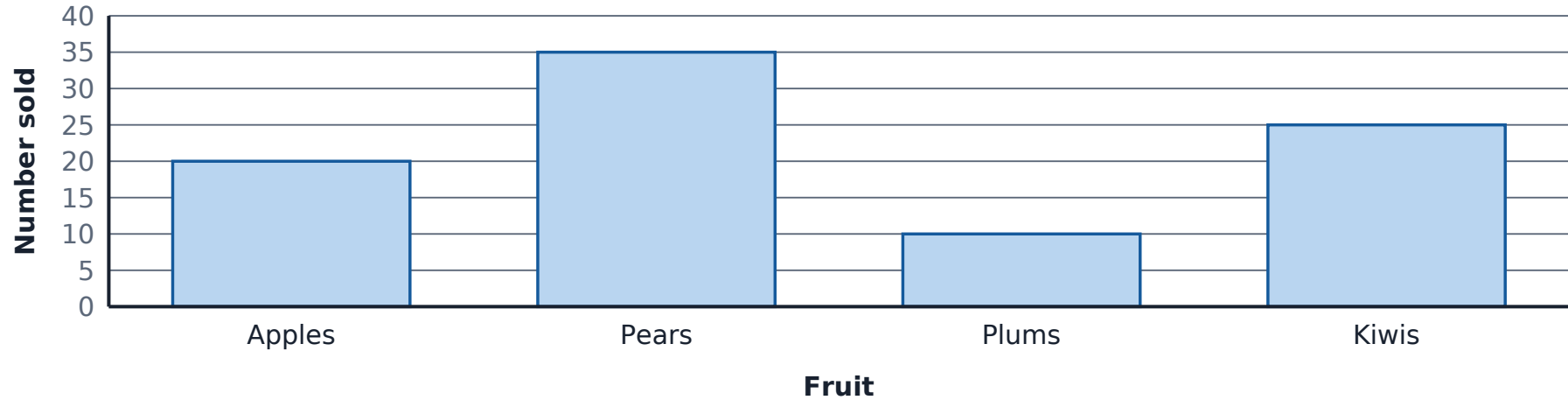
B. False

How many fewer plums than pears were sold?



Plums: _____

How many apples and kiwis were sold in all?



Fruit: _____

SET 53 • 5.G.A.2

BACKWARD 3

Fractions and whole numbers on a number line • Grade 3 • Number & Operations–Fractions • 3.NF.A.2

What number is at point A?



A = _____

True or false?

$2\frac{1}{2}$ is halfway between 2 and 3 on a number line.

A. True

B. False

Which point is at $1\frac{1}{2}$?



A. Point A

B. Point B

C. Point C

D. Point D

What number is at point B?



B = _____

True or false?

On a number line, $5\frac{1}{2}$ is between 6 and 7.

A. True

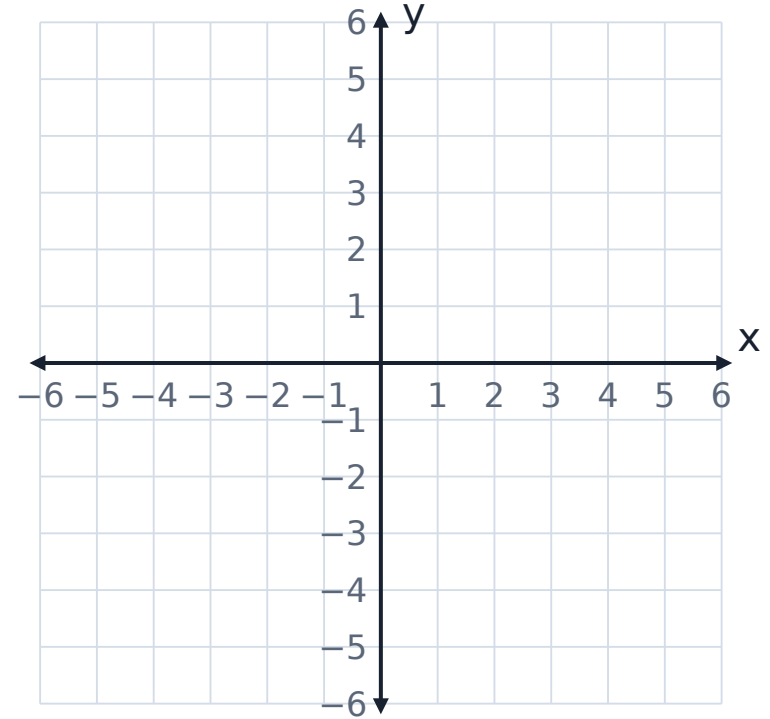
B. False

SET 53 • 5.G.A.2

FORWARD 1

Reflect points across the axes • Grade 6 • The Number System • 6.NS.C.6.b

Plot and label $A(2, 4)$. Then plot its reflection across the x -axis and label it B .



The point $(5, 2)$ is reflected across the x -axis.
What are the coordinates of its image?

Image: _____

The point $(-3, 5)$ is reflected across the x -axis.

What are the coordinates of its image?

- A.** $(3, -5)$
- B.** $(3, 5)$
- C.** $(-3, -5)$
- D.** $(5, -3)$

The point $(-4, 1)$ is reflected across the y -axis.
What are the coordinates of its image?

Image: _____

True or false?

The points $(3, 5)$ and $(-3, 5)$ are reflections of each other across the x-axis.

A. True

B. False

SET 53 • 5.G.A.2

FORWARD 2

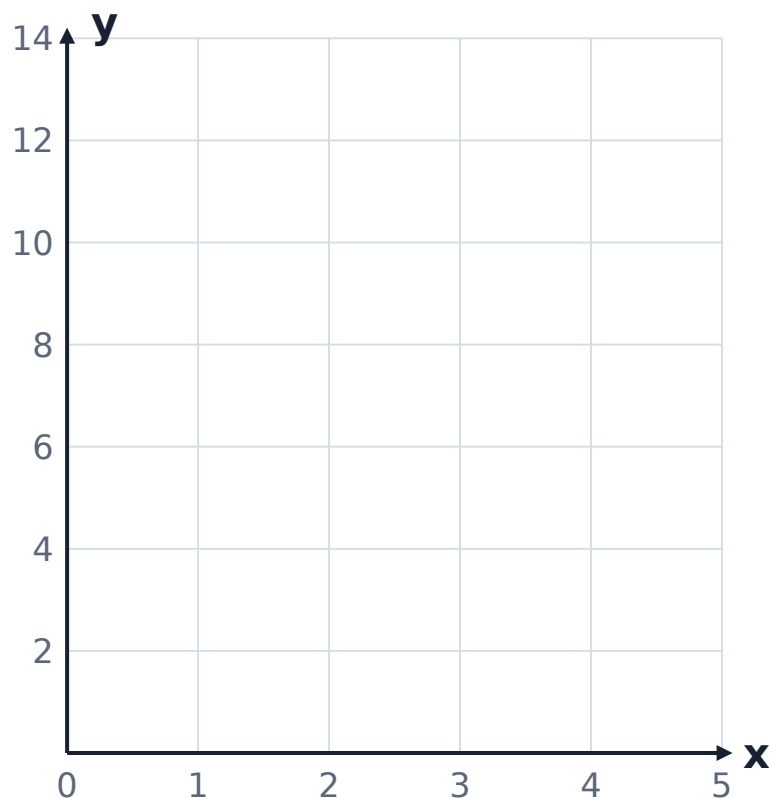
Graph pairs to decide whether a relationship is proportional • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

Graph the pairs (1, 3), (2, 6),
and (4, 12).

Is the relationship
proportional? Explain using
your graph.

Answer: _____

Reason: _____

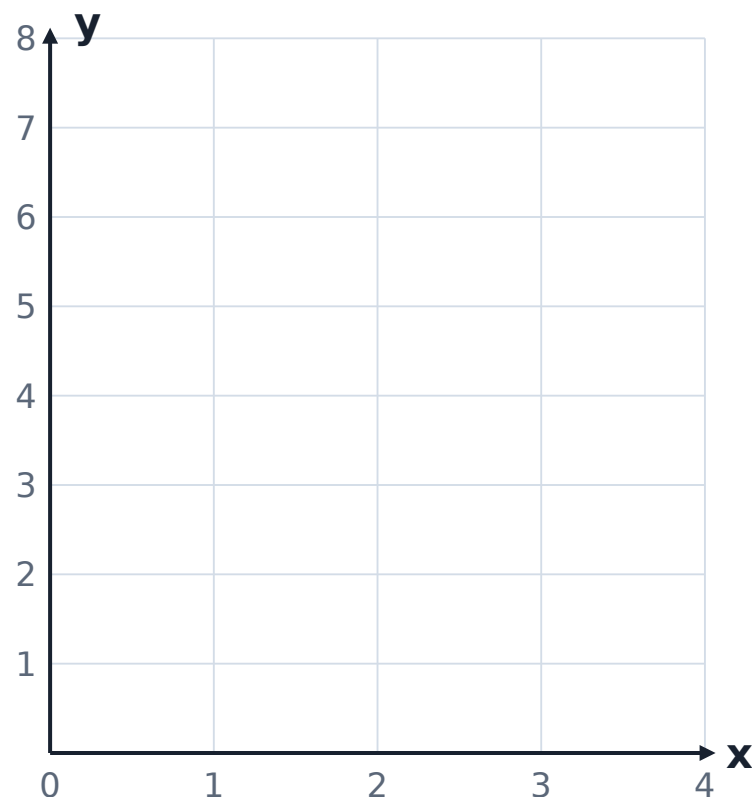


Graph the pairs $(0, 2)$, $(1, 4)$, and $(2, 6)$.

Is the relationship proportional? Explain using your graph.

Answer: _____

Reason: _____



Which graph shows a proportional relationship?

- A.** A curve through $(0, 0)$
- B.** Points that are not on a line
- C.** A straight line through $(0, 3)$
- D.** A straight line through $(0, 0)$

True or false?

The points (2, 1), (4, 2), and (6, 3) show a proportional relationship.

- A. True
- B. False

Graph the pairs from the table.

Is the relationship proportional? Explain using your graph.

x	1	2	3
y	4	8	12



Answer: _____

Reason: _____

Solve real-world problems by graphing points and interpreting coordinates

MAIN	Grade 5 main question	5.G.A.2
BACKWARD 2	Distance between whole numbers on a number line	2.MD.B.6
BACKWARD 3	Solve problems with a scaled bar graph	3.MD.B.3
FORWARD 1	Find distances in all four quadrants	6.NS.C.8
FORWARD 2	Use a scale on a map grid	7.G.A.1

SET 54 • 5.G.A.2

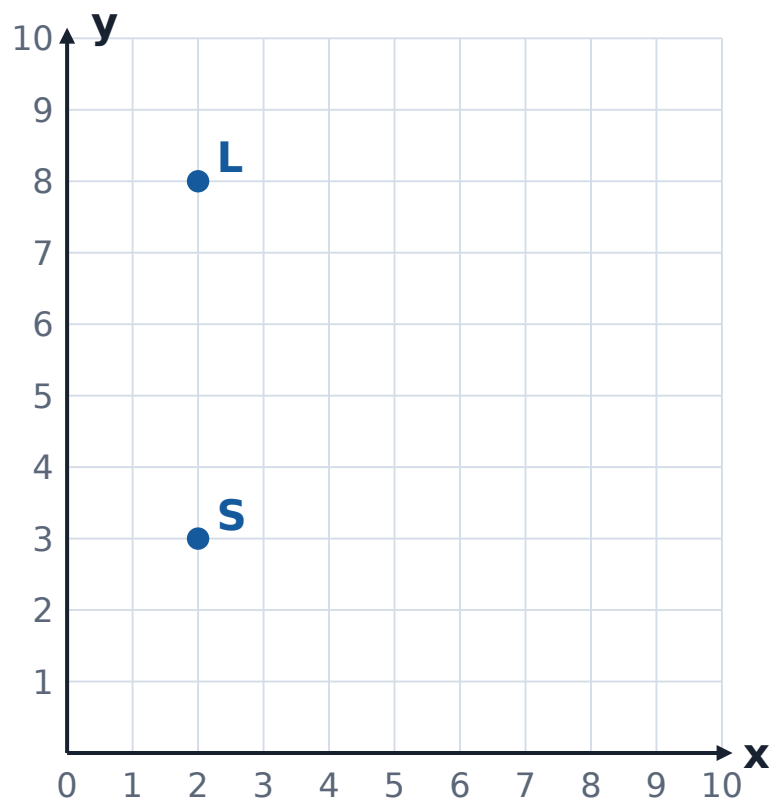
MAIN

Grade 5 main question • Grade 5 • Geometry • 5.G.A.2

The grid is a map. Each unit is 1 block.

How many blocks is it from the school S to the library L?

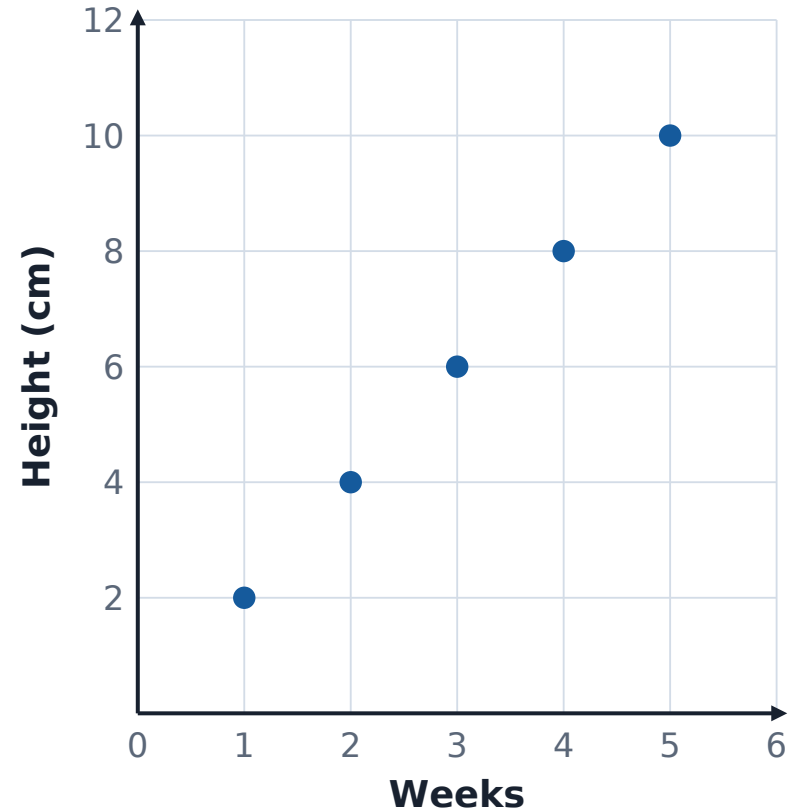
Blocks: _____



The graph shows the height of a plant.

How tall was the plant after 4 weeks?

Height: _____



The point (3, 12) is on a graph of hours worked (x) and dollars earned (y).

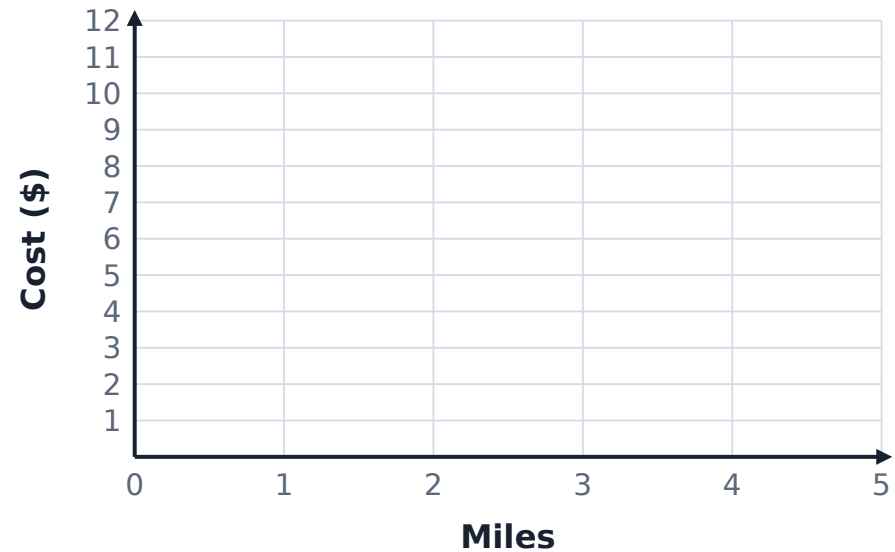
What does the point mean?

- A.** Working 3 hours earns \$12.
- B.** Working 12 hours earns \$3.
- C.** The pay starts at \$12.
- D.** Each hour earns \$3.

The table shows the cost of a taxi ride.

Graph the points.

Miles	1	2	3	4
Cost (\$)	5	7	9	11



On a map grid, the park is at (1, 6) and the pool is at (7, 6). Each unit is 100 meters.

How far apart are the park and the pool?

Distance: _____

SET 54 • 5.G.A.2

BACKWARD 2

Distance between whole numbers on a number line • Grade 2 • Measurement & Data • 2.MD.B.6

What is the distance from 2 to 7 on a number line?

Distance: _____

True or false?

The distance from 3 to 10 is 13 units.

A. True

B. False

How many units apart are 1 and 8 on a number line?

A. 9

B. 8

C. 1

D. 7

What is the distance from 6 to 6 on a number line?

Distance: _____

How many units apart are points A and B?



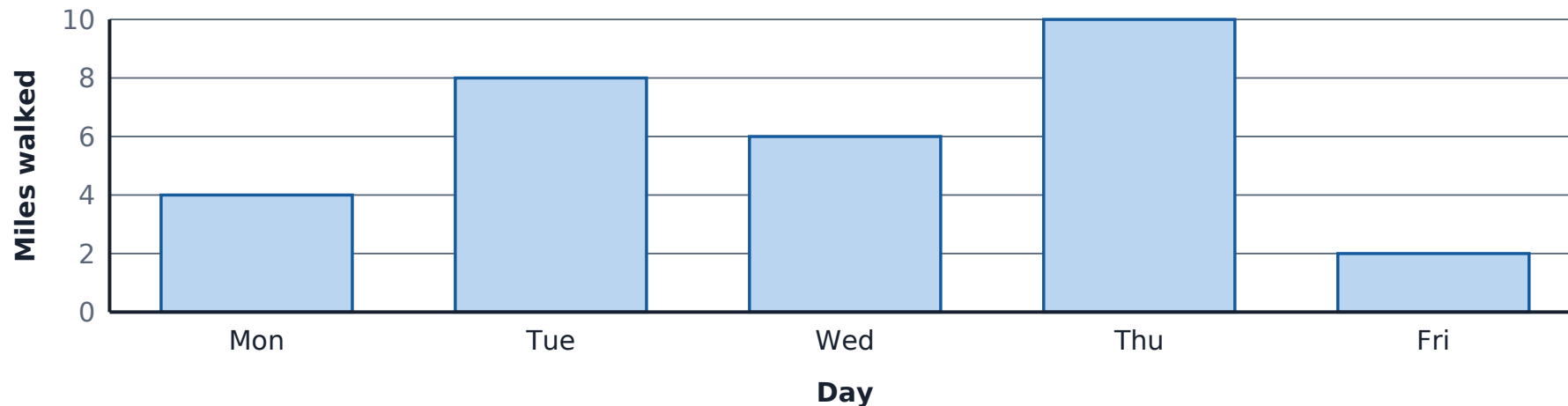
Units: _____

SET 54 • 5.G.A.2

BACKWARD 3

Solve problems with a scaled bar graph • Grade 3 • Measurement & Data •
3.MD.B.3

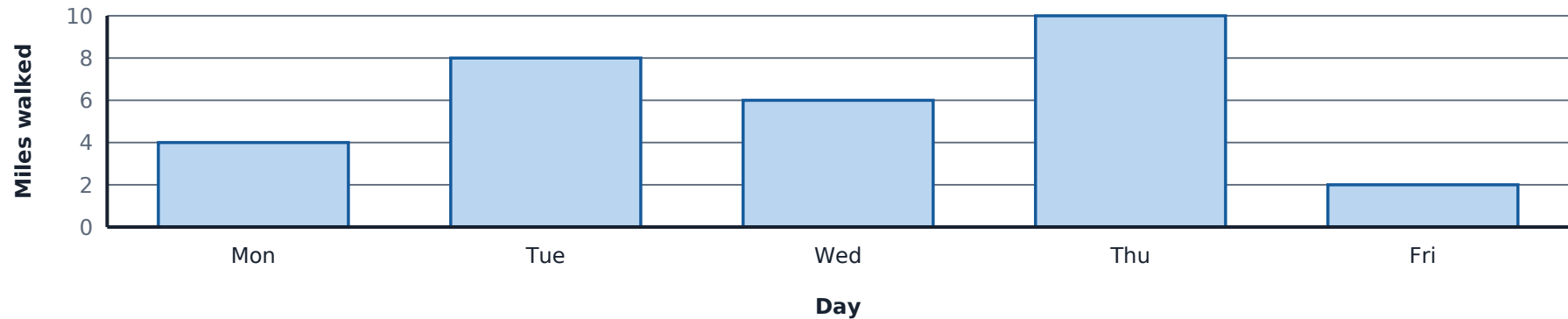
How many miles were walked on Thursday?



Miles: _____

True or false?

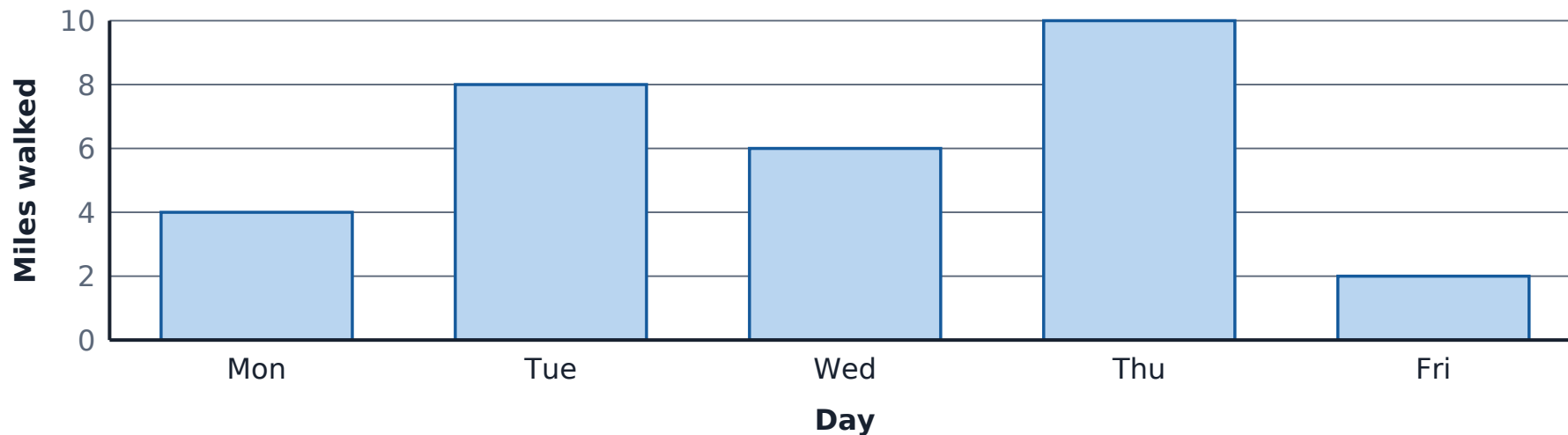
The same number of miles were walked on Monday and Wednesday.



A. True

B. False

On which day were 8 miles walked?



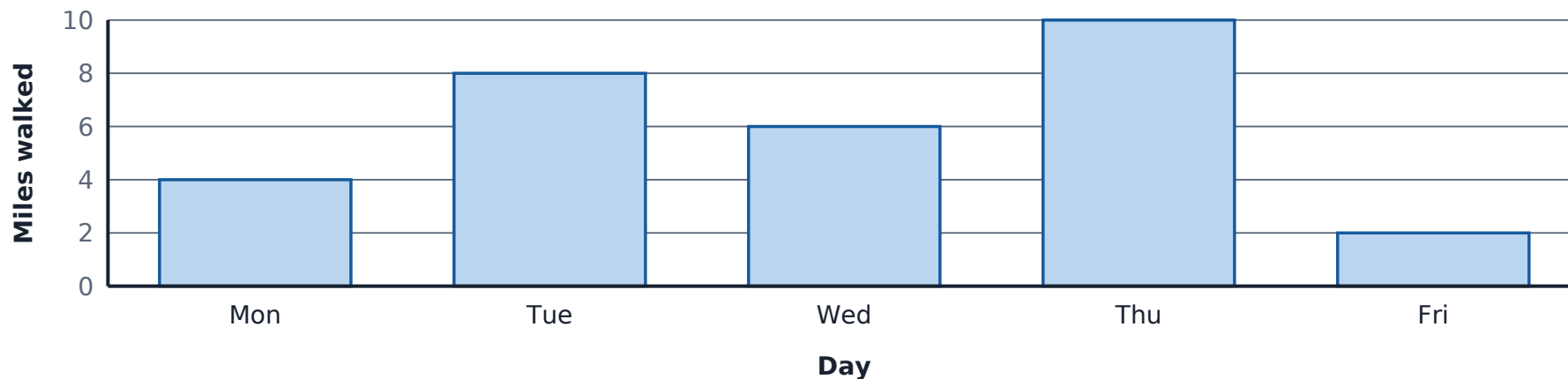
A. Thursday **B.** Friday **C.** Monday **D.** Tuesday

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 54 • BACKWARD 3 • Q3/5

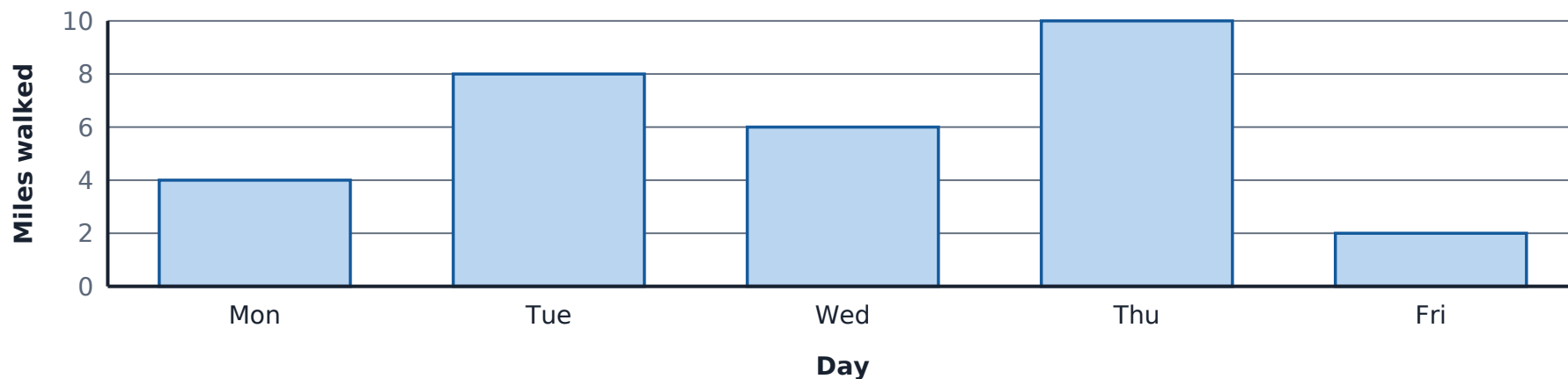
How many more miles were walked on Thursday than on Friday?



Miles: _____

True or false?

Each grid line on the scale stands for 2 miles.



A. True

B. False

SET 54 • 5.G.A.2

FORWARD 1

Find distances in all four quadrants • Grade 6 • The Number System • 6.NS.C.8

Find the distance between $(-4, 2)$ and $(5, 2)$.

Distance: _____

Find the distance between $(3, -6)$ and $(3, 1)$.

Distance: _____

On a map, the store is at $(-2, -3)$ and the bank is at $(-2, 4)$. Each unit is 1 block.

How far apart are they?

A. 7 blocks

B. 1 block

C. 4 blocks

D. 6 blocks

A rectangle has vertices at $(-3, 1)$, $(2, 1)$, $(2, -2)$, and $(-3, -2)$.

What is its perimeter?

Perimeter: _____

True or false?

The distance between $(-5, 0)$ and $(5, 0)$ is 0.

A. True

B. False

SET 54 • 5.G.A.2

FORWARD 2

Use a scale on a map grid • Grade 7 • Geometry • 7.G.A.1

On a map grid, each unit is 2 miles. Two towns are at (1, 3) and (6, 3).

What is the actual distance between them?

Distance: _____

On a floor plan, each unit is $1\frac{1}{2}$ feet. A wall goes from (0, 0) to (0, 8).

How long is the real wall?

Length: _____

A scale drawing uses 1 grid unit = 5 m. A path on the grid is 7 units long.

How long is the real path?

A. 12 m

B. 35 m

C. 2 m

D. 7 m

A garden plan has corners at $(0, 0)$, $(6, 0)$, $(6, 4)$, and $(0, 4)$. Each unit is 3 feet.

What are the real length and width?

Length and width: _____

True or false?

If each grid unit is 10 km, the points (2, 5) and (2, 9) are 4 km apart.

- A. True
- B. False

Attributes of a category belong to all of its subcategories

MAIN	Grade 5 main question	5.G.B.3
BACKWARD 1	Quadrilaterals and their attributes	3.G.A.1
BACKWARD 2	Parallel lines, perpendicular lines, and angles	4.G.A.1
BACKWARD 3	Classifying shapes by sides and angles	4.G.A.2
FORWARD 1	Draw polygons in the coordinate plane	6.G.A.3
FORWARD 2	Draw shapes with given conditions	7.G.A.2

SET 55 • 5.G.B.3

MAIN

Grade 5 main question • Grade 5 • Geometry • 5.G.B.3

True or false?

Every parallelogram has opposite sides of equal length. A rhombus is a parallelogram, so every rhombus has opposite sides of equal length.

A. True

B. False

All parallelograms have two pairs of parallel sides.

Which shape must also have two pairs of parallel sides?

- A.** Kite
- B.** Pentagon
- C.** Triangle
- D.** Rhombus

A rhombus is a quadrilateral with 4 sides of equal length.

Is every square a rhombus? Explain.

Answer: _____

Explanation: _____

True or false?

All quadrilaterals have 4 right angles.

A. True

B. False

Which statement is true?

- A.** Every parallelogram is a square.
- B.** Every square is a rectangle.
- C.** Every quadrilateral is a rectangle.
- D.** Every rectangle is a square.

SET 55 • 5.G.B.3

BACKWARD 1

Quadrilaterals and their attributes • Grade 3 • Geometry • 3.G.A.1

How many sides does every quadrilateral have?

Sides: _____

True or false?

Triangles, rectangles, and squares are all quadrilaterals.

A. True

B. False

Which shape is NOT a quadrilateral?

- A.** Rectangle
- B.** Square
- C.** Rhombus
- D.** Triangle

True or false?

A square has 4 sides of equal length.

A. True

B. False

Name a quadrilateral that always has 4 right angles.

Shape: _____

SET 55 • 5.G.B.3

BACKWARD 2

Parallel lines, perpendicular lines, and angles • Grade 4 • Geometry • 4.G.A.1

True or false?

Parallel lines are always the same distance apart.

- A. True
- B. False

Which angle measure is a right angle?

- A.** 90°
- B.** 180°
- C.** 45°
- D.** 120°

What do we call two lines that meet to form right angles?

Answer: _____

True or false?

An obtuse angle is smaller than a right angle.

A. True

B. False

How many pairs of parallel sides does a rectangle have?

A. 4

B. 0

C. 2

D. 1

SET 55 • 5.G.B.3

BACKWARD 3

Classifying shapes by sides and angles • Grade 4 • Geometry • 4.G.A.2

True or false?

A shape with no parallel sides can still be a quadrilateral.

A. True

B. False

Which kind of triangle has a right angle?

- A.** An equilateral triangle
- B.** An acute triangle
- C.** A right triangle
- D.** An obtuse triangle

How many right angles does a square have?

Right angles: _____

True or false?

A rhombus always has 4 right angles.

A. True

B. False

Which shape always has two pairs of parallel sides?

A. Rectangle

B. Kite

C. Triangle

D. Pentagon

SET 55 • 5.G.B.3

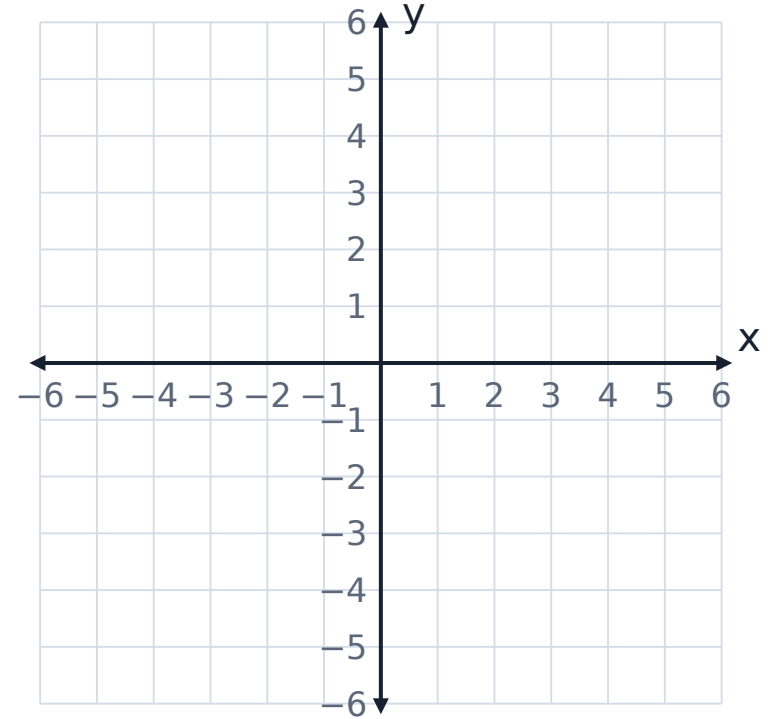
FORWARD 1

Draw polygons in the coordinate plane • Grade 6 • Geometry • 6.G.A.3

Plot $A(-2, 1)$, $B(3, 1)$, $C(3, -3)$, and $D(-2, -3)$. Connect them in order.

What is the most specific name for the polygon?

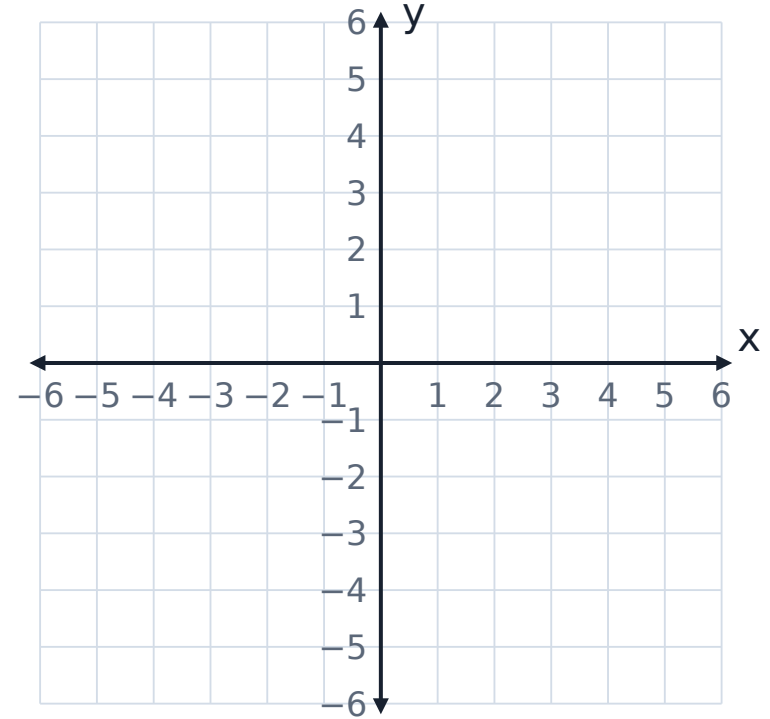
Most specific name: _____



Plot $E(0, 3)$, $F(3, 0)$, $G(0, -3)$, and $H(-3, 0)$. Connect them in order.

What is the most specific name for the polygon?

Most specific name: _____



The points $(-1, 2)$, $(4, 2)$, and $(4, -1)$ are three vertices of a rectangle.

What is the fourth vertex?

A. $(1, -1)$

B. $(-1, -1)$

C. $(-1, 4)$

D. $(4, 2)$

What is the length of the side from $(-2, 5)$ to $(4, 5)$?

Length: _____

True or false?

The points $(0, 0)$, $(4, 0)$, $(4, 2)$, and $(0, 2)$ are the vertices of a square.

A. True

B. False

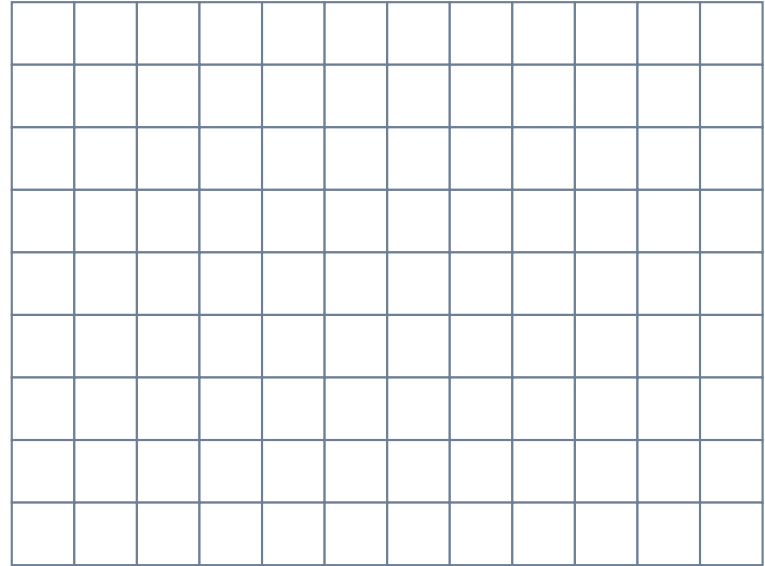
SET 55 • 5.G.B.3

FORWARD 2

Draw shapes with given conditions • Grade 7 • Geometry • 7.G.A.2

Draw a triangle with sides of 3 units, 4 units, and 5 units. Put the 3-unit and 4-unit sides along grid lines.

Classify the triangle by its angles.



Kind of triangle: _____

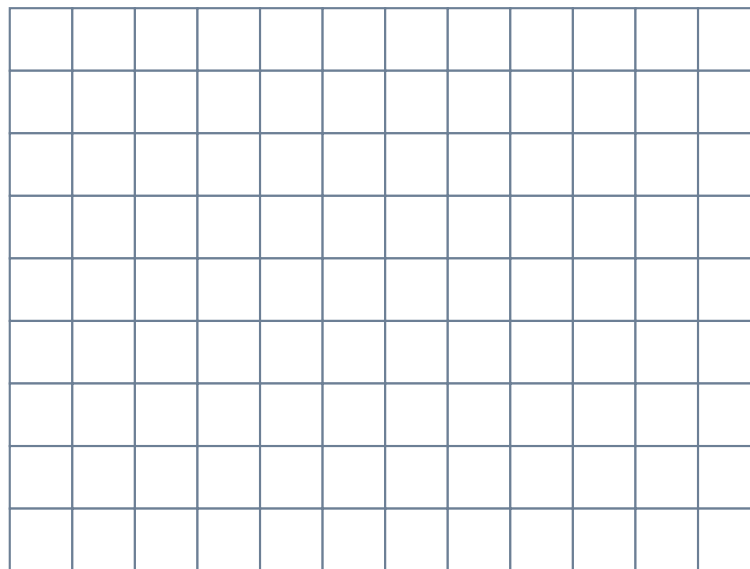
Can a triangle have sides of 2 cm, 2 cm, and 5 cm?
Explain.

Answer: _____

Explanation: _____

Draw a triangle with angles of 90° , 45° , and 45° . Make its two shorter sides 4 units long, along grid lines.

Could a triangle with these same angles be drawn in a different size?



Answer: _____

Can a triangle have angles of 100° , 50° , and 40° ?
Explain.

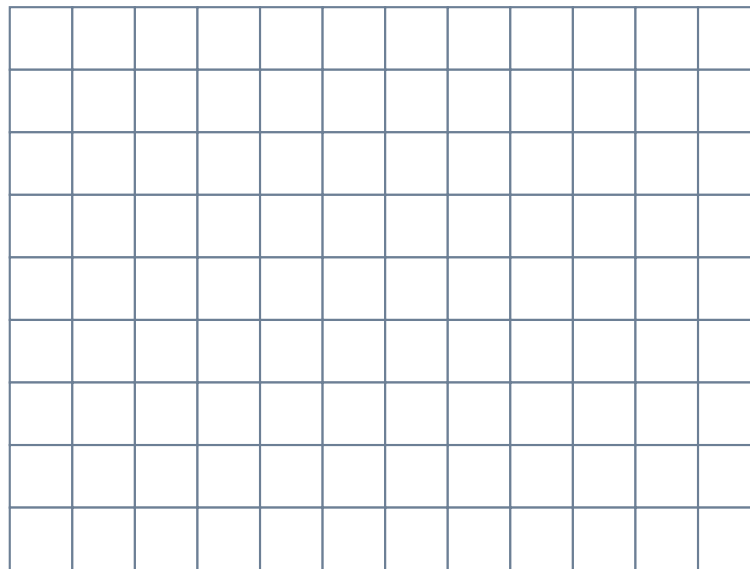
Answer: _____

Explanation: _____

Draw a quadrilateral that has exactly one pair of parallel sides.

What is the name of this kind of quadrilateral?

Name: _____



Classify quadrilaterals in a hierarchy based on their properties

MAIN	Grade 5 main question	5.G.B.4
BACKWARD 1	Quadrilaterals	3.G.A.1
BACKWARD 2	Classifying by parallel sides and right angles	4.G.A.2
FORWARD 1	Area of parallelograms, rhombuses, and trapezoids	6.G.A.1
FORWARD 2	Draw quadrilaterals and triangles with given conditions	7.G.A.2

SET 56 • 5.G.B.4

MAIN

Grade 5 main question • Grade 5 • Geometry • 5.G.B.4

Which list goes from the most general category to the most specific?

- A.** Square, rectangle, parallelogram, quadrilateral
- B.** Rectangle, square, quadrilateral, parallelogram
- C.** Quadrilateral, parallelogram, rectangle, square
- D.** Parallelogram, quadrilateral, square, rectangle

True or false?

A square is a rectangle, a rhombus, and a parallelogram.

A. True

B. False

Which of these categories does every rhombus belong to?

quadrilateral, parallelogram, rectangle, square

Categories: _____

A shape has 4 sides of equal length and 4 right angles.

What is the most specific name for it?

- A.** Rectangle
- B.** Rhombus
- C.** Square
- D.** Quadrilateral

True or false?

Every parallelogram is a rhombus.

A. True

B. False

SET 56 • 5.G.B.4

BACKWARD 1

Quadrilaterals • Grade 3 • Geometry • 3.G.A.1

True or false?

A rectangle is a quadrilateral.

A. True

B. False

Which shape always has 4 sides of equal length?

- A.** Triangle
- B.** Rectangle
- C.** Rhombus
- D.** Kite

How many angles does a quadrilateral have?

Angles: _____

True or false?

Every quadrilateral is a square.

A. True

B. False

Name a shape that is a quadrilateral but is not a parallelogram.

Shape: _____

SET 56 • 5.G.B.4

BACKWARD 2

Classifying by parallel sides and right angles • Grade 4 • Geometry • 4.G.A.2

True or false?

A quadrilateral with 2 pairs of parallel sides is a parallelogram.

A. True

B. False

Which shape always has 4 right angles?

- A.** Rectangle
- B.** Parallelogram
- C.** Kite
- D.** Rhombus

How many pairs of parallel sides does a parallelogram have?

Pairs: _____

True or false?

The sides of a square that meet at a corner are perpendicular.

- A. True
- B. False

Which description fits every rhombus?

- A.** No parallel sides
- B.** 3 sides
- C.** 4 sides of equal length
- D.** 4 right angles

SET 56 • 5.G.B.4

FORWARD 1

Area of parallelograms, rhombuses, and trapezoids • Grade 6 • Geometry •
6.G.A.1

A parallelogram has a base of 10 cm and a height of 6 cm.

What is its area?

Area: _____

A rhombus has a base of 7 in. and a height of 5 in.
What is its area?

Area: _____

A trapezoid has bases of 4 m and 10 m and a height of 6 m. It is cut into a 4-m by 6-m rectangle and two triangles, each with a base of 3 m and a height of 6 m.

What is the area of the trapezoid?

- A.** 84 square meters
- B.** 42 square meters
- C.** 60 square meters
- D.** 24 square meters

A trapezoid has bases of 6 ft and 10 ft and a height of 4 ft.

What is its area?

Area: _____

True or false?

A parallelogram has a greater area than a rectangle with the same base and the same height.

A. True

B. False

SET 56 • 5.G.B.4

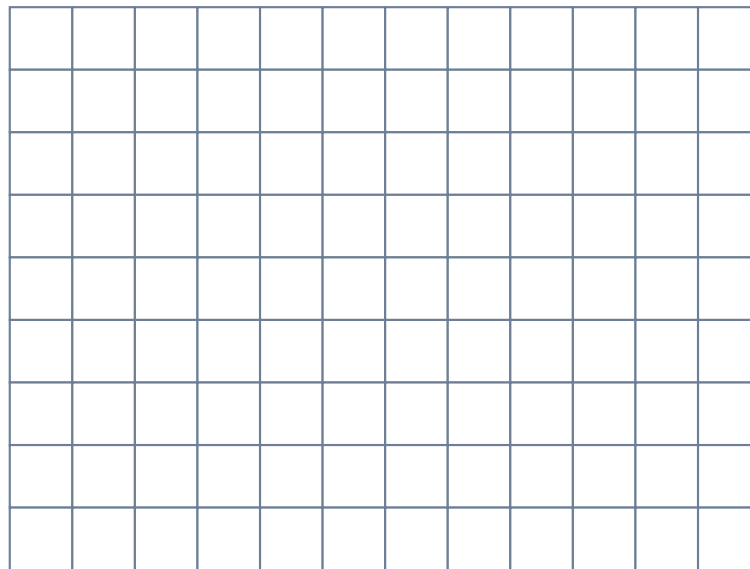
FORWARD 2

Draw quadrilaterals and triangles with given conditions • Grade 7 • Geometry •
7.G.A.2

Draw a rectangle that is not a square.

Write its side lengths.

Side lengths: _____

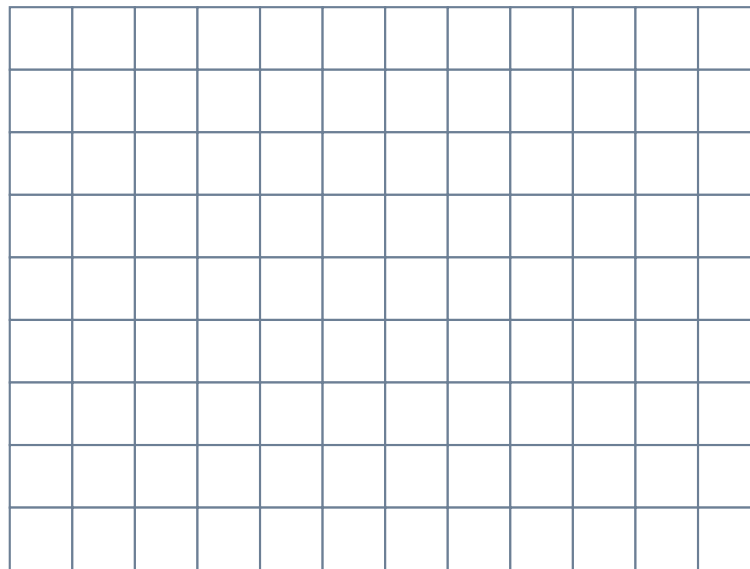


Draw a rhombus that is not a square, with every side 5 units long.

(Hint: a side can go 3 units across and 4 units up.)

How do you know it is not a square?

Reason: _____



Can a quadrilateral have exactly 3 right angles? Explain.

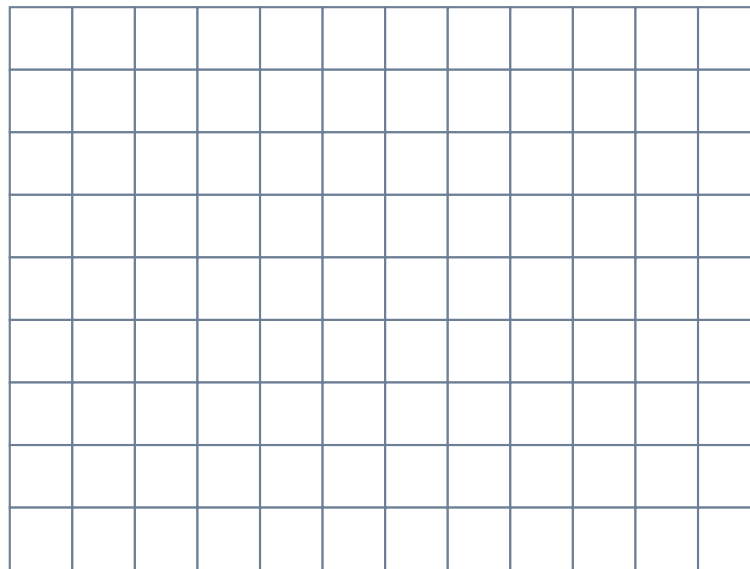
Answer: _____

Explanation: _____

Draw a parallelogram that is not a rectangle, with a base of 6 units and a height of 3 units.

What is its area?

Area: _____



A triangle has sides of 6 cm, 6 cm, and 6 cm. Can triangles of different shapes be drawn with these sides? Explain.

Answer: _____

Explanation: _____

Classify triangles by their sides and angles

MAIN	Grade 5 main question	5.G.B.4
BACKWARD 1	Right, acute, and obtuse triangles	4.G.A.2
BACKWARD 2	Kinds of angles	4.G.A.1
BACKWARD 3	Triangles	2.G.A.1
FORWARD 1	Area of triangles	6.G.A.1
FORWARD 2	Write and solve equations for unknown angles	7.G.B.5

SET 57 • 5.G.B.4

MAIN

Grade 5 main question • Grade 5 • Geometry • 5.G.B.4

A triangle has sides of 6 cm, 6 cm, and 6 cm.

Which name fits it?

- A.** Obtuse
- B.** Scalene
- C.** Equilateral
- D.** Right

A triangle has angles of 90° , 50° , and 40° .
Classify it by its angles.

Classification: _____

True or false?

An isosceles triangle has at least two sides of equal length. Every equilateral triangle is also an isosceles triangle.

- A.** True
- B.** False

A triangle has sides of 3, 5, and 7 units and an angle of 120° .

Which name fits it?

A. Acute equilateral

B. Right scalene

C. Right isosceles

D. Obtuse scalene

A triangle has angles of 70° , 70° , and 40° .
Classify it by its angles and by its sides.

By angles: _____

By sides: _____

SET 57 • 5.G.B.4

BACKWARD 1

Right, acute, and obtuse triangles • Grade 4 • Geometry • 4.G.A.2

True or false?

A right triangle has exactly one right angle.

A. True

B. False

Which angle measure is obtuse?

- A.** 10°
- B.** 45°
- C.** 130°
- D.** 90°

A triangle has one angle greater than 90° .

What kind of triangle is it?

Kind: _____

True or false?

In an acute triangle, all three angles are less than 90° .

A. True

B. False

Which angle measure is acute?

A. 100°

B. 180°

C. 60°

D. 90°

SET 57 • 5.G.B.4

BACKWARD 2

Kinds of angles • Grade 4 • Geometry • 4.G.A.1

True or false?

A right angle measures 90° .

A. True

B. False

What kind of angle measures 35° ?

- A.** Right
- B.** Straight
- C.** Obtuse
- D.** Acute

What kind of angle measures 150° ?

Kind: _____

True or false?

Two right angles together make 360° .

A. True

B. False

What kind of angle measures exactly 90° ?

Kind: _____

SET 57 • 5.G.B.4

BACKWARD 3

Triangles • Grade 2 • Geometry • 2.G.A.1

How many sides does a triangle have?

Sides: _____

True or false?

A triangle has 3 angles.

A. True

B. False

Which shape has 3 sides?

- A.** Square
- B.** Hexagon
- C.** Pentagon
- D.** Triangle

True or false?

All triangles have sides of equal length.

A. True

B. False

How many vertices (corners) does a triangle have?

Vertices: _____

SET 57 • 5.G.B.4

FORWARD 1

Area of triangles • Grade 6 • Geometry • 6.G.A.1

A right triangle has legs that are 6 cm and 8 cm long.
What is its area?

Area: _____

A triangle has a base of 10 in. and a height of 7 in.
What is its area?

Area: _____

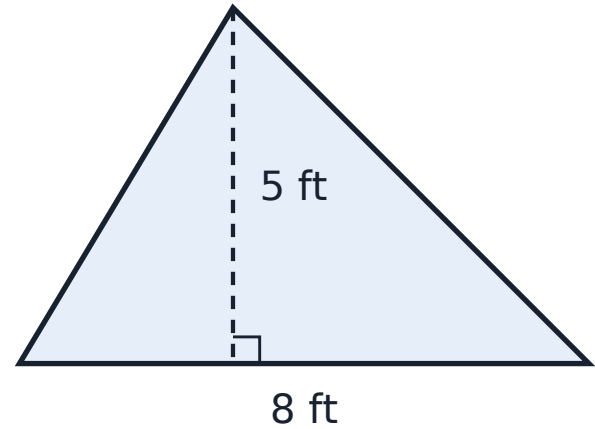
An obtuse triangle has a base of 5 m. Its height, measured outside the triangle, is 4 m.

What is its area?

- A.** 20 square meters
- B.** 40 square meters
- C.** 10 square meters
- D.** 9 square meters

Find the area of the triangle.

Area: _____



True or false?

Two triangles with the same base and the same height have the same area.

A. True

B. False

SET 57 • 5.G.B.4

FORWARD 2

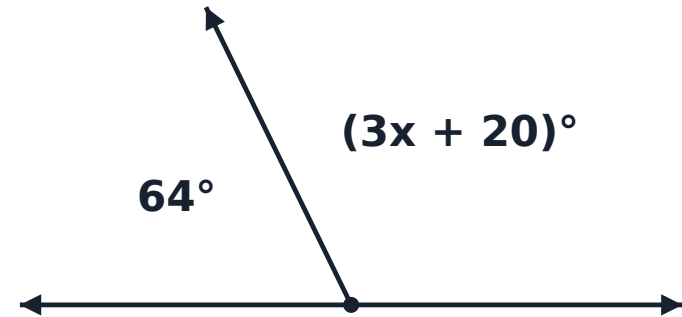
Write and solve equations for unknown angles • Grade 7 • Geometry • 7.G.B.5

The two angles form a straight line.

Write an equation and solve for x .

Equation: _____

$x =$ _____

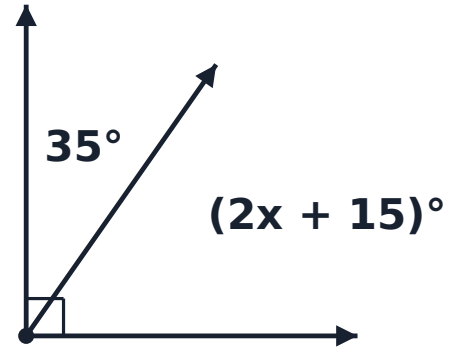


The two angles together form a right angle.

Write an equation and solve for x .

Equation: _____

$x =$ _____

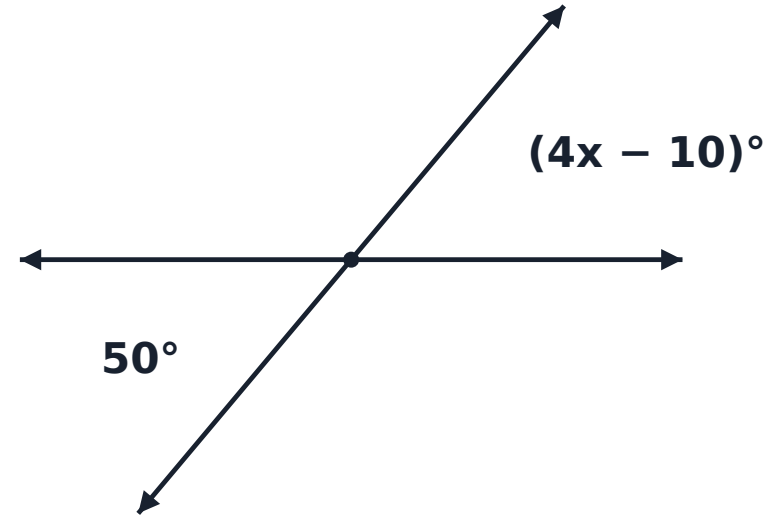


Two lines cross. The marked angles are vertical angles.

Write an equation and solve for x .

Equation: _____

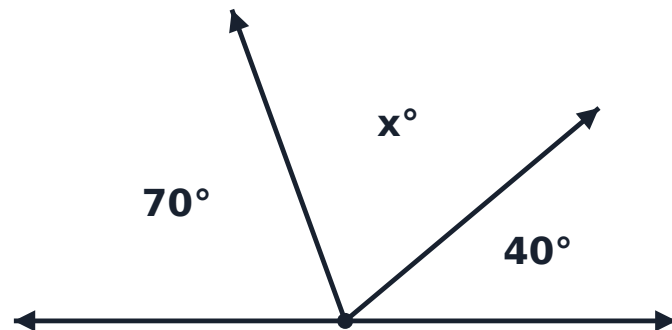
$x =$ _____



Three angles together form a straight line.

Which equation can be used to find x ?

- A.** $x = 40 \times 70$
- B.** $40 + x + 70 = 180$
- C.** $40 + x = 70$
- D.** $40 + x + 70 = 90$



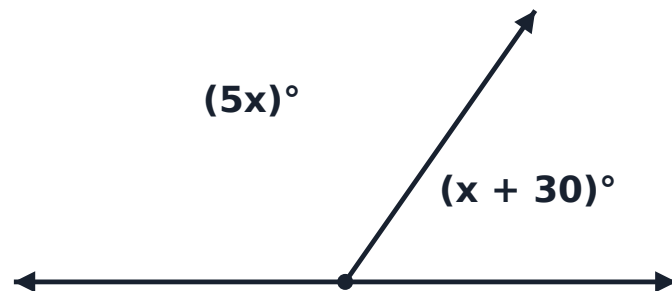
Angle ABC and angle CBD together form a straight line. Angle ABC is $(5x)^\circ$ and angle CBD is $(x + 30)^\circ$.

Write an equation, solve for x , and find both angles.

Equation: _____

$x =$ _____

Angles: _____



Place triangles in a hierarchy of categories and subcategories

MAIN	Grade 5 main question	5.G.B.4
BACKWARD 1	Classify triangles by their angles	4.G.A.2
BACKWARD 2	Shapes in different categories share attributes	3.G.A.1
FORWARD 1	Area of special triangles	6.G.A.1
FORWARD 2	Decide whether conditions give one triangle, more than one, or no...	7.G.A.2

SET 71 • 5.G.B.4

MAIN

Grade 5 main question • Grade 5 • Geometry • 5.G.B.4

Which list goes from the most general category to the most specific?

- A.** Triangle, isosceles triangle, equilateral triangle
- B.** Isosceles triangle, triangle, equilateral triangle
- C.** Triangle, equilateral triangle, isosceles triangle
- D.** Equilateral triangle, isosceles triangle, triangle

True or false?

Every right isosceles triangle belongs to both the category of right triangles and the category of isosceles triangles.

A. True

B. False

Isosceles means at least two sides of equal length.

A triangle has sides of 7, 7, and 7 units. Which of these categories does it belong to?

triangle, isosceles, equilateral, scalene

Categories: _____

Isosceles means at least two sides of equal length.

Which statement is true?

- A.** Some isosceles triangles are equilateral.
- B.** No right triangle is isosceles.
- C.** Every isosceles triangle is equilateral.
- D.** Some scalene triangles are isosceles.

Explain why a triangle cannot be both a right triangle and an obtuse triangle.

Explanation: _____

SET 71 • 5.G.B.4

BACKWARD 1

Classify triangles by their angles • Grade 4 • Geometry • 4.G.A.2

True or false?

A triangle with angles of 30° , 60° , and 90° is a right triangle.

A. True

B. False

A triangle has angles of 110° , 40° , and 30° .

What kind of triangle is it?

- A.** Equilateral
- B.** Acute
- C.** Right
- D.** Obtuse

A triangle has angles of 50° , 60° , and 70° .
Is it acute, right, or obtuse?

Answer: _____

True or false?

A triangle can have two obtuse angles.

A. True

B. False

Which three angles could be the angles of an acute triangle?

- A.** 100° , 50° , 30°
- B.** 120° , 30° , 30°
- C.** 80° , 60° , 40°
- D.** 90° , 45° , 45°

SET 71 • 5.G.B.4

BACKWARD 2

Shapes in different categories share attributes • Grade 3 • Geometry • 3.G.A.1

True or false?

Rhombuses, rectangles, and squares are all quadrilaterals.

A. True

B. False

What do all quadrilaterals have in common?

- A.** 4 right angles
- B.** 4 sides
- C.** 2 pairs of parallel sides
- D.** 4 sides of equal length

Name a quadrilateral that is not a rectangle.

Shape: _____

True or false?

Every quadrilateral is a rhombus.

A. True

B. False

A square is also a ____.

- A.** rectangle
- B.** pentagon
- C.** hexagon
- D.** triangle

SET 71 • 5.G.B.4

FORWARD 1

Area of special triangles • Grade 6 • Geometry • 6.G.A.1

A right isosceles triangle has legs that are 6 cm long.
What is its area?

Area: _____

True or false?

A triangle with a base of 4 in. and a height of $3\frac{1}{2}$ in. has an area of 7 square inches.

- A.** True
- B.** False

A square with sides of 8 m is cut along a diagonal into two triangles.

What is the area of each triangle?

- A.** 64 square meters
- B.** 8 square meters
- C.** 32 square meters
- D.** 16 square meters

An isosceles triangle has a base of 10 ft and a height of 12 ft.

What is its area?

Area: _____

True or false?

A right triangle with legs of 5 units and 9 units has an area of 45 square units.

A. True

B. False

SET 71 • 5.G.B.4

FORWARD 2

Decide whether conditions give one triangle, more than one, or none • Grade 7
• Geometry • 7.G.A.2

How many differently shaped triangles can be drawn with sides of 5 cm, 5 cm, and 8 cm?

- A.** Exactly two
- B.** More than one
- C.** Exactly one
- D.** None

True or false?

Triangles of many different sizes can be drawn with angles of 60° , 60° , and 60° .

- A. True
- B. False

Can a triangle have sides of 3 m, 4 m, and 8 m? Explain.

Answer: _____

Explanation: _____

True or false?

A triangle can have angles of 90° , 60° , and 40° .

A. True

B. False

Can a right triangle also be isosceles? If it can, give its angle measures.

Answer: _____

Angles: _____